Facilitator's Book

Teaching reference for the 40-hour CDF intensive

SpaceCDF -- AI-Assisted Concurrent Design Facility
University of Ottawa - SEDTI
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DRAFT

Facilitator's Book

Session 1.1: Introduction to Space Mission Design

Duration: 3 hours (Monday AM)

Prerequisites: None (engineering background assumed)

References:

- NASA, Systems Engineering Handbook Rev 2 (SP-2016-6105), 2016
- NASA, NPR 7123.1D -- Systems Engineering Processes and Requirements, 2020
- ECSS, ECSS-M-ST-10C Rev.1 -- Space Project Management, 2009
- ESA, "20 Years of Concurrent Design at ESA", 2018

Learning Objectives

By the end of this session, participants will be able to:

1. Describe the purpose and structure of a Concurrent Design Facility (CDF) and contrast it with sequential design
2. Explain the System-V model and its two sides (decomposition + integration)
3. List the 17 common technical processes defined in NPR 7123.1D and explain their recursive application
4. Map NASA lifecycle phases (Pre-A through F) to ECSS phases and identify key review gates
5. Identify the role of each CDF engineering position and their parameter ownership

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1. What is a Concurrent Design Facility? (30 min)

1.1 The Problem with Sequential Design

Begin by asking the group: "How is spacecraft design traditionally done?"

Traditional spacecraft design follows a sequential (waterfall) approach: one discipline completes its work, documents it, and passes the design to the next discipline. This method dominated the space industry from the 1960s through the 1990s. Its limitations are well documented:

- Interface mismatches discovered late in integration, requiring costly redesign
- Budget overruns in mass, power, and cost not caught until formal reviews
- Long iteration cycles -- months between design reviews with limited inter-discipline feedback
- Knowledge silos -- critical design rationale locked in individual engineers' notes
- Rework costs -- the cost of correcting an error increases by roughly an order of magnitude per lifecycle phase (the "1-10-100 rule")

The aerospace industry's response was concurrent engineering (CE), also called Integrated Product Development (IPD) or Integrated Concurrent Engineering (ICE). The core idea: bring all disciplines together to work simultaneously on a shared parametric model, resolving conflicts in real-time rather than discovering them in review.

> *Industry Practice: Boeing's Phantom Works implemented concurrent engineering for the X-32 Joint Strike Fighter demonstrator in the late 1990s, reducing design iteration time from weeks to hours. JPL's Team X has been performing rapid mission assessments since 1995, completing Pre-Phase A studies in as little as one week. ESA's CDF, established in November 1998 at ESTEC (Noordwijk, Netherlands), became the most widely replicated model for space-specific concurrent design.*

1.2 ESA's CDF -- The Reference Model

ESA's Concurrent Design Facility pioneered the approach for space missions and has been widely replicated at space agencies worldwide (DLR, CNES, JAXA, CSA, and over 60 universities).

Parameter	Value
Established	November 1998, ESTEC, Noordwijk, NL
Team size	15--25 domain specialists per study
Session format	3--4 hour focused design sessions, typically 2 per week
Study duration	3--6 weeks for a complete Phase 0/A mission assessment
Tooling	Originally Excel-based IDM; now Open Concurrent Design Tool (OCDT)
Output	Complete mission feasibility assessment: mass, power, cost, risk, schedule
Track record	200+ studies completed by 20th anniversary (2018)
Cost savings	Studies completed in ~1/5 the time and ~1/3 the cost of traditional approach

[Source: ESA CDF official documentation; Bandecchi et al., "The ESA/ESTEC Concurrent Design Facility", ESA Bulletin No. 107, 2001]

1.3 Key Principles of Concurrent Design

Shared parametric model. All disciplines read from and write to a common data model. When the power engineer changes the solar array area, the structures engineer immediately sees the mass impact, and the thermal engineer sees the changed radiator view factor. This eliminates the "frozen interface document" problem.

Scoped parameter ownership. Each engineering position "owns" a defined set of parameters. Only the power engineer can modify EPS parameters; only the AOCS engineer can modify attitude control parameters. This prevents conflicting edits while maintaining concurrent access.

Real-time conflict detection. When parameter changes create conflicts (e.g., mass budget exceeded, link margin negative), the system flags them immediately. The systems engineer arbitrates.

Design convergence through iteration. A CDF study typically converges through 3--5 major iterations, with each session refining the design toward closure. The parametric model propagates changes automatically, so the team sees the system-level impact of every decision.

1.4 SpaceCDF -- A Web-Based CDF

SpaceCDF implements the CDF concept as a web-based tool accessible from any browser:

Feature	Implementation
Shared model	Real-time synchronisation via WebSocket; all participants see live updates
Parameter ownership	15 engineering positions with scoped editing rights
Design convergence	20 automated design agents that propagate parametric relationships
Conflict detection	Automated constraint engine with 187 inter-parameter connections
Document generation	ECSS-compliant exports (Word, PDF) with full traceability
Review gates	Automated gate criteria evaluation for MCR, SRR, PDR, CDR

Discussion prompt: What advantages does concurrent design offer over sequential? What risks does it introduce (e.g., groupthink, premature convergence)?

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2. The System-V Model (40 min)

2.1 Origin and Purpose

The "Vee" (V) model is the foundational framework for systems engineering. It appears in NASA SEH Section 2.3 (Figure 2.3-1), in ECSS-E-ST-10C, and in ISO/IEC 15288. It is not a project schedule -- it is a logical model showing the relationship between decomposition (breaking the problem down) and integration (building and verifying the solution).

The V-model answers two fundamental questions:

1. Left side (top-down): How do we decompose the mission need into buildable components?
2. Right side (bottom-up): How do we verify that what we built satisfies the original need?

The key insight is horizontal traceability: every element on the left side has a corresponding verification or validation activity on the right side, connected at the same level of abstraction.

2.2 V-Model Diagram

Figure: System-V Model (after NASA SEH Figure 2.3-1)

(See digital/markdown version for interactive diagram)

2.3 Left Side: Top-Down Decomposition

The left side of the V answers "how do we break the problem down into solvable pieces?"

Level	Activity	Key Question	Output
1. Mission Need	Define the problem and concept of operations	What problem are we solving? For whom?	ConOps, problem statement, MoEs
2. Stakeholder Requirements	Capture needs of all stakeholders	What must the system achieve? (WHAT, not HOW)	Stakeholder requirements baseline
3. System Architecture	Define segments, elements, interfaces	How is the system structured?	System architecture, ICDs
4. Subsystem Design	Design each subsystem to meet allocated requirements	How does each subsystem work?	Subsystem specifications
5. Component Selection	Choose or design lowest-level components	Which hardware/software fulfils each need?	Component specifications, BoM

2.4 Right Side: Bottom-Up Integration

The right side answers "how do we verify that what we built satisfies the original need?"

Level	Activity	Key Question	Verification Method
5. Component Verification	Test each component against its spec	Does each part work?	Inspection, test
4. Subsystem Verification	Integrate and test each subsystem	Do subsystems meet allocated requirements?	Test, analysis
3. Integration & Test	Assemble the system and test interfaces	Do the subsystems work together?	Test, demonstration
2. System Verification	Verify system against requirements	Does the system meet all requirements?	Test, analysis, inspection, demonstration
1. Mission Validation	Validate against the original need	Does it solve the original problem?	Operations, user acceptance

2.5 Horizontal Traceability

The orange dashed arrows in the diagram represent horizontal traceability -- the principle that every element on the left must have a corresponding verification/validation activity on the right:

- Each requirement (left) has a verification method (right): test, analysis, inspection, or demonstration (TAID)
- Each design decision (left) has a test or analysis (right) to confirm it works

- This traceability is captured in the Requirements Verification Matrix (RVM)

> *Key Equation: The traceability completeness metric is:*
>
> *$T_c = (N_{verified})/(N_{total\_requirements}) \times 100\%$*
>
> *A target of $T_c = 100\%$ is required at CDR. At PDR, $T_c >= 90\%$ is typical.*

2.6 Iteration and the SE Engine

The V is not a single pass. Real design iterates -- requirements change as design constraints are discovered, and design parameters feed back to refine requirements. This iterative loop is captured in the concept of the "SE Engine" (NASA SEH Section 2.1): the 17 common technical processes applied recursively at every level of the system hierarchy.

[Source: NASA SEH Section 2.3, Figure 2.3-1; INCOSE Systems Engineering Handbook, 4th Edition, Section 3.2]

Exercise: On a whiteboard, draw the V-model from memory and label each level. Add the horizontal traceability arrows and name one verification method for each level.

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3. The 17 Common Technical Processes (30 min)

3.1 Overview

NASA's NPR 7123.1D defines 17 processes that apply recursively at every level of the system hierarchy. They are grouped into three categories, collectively called the "SE Engine". These processes are not sequential -- they execute concurrently and iteratively, which is precisely what a CDF facilitates.

[Source: NPR 7123.1D Chapter 3; NASA SEH Section 2.1, Figure 2.1-1]

3.2 System Design Processes (1--4)

These processes decompose the problem into a solution. They correspond to the left side of the V-model.

#	Process	Key Activity	Key Output	NASA SEH Reference
1	Stakeholder Expectations Definition	Elicit needs, define ConOps, establish MoEs	Stakeholder requirements baseline	Section 4.1
2	Technical Requirements Definition	Write "shall" statements, define MoPs and TPMs	Technical requirements baseline	Section 4.2
3	Logical Decomposition	Functional analysis, behavioural modelling, derived requirements	Functional architecture	Section 4.3

4	Design Solution Definition	Trade studies, select among alternatives, produce baseline design	Design solution baseline	Section 4.4

3.3 Product Realisation Processes (5--9)

These processes build, verify, and deliver the solution. They correspond to the right side of the V-model.

#	Process	Key Activity	Key Output	NASA SEH Reference
5	Product Implementation	Make, buy, code, or reuse lowest-level products	Hardware/software products	Section 5.1
6	Product Integration	Assemble per integration plan, verify interfaces	Integrated system	Section 5.2
7	Product Verification	Confirm product meets technical requirements (TAID)	Verification evidence	Section 5.3
8	Product Validation	Confirm product meets stakeholder expectations	Validation evidence	Section 5.4
9	Product Transition	Deliver, deploy, hand over to operations	Operational system	Section 5.5

3.4 Technical Management Processes (10--17)

These processes manage the engineering work and apply throughout the lifecycle.

#	Process	Key Activity	Governing Document
10	Technical Planning	SEMP, subsidiary plans, WBS	NPR 7123.1D Section 3.10
11	Requirements Management	Baselining, traceability, change control	NPR 7123.1D Section 3.11
12	Interface Management	ICDs, IRDs, internal + external interfaces	NPR 7123.1D Section 3.12
13	Technical Risk Management	Identification, assessment, mitigation per NPR 8000.4	NPR 8000.4B
14	Configuration Management	Baselines, CM plan, CCBs	NPR 7120.5F Chapter 4
15	Technical Data Management	Data rights, retention, dissemination	NPR 7123.1D Section 3.15
16	Technical Assessment	TPMs, reviews, EVM, health checks	NPR 7123.1D Section 3.16
17	Decision Analysis	Structured alternative selection (trade studies)	NASA SEH Section 6.8

3.5 Recursion Across the System Hierarchy

A critical property of the 17 processes is that they apply recursively at every level of the system hierarchy:

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Mission Level:     17 processes applied (mission-level requirements, ConOps, validation)
  System Level:    17 processes applied (system requirements, architecture, verification)
    Subsystem Level: 17 processes applied (subsystem requirements, design, testing)
      Component Level: 17 processes applied (component specs, procurement, acceptance)
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At each level, the same processes execute but with different scope, detail, and duration. In a CDF environment, multiple levels are often worked concurrently -- the systems engineer manages the mission and system levels while subsystem engineers work their respective domains.

> *Industry Practice: On the James Webb Space Telescope (JWST), requirements management (Process 11) was applied at five levels of the system hierarchy simultaneously: Observatory, Optical Telescope Element, Integrated Science Instrument Module, individual instruments (NIRSpec, MIRI, NIRCam, FGS), and components (detectors, mechanisms). Each level had its own requirements baseline, traceability matrix, and verification plan. The total requirements count exceeded 10,000.*

Exercise: Map each of the 17 processes to a feature in SpaceCDF. Which processes does the tool support directly? Which require human judgment? Complete Part A of Worksheet 1.1.

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4. Lifecycle Phases and Review Gates (30 min)

4.1 NASA Lifecycle Phases

NASA defines seven lifecycle phases for space flight projects, governed by NPR 7120.5F. Each phase has specific objectives, activities, and exit criteria evaluated at formal review gates called Key Decision Points (KDPs).

[Source: NPR 7120.5F Chapter 2; NASA SEH Chapter 3]

4.2 Lifecycle Phase Diagram

Figure: NASA Space Flight Project Lifecycle (NPR 7120.5F)

(See digital/markdown version for interactive diagram)

4.3 Phase Details

Phase	Name	Primary Activity	Exit Review	Key Outputs
Pre-A	Concept Studies	Identify need, explore feasibility, develop ConOps	MCR	Mission concept, preliminary ConOps, feasibility assessment
A	Concept & Technology Development	Develop requirements, mature technology to TRL 4+	SRR/SDR	Requirements baseline, technology development plan
B	Preliminary Design & Tech Completion	Preliminary design, close preliminary budgets, TRL 6+	PDR	Preliminary design, budget allocations, test plans
C	Final Design & Fabrication	Detailed design, build hardware, TRL 8	CDR	Detailed drawings, as-built documentation, flight hardware
D	Assembly, Integration & Test, Launch	System AIT, environmental testing, launch	TRR, ORR,	Tested flight system, launch readiness
E	Operations & Sustainment	Operate the mission, produce data products	PLAR	Science data, operational lessons
F	Closeout	Decommission, passivate, dispose, lessons learned	DR	Disposal confirmation, final report

Key Decision Points (KDPs): These are go/no-go authority decisions between phases, lettered A through F.

They are distinct from technical reviews -- KDPs are management decisions informed by technical review outcomes. For projects exceeding $250M (USD), KDP-C requires a Joint Confidence Level (JCL) analysis per NPR 7120.5F.

4.4 ECSS Lifecycle Phases

ECSS defines a similar but not identical lifecycle. The phase letters conveniently align, but the entry/exit criteria and review content differ.

[Source: ECSS-M-ST-10C Rev.1, Section 5.3]

ECSS Phase	Name	NASA Equivalent	Key Difference
0	Mission Analysis / Needs Identification	Pre-A	ECSS Phase 0 includes formal mission analysis
A	Feasibility	A	ECSS-A focuses on demonstrating feasibility
B	Preliminary Definition (B1/B2)	B	ECSS splits into B1 (system) and B2 (detailed)
C	Detailed Definition	C	Similar scope
D	Qualification & Production	D	ECSS-D emphasises qualification models
E	Utilisation	E	Similar scope
F	Disposal	F	ECSS-F explicitly addresses passivation

4.5 Review Gates -- What Each Checks

Review	Full Name	Key Question	Required Evidence
MCR	Mission Concept	Is the mission need justified? Is space the right answer?	Problem statement, stakeholder analysis, alternatives analysis, ConOps draft
SRR	System Requirements	Are requirements complete, consistent, traceable, and verifiable?	Requirements baseline, traceability matrix, risk register, preliminary ConOps
SDR	System Design Review	Does the system architecture satisfy requirements?	System architecture, functional decomposition, interface definitions
PDR	Preliminary Design Review	Does the preliminary design meet all requirements?	Design description, budget status (mass/power/cost), risk mitigation plans
CDR	Critical Design Review	Is the design complete and ready for fabrication?	Detailed drawings, analysis results, test plans, all budgets closed with positive margins
TRR	Test Readiness Review	Is the system ready for formal testing?	Test procedures, facility readiness, acceptance criteria
FRR	Flight Readiness Review	Is everything ready for launch?	All tests passed, waivers documented, launch procedures verified

> *Industry Practice: RADARSAT-2 (MDA/CSA, launched 2007) went through all NASA-equivalent review gates over a 6-year development. The CDR alone involved over 400 review items and 50 reviewers across 3 weeks. The mission exceeded its 7-year design life, operating for over 15 years -- a testament to thorough verification at each gate.*

Exercise: In SpaceCDF, go to the Gate Review tab and examine the MCR exit criteria. Which criteria are auto-evaluated by the tool? Which require manual review by the facilitator?

5. CDF Engineering Positions (20 min)

5.1 Position Overview

In a CDF study, each engineering position owns a set of parameters and is responsible for their domain's design decisions. SpaceCDF supports 15 positions:

Position	Responsibility	Key Owned Parameters	Primary Interfaces
Systems Engineer	Overall architecture, budgets, margins, conflict resolution	Mass margin, power margin, system-level budgets	All positions
Mission Analyst	Orbit design, coverage analysis, ground station access	Altitude, inclination, RAAN, eclipse fraction, contact time	Comms, Payload, Ground
Payload Lead	Instrument performance, data generation rates	GSD, spectral bands, data rate, FOV, SNR	Mission Analyst, Power, AOCS, Comms
Power Engineer	Solar arrays, batteries, EPS architecture, duty cycling	SA area, battery capacity, bus voltage, DoD	All (power is a universal constraint)
AOCS Engineer	Attitude sensors, actuators, pointing modes	Pointing accuracy, slew rate, wheel momentum, sensor suite	Payload, Structures, Propulsion
Thermal Engineer	Temperature control, radiators, heaters, coatings	Max/min temps, radiator area, heater power, thermal margin	Structures, Power, Payload
Comms Engineer	Link budget, transponder, antenna, spectrum licensing	Link margin, data rate, frequency band, EIRP, G/T	Mission Analyst, Payload, Ground
Propulsion Engineer	Delta-V budget, thruster selection, propellant mass	Isp, propellant mass, total impulse, thrust vector	Mission Analyst, AOCS, Structures
Structures Engineer	Primary structure, mechanisms, launch loads	Structure mass, natural frequency, margin of safety	All (mass is a universal constraint)
Cost Engineer	WBS, cost estimating relationships, schedule	Total cost, per-subsystem cost, launch cost, risk-adjusted cost	Systems Engineer, all subsystems
Compliance Engineer	Standards, frequency licensing, export control	Standard applicability, ITAR/EAR status, filing status	All
User Representa	End-user needs, data product requirements	Data format, latency, accessibility, coverage	Payload, Comms, Ground
Mission Operations	Ground ops concept, staffing, automation	Contact schedule, automation level, anomaly procedures	Comms, Ground, Systems
Ground Segment	Ground stations, data processing pipeline	Station network, processing latency, archive capacity	Comms, Mission Ops, User Rep
Software Engineer	Flight software, FDIR, TC/TM interfaces	FSW architecture, command dictionary, FDIR rules	OBC, AOCS, Comms, Payload

5.2 Interface Conflicts

The most productive CDF sessions are those where interface conflicts are identified and resolved. Common conflict patterns:

Conflict	Positions Involved	Resolution Approach
Mass budget exceeded	Systems, Structures, all subsystems	Re-allocate margins, descope payload, select lighter components
Power budget exceeded	Systems, Power, Payload, Comms	Adjust duty cycles, increase SA area, reduce payload power
Pointing requirement vs actuator capability	Payload, AOCS, Structures	Relax pointing requirement, add fine steering mirror, improve isolation
Downlink capacity vs data generation	Comms, Payload, Ground	Add ground stations, increase TX power, reduce data rate, add compression

Volume constraint vs thermal rejection	Structures, Thermal, Payload	Redesign radiator location, add deployable radiator, reduce heat dissipation

Discussion prompt: Which positions would interact most frequently in your mission? Where do you expect the most difficult trade-offs?

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6. Session Summary

Topic	Key Takeaway
CDF	Concurrent multi-discipline design resolves conflicts in real-time; SpaceCDF implements this as a web-based tool
System-V	Left side decomposes (need to components); right side integrates (build to validate); horizontal traceability links them
17	Recursive at every system level; grouped as Design (1--4), Realisation (5--9), Management (10--17)
Lifecycle Phases	Pre-A through F with KDP gates; ECSS phases approximately align; SpaceCDF covers Pre-A through C
Review	Each gate answers a specific question with required evidence; auto-evaluated where possible in SpaceCDF
Positions	Each owns a parameter domain; conflicts arise at interfaces; the systems engineer arbitrates

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1U Worked Example: UniSat-1

Throughout this course, we use a second running example alongside the 3U EO CubeSat: UniSat-1, a 1U CubeSat technology demonstrator designed by a university team. This is the simplest realistic spacecraft design.

Mission: Demonstrate a novel MEMS-based magnetometer for space weather monitoring from LEO.

Why 1U? The 1U form factor (100 x 100 x 113.5 mm, up to 1.33 kg) is the smallest standard CubeSat and the entry point for many university and educational missions. It forces extreme design discipline -- every gram, every milliwatt, and every cubic centimetre matters.

Parameter	Value
Form factor	1U (100 x 100 x 113.5 mm)
Mass limit	1.33 kg (CDS Rev 14)
Target mass	1.0 kg
Orbit	400 km circular, 51.6 deg (ISS rideshare)
Design lifetime	6 months
Payload	MEMS magnetometer (50 g, 0.2 W, < 1 kbps)
Comms	UHF 437 MHz, 9600 bps
Power	~2 W orbit average (body-mounted solar cells)
AOCS	Passive magnetic (permanent magnet + hysteresis rods)
Propulsion	None (natural deorbit in ~1 year)
Estimated cost	50--200 kEUR
Development time	6--12 months

UniSat-1 illustrates that a meaningful space mission can be accomplished with just five subsystems (EPS, OBC, Comms, Structure, Payload), no active attitude control, no propulsion, and no thermal hardware beyond surface coatings. As the course progresses, each session will show how the same design processes apply to UniSat-1, but with radically simpler solutions at every step.

Discussion prompt: How does the CDF process differ when the team has only 5 subsystems instead of 8--9? Which engineering positions are still needed, and which can be combined?

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7. Tool Exercise (15 min)

1. Open SpaceCDF and navigate through the workflow steps (Need -> Concept -> Requirements -> Design)
2. On the Design Dashboard, identify which KPI cards correspond to which engineering positions
3. Go to the Positions tab and review the key questions for your assigned position
4. Go to the Gate Review tab and examine the MCR exit criteria

Complete Worksheet 1.1: Map the 17 processes to SpaceCDF features and identify lifecycle phases for given activities.

References

1. NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016
2. NASA, NPR 7123.1D -- Systems Engineering Processes and Requirements, 2020
3. NASA, NPR 7120.5F -- NASA Space Flight Program and Project Management Requirements, 2021
4. ECSS, ECSS-M-ST-10C Rev.1 -- Space Project Management, 2009
5. INCOSE, Systems Engineering Handbook, 4th Edition, 2015
6. Bandecchi, M. et al., "The ESA/ESTEC Concurrent Design Facility", ESA Bulletin No. 107, 2001
7. Wertz, J.R. et al., Space Mission Engineering: The New SMAD (SMAD4), Microcosm Press, 2011

Session 1.2: The Canadian Space Ecosystem & Regulatory Environment

Duration: 4 hours (Monday PM + Tuesday AM)

Prerequisites: Session 1.1

References:

- Government of Canada, Canadian Space Agency Act (S.C. 1990, c. 13)

- Government of Canada, Remote Sensing Space Systems Act (RSSSA, S.C. 2005, c. 45)
- Government of Canada, Radiocommunication Act (R.S.C., 1985, c. R-2)
- ISED Canada, Spectrum Management and Telecommunications
- Government of Canada, Export and Import Permits Act (EIPA)
- NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016
- ITU, Radio Regulations, Edition 2020

Learning Objectives

By the end of this session, participants will be able to:

1. Describe the Canadian Space Agency's mandate and organisational structure
2. Map the Canadian space industry ecosystem -- prime contractors, SMEs, universities, and funding programs
3. Identify the regulatory bodies that govern space activities in Canada (CSA, ISED, GAC, DND)
4. Explain the RSSSA licensing process for remote sensing satellites
5. Describe the spectrum licensing process through ISED and ITU coordination
6. Identify Canadian export control obligations under EIPA, ITAR, and EAR

Part 1: The Canadian Space Ecosystem (Monday PM -- 2 hours)

1. The Canadian Space Agency (30 min)

1.1 Mandate and History

The Canadian Space Agency (CSA) was established by the Canadian Space Agency Act (S.C. 1990, c. 13) with the mandate to "promote the peaceful use and development of space, to advance the knowledge of space through science and to ensure that space science and technology provide social and economic benefits for Canadians."

Milestone	Year	Significance
Alouette 1	1962	Canada becomes the 3rd nation to build and design a satellite
Hermes (CTS)	1976	World's first high-power direct broadcast satellite (CSA/NASA)
Canadarm	1981	Robotic arm for Space Shuttle; established Canada's space robotics dominance
RADARSAT-1	1995	First Canadian Earth observation satellite; C-band SAR
CSA established	1990	Canadian Space Agency Act creates the agency
Canadarm2	2001	ISS robotic arm; continues robotics legacy
RADARSAT-2	2007	Commercial SAR satellite (MDA); RSSSA applies
RADARSAT Constellation Mission	2019	Three C-band SAR satellites; maritime surveillance, disaster management
Canadarm3	In dev.	For the Lunar Gateway; next-generation autonomous robotics

Lunar Gateway participation

2019+

Canada committed as partner; astronaut flights planned

[Source: CSA, "Canada's Space Milestones", www.asc-csa.gc.ca]

1.2 CSA Organisational Focus Areas

The CSA is headquartered in Saint-Hubert, Quebec, and organises its activities around three strategic priorities:

Priority	Activities	Key Programs
Space Exploration	ISS operations, Lunar Gateway, Canadarm3, astronaut program	Lunar Exploration Accelerator Program (LEAP)
Space Utilisation	Earth observation, satellite communications, GNSS augmentation	RADARSAT, SmartEarth, eHealth
Space Science & Technology	Astronomy, planetary science, technology development	CASTOR, JWST/NIRISS, space health

1.3 Canadian Space Strategy (2019)

The Exploration, Imagination, Innovation: A New Space Strategy for Canada (2019) committed $2.05B over 24 years and established five priorities:

1. Ensure astronaut health and well-being in deep space
2. Develop and use advanced AI and robotics in space
3. Connect rural and remote Canadians through next-generation satellite communications
4. Use space data and technology to manage natural resources sustainably
5. Position the Canadian space sector for the future

> *Industry Practice: The $150M Lunar Exploration Accelerator Program (LEAP) specifically targets Canadian SMEs and universities to develop technologies for lunar surface science, health, AI, and robotics. This "pull-through" model -- where a strategic government investment creates opportunities for industry -- is a hallmark of the Canadian space ecosystem.*

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2. Canadian Space Industry Map (30 min)

2.1 Industry Structure

The Canadian space industry generated approximately $2.7B in revenue (2022) and employs ~12,000 people. It is structured in tiers:

Tier	Role	Key Players	Revenue Share
Tier 1: Primes	System integration, large platforms, constellation operations	MDA Space, Telesat, SFL (U of T)	~60%
Tier 2: Subsystem/Compon	Subsystem design, specialised components, ground systems	Honeywell Canada, ABB, Magellan Aerospace, NGC Aerospace	~25%
Tier 3: SMEs & Startups	Niche technologies, applications, data analytics	GHGSat, NorthStar, Kepler Communications, Wyvern	~10%
Tier 4: Academia	R&D, training, technology incubation	U of T (SFL), York U, UBC, U of Calgary, Western	~5%

2.2 Key Canadian Space Capabilities

Canada has world-leading capabilities in several domains:

Capability	Key Player(s)	Notable Missions	Global Standing
Space robotics	MDA Space	Canadarm, Canadarm2, Dextre, Canadarm3	World leader
SAR Earth observation	MDA Space, CSA	RADARSAT-1/2/RCM	Top 3 globally
Satellite communications	Telesat	Anik fleet, Telesat Lightspeed (LEO constellation)	Pioneer in GEO DBS
Small satellite systems	SFL (U of T)	MOST, NEOSSat, GHGSat-D, BRITE	Top university lab globally
GHG monitoring	GHGSat	GHGSat constellation (methane)	World's only commercial GHG monitoring constellation
Optical instruments	ABB, Honeywell	JWST/FGS-NIRISS, OSIRIS-REx laser altimeter	Tier 1 instrument builder
Star trackers/AOCS	NGC Aerospace/Honeywell	Numerous missions	Widely used sensors

2.3 Funding Programs

Canadian space activities are funded through several mechanisms:

Program	Administered By	Scope	Typical Value
Space Technology Development Program (STDP)	CSA	TRL advancement for flight heritage	100K--3M per project
Class Grants and Contributions	CSA	Space science, awareness, STEM	Variable
LEAP	CSA	Lunar technology development	Up to $3M per project
IRAP	NRC	SME innovation support	50K--1M
SIF	ISED	Strategic innovation projects	10M--100M+
DRDC contracts	DND	Dual-use technology	Classified--$10M
NSERC / CFI	Federal	Academic research infrastructure	50K--5M

> *Industry Practice: GHGSat (Montreal) progressed from a CSA-funded technology demonstrator (GHGSat-D "Claire", launched 2016) to a commercial constellation of 12+ satellites monitoring methane emissions from individual industrial facilities. This trajectory -- government seed funding enabling commercial scale-up -- is the canonical Canadian space industry growth model.*

Discussion prompt: For your mission concept, which industry tier would you engage? What CSA funding programs might apply? What capabilities would need to be sourced internationally?

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3. Comparison with Other National Models (20 min)

Understanding Canada's space ecosystem requires comparing it with partner agencies:

Parameter	Canada (CSA)	USA (NASA)	Europe (ESA)	Japan (JAXA)
Annual budget (approx.)	~$300M CAD	~$25B USD	~$7.5B EUR	~$1.5B USD
Staff	~670	~18,000	~2,300	~1,500

Launch capability	None (uses partners)	SLS, commercial (SpaceX, ULA)	Ariane 6, Vega-C	H3, Epsilon
Key strength	Robotics, SAR, instruments	Full spectrum	Science, EO, navigation	Rockets, science, ISS (Kibo)
Industry model	Government seed + commercial scale	Government prime + commercial	Juste retour (geographic return)	Government-led + commercial

Canada's lack of indigenous launch capability means all Canadian satellites launch on foreign vehicles, which has implications for export control (Section 5 below).

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4. Canadian Space in the International Context (20 min)

4.1 International Partnerships

Canada participates in space through multilateral and bilateral agreements:

Partnership	Canadian Contribution	Canadian Benefit
ISS	Canadarm2, Dextre, SPDM	Astronaut access, research time, technology demonstrations
Lunar Gateway	Canadarm3	Astronaut flights, lunar science access
Copernicus	Data access agreement	Free access to Sentinel data for Canadian users
COSPAS-SARS	Canadian ground stations, MEOSAR payloads	Search and rescue capability for Canadian territory
Five Eyes	Intelligence sharing, dual-use technology	Access to allied space intelligence capabilities

4.2 The Canadian Niche Strategy

Canada has historically pursued a "niche excellence" strategy: rather than attempting full-spectrum space capability, Canada develops world-class expertise in specific domains (robotics, SAR, instruments) and trades access to these capabilities for partnership in larger programs.

> *Key Equation: The "partnership leverage ratio" -- informal but useful -- describes how Canada trades capability for access:*
>
> *L = (V_access)/(C_contribution)*
>
> *Where V_access is the value of program access obtained and C_contribution is the cost of Canada's contribution. For the ISS, Canada's Canadarm2/Dextre contribution (~2B CAD over 30 years) provides access to a150B+ program, yielding L ~= 75. This is the highest leverage ratio of any ISS partner.*

Part 2: Regulatory Environment (Tuesday AM -- 2 hours)

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5. Regulatory Framework Overview (15 min)

Space activities in Canada are governed by multiple federal bodies. There is no single "Space Act" -- instead, regulation is distributed across several statutes:

Regulatory Body	Statute	What It Regulates	Relevance to Mission Design
CSA	Canadian Space Agency Act	Space program coordination, policy	Mission approval, funding
ISED	Radiocommunication Act	Radio spectrum allocation, licensing	Frequency licensing for all RF transmissions
ISED	Remote Sensing Space Systems Act (RSSSA)	Operation of remote sensing satellites	Licensing for any satellite that images Earth
GAC	Export and Import Permits Act (EIPA)	Export control of strategic goods	Export permits for satellite technology
GAC	Controlled Goods Program	Access to controlled (ITAR/CG) goods	Personnel security clearances
DND/DRDC	National Defence Act	Dual-use technology, military space	Classification, security requirements
Transport Canada	(No current statute)	Launch operations (future)	Canada has no domestic launch regulation yet

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6. Remote Sensing Space Systems Act (RSSSA) (30 min)

6.1 Purpose and Scope

The RSSSA (S.C. 2005, c. 45) regulates the operation of remote sensing satellites from Canada. It was enacted primarily in response to the commercialisation of RADARSAT-2 (the first Canadian commercial EO satellite) and addresses national security and foreign policy concerns related to Earth imagery.

[Source: Government of Canada, RSSSA, https://laws-lois.justice.gc.ca/eng/acts/r-5.4/]

6.2 Who Needs a Licence?

An RSSSA licence is required for any person or entity that:

- Operates a remote sensing space system from Canada, or
- Controls the collection or distribution of remote sensing data from a Canadian system

This applies regardless of where the satellite is manufactured or launched. If it is operated from Canadian soil, RSSSA applies.

6.3 Licence Categories and Conditions

Licence Type	Scope	Typical Conditions
System Operating Licence	Operate a remote sensing satellite system	Data handling plan, priority access for Canadian government, shutter control provisions
Interoperability Arrangements	Share data with foreign entities	Approval for each foreign government or commercial partner

Key conditions typically imposed:

Condition	Description	Rationale
Priority access	Government of Canada can request priority imaging of any area	National security
Shutter control	Government can restrict imaging of specified areas	National security, diplomatic relations
Data disposition	Approved plan for data storage, distribution, and destruction	Prevent unauthorised access
System disposal	Plan for end-of-life decommissioning	Space debris mitigation
Raw data protection	Restrictions on distribution of unprocessed data	Prevent circumvention of controls

6.4 Implications for CubeSat and University Missions

Even small CubeSat missions with cameras must comply with RSSSA if they produce images of Earth. Key considerations:

Parameter	Threshold	Implication
GSD	No formal threshold in the Act	All resolutions require licensing
Orbit type	Any orbit imaging Earth	LEO, MEO, GEO all in scope
Data distribution	Any distribution of imagery	Must have approved data handling plan
Foreign participation	Any foreign access to data	Requires interoperability arrangement

> *Industry Practice: MDA's RADARSAT-2 operates under an RSSSA licence that includes shutter control provisions. During the 2010 Haiti earthquake response, the Government of Canada exercised priority access to direct RADARSAT-2 to image the disaster zone within hours. This demonstrates both the operational utility and the regulatory mechanism of RSSSA.*

Discussion prompt: Does your mission concept involve any remote sensing of Earth? If so, what RSSSA conditions might apply?

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7. Spectrum Management and Licensing (40 min)

7.1 Why Spectrum Licensing Matters

Every satellite that transmits or receives radio signals requires spectrum authorisation. This is not optional -- unauthorised RF transmission is illegal in all ITU member states and can result in harmful interference to other systems.

The process involves two levels:

1. National: ISED (Innovation, Science and Economic Development Canada) issues the radio licence
2. International: ITU (International Telecommunication Union) coordinates the frequency filing to prevent interference with other national administrations

7.2 The ITU Framework

The International Telecommunication Union allocates spectrum globally through the Radio Regulations (RR), updated at World Radiocommunication Conferences (WRC) every 3--4 years.

ITU Concept	Description	Relevance
Frequency allocation	Spectrum divided among radio services (Fixed Satellite, Mobile, Earth Exploration Satellite, etc.)	Determines which bands are available for your mission type
ITU Regions	Region 1 (Europe/Africa), Region 2 (Americas), Region 3 (Asia/Pacific)	Canada is in Region 2; allocations may differ by region
Coordination	Bilateral/multilateral agreements to avoid interference	Required before a new satellite system can operate
Notification	Formal filing with the ITU Radiocommunication Bureau (BR)	Records the satellite system in the Master International Frequency Register (MIFR)

7.3 Common Satellite Frequency Bands

Band	Frequency	Typical Use	Key Characteristics
UHF	400--450 MHz	CubeSat TT&C, AIS, ADS-B	Simple antennas, low data rate (1--9.6 kbps), crowded
S-band	2.0--2.3 GHz	TT&C (uplink + downlink)	Standard for housekeeping; moderate antenna gain required
X-band	8.025--8.4 GHz	EO payload data downlink	Higher data rates (10--150 Mbps); requires tracking antenna on ground
Ka-band	26.5--40 GHz	High-throughput data downlink, inter-satellite links	Very high data rates (100+ Mbps); rain attenuation significant
L-band	1.5--1.6 GHz	GNSS, mobile satellite service	Navigation, SAR beacons
C-band	3.7--4.2 / 5.9--6.4 GHz	Legacy GEO comms (DBS), SAR (RADARSAT)	Being repurposed for 5G in some countries

7.4 The Licensing Process (Canada)

The ISED spectrum licensing process for a satellite system follows these steps:

Step	Activity	Duration	Documents
1	Pre-application consultation with ISED	2--4	Preliminary technical description
2	Formal application submission	--	ISED application form, technical annex
3	ISED technical review	3--6 months	Interference analysis, coordination requirements
4	ITU filing (Advance Publication Information -- API)	--	ITU API filing (submitted by ISED on Canada's behalf)
5	ITU coordination with affected administrations	2--7	Coordination agreements
6	ITU notification and recording in MIFR	--	MIFR entry
7	ISED licence issuance	--	Radio licence

> *Key Equation: The link budget equation governs whether a given frequency allocation provides sufficient performance. The received signal power is:*
>
> *P_r = P_t + G_t + G_r - L_fs - L_atm - L_misc*
>
> *Where all values are in dB:*
>
> *- P_t = transmit power (dBW)*
>
> *- G_t = transmit antenna gain (dBi)*
>
> *- G_r = receive antenna gain (dBi)*
>
> *- L_fs = 20log_10(4pi d / lambda) = free-space path loss (dB)*

> *- L_atm = atmospheric loss (dB)*
>
> *- L_misc = miscellaneous losses (pointing, polarisation, etc.)*
>
> *The link margin is then:*
>
> *M = P_r - P_r,min = (E_b/N_0)_received - (E_b/N_0)_required*
>
> *A positive margin of at least 3 dB is typically required; 6 dB for critical links.*

7.5 CubeSat-Specific Considerations

Many CubeSat operators use amateur radio frequencies (VHF/UHF) during development but must transition to commercial bands for operational missions, especially if there is any commercial data distribution.

Approach	Licensing Path	Limitations
Amateur radio (VHF/UHF)	IARU coordination; national amateur licence	No commercial use; data must be freely available; limited power/bandwidth
Commercial UHF	ISED commercial licence; ITU coordination	Low data rate; crowded band; longer licensing timeline
S-band TT&C	ISED licence; SFCG coordination	Standard path; moderate timeline (6--12 months)
X-band payload	ISED licence; ITU coordination	Higher data rate; requires tracking ground station

> *Industry Practice: Kepler Communications (Toronto) initially operated its first pathfinder satellites on Ku-band for store-and-forward IoT data relay. Obtaining Ku-band spectrum rights required coordination with incumbent GEO operators, a process that took over 2 years. The lesson: begin spectrum licensing as early as Phase A.*

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8. Export Control (30 min)

8.1 The Export Control Landscape

Satellite technology is controlled goods in most countries due to dual-use (civil/military) applications. Canadian space companies and universities must navigate three overlapping regimes:

Regime	Administered By	Scope	Key List
EIPA (Canada)	Global Affairs Canada (GAC)	Canadian-origin goods and technology	Export Control List (ECL), Groups 1--7
ITAR (USA)	US State Department (DDTC)	US-origin defence articles and services	US Munitions List (USML), Category XV (spacecraft)
EAR (USA)	US Commerce Department (BIS)	US-origin dual-use goods and technology	Commerce Control List (CCL), ECCN 9Axx

8.2 Canadian Export Controls (EIPA)

The Export and Import Permits Act requires an export permit for goods on Canada's Export Control List (ECL). Space-relevant ECL groups:

ECL	Description	Examples
Group 1	Dual-use goods (Wassenaar Arrangement)	Radiation-hardened electronics, space-qualified components
Group 2	Munitions (defence goods)	Military satellite systems, encrypted comms
Group 4	Nuclear-related (NSG)	RTGs, nuclear propulsion components
Group 6	Missile technology (MTCR)	Complete rocket systems, guidance systems, propulsion
Group 7	Chemical/biological weapons	Not typically space-relevant

8.3 US Export Controls (ITAR and EAR)

Any mission using US-origin components (which includes nearly all commercial space electronics) must comply with US export controls:

ITAR (International Traffic in Arms Regulations):

- USML Category XV covers spacecraft and related articles
- Requires a Technical Assistance Agreement (TAA) or Manufacturing License Agreement (MLA)
- Applies to US-origin defence articles regardless of where they are located
- Violation penalties: up to $1M per violation and 20 years imprisonment

EAR (Export Administration Regulations):

- Controls dual-use items (commercial space components, software, technology)
- Uses Export Control Classification Numbers (ECCNs)
- Requires a licence for export to most countries (except Canada for many items under bilateral agreements)
- The US-Canada Defence Production Sharing Agreement provides some exemptions

8.4 Practical Implications for Mission Design

Design Decision	Export Control Implication
Using US-origin reaction wheels	EAR licence may be required; ITAR if military-grade
Collaborating with a non-Five Eyes university	TAA required for ITAR-controlled technical data
Launching on a non-US rocket	ITAR Technology Transfer Agreement for US components on foreign launcher
Open-sourcing satellite software	Must verify no controlled algorithms (e.g., encryption above EAR thresholds)
Hosting a foreign payload	May require export permit under EIPA and/or ITAR approval
Publishing detailed design in a paper	Technical data review required if ITAR-controlled components discussed

> *Industry Practice: MDA's export of RADARSAT-2 ground segment technology to international partners required extensive EIPA permitting and ITAR compliance review (since US-origin components were embedded in the system). The process took 18+ months and required dedicated compliance staff. Even university CubeSat projects using US-origin COTS components must track their ECCN classifications.*

Discussion prompt: Does your mission concept involve any US-origin components? Any international partners? What export control obligations might apply?

Session Summary

Topic	Key Takeaway
CSA mandate	Promote peaceful use of space; niche excellence strategy (robotics, SAR, instruments)
Industry structure	4-tier ecosystem; government seed funding enables commercial scale-up
RSSSA	Any remote sensing satellite operated from Canada requires a licence; includes shutter control
Spectrum licensing	Two-level process (ISED national + ITU international); start in Phase A; 2+ year timeline
Export control	Three overlapping regimes (EIPA, ITAR, EAR); nearly all missions encounter US controls
Key lesson	Regulatory compliance is not optional -- it shapes design decisions from the earliest phases

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References

1. Government of Canada, Canadian Space Agency Act (S.C. 1990, c. 13)
2. Government of Canada, Remote Sensing Space Systems Act (S.C. 2005, c. 45)
3. Government of Canada, Radiocommunication Act (R.S.C., 1985, c. R-2)
4. Government of Canada, Export and Import Permits Act (R.S.C., 1985, c. E-19)
5. ISED, Spectrum Management and Telecommunications
6. ITU, Radio Regulations, Edition 2020
7. CSA, Exploration, Imagination, Innovation: A New Space Strategy for Canada, 2019
8. CSA, State of the Canadian Space Sector Report, 2022
9. US Department of State, ITAR (22 CFR 120-130)
10. US Department of Commerce, EAR (15 CFR 730-774)

Session 1.3: International Standards for Space Systems

Duration: 4 hours (Tuesday PM + Wednesday)

Prerequisites: Sessions 1.1--1.2

References:

- ECSS, ECSS System -- Description, Implementation and General Requirements (ECSS-S-ST-00C), 2020
- NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016
- CDS, Interface Definition Document for Launch Vehicle Adapters, Rev 14, 2023
- IADC, Space Debris Mitigation Guidelines (IADC-02-01 Rev 3), 2021
- ISO, ISO 24113:2023 -- Space Debris Mitigation Requirements, 2023
- UNOOSA, COPUOS Guidelines for the Long-term Sustainability of Outer Space Activities, 2019
- ESA, Space Debris Mitigation Compliance Verification Guidelines (ESSB-HB-U-002), 2023

Learning Objectives

By the end of this session, participants will be able to:

1. Describe the ECSS standard framework and its hierarchical structure (Management, Engineering, Product Assurance, Sustainability)
2. Compare ECSS and NASA standards and identify equivalences
3. Explain the CDS Rev 14 interface standard and its role in launch vehicle integration
4. Apply space debris mitigation guidelines (IADC, ISO 24113, FCC 5-year rule) to mission design
5. Describe the role of COPUOS and the Outer Space Treaty framework
6. Use SpaceCDF's compliance tracking features

Part 1: The ECSS Framework (Tuesday PM -- 2 hours)

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1. Why Standards Matter in Space Engineering (20 min)

1.1 The Cost of Non-Compliance

Space missions operate in an environment where failures are catastrophic, repair is impossible, and the consequences of interference or debris affect all space users. Standards exist to:

1. Ensure mission safety and reliability -- by codifying lessons learned from decades of space operations
2. Enable interoperability -- by defining common interfaces, data formats, and protocols
3. Reduce cost -- by providing proven design approaches rather than re-inventing solutions
4. Satisfy regulatory requirements -- by demonstrating compliance with national and international obligations
5. Facilitate technology transfer -- by establishing a common engineering language

> *Industry Practice: The loss of the Mars Climate Orbiter (1999) is the canonical example of what happens without rigorous interface standards. Lockheed Martin's ground software produced thruster force data in pound-force seconds, while NASA's navigation software expected Newton-seconds. The spacecraft entered the Martian atmosphere at 57 km altitude instead of the planned 226 km and was destroyed. Total loss: $327.6M. This failure directly led to NASA's strengthening of Process 12 (Interface Management) in NPR 7123.1D. The lesson: standards and interface control are not bureaucratic overhead -- they are mission-critical.*

1.2 The Major Standard Frameworks

Three major standard frameworks govern space activities globally:

ECSS (European Cooperation for Space Standardization)	Full lifecycle, all disciplines	ESA, European industry, CSA (partial)	144 standards + 56 handbooks
NASA Technical Standards	NASA programs and projects	NASA centres, US contractors	NPR/NPD/NASA-STD series
ISO TC 20/SC 14	Space systems and operations	International (reference standard)	~40 standards

These frameworks are not mutually exclusive. Many missions (including Canadian ones) adopt a tailored combination: ECSS for systems engineering and product assurance, NASA standards for specific technical domains, and ISO for debris mitigation.

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2. ECSS Standard Hierarchy (40 min)

2.1 Structure

The ECSS system organises standards into four branches, each with three levels:

Figure: ECSS Standard Hierarchy

(See digital/markdown version for interactive diagram)

2.2 The Four Branches

Branch	Code	Scope	Example Standard
Management	M	Project management, planning, reviews, configuration management, risk management	ECSS-M-ST-10C Rev.1 (Space Project Management)
Engineering	E	Technical disciplines: systems, structures, thermal, EPS, comms, AOCS, software, ground	ECSS-E-ST-10C (Systems Engineering General Requirements)
Product Assurance	Q	Quality, reliability, safety, EEE components, materials, contamination	ECSS-Q-ST-10C (Product Assurance Management)
Sustainability	U	Space debris mitigation, end-of-life disposal, space environment protection	ECSS-U-AS-10C Rev.2 (adoption of ISO 24113)

2.3 Document Types

Type	Code	Meaning	Obligation
Standard	ST	Contains "shall" requirements -- mandatory when invoked	Must comply or formally request a waiver
Handbook	HB	Contains guidance, best practices, worked examples	Advisory -- not mandatory but strongly recommended

Technical Memorandum	TM	Contains background information, state of the art	Informative only

2.4 Tailoring

A critical concept in ECSS (and all standard frameworks) is tailoring -- the process of selecting which standards and which requirements within those standards apply to a given project.

ECSS-S-ST-00C defines three tailoring levels:

Tailoring Level	Description	Typical Application
Full application	All requirements of the invoked standard apply	Large ESA missions (Sentinel, JUICE)
Partial application	Selected clauses apply; others are waived with rationale	Medium missions, CSA projects
Not applicable	Standard is not invoked for this project	CubeSat missions, technology demonstrators

The tailoring rationale is documented in the Product Assurance and Safety Plan (PASP) and the Systems Engineering Management Plan (SEMP).

> *Key Equation: The cost of standards compliance scales non-linearly with mission class. A rough relationship observed in ESA studies:*
>
> *C_compliance ~= k * M_SC^0.7 * N_standards^0.3*
>
> *Where M_SC is the spacecraft dry mass (kg), N_standards is the number of invoked standards, and k is a constant dependent on the organisation's maturity. For a 10 kg CubeSat invoking 5 standards, compliance costs are roughly 5--10% of mission cost. For a 1000 kg satellite invoking 40 standards, compliance costs can reach 15--20%.*

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3. Key ECSS Engineering Standards (30 min)

3.1 Systems Engineering (ECSS-E-ST-10C)

This is the master engineering standard, equivalent to NASA's NPR 7123.1D. It defines:

- System engineering processes and activities
- Requirements engineering (writing, verification, traceability)
- Functional analysis and decomposition
- Interface management
- Configuration management (technical aspects)

ECSS-E-ST-10C Concept	NASA Equivalent	Key Difference
System requirements specification	Technical requirements baseline	ECSS requires a formal Requirements Specification Document (RSD)
Functional analysis	Logical decomposition (Process 3)	Similar scope

Verification matrix	Requirements verification matrix	ECSS uses a formal DRD (Document Requirements Definition)
Design justification file	Design solution baseline	ECSS requires a DJF that traces every design decision

3.2 Key Subsystem Standards

Standard	Discipline	Key Requirements	NASA Equivalent
ECSS-E-ST-20	Electrical & Electronic	EPS design, grounding, EMC, harness	NASA-STD-4003A (EEE),
ECSS-E-ST-31	Thermal Control	Thermal design, analysis, testing	NASA-STD-5001B (Structural Design)
ECSS-E-ST-32	Structures	Structural design, factors of safety, testing	NASA-STD-5001B, GEVS
ECSS-E-ST-33-01C	Mechanisms	Mechanism design, testing, lubrication	NASA-STD-5017
ECSS-E-ST-35	Propulsion	Propulsion system design, testing	--
ECSS-E-ST-40	Software	SW development, verification, FDIR	NASA-STD-8739.8
ECSS-E-ST-50	Communications	Comms system design, link budget, protocols	CCSDS standards
ECSS-E-ST-60	Control	AOCS design, pointing, navigation	--
ECSS-E-ST-70C	Ground Systems	Ground segment design, operations	CCSDS standards

3.3 Structural Design Requirements (Example)

To illustrate the depth of ECSS standards, consider the structural design requirements from ECSS-E-ST-32C:

Requirement	Value	Rationale
Factor of Safety (FoS) -- Yield	>= 1.25	Prevent permanent deformation under limit loads
Factor of Safety -- Ultimate	>= 1.5	Prevent structural failure under ultimate loads
Qualification loads	1.25 x limit loads	Demonstrate margin beyond expected environment
First natural frequency (axial)	> 25 Hz typical (launcher-dependent)	Avoid coupling with launcher modes
First natural frequency (lateral)	> 10 Hz typical (launcher-dependent)	Avoid coupling with launcher modes

> *Key Equation: The margin of safety (MoS) is defined as:*
>
> *MoS = (sigma_allowable)/(FoS x sigma_applied) - 1*
>
> *A positive MoS (MoS > 0) indicates the structure meets the requirement. A MoS of 0.0 means the structure exactly meets the requirement with no margin to spare.*
>
> *Example: If the allowable stress is 280 MPa, the applied stress is 150 MPa, and the FoS is 1.5:*
>
> *MoS = (280)/(1.5 x 150) - 1 = (280)/(225) - 1 = 0.244*

---

4. NASA Standards Comparison (20 min)

4.1 NASA Technical Standards Architecture

NASA's standards are organised differently from ECSS. Key document types:

Type	Code	Example	Obligation
NASA Policy Directive	NPD	NPD 8700.1 (Safety and Mission	Mandatory for all NASA programs
NASA Procedural	NPR	NPR 7123.1D (SE Processes)	Mandatory; defines processes and requirements
NASA Technical Standard	NASA-S	NASA-STD-5001B (Structural Design)	Mandatory when invoked
NASA Handbook	NASA-H	NASA-HDBK-4001 (EEE Parts)	Advisory guidance
Special Publication	SP	SP-2016-6105 (SEH)	Reference text

4.2 Cross-Reference Table

Domain	ECSS Standard	NASA Standard	ISO Standard
Systems Engineering	ECSS-E-ST-10C	NPR 7123.1D, SP-2016-6105	ISO 15288
Project Management	ECSS-M-ST-10C	NPR 7120.5F	ISO 21500
Configuration Management	ECSS-M-ST-40C	NPR 7120.5F Ch. 4	ISO 10007
Risk Management	ECSS-M-ST-80C	NPR 8000.4B	ISO 31000
Structural Design	ECSS-E-ST-32C	NASA-STD-5001B	--
Debris Mitigation	ECSS-U-AS-10C	NASA-STD-8719.14A	ISO 24113
Cleanliness	ECSS-Q-ST-70-01C	NASA-SN-C-0005	ISO 14644
Software	ECSS-E-ST-40C	NASA-STD-8739.8	ISO 12207

Part 2: Launch Interfaces, Debris Mitigation & International Law (Wednesday -- 2 hours)

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5. CDS Rev 14 -- Launch Vehicle Interface Standard (30 min)

5.1 What is the CDS?

The Cubesat Design Specification (CDS), maintained by the California Polytechnic State University (Cal Poly) and updated by the CubeSat community, defines the mechanical, electrical, and operational interfaces between CubeSats and their deployment systems (P-PODs, ISIPOD, etc.).

[Source: CDS Rev 14, 2022, available via Cal Poly CubeSat Program]

5.2 Key CDS Requirements

Parameter	CDS Rev 14 Requirement	Rationale
Unit dimensions	100 x 100 x 113.5 mm per U	Standard deployer rail spacing
Mass per U	<= 2.0 kg (with waiver up to 2.66 kg for some deployers)	Deployer spring mechanism limits

Rail material	Hard-anodised aluminium (7075 or 6061-T6)	Deployer contact surface compatibility
Centre of gravity	Within 2 cm of geometric centre	Deployment dynamics, tumble rate control
Deployables	Must be constrained during launch; no protrusion beyond CubeSat envelope	Protect adjacent payloads in deployer
Separation springs	Prohibited on CubeSat (deployer provides)	Standardised deployment mechanism
RF silence	No RF transmission until 30 minutes after deployment	Avoid interference with launch vehicle
Deployment switches	Minimum 1 per deployable; 2 for redundancy	Prevent premature deployment
Battery charge state	Fully charged (recommended); charging from deployer not available	No power interface with deployer
Propulsion	If present: must be inhibited by 3 independent inhibits; no toxic propellants	Safety of primary payload and deployer

5.3 Form Factors

Form Factor	Dimensions (mm)	Mass Limit	Typical Applications
1U	100 x 100 x 113.5	2.0 kg	Technology demonstrators, IoT nodes
1.5U	100 x 100 x 170.2	3.0 kg	Enhanced technology demonstrators
2U	100 x 100 x 227.0	4.0 kg	Simple instruments, store-and-forward
3U	100 x 100 x 340.5	6.0 kg	Standard EO/science missions
6U	100 x 226.3 x 340.5	12.0 kg	Advanced EO, communications
12U	226.3 x 226.3 x 340.5	24.0 kg	High-performance missions
16U	226.3 x 226.3 x 454.0	32.0 kg	Near-microsatellite capability

5.4 Rideshare Launch Interfaces

For non-CubeSat smallsats (microsatellites 10--200 kg), the interface standard depends on the launch provider:

Launch Provider	Interface Standard	Adapter Type	Key Document
SpaceX (Rideshare)	ESPA-class	15" or 24" ESPA port	SpaceX Rideshare User's Guide
Rocket Lab (Electron)	Custom separation system	Rocket Lab-provided	Electron Payload User's Guide
Arianespace (Vega-C)	ASAP-S	Multi-payload adapter	Vega-C User Manual
ISRO (PSLV)	Custom adapter	ISRO-provided	PSLV User Manual

> *Industry Practice: Planet Labs' SuperDove constellation (150+ satellites) uses a standardised 3U-plus form factor that is CDS-compliant but extends the standard with a custom deployer arrangement for batch deployment. Each SuperDove satellite (mass ~5.8 kg) carries a multispectral imager with 8 bands and ~3m GSD. The standardisation of the bus design enabled a manufacturing rate of 2+ satellites per week -- only possible because the CDS provides a stable interface baseline.*

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6. Space Debris Mitigation (40 min)

6.1 The Debris Problem

As of 2025, there are approximately:

- 36,500+ tracked objects larger than 10 cm in Earth orbit
- 1,000,000+ estimated objects 1--10 cm (untracked, lethal to spacecraft)
- 130,000,000+ estimated objects 1 mm -- 1 cm (can damage components)

The debris population is growing due to collisions (the Kessler Syndrome), anti-satellite tests (e.g., Chinese ASAT 2007: 3,500+ tracked fragments), and the rapid growth of mega-constellations.

[Source: ESA Space Debris Office, Annual Report 2024; NASA ODPO Orbital Debris Quarterly News]

6.2 Debris Mitigation Guidelines Hierarchy

Level	Document	Scope	Status
International voluntary	IADC Space Debris Mitigation Guidelines (Rev 3, 2021)	Global best practice	Advisory; adopted by COPUOS
International standard	ISO 24113:2023	Requirements for debris mitigation	Standard; invoked by many agencies
European	ECSS-U-AS-10C Rev.2	ECSS adoption notice for ISO 24113	Mandatory for ESA
US regulation	FCC 47 CFR 25.114(d)(14)	Post-mission disposal for US-licensed satellites	Legally binding for FCC licensees
NASA standard	NASA-STD-8719.14A	NASA process for limiting orbital debris	Mandatory for NASA missions
ESA requirement	ESA/ADMIN/IPOL(2023)2	Clean Space requirements	Mandatory for ESA missions (2023+)

6.3 Key Debris Mitigation Requirements

Requirement	Source	Value	Notes
Post-mission disposal	IADC, ISO 24113	<= 25 years	Voluntary guideline; being tightened
Post-mission disposal	FCC (2024+)	<= 5 years	Legally binding for FCC-licensed sats
Probability of successful disposal	ISO 24113	>= 0.9	Must demonstrate reliability of deorbit mechanism
Casualty risk on re-entry	NASA-STD-8719.14A	<= 1:10,000 per event	Drives material selection and design-for-demise
Collision avoidance probability	IADC	< 10^-4 per year (cumulative)	Drives orbit selection and manoeuvre capability
Passivation	ISO 24113, ECSS-U-AS-10C	All stored energy sources depleted at EOL	Batteries, pressure vessels, wheels, RF

6.4 Post-Mission Disposal Options

Method	Applicable Orbit	Mechanism	Time to Re-entry
Atmospheric drag (natural)	LEO < 600 km	Natural orbital decay	Months to years
Atmospheric drag (augmented)	LEO < 700 km	Drag sail, drag tether	Weeks to years
Propulsive deorbit	LEO	Thrusters lower perigee to ~200 km	Days
Graveyard orbit	GEO	Raise orbit ~300 km above GEO	Indefinite (not re-entry)
Heliocentric disposal	Beyond GEO	Escape Earth orbit	N/A

> *Key Equation: The orbital lifetime of a satellite in LEO due to atmospheric drag is approximately:*

> *\tau ~= (C_D * A * rho * a^2)/(2m) * (complex integral)*
>
> *A simpler rule-of-thumb for circular orbits:*
>
> *\tau_years ~= (h - 200)/(30) * (m / A)/(50)*
>
> *Where h is altitude in km, m is mass in kg, and A is the cross-sectional area in m^2. This is very approximate; actual lifetime depends on solar activity (which modulates atmospheric density at high altitudes), drag coefficient, and orbit eccentricity. Use NASA's DAS (Debris Assessment Software) or ESA's DRAMA tool for accurate predictions.*
>
> *More precisely, the ballistic coefficient is:*
>
> *B = (m)/(C_D * A)*
>
> *Lower B (lighter, larger area) means faster re-entry. A 3U CubeSat (m ~= 4 kg, A ~= 0.03 m^2, C_D ~= 2.2) at 400 km has B ~= 61 kg/m^2 and will re-enter within ~1--3 years depending on solar cycle.*

6.5 Practical Implications for Mission Design

Design Decision	Debris Mitigation Impact
Orbit altitude selection	Altitudes > 600 km require active deorbit; > 700 km strongly discouraged for non-manoeuvrable spacecraft
Propulsion system	Required for altitudes > 500--600 km to meet 25-year (or 5-year FCC) rule
Passivation design	Must design battery disconnect, pressure relief, wheel spin-down circuits
Material selection	Aluminium structures preferred for design-for-demise; titanium and carbon fibre survive re-entry
Collision avoidance	Manoeuvre capability or conjunction assessment service (e.g., 18th SDS, ESA SSA) needed

> *Industry Practice: OneWeb (648 satellites at 1200 km) carries propulsion on every satellite specifically for end-of-life deorbit, since natural decay from 1200 km would take centuries. Each satellite carries sufficient propellant for multiple collision avoidance manoeuvres plus a complete deorbit burn to lower perigee below 300 km. The deorbit operation takes approximately 3 months per satellite.*

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7. International Space Law and COPUOS (30 min)

7.1 The UN Space Treaties

The legal framework for space activities is established by five UN treaties negotiated under the Committee on the Peaceful Uses of Outer Space (COPUOS):

Treaty	Year	Key Provisions	Ratification
Outer Space Treaty (OST)	1967	Space is free for exploration; no national appropriation; states responsible for national activities; liability for damage	114 parties (incl. Canada)
Rescue Agreement	1968	Return astronauts and space objects	99 parties
Liability Convention	1972	Launching state liable for damage on Earth (absolute) and in space (fault-based)	98 parties
Registration Convention	1976	States must register space objects with UN	72 parties
Moon Agreement	1979	Moon and celestial bodies are "common heritage of mankind"	18 parties (NOT US, Russia, China)

7.2 Key Legal Principles for Mission Design

Principle	Source	Implication
State responsibility	OST Art. VI	The Government of Canada is internationally responsible for all Canadian space activities, including private/university missions
Authorisation and supervision	OST Art. VI	Canada must authorise and continuously supervise all non-governmental space activities (this is why RSSSA exists)
Liability	Liability Convention	Canada (as launching state) is liable for damage caused by Canadian satellites; absolute liability on Earth, fault-based in space
Registration	Registration Convention	Canada must register all space objects with the UN; CSA maintains the Canadian registry
Non-contamination	OST Art. IX	Must avoid harmful contamination of space and celestial bodies (planetary protection)
Due regard	OST Art. IX	Must conduct activities with "due regard" for other states' interests (debris mitigation)

7.3 COPUOS Long-Term Sustainability Guidelines (2019)

In 2019, COPUOS adopted 21 guidelines for the long-term sustainability of outer space activities. These are voluntary but politically significant:

Guideline Category	Count	Key Points
Policy and regulatory	7	Adopt national regulatory frameworks, register space objects, share SSA data
Safety of operations	4	Conjunction assessment, collision avoidance, re-entry risk assessment
International cooperation	4	Share debris mitigation best practices, coordinate spectrum use
Scientific and technical	6	Improve debris models, develop removal technology, research space weather effects

[Source: A/74/20, Report of COPUOS, 2019, Annex II]

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8. SpaceCDF Compliance Features (20 min)

SpaceCDF tracks compliance through the Compliance Engineer position:

Feature	What It Tracks	Automation Level
Standard applicability matrix	Which ECSS/NASA/ISO standards apply to this mission	Semi-automated (suggests based on mission type)
Debris mitigation compliance	Post-mission lifetime, passivation plan, casualty risk	Automated (calculates from orbit and mass)
RSSSA checklist	Licence requirements for Canadian remote sensing	

missions

Manual (checklist with guidance)

Spectrum filing status	ISED/ITU filing progress	Manual (status tracking)
Export control classification	ECCN/USML classification of key components	Manual (per-component)
CDS compliance	Mechanical/electrical interface compliance with CDS Rev 14	Semi-automated (checks mass, dimensions, CG)

Exercise: In SpaceCDF, navigate to the Compliance panel. Review the debris mitigation compliance status for your mission. What orbit altitude would be needed to comply with the FCC 5-year rule without propulsion? Use the tool's orbital lifetime calculator.

Session Summary

Topic	Key Takeaway
ECSS framework	Four branches (M, E, Q, U); three document types (ST, HB, TM); tailoring is essential
NASA standards	Organised as NPD/NPR/NASA-STD/NASA-HDBK; broadly equivalent to ECSS
CDS Rev 14	Defines CubeSat mechanical/electrical/operational interfaces; compliance is required for rideshare launch
Debris mitigation	IADC 25-year rule; FCC 5-year rule (2024+); passivation required; drives orbit and propulsion design
International law	OST establishes state responsibility; Canada liable for all Canadian space activities
COPUOS	21 sustainability guidelines (2019); voluntary but increasingly influential
Compliance in SpaceCDF	Tracked through the Compliance Engineer position; automated where possible

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References

1. ECSS, ECSS-S-ST-00C -- Space Standardization, 2020
2. ECSS, ECSS-E-ST-10C -- System Engineering General Requirements, 2009
3. ECSS, ECSS-E-ST-32C Rev.1 -- Structural General Requirements, 2008
4. ECSS, ECSS-U-AS-10C Rev.2 -- Adoption Notice of ISO 24113, 2023
5. ISO, ISO 24113:2023 -- Space Systems: Space Debris Mitigation Requirements, 2023
6. IADC, IADC-02-01 Rev 3 -- Space Debris Mitigation Guidelines, 2021
7. Cal Poly, CubeSat Design Specification (CDS) Rev 14, 2022
8. NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016
9. NASA, NASA-STD-8719.14A -- Process for Limiting Orbital Debris, 2019
10. UNOOSA, A/74/20 -- Report of COPUOS, 2019
11. UNOOSA, Outer Space Treaty, 1967
12. FCC, 47 CFR 25.114 -- Satellite Applications

Session 1.4: Mission Needs, Stakeholder Analysis &

Trade Studies

Duration: 6 hours (Thursday + Friday)

Prerequisites: Sessions 1.1--1.3

References:

- NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016 -- Sections 4.1, 4.4, 6.8
- NASA, NPR 7123.1D -- Sections 3.2.1 (Process 1), 3.2.4 (Process 4), 3.5.8 (Process 17)
- ECSS, ECSS-E-ST-10C -- Section 5.1 (Requirements Engineering)
- ECSS, ECSS-M-ST-10C Rev.1 -- Section 5.3 (Review Process)
- Wertz, J.R. et al., Space Mission Engineering: The New SMAD (SMAD4), Microcosm Press, 2011 -- Chapters 1--3
- NASA SEH Appendix C -- How to Write a Good Requirement
- NASA SEH Appendix S -- ConOps Outline

Learning Objectives

By the end of this session, participants will be able to:

1. Write a clear problem statement that defines the mission need without prescribing a solution
2. Identify and categorise mission stakeholders using a structured analysis matrix
3. Define mission objectives with measurable success criteria and trace them through the MoE/MoP/TPM hierarchy
4. Structure and execute a trade study using weighted scoring methods (Decision Analysis -- Process 17)
5. Evaluate space vs non-space alternatives objectively and document the decision rationale
6. Design a Concept of Operations (ConOps) including mission phases, operational modes, and data flow
7. Use SpaceCDF Steps 1--2 and the Trade Studies tab to capture and evaluate mission concepts

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Part 1: Mission Need & Stakeholder Analysis (Thursday AM -- 2 hours)

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1. The Problem Statement (30 min)

1.1 Why Start with the Problem?

NASA SEH Section 4.1.1 states: "Clearly define the problem before solving it."

The single most common failure mode in mission design is not technical -- it is solving the wrong problem. Teams jump to solutions ("we need a 6U CubeSat at 500 km") before articulating the actual need ("agricultural monitoring in sub-Saharan Africa requires 10 m resolution multispectral imagery every 5 days"). This premature commitment to a solution:

- Closes off potentially superior alternatives (commercial data purchase, aerial surveys, ground sensors)
- Introduces unnecessary constraints that drive up cost and risk
- Makes it impossible to evaluate mission success (success against what?)
- Violates the fundamental SE principle of WHAT before HOW

[Source: NASA SEH Section 4.1; NASA SEH Appendix C -- "How to Write a Good Requirement"]

1.2 Structure of a Good Problem Statement

A rigorous problem statement answers four questions:

Question	Content	Example
What is the problem?	The capability gap -- what is missing or inadequate	"Lack of timely, affordable multispectral imagery at sufficient resolution and revisit"
Who is affected?	Stakeholders and end users who suffer from the gap	"Agricultural monitoring agencies in sub-Saharan Africa"
What is the impact?	Consequence of not solving it -- quantified if possible	"Delayed or inaccurate crop assessments affecting food aid allocation for 300M people"
What constraints exist?	Budget, schedule, political, regulatory, or technical limitations	"Budget: <$10M total mission cost; timeline: 3 years to first data"

1.3 Good vs Bad Problem Statements

Example -- GOOD:

> *"Agricultural monitoring agencies in sub-Saharan Africa lack timely, affordable access to multispectral imagery at sufficient resolution (<= 10 m) and revisit rate (<= 5 days) to support crop yield prediction and food security planning. Current Sentinel-2 data provides 10 m resolution but only 5-day revisit at the equator, and cloud cover reduces usable observations to approximately 60%. The consequence is delayed or inaccurate crop assessments affecting food aid allocation for 300 million people. Budget constraint: total mission cost < \10$M; first data delivery within 3 years."*

Example -- BAD:

> *"We need a 6U CubeSat with a multispectral imager at 500 km SSO."*

The bad example prescribes a solution (6U CubeSat, SSO), specifies a design parameter (500 km), and says nothing about the actual problem. It is a solution statement, not a problem statement.

1.4 The WHAT vs HOW Principle

At the mission need level, everything should describe WHAT is needed, not HOW to achieve it:

WHAT (correct at this stage)	HOW (premature at this stage)
"10 m resolution imagery"	"Use a 15 cm aperture telescope"

"Daily revisit at equator"	"Deploy a Walker delta constellation of 12 satellites"
"Data within 6 hours of acquisition"	"Use X-band downlink at 150 Mbps to 3 ground stations"
"Total cost under $10M"	"Use COTS components exclusively"
"5-year operational lifetime"	"Select radiation-hardened components"

The HOW column is not wrong -- these are all reasonable design decisions. But they belong in later phases (Phase A/B), not in the problem statement. Fixing the HOW prematurely eliminates design freedom and may lead to suboptimal solutions.

Discussion prompt: Why is it harmful to specify HOW at this stage? What design options does it close off?

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2. Stakeholder Identification & Analysis (30 min)

2.1 Who is a Stakeholder?

NASA SEH Section 4.1.2 defines stakeholders as "all parties who have a legitimate interest in the system throughout its lifecycle." This is broader than just the end user -- it includes everyone who affects or is affected by the mission.

2.2 Stakeholder Categories for Space Missions

Category	Examples	Typical Needs	Typical Constraints
End Users	Scientists, farmers, shipping companies, emergency responders	Data quality, resolution, latency, format, accessibility	Data rights, training, infrastructure
Operators	Mission control centre, ground station operators	Operability, automation level, staffing, training	Budget for operations, personnel availability
Sponsors / Funders	Space agency (CSA), university, commercial investor, DND	Return on investment, schedule adherence, risk profile	Budget cap, political timelines, reporting requirements
Regulatory	ISED, ITU, FCC, export control (GAC)	Compliance with spectrum, debris, RSSSA, export regulations	Licensing timelines, filing requirements
Launch Provider	SpaceX, Rocket Lab, ISRO, Arianespace	CDS compliance, mass/volume limits, schedule, payment	Interface requirements, launch window, manifesting
Ground Segment	KSAT, SSC, SatNOGS, own stations	Frequency compatibility, data volume, contact time	Station availability, geographic coverage
Data Consumer	Archives (e.g., EODMS), APIs, partner agencies	Data format (NetCDF, GeoTIFF), metadata standards, timeliness	Format standards, distribution agreements
General Public	Taxpayers (for government-funded missions)	Value for money, transparency, societal benefit	Political expectations, public engagement
Orbital Environment	Other satellite operators, debris community	Debris mitigation, spectrum cleanliness, collision avoidance	IADC guidelines, ISO 24113, FCC 5-year rule

2.3 Stakeholder Analysis Matrix

For each stakeholder, capture five attributes:

Attribute	Description	Scale
Name / Role	Who are they?	Text

Needs

What do they require from the mission?

Text (specific, measurable where possible)

Constraints	What limitations do they impose?	Text (mandatory vs desirable)
Priority	How critical is satisfying this stakeholder?	Primary (must satisfy) / Secondary (should satisfy) / Tertiary (nice to have)
Influence	How much power do they have over the mission?	High / Medium / Low

> *Industry Practice: For the RADARSAT Constellation Mission (RCM), the stakeholder analysis identified over 20 distinct stakeholder groups, including DND (maritime surveillance), Environment Canada (sea ice monitoring), Agriculture and Agri-Food Canada (crop monitoring), Natural Resources Canada (forestry), Public Safety Canada (disaster response), and international partners. Each stakeholder had different priority imaging modes, coverage requirements, and data latency needs. The systems engineering challenge was to design a 3-satellite constellation that satisfied all these needs within a single system architecture -- a classic example of multi-stakeholder optimisation.*

Exercise: Complete Part A of Worksheet 1.4 -- identify at least 4 stakeholders for your team's mission and fill in the analysis matrix.

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3. Mission Objectives and the MoE/MoP/TPM Hierarchy (30 min)

3.1 From Need to Objectives

Objectives translate the problem statement into specific, testable goals. Each objective must have:

Attribute	Description	Example
Text	Clear statement of what the mission will achieve	"Provide 10 m GSD multispectral imagery over the target region"
Priority	Primary (mission fails without) or secondary (desirable)	Primary
Measurable criterion	How you know the objective is met -- a number with a unit	"GSD <= 10 m at nadir, 4+ spectral bands, revisit <= 5 days"
Type	Category of objective	Observation, communication, navigation, science, technology demonstration

3.2 Writing Good Objectives

Good Objective	Why It Is Good
"Provide 10 m GSD multispectral imagery with 4+ bands for the target region (30S--30N), with <= 5-day revisit and <= 24 h data latency"	Specific (10 m, 4 bands, geographic scope), measurable (all criteria have numbers + units), relevant to agriculture, achievable with CubeSat technology
"Achieve 99.5% AIS ship detection probability in the North Atlantic within 30 minutes of ship transmission"	Specific (AIS, North Atlantic), measurable (99.5%, 30 min), relevant to maritime safety

Bad Objective	Why It Is Bad
"Take pictures from space"	Not specific, not measurable, no criterion for success
"Build a 3U CubeSat"	This is a solution, not an objective -- it says nothing about what the mission should accomplish
"Demonstrate new technology"	What technology? What does "demonstrate" mean? How do you know it succeeded?

3.3 The MoE / MoP / TPM Hierarchy

These three measures form a chain from operational need to tracked design parameter:

[Source: NASA SEH Section 4.1.4, Section 4.2.4, Section 6.7.3]

Measure	What It Measures	Who Sets It	Example	When Evaluated
MoE (Measure of Effectiveness)	How well the system satisfies the operational need	Users / stakeholde	"% of crop assessments delivered within 5 days of acquisition"	Phase E (operations)
MoP (Measure of Performance)	Technical performance of the system	Systems engineer	"GSD <= 10 m at nadir"	Phase C/D (verification)
TPM (Technical Performance Measure)	Design parameter tracked over time	Design team	"Current best estimate of imager mass vs allocation (1.5 kg target)"	All phases (continuous)

The hierarchy flows downward:

```
Stakeholder Need
  -> MoE (operational effectiveness)
    -> Mission Objective
      -> MoP (technical performance)
        -> Technical Requirement ("shall" statement)
          -> TPM (tracked design parameter)
```

> *Key Equation: Measures of Effectiveness often involve probability and time, connecting to system-level performance:*
>
> *MoE = P_detection x P_data_delivery x f(T_latency)*
>
> *For an Earth observation mission:*
>
> *MoE_coverage = (A_imaged_per_revisit)/(A_total_target) x (1 - P_cloud)*
>
> *Where A_imaged_per_revisit depends on swath width and orbit ground track spacing, A_total_target is the target area, and P_cloud is the cloud cover probability.*

Exercise: For one of your objectives, trace the full hierarchy from stakeholder need through MoE, MoP, and TPM. Complete Part B of Worksheet 1.4.

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4. From Need to Design: The Flow (20 min)

Show how the mission need flows through to design decisions without the need itself prescribing the design:

Figure: From Mission Need to Design Decision

(See digital/markdown version for interactive diagram)

At each level in this hierarchy, the design team makes decisions informed by analysis and trade studies. The tool supports this process with parametric calculations and automated constraint checking, but the team retains decision authority.

Part 2: Trade Study Methodology (Thursday PM -- 2 hours)

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5. Decision Analysis Framework (40 min)

5.1 Process 17: Decision Analysis

NASA SEH Process 17 (Decision Analysis) provides the framework for structured alternative evaluation. Every major design decision should follow this process to ensure decisions are auditable, traceable, and defensible.

[Source: NASA SEH Section 6.8, Process 17: Decision Analysis]

5.2 The Five Elements of a Rigorous Trade Study

Element	Description	Pitfall to Avoid
1. Decision statement	What are we deciding? Framed as a question.	Too broad ("What should we build?") or too narrow ("Which reaction wheel?") for the current phase
2. Alternatives	What are the options? Minimum 3, including "do nothing"	Straw-man alternatives designed to lose; failure to include obvious options
3. Criteria	What matters? Performance, cost, schedule, risk, compliance	Missing a critical criterion; criteria that overlap (double-counting)
4. Weightings	How important is each criterion? Normalised to sum to 1.0	All criteria weighted equally (means nothing matters more than anything else); weights set after seeing scores
5. Scoring	How well does each alternative perform?	Inconsistent scoring scale; anchoring bias; group conformity

5.3 Weighting Methods

Method	Procedure	Best When	Limitations
Pairwise comparison	Compare criteria two at a time; count wins. Weight = wins / total comparisons	Small number of criteria (< 8)	Circular preferences possible; does not capture magnitude
Swing weighting	For each criterion, assess the value of moving from worst to best case. Normalise.	Criteria have different units and scales	Requires clear understanding of worst/best cases
Direct assignment	Team discusses and agrees on weights	Quick preliminary assessment	Subjective; dominated by vocal team members
AHP (Analytic Hierarchy Process)	Pairwise comparison on a 1--9 ratio scale; compute eigenvector	Complex decisions with many stakeholders	Mathematically complex; consistency check needed

> *Key Equation (Pairwise Comparison): For n criteria, there are (n(n-1))/(2) pairwise comparisons. The raw weight of criterion i is:*
>
> *w_i^raw = Sum_j != i p_ij*
>
> *Where p_ij = 1 if criterion i is preferred over j, 0.5 for a tie, and 0 if j is preferred. The normalised weight is:*
>
> *w_i = (w_i^raw)/(Sum_k=1^n w_k^raw)*

5.4 Scoring Methods

Type	Description	Example	When to Use
Quantitative	Numeric value normalised to 0--1	Mass: 5 kg scores 0.8, 10 kg scores 0.4	When hard numbers exist
Qualitative	Verbal rating mapped to number	"Excellent" = 1.0, "Good" = 0.75, "Fair" = 0.5, "Poor" = 0.25	When only expert judgment is available
Threshold (go/no-go)	Pass/fail gate applied before scoring	"TRL >= 6 required" -- below threshold is eliminated	Non-negotiable requirements

5.5 Weighted Score Calculation

For each alternative a across n criteria:

> *Key Equation:*
>
> *S(a) = Sum_c=1^n w_c * s(a,c)*
>
> *Where w_c is the normalised weight of criterion c and s(a,c) is the normalised score (0--1) of alternative a on criterion c.*
>
> *For "higher is better" criteria:*
>
> *s(a,c) = (v(a,c) - v_min(c))/(v_max(c) - v_min(c))*

> *For "lower is better" criteria:*
>
> *$s(a,c) = 1 - (v(a,c) - v\_min(c))/(v\_max(c) - v\_min(c))$*

5.6 Sensitivity Analysis

A trade study result is only meaningful if it is robust -- i.e., the ranking does not change with small perturbations in weights or scores. Sensitivity analysis tests this:

Method	Procedure	What It Reveals
Weight perturbation	Vary each weight by +/- 20%, re-normalise, re-score	Which criteria are "swing" criteria that could flip the outcome
Score perturbation	Vary each score by +/- 1 level, re-calculate	Which scores are the most uncertain and impactful
Threshold analysis	For each criterion, find the weight at which the 2nd-place alternative overtakes the 1st	How much the weights would need to change to reverse the decision

> *Industry Practice: For the Iridium NEXT constellation (66 operational + 6 on-orbit spares), the initial trade study for the constellation architecture compared Walker Delta, Walker Star, and hybrid configurations across 12 criteria including global coverage, revisit time, inter-satellite link geometry, launch cost, and orbital debris risk. Sensitivity analysis revealed that the ranking was robust for all weight perturbations up to +/- 30%, giving high confidence in the selected Walker Star configuration at 780 km altitude.*

Exercise: Practice the pairwise comparison method by weighting 4 criteria for a lunch restaurant choice: taste, cost, healthiness, speed. Then score 3 options. This builds the skill on a low-stakes example before applying it to mission design.

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6. Space vs Non-Space Alternatives (30 min)

6.1 The Most Important Trade Study

Before committing to building a satellite, teams must honestly evaluate whether existing solutions meet the need. This is the most important trade study in the mission lifecycle because it determines whether the entire project should proceed.

6.2 Alternative Categories

Category	Examples	Strengths	Weaknesses
Existing free data	Copernicus Sentinel-2, Landsat 8/9	Zero acquisition cost, proven quality, long archive	Fixed resolution/revisit, no tasking control
Commercial data purchase	Planet (SuperDove), Maxar (WorldView), ICEYE (SAR)	High resolution, fast tasking, no development risk	Ongoing cost, data rights limitations, vendor dependency
Aerial (drones/aircraf	Survey drones, P-3 Orion, Twin Otter	Very high resolution (< 1 m), flexible scheduling	Limited area coverage, weather dependent, regulatory constraints
Ground			

sensors

IoT networks, weather stations,

tide gauges

Continuous monitoring, low per-unit

cost	
	Point measurements only, no spatial

coverage

Dedicated satellite	New CubeSat, SmallSat, or microsatellite	Full control, custom instrument, IP ownership	High cost ($2--50M), development risk, 2--5 year schedule
Constellation	Multiple dedicated satellites	Global coverage, short revisit (< 1 day)	Much higher cost, operational complexity
Hosted payload	Payload on another operator's bus	Lower cost, shared bus risk	Compromised orbit, limited control, schedule dependency

6.3 Decision Criteria: When is Space Justified?

Space is likely NOT the right answer when:

- Existing free data (Sentinel-2, Landsat) meets resolution and revisit needs
- Coverage requirement is local (drones or aircraft are cheaper and higher resolution)
- Data rate is very low (ground sensors with cellular/satellite IoT backhaul suffice)
- Budget does not support minimum viable satellite cost (~$2M for a basic 3U CubeSat)
- Technology readiness is too low for the available schedule

Space is likely justified when:

- Global or wide-area coverage is needed simultaneously
- Persistent monitoring is required (24/7 or very frequent revisit < 5 days)
- User needs data ownership and control over acquisition scheduling
- No existing service provides the required measurement type (e.g., specific wavelength, polarisation)
- Regulatory or sovereignty requirements demand national control over the sensor

6.4 Constellation Sizing

For missions requiring short revisit, the number of satellites drives cost:

> *Key Equation: For a Walker Delta constellation at altitude h with half-swath angle η, the number of orbital planes P and satellites per plane S needed for revisit time T_{rev} is approximately:*
>
> $$P \times S \geq \frac{2\pi R_E}{v_{ground} \times T_{rev} \times \tan(\eta)}$$
>
> *Where $v_{ground} \approx 7.1$ km/s (ground track velocity at 500 km) and η is related to the instrument swath width W by $\eta = W / (2 R_E)$ for small angles.*
>
> *More practically, constellation cost scales sub-linearly due to learning curves:*
>
> $$C_{total} = C_1 \times \sum_{i=1}^{N} i^{\log_2(L)}$$
>
> *Where C_1 is the first unit cost, N is the number of satellites, and L is the learning curve factor (typically 0.90--0.95 for small batches, 0.85 for large batches like Planet's SuperDove).*

7. Documenting the Trade Decision (20 min)

Every trade study must produce an auditable decision record. NASA SEH Process 17 requires documentation of:

Element	Content	Purpose
Decision statement	What was decided	Clarity
Alternatives considered	All options evaluated, including rejected ones	Completeness
Criteria and weights	What mattered and how much	Transparency
Scoring rationale	Why each alternative received its score	Auditability
Result and recommendation	The selected alternative and its total score	Decision record
Sensitivity analysis	Robustness of the result	Confidence
Risks of selected option	What could go wrong with the chosen approach	Risk awareness
Responsible person and date	Who decided and when	Accountability

> *Industry Practice: The Canadian Hydrographic Service evaluated commercial SAR data (ICEYE, Capella) versus a dedicated satellite for Arctic maritime surveillance. The trade study documented 8 alternatives across 11 criteria, with weights set by a stakeholder panel including DND, Transport Canada, and CCG. The decision to use RCM data supplemented by commercial SAR tasking -- rather than building a new satellite -- saved an estimated $100M while meeting 90% of the maritime domain awareness requirements.*

Part 3: Concept of Operations (Friday -- 2 hours)

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8. Mission Architecture (30 min)

8.1 Three Segments of a Space Mission

Every space mission comprises three segments that must be designed together:

Segmen	Components	Key Design Drivers
Space Segment	Platform (bus): EPS, AOCS, OBC, thermal, structure, propulsion. Payload: instrument(s). Communications: TT&C + payload data link.	Mass, power, volume, orbit, lifetime
Ground Segment	Ground Operations: commanding, telemetry, orbit determination. Payload Data Centre: data reception, processing (L0-L1-L2-L3), archive.	Contact time, data volume, processing capacity
User Segment	Data products and services. APIs, portals, archives. Training and documentation.	Latency, format, accessibility, user capacity

8.2 Data Interfaces Between Segments

Interface	Direction	Band/Protocol	Content
TM/TC	Space <-> Ground Ops	S-band (typical)	Housekeeping telemetry, telecommands
Payload data	Space -> Data Centre	X-band or Ka-band	Science/imagery data (high volume)

Orbit/TLE

Ground Ops -> Data Centre

Network

Geolocation metadata for data products

Data products	Data Centre -> Users	Internet/API	Processed imagery (L2/L3), analytics

8.3 Interactive Architecture Diagram

SpaceCDF provides a drag-and-drop architecture diagram editor in the ConOps tab:

Symbol	Type	Represents
Satellite (blue)	satellite	Space segment (spacecraft + payload)
Ground Station (green)	groundStation	Ground receiving station with antenna
Processing (cyan)	processing	Data processing, MCC, archive
User (amber)	user	End user / data consumer
Sensor (orange)	sensor	Ground sensor, IoT device, in-situ instrument
GNSS/External (purple)	gnss	External system (GNSS, relay sat, other constellation)

Exercise: In the ConOps tab, build your mission architecture diagram. Add all segments, label all connections with data type and frequency band.

---

9. Mission Phases and Operational Modes (30 min)

9.1 Operational Mission Phases

Phase	Duration (typical CubeSat)	Activities	Key Risks
LEOP	1--3 days	Deployment, antenna deploy, first contact, initial health check	Deployment failure, tumbling, no contact
Commissioning	2--4 weeks	Subsystem checkout, calibration, first light, orbit determination	Anomalies, calibration issues
Nominal Ops	Months to years	Routine acquisition, downlink, orbit maintenance	Component degradation, anomalies
Extended Ops	Beyond design life	Continued operations with degraded performance	Solar array degradation, propellant depletion
Disposal	Days to months	Passivation, deorbit manoeuvre or natural decay	Failure of deorbit system

9.2 Operational Modes and Power Budgets

Each mode defines which subsystems are active, the pointing configuration, power demand, and data flow:

Mode	Subsystems Active	Pointing	Power (3U typical)	Data Flow
Safe	EPS, OBC, TTC (beacon), AOCS (coarse)	Sun-pointing	~1--2 W	Beacon only
Idle	EPS, OBC, AOCS (standby), TTC (beacon)	Inertial hold	~2 W	Health TM periodic
Science/Imaging	+ Payload, AOCS (fine pointing)	Nadir or target	~6 W	Instrument -> OBC storage
Downlink	+ TTC (full TX power)	Ground station track	~8 W	OBC -> TX -> GS
Eclipse	EPS (battery), OBC, TCS (heaters), AOCS	Inertial hold	~3 W (battery)	None
Orbit Maintenance	+ Propulsion	Thrust direction	~7 W	Manoeuvre TM

9.3 Duty Cycling and Power Analysis

CubeSats have limited power generation (7--25 W for a 3U with deployable panels). Not all modes can run simultaneously. The orbit timeline determines what can happen when:

Typical 95-minute orbit at 500 km SSO:

- 60 min sunlight, 35 min eclipse
- ~10 min imaging per orbit (~10% duty cycle)
- ~8 min downlink per pass (1--2 passes/day over a single ground station)
- ~42 min idle
- 35 min eclipse (battery-powered)

> *Key Equations -- Power Budget:*
>
> *Orbit-average power:*
>
> $P_{avg} = \sum_i P_{mode,i} \times DC_i$
>
> *Where DC_i is the duty cycle (fraction of orbit) for each mode.*
>
> *Solar array sizing:*
>
> $P_{SA} = P_{sunlight} + (P_{eclipse} \times t_{eclipse})/(t_{sunlight} \times \eta_{charge})$
>
> *Where $\eta_{charge} \approx 0.9$ (battery charge efficiency).*
>
> *Battery sizing:*
>
> $E_{battery} = (P_{eclipse} \times t_{eclipse})/(DoD \times \eta_{discharge})$
>
> *Where DoD is the maximum depth of discharge (typically 0.2--0.3 for long life, up to 0.5 for short missions) and $\eta_{discharge} \approx 0.95$.*

[Source: Wertz et al., SMAD4, Chapter 11; ECSS-E-ST-20C power budget methodology]

---

10. Data Flow Pipeline (20 min)

10.1 End-to-End Data Flow

```
Instrument -> Onboard Storage -> Downlink -> Ground Reception
  -> Processing (L0 -> L1 -> L2) -> Archive -> User Delivery
```

Stage	Key Parameter	Sized By
Data generation	GB/day	Payload data rate x imaging duty cycle
Onboard storage	GB	Must hold >= 1 day of data (2 x for margin)
Downlink per pass	GB/pass	Link data rate x contact time per pass
Processing	hours	Algorithm complexity, compute infrastructure
User delivery	hours	Archive API, network bandwidth

10.2 Data Budget Balance

> *Key Equation: For the system to be sustainable, daily downlink capacity must equal or exceed daily data generation:*
>
> *C_downlink = R_data x N_passes x t_contact x eta_protocol >= G_daily*
>
> *Where:*
>
> *- R_data = data link rate (Mbps)*
>
> *- N_passes = number of ground station passes per day*
>
> *- t_contact = average contact time per pass (seconds)*
>
> *- eta_protocol = protocol overhead factor (~0.8 for CCSDS framing)*
>
> *- G_daily = daily data generation (Mb)*
>
> *If C_downlink < G_daily, data accumulates on board and storage eventually fills. Solutions:*
>
> *- Increase R_data (higher TX power, higher frequency band, better antenna)*
>
> *- Increase N_passes (more ground stations, polar ground station for SSO)*
>
> *- Increase t_contact (higher altitude increases contact time but worsens resolution)*
>
> *- Decrease G_daily (lower duty cycle, on-board compression, selective downlink)*

SpaceCDF exercise: Check the Data Budget on the Dashboard. Does your design balance? If not, which parameter would you change first?

---

11. ConOps Tool Exercise (20 min)

1. Navigate to the ConOps tab in SpaceCDF
2. Review and edit the mission architecture diagram -- add all segments and connections
3. Edit the mission phases: adjust durations for your mission type
4. Review the operational modes: are the right modes defined? Adjust power values.
5. Check the data flow pipeline: does downlink capacity balance data generation?

Complete Worksheet 1.4, Parts E and F: Document your ConOps outline and calculate the orbit-average power budget.

1U Worked Example: UniSat-1

Trade Study: Why 1U instead of 2U or 3U for UniSat-1?

The UniSat-1 team must justify the choice of a 1U form factor for their MEMS magnetometer technology demonstration. This is a classic Process 17 (Decision Analysis) exercise.

Decision statement: "What CubeSat form factor best supports a MEMS magnetometer technology demonstration within the university's budget and schedule constraints?"

Alternatives:

Alternative	Mass Limit	Internal Volume	Typical Cost	Dev Time
1U	1.33 kg	~1000 cm^3	50--200 kEUR	6--12 months
2U	2.66 kg	~2000 cm^3	100--400 kEUR	12--18 months
3U	4.0 kg	~3000 cm^3	200--800 kEUR	18--24 months

Criteria and scoring:

Criterion	Weigh	1U	2U	3U	Rationale
Cost	0.35	1.0	0.5	0.2	University budget is 150 kEUR total
Schedule	0.25	1.0	0.6	0.3	Must launch within 12 months
Payload fits	0.20	0.8	1.0	1.0	MEMS sensor is 50 g, 0.2 W -- fits easily in 1U
Design simplicity	0.10	1.0	0.7	0.5	Smaller team, fewer subsystems
Data return	0.10	0.5	0.7	1.0	More volume allows better comms, but 9600 bps is sufficient for < 1 kbps payload
Weighted Total		0.90	0.63	0.41	

Result: 1U wins decisively. The MEMS magnetometer payload (50 g, 0.2 W, < 1 kbps) has no need for the extra volume, mass, or power that 2U/3U would provide. The additional cost and schedule of a larger bus are unjustified.

Sensitivity check: Even if cost weight drops from 0.35 to 0.15 (and schedule from 0.25 to 0.15, redistributing to data return), 1U still wins (0.82 vs 0.68 vs 0.51). The result is robust.

Key lesson: Do not over-design the bus for a simple payload. The 1U form factor imposes healthy constraints that force the team to focus on the mission objective rather than adding unnecessary capability.

Session Summary

Topic	Key Takeaway
Problem statement	Defines WHAT is needed, not HOW to solve it; answers what, who, impact, constraints
Stakeholder analysis	Identify all parties with legitimate interest; capture needs, constraints, priority, influence
Objectives	Must be specific, measurable, and traceable to stakeholder needs via MoE/MoP/TPM chain
Trade study structure	5 elements: decision, alternatives, criteria, weights, scores; must include sensitivity analysis
Space vs non-space Decision	Existing services may already meet the need -- evaluate honestly before committing to build

documentation

Every trade decision must have auditable rationale per Process 17

ConOps	Three segments (space, ground, user); operational modes drive power budget via duty cycling
Data pipeline	Daily downlink capacity must equal or exceed daily data generation

---

References

1. NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016
2. NASA, NPR 7123.1D -- Systems Engineering Processes and Requirements, 2020
3. ECSS, ECSS-E-ST-10C -- System Engineering General Requirements, 2009
4. Wertz, J.R. et al., Space Mission Engineering: The New SMAD (SMAD4), Microcosm Press, 2011
5. NASA SEH Appendix C -- How to Write a Good Requirement
6. NASA SEH Appendix S -- ConOps Outline
7. Saaty, T.L., "The Analytic Hierarchy Process", McGraw-Hill, 198090057-I)

Session 2.1: The System-V and Requirements Engineering

Duration: 2 hours

Prerequisites: Week 1 complete (mission need, stakeholder analysis, ConOps defined)

SpaceCDF Tab: Requirements

References

- NASA, Systems Engineering Handbook (NASA/SP-2016-6105 Rev 2), 2016, Ch. 4.2 & Appendix C
- ECSS, ECSS-E-ST-10-06C: Technical Requirements Specification, 2009, Sec. 5
- ECSS, ECSS-E-ST-10C Rev.1: Space Engineering -- System Engineering General Requirements, 2017, Sec. 5.2
- Wertz, Everett & Puschell, Space Mission Engineering: The New SMAD, 2011, Ch. 3--4
- INCOSE, Systems Engineering Handbook, 5th ed., 2023, Ch. 2.3.5

Learning Objectives

By the end of this session, participants will be able to:

1. Explain the System-V lifecycle model and how requirements flow down the left side while verification flows up the right side
2. Write requirements that satisfy the SMART criteria and express WHAT, not HOW
3. Classify requirements by type: functional, performance, interface, constraint, and environmental
4. Construct a three-level requirement hierarchy (mission, system, subsystem) with bidirectional traceability
5. Assign verification methods (ATID) to requirements and justify each choice
6. Use SpaceCDF's Requirements tab to generate, validate, and manage requirements

---

1. The System-V Model (20 min)

Teaching Notes

The System-V (or "Vee model") is the canonical systems engineering lifecycle. It connects the decomposition of requirements on the left branch to the integration and verification on the right branch. Every horizontal level on the V represents a matched pair: requirements at that level are verified by the corresponding test or analysis campaign at the same level on the right side.

[Source: NASA SEH Fig. 3-1; ECSS-E-ST-10C Rev.1 Fig. 5-1; INCOSE SEH 5th ed. Fig. 2.7]

The V-Model Structure

Figure: Stakeholder Needs

(See digital/markdown version for interactive diagram)

Key insight: The V is not a timeline -- it is a correspondence map. The left branch asks "what must be done?" at increasing levels of detail. The right branch asks "did we do it?" at corresponding levels. Each horizontal row is a matched pair.

Phase Mapping to the V

V-Level	Left Branch (Decomposition)	Right Branch (Verification)	NASA Phase
Top	Stakeholder needs & objectives	Operational validation (does the mission deliver value?)	Pre-A / E

Mission	Mission-level requirements	Mission-level V&V (commissioning, in-orbit checkout)	A / E
System	System-level requirements	System integration & test (end-to-end)	B / D
Subsyst	Subsystem requirements	Subsystem qualification & acceptance test	C / D
Bottom	Implementation (detailed design, build)	Component acceptance	C-D

Real Mission Example: Planet SuperDove

Planet's SuperDove constellation (2020--present) illustrates the V-model in commercial practice:

Level	Requirement (Left)	Verification (Right)
Stakeholder	"Daily global coverage at <5 m for agricultural analytics"	In-orbit validation: coverage maps, user feedback
Mission	"Revisit <= 1 day at +-60deg latitude"	Constellation coverage simulation confirmed
System	"GSD <= 3.7 m at 475 km SSO"	End-to-end imaging test from orbit
Subsystem	"Telescope aperture >= 90 mm, 8 spectral bands"	Instrument calibration on ground + in-orbit

[Source: Planet Labs, "Planet Imagery Product Specifications," 2023]

---

2. What is a Requirement? (15 min)

Teaching Notes

[Source: NASA SEH Sec. 4.2; ECSS-E-ST-10-06C Sec. 5.2]

A requirement is a formal, verifiable statement of what the system must do or how well it must perform. It is expressed as a "shall" statement with a measurable threshold.

The Shall Convention

Requirements use contractually precise language:

Keyword	Meaning	Contractual Force
"Shall"	Mandatory requirement -- must be met	Binding
"Should"	Goal or guideline -- desired but not mandatory	Advisory
"Will"	Statement of fact or intent -- not a requirement	Informational

Example:

> *"The system shall achieve a ground sample distance of 10 m or better at nadir from the operational orbit."*

This is testable: GSD can be computed from optical geometry and verified from calibration imagery.

Requirement Types

Type	Definition	Example
Functional	What the system must do (a capability)	"The system shall acquire multispectral imagery of the target area"
Performanc	How well it must do it (quantified)	"The system shall achieve GSD <= 10 m at nadir"
Interface	Boundary agreements between elements	"The EPS shall provide 5.0 V +/- 0.25 V regulated bus to all subsystems"
Constraint	Non-negotiable external limits	"The system total mass shall not exceed 6.0 kg"
Environmental	Survival conditions	"The system shall survive launch loads of 9 g axial and 4 g lateral"

[Source: NASA SEH Appendix C, Table C-1; ECSS-E-ST-10-06C Sec. 5.2.2]

Requirements vs. Design Choices

Requirement (WHAT)	Design Choice (HOW)
"The system shall achieve GSD <= 10 m"	"Use a 150 mm aperture telescope"
"The system shall survive launch loads"	"Use aluminium 7075-T6 structure"
"The system shall provide >= 3 dB link margin"	"Use S-band with 2 W transmitter"
"The system shall deorbit within 5 years of EOL"	"Include cold gas propulsion system"
"The system shall operate for >= 3 years"	"Use rad-tolerant components"

Key principle: Requirements constrain the solution space without specifying the solution. This preserves design freedom for trade studies.

---

3. The SMART Framework (20 min)

Teaching Notes

While NASA Appendix C provides a detailed quality checklist, the SMART acronym serves as a rapid quality screen:

Letter	Meaning	Test Question
S	Specific	Does it address exactly one concern? Is it unambiguous?
M	Measura	Does it have a numeric threshold with units? Can it be verified?
A	Achievab	Is it technically feasible with current or near-term technology?
R	Relevant	Does it trace to a stakeholder need or higher-level objective?
T	Traceable	Can you identify its parent (where it came from) and its children (how it will be verified)?

NASA SEH Appendix C: Full Quality Checklist

[Source: NASA SEH Appendix C -- "How to Write a Good Requirement"]

The full NASA checklist adds criteria beyond SMART:

1. Single requirement per statement -- no compound "and" requirements
2. Positive form -- state what the system SHALL DO, not what it shall NOT do
3. Active voice -- "The system shall..." not "It is required that..."

4. No implementation -- avoid naming specific hardware, software, or methods
5. Verifiable -- must be provable by analysis, test, inspection, or demonstration
6. No TBDs -- every threshold must have a value (even if it changes later)
7. Consistent -- no contradictions with other requirements in the set
8. Bounded -- tolerance or range specified where appropriate

Common Anti-Patterns

Anti-Pattern	Problem	Better Version
"The spacecraft shall operate at 500 km altitude"	Prescribes orbit (design choice)	"The system shall provide global coverage with revisit <= 5 days at +/-60 deg latitude"
"The spacecraft shall use triple-junction GaAs solar cells"	Prescribes technology	"The EPS shall generate >= 15 W EOL with <= 0.5 m^2 array area"
"The system shall use AX.25 protocol"	Prescribes implementation	"The TTC system shall provide reliable command reception with BER <= 10^-6"
"The spacecraft shall be a 3U CubeSat"	Prescribes form factor	"The system shall have total mass <= 6 kg and fit within a standard 3U deployer envelope"

Exception: Interface requirements ARE specific because they define contractual boundaries:

> *"The EPS shall provide 28 V +/- 2 V regulated bus voltage to all subsystems."*

This is acceptable because bus voltage is a negotiated interface agreement, not a unilateral design choice.

---

4. Requirement Hierarchy and Traceability (25 min)

Teaching Notes

Requirements exist at multiple levels, each decomposed from the level above. Every requirement must trace bidirectionally.

Hierarchy Structure

```
Stakeholder Need: "Timely agricultural monitoring for food security"
  |
  v derives
Mission Requirement (MR):
  MR-001: "The system shall provide multispectral imagery with
           GSD <= 10 m and revisit <= 5 days over target region"
  |
  v decomposes into
System Requirements (SR):
  SR-PL-001:   "The payload shall achieve GSD <= 10 m at nadir
                from the operational orbit"
  SR-AOCS-001: "The AOCS shall provide pointing accuracy <= 0.1 deg
                (3-sigma) during imaging"
```

```
  SR-LINK-001: "The comms system shall downlink >= 5 GB/day"
  |
  v derives
Subsystem Requirements (SSR):
  SSR-PL-001a:   "The telescope aperture shall be >= 80 mm"
  SSR-AOCS-001a: "The star tracker shall provide <= 5 arcsec
                  attitude knowledge (3-sigma)"
  SSR-LINK-001a: "The X-band transmitter shall provide >= 2 W RF power"
```

Level Definitions

Level	Scope	Written By	Verified By
Mission	What the mission achieves (MoE-derived)	Systems engineer + stakeholders	Operational validation
System	What the system must do (MoP-derived)	Systems engineer	System-level integration & test
Subsyst	What each subsystem must provide	Subsystem engineers	Subsystem qualification & acceptance
Component	What each component must meet	Component engineers	Component acceptance testing

Traceability Rules

Every requirement must trace in three directions:

1. Upward: to its parent requirement or objective (WHY does this exist?)
2. Downward: to child requirements or design parameters (HOW is it decomposed?)
3. Horizontally: to a verification method (HOW will it be confirmed?)

This three-axis traceability is captured in the Requirements Traceability Matrix (RTM).

[Source: NASA SEH Sec. 6.2 (Process 11: Requirements Management); ECSS-E-ST-10C Sec. 5.2]

Splitting Compound Requirements

A compound requirement that addresses multiple concerns must be split for independent verification:

Compound (Poor)	Split (Correct)
"The system shall provide 10 m GSD with 5-day revisit and 24 h latency"	MR-001: "GSD <= 10 m" + MR-002: "Revisit <= 5 days" + MR-003: "Latency <= 24 h"
"The EPS shall provide positive power margin in all modes including eclipse"	SR-PWR-001: "Positive margin in sunlit modes" + SR-PWR-002: "Positive margin in eclipse" + SR-PWR-003: "Battery DOD <= 30% in worst-case eclipse"

Rationale: Each split requirement can be verified independently. If GSD passes but revisit fails, the team knows exactly what to fix without ambiguity.

---

5. Verification Methods -- ATID (20 min)

Teaching Notes

[Source: ECSS-E-ST-10-02C Rev.1 Sec. 5.3; NASA SEH Sec. 5.3]

Every requirement must have an assigned verification method. The four standard methods (sometimes called ATRI or ATID) are:

Method	Code	What It Proves	When Used
Analysis	A	Requirement satisfied by mathematical model, simulation, or similarity	Phase B--C; when testing is impractical or cost-prohibitive
Test	T	Requirement satisfied by physical measurement	Phase C--D; when proof requires hardware in a controlled environment
Inspection	I	Requirement confirmed by visual or physical examination	Physical characteristics (dimensions, labels, markings, mass)
Demonstration	D	Requirement confirmed by operational exercise	Functional requirements proven by executing procedures

Verification Method Selection Guide

Requirement Type	Typical Method	Rationale
Mass <= X kg	I (weigh it)	Direct physical measurement
Pointing <= Y deg	A (simulation) + T (TVAC/HITL)	Analysis first, confirmed by test
Link margin >= 3 dB	A (link budget analysis)	Full test requires satellite in orbit
Survival at launch loads	T (vibration test)	Must physically prove structural integrity
Data latency <= 24 h	D (end-to-end ops exercise)	Pipeline is procedural -- demonstrate it works
Operating temperature range	T (thermal vacuum test)	Thermal environment must be simulated
Software FDIR logic	D (fault injection test)	Demonstrate correct autonomous response

Verification by Project Phase

Phase	Verification Activities
B	Analysis verification (models, simulations, budgets, trade studies)
C	Design verification (detailed analysis, breadboard testing, qualification)
D	Acceptance testing (flight hardware), system integration, environmental test
E	In-orbit validation (commissioning, calibration, operational checkout)

---

6. Worked Example: 3U Earth Observation CubeSat (10 min)

> *Worked Example -- Deriving Requirements for a 3U EO CubeSat*
>
> *Stakeholder need: "Monitor crop health across the Canadian prairies with weekly updates."*
>
> *Step 1 -- Mission requirement:*

> *MR-001: "The system shall provide NDVI-capable imagery with GSD <= 10 m and revisit <= 7 days over 49--54 deg N, 100--115 deg W."*
>
> *Step 2 -- System requirements (decomposed):*
>
> *- SR-PL-001: "The payload shall acquire imagery in at least RED (630--690 nm) and NIR (760--900 nm) bands."*
>
> *- SR-AOCS-001: "The AOCS shall provide pointing accuracy <= 0.5 deg (3-sigma) during imaging."*
>
> *- SR-LINK-001: "The comms system shall downlink >= 2 GB per day to a Canadian ground station."*
>
> *- SR-PWR-001: "The EPS shall provide positive power margin in all operational modes."*
>
> *Step 3 -- Subsystem requirements (derived):*
>
> *- SSR-PL-001a: "The telescope aperture shall be >= 80 mm." [Derived from GSD + altitude]*
>
> *- SSR-AOCS-001a: "The star tracker shall provide <= 30 arcsec attitude knowledge." [Derived from pointing budget]*
>
> *- SSR-LINK-001a: "The X-band transmitter shall provide >= 2 W RF output power." [Derived from link budget]*
>
> *Step 4 -- Verification assignment:*
>
> | Requirement | Method | Phase | Rationale |
> |-------------|--------|-------|-----------|
> | MR-001 (GSD + revisit) | A + D | B + E | Analysis during design; demonstrated from orbit |
> | SR-PL-001 (spectral bands) | T | D | Spectral calibration in lab |
> | SR-AOCS-001 (pointing) | A + T | B + D | Simulation then hardware-in-loop test |
> | SR-LINK-001 (downlink) | A | B | Link budget analysis |
> | SSR-PL-001a (aperture) | I | D | Physical measurement |

1U Worked Example: UniSat-1

> *Worked Example -- Deriving Requirements for a 1U Technology Demonstrator*
>
> *Stakeholder need: "Demonstrate the feasibility of a MEMS-based magnetometer for space weather monitoring in LEO."*
>
> *Step 1 -- Mission requirements:*
>
> | ID | Requirement | Type | Rationale |

|----|------------|------|-----------|
MR-001	"The system shall measure the local magnetic field vector with resolution <= 10 nT at 1 Hz sampling rate for a minimum of 6 months."	Performance	Minimum science return for technology validation
MR-002	"The system total mass shall not exceed 1.33 kg."	Constraint	CDS Rev 14, 1U mass limit
MR-003	"The system shall downlink at least 1 MB of magnetometer data per day."	Performance	Sufficient for statistical analysis of sensor performance
MR-004	"The system shall operate for a minimum of 6 months in LEO."	Performance	Minimum mission duration for seasonal variation coverage

Step 2 -- System requirements (decomposed):

- SR-PWR-001: "The EPS shall provide orbit-average power >= 2 W to all subsystems."
- SR-PWR-002: "The battery shall provide >= 10 Wh capacity."
- SR-LINK-001: "The comms system shall provide a downlink data rate >= 9600 bps."
- SR-LINK-002: "The comms system shall achieve link margin >= 3 dB at 10 deg minimum elevation."
- SR-OBC-001: "The OBC shall consume <= 0.5 W average power."
- SR-STR-001: "The structure shall comply with CDS Rev 14, 1U envelope (100 x 100 x 113.5 mm)."

Step 3 -- Subsystem requirements (derived):

- SSR-PWR-001a: "Body-mounted solar cells shall generate >= 2 W orbit-average power accounting for eclipse and geometry." [Derived from power budget]
- SSR-LINK-001a: "The UHF transmitter shall provide >= 0.5 W RF output at 437 MHz." [Derived from link budget]

Step 4 -- Verification assignment:

Requirement	Method	Phase	Rationale
MR-001 (magnetometer performance)	T + D	D + E	Ground calibration then in-orbit demonstration
MR-002 (mass <= 1.33 kg)	I	D	Weigh the flight unit
SR-PWR-001 (orbit-avg power)	A	B	Power budget analysis
SR-LINK-001 (9600 bps downlink)	A + D	B + E	Link budget analysis; demonstrated from orbit
SR-STR-001 (CDS compliance)	I	D	Physical measurement against CDS template

Key difference from 3U: The 1U requirement set is much smaller (~15--20 requirements vs ~40--60 for a 3U EO mission). There are no pointing requirements, no imaging requirements, and no propulsion requirements. This makes the verification campaign significantly simpler and cheaper.

7. SpaceCDF Exercise (30 min)

Instructions

1. Navigate to the Requirements tab in SpaceCDF
2. Click "Generate from Objectives" -- the AI agent generates SMART requirements from your mission objectives and ConOps
3. For each generated requirement:
- Review the SMART validation badges (green = pass, amber = warning, red = fail)
- Check: does it say WHAT not HOW? Is it a single concern?
- Accept, Edit, or Reject each one
4. Use the Level filter (Mission / System / Subsystem) to view by hierarchy level
5. Use the Type filter (Functional / Performance / Interface / Constraint) to classify
6. Navigate to the V&V Matrix sub-panel to assign verification methods (A/T/I/D) to each accepted requirement

Exercise Tasks

1. Generate requirements for your team's mission
2. Identify at least one requirement that specifies HOW (implementation) -- rewrite it as WHAT
3. Split any compound requirements into individual testable statements
4. Classify 5 requirements by type (functional, performance, interface, constraint, environmental)
5. For 5 key requirements, assign verification method (A/T/I/D) and target phase (B/C/D/E)
6. Complete Worksheet 2.1

Key Equations Reference

> *There are no equations in requirements engineering per se, but the following relationship governs traceability completeness:*
>
> *$Coverage = \fracN_verified requirementsN_total requirements x 100%$*
>
> *Target: 100% coverage -- every requirement must have at least one assigned verification method and one parent traceability link.*

Session Summary

Topic	Key Takeaway
System-V model	Left branch decomposes requirements; right branch integrates and verifies at matching levels
Shall statements	Formal, verifiable, single-concern statements
WHAT not HOW	Requirements preserve design freedom; design choices come later in trade studies
SMART	Specific, Measurable, Achievable, Relevant, Traceable
Requirement types	Functional, performance, interface, constraint, environmental
Hierarchy	Mission -> System -> Subsystem with bidirectional traceability (up, down, horizontal)
Splitting	One concern per requirement for independent verification
ATID	Analysis, Test, Inspection, Demonstration -- assigned per requirement per phase

Session 2.2: Functional Decomposition and Allocation

Duration: 2 hours

Prerequisites: Session 2.1 (requirements defined and validated)

SpaceCDF Tab: Functions

References

- NASA, Systems Engineering Handbook (NASA/SP-2016-6105 Rev 2), 2016, Sec. 4.3 (Process 3: Logical Decomposition)
- ECSS, ECSS-E-ST-10C Rev.1: System Engineering General Requirements, 2017, Sec. 5.3
- INCOSE, Systems Engineering Handbook, 5th ed., 2023, Ch. 2.3.5.3
- Wertz, Everett & Puschell, Space Mission Engineering: The New SMAD, 2011, Ch. 2
- Blanchard & Fabrycky, Systems Engineering and Analysis, 5th ed., 2010, Ch. 3

Learning Objectives

By the end of this session, participants will be able to:

1. Explain the purpose of functional decomposition in bridging requirements to design
2. Decompose mission objectives into a hierarchical function tree (3+ levels)
3. Categorise functions by type (observe, communicate, navigate, point, power, protect, store, process, support)
4. Allocate functions to subsystem domains, including multi-allocation for shared responsibilities
5. Derive requirements from functional analysis
6. Identify and resolve coverage gaps (leaf functions without linked requirements)
7. Use SpaceCDF's function tree editor to build and validate a function architecture

1. What is Functional Decomposition? (20 min)

Teaching Notes

[Source: NASA SEH Sec. 4.3 -- Process 3: Logical Decomposition]

Functional decomposition answers: "What must the system DO to satisfy the requirements?"

It creates a bridge between requirements (WHAT the system must achieve) and physical design (HOW it will be built). Functions describe actions the system must perform, without specifying the physical hardware or software that will perform them.

The Decomposition Flow

Figure: Objectives & Requirements

(See digital/markdown version for interactive diagram)

Function Types

Every spacecraft function falls into one of these categories:

Type	Verb	Description	Example
Observe	Sense/measure	Acquire data from the environment	Capture multispectral imagery, receive AIS signals
Communicate	Transfer	Move information between elements	Downlink science data, uplink telecommands
Navigate	Determine/control position	Know or change the orbit	Perform orbit maintenance manoeuvre
Point	Orient	Control spacecraft attitude	Nadir pointing during imaging, target tracking
Power	Generate/store/distribute	Provide electrical energy	Solar power generation, battery charge management
Protect	Maintain environment	Keep components within limits	Thermal control, radiation shielding
Store	Record	Buffer data for later use	Onboard solid-state data recording
Process	Transform	Convert data between forms	On-board image compression, L0 framing
Support	Provide structure	Mechanical integrity	Survive launch loads, deploy mechanisms

Universal Functions

Regardless of mission type, every spacecraft must perform these five functions:

1. Generate electrical power (solar, RTG, battery)
2. Maintain thermal environment (passive/active thermal control)

3. Survive the launch environment (structural loads, vibration, shock)
4. Communicate with ground for TTC (telemetry, tracking, command)
5. Dispose of the spacecraft at end of life (deorbit, graveyard, passivation)

SpaceCDF auto-generates these universal functions for every mission.

---

2. Mission-Type-Specific Function Trees (20 min)

Teaching Notes

Different mission types produce fundamentally different primary function trees. The universal functions remain the same; it is the mission-specific branch that differentiates.

Earth Observation (Optical/SAR)

```
F-001: Acquire imagery of target area
  +-- F-002: Point instrument at target
  +-- F-003: Capture image data (expose detector / SAR pulse)
  +-- F-004: Store acquired data onboard
  +-- F-005: Downlink data to ground station
```

Real example -- Planet SuperDove: F-001 decomposes into 8-band pushbroom acquisition, on-board radiometric correction, lossless compression, and X-band downlink. Each sub-function traces to specific SuperDove requirements (3.7 m GSD, 8 spectral bands, 200 Mbps downlink).

Communications Relay (Store-and-Forward or Bent-Pipe)

```
F-001: Relay communications between users
  +-- F-002: Receive uplink signal from user terminal
  +-- F-003: Process and route data (store-and-forward or bent-pipe)
  +-- F-004: Transmit downlink signal to destination terminal or gateway
```

Real example -- Astrocast (IoT): F-001 decomposes into L-band receive (from IoT devices), on-board message deduplication and store-and-forward, and UHF/S-band downlink to gateway stations. Each message is <= 160 bytes.

[Source: Astrocast, "Astrocast Network Overview," astrocast.com]

AIS/IoT Receiver (Passive)

```
F-001: Receive and process signals of interest
  +-- F-002: Receive AIS/IoT signals (passive -- no uplink transmission)
  +-- F-003: Decode and validate messages
```

```
    +-- F-004: Store processed data onboard
    +-- F-005: Downlink to ground for distribution
```

Ground Segment Functions

Missions that include ground-side processing need ground-domain functions:

```
  F-G01: Receive satellite data at ground station
  F-G02: Process data pipeline (L0 -> L1 -> L2 products)
  F-G03: Archive and distribute data products to users
  F-G04: Operate mission control centre (planning, commanding, monitoring)
```

SpaceCDF supports allocation to ground domains: ground_station, ground_processing, ground_sensor.

---

3. Allocation to Subsystems (25 min)

Teaching Notes

Each function must be allocated to one or more responsible subsystem domains. This allocation defines system boundaries and, critically, identifies interfaces wherever a function is shared.

Allocation Rules

Function	Typical Allocation	Multi-allocation?
Acquire imagery	Payload	No (single owner)
Point at target	AOCS	No
Store data	OBC/Data Handling	No
Downlink data	Comms + AOCS	Yes -- comms for RF chain, AOCS for antenna pointing
Generate power	EPS	No
Maintain thermal	Thermal	No
Relay communications	Payload + Comms	Yes -- boundary depends on architecture
Survive launch	Structure	No
Dispose at EOL	Propulsion (or Ops)	No (or multi if drag-sail)

Multi-Allocation and Interface Identification

When a function is allocated to more than one subsystem, it creates an interface that must be explicitly managed.

Example -- "Downlink data to ground station":

Responsible Subsystem	Contribution	Interface Created

Comms (Link)	Transponder, antenna, modulation, RF chain	Comms <-> AOCS (antenna pointing)
AOCS	Antenna pointing towards ground station during pass	Comms <-> Data Handling (packet routing)
Data Handling	Data packaging, prioritisation, CCSDS framing	Data Handling <-> AOCS (pass scheduling)

SpaceCDF supports multi-allocation by entering comma-separated domains.

Discussion prompt: For each multi-allocated function in your design, where should the system boundary be drawn? Who "owns" the function, and who is a "contributor"?

Derived Requirements from Functions

Each allocated function generates derived requirements -- requirements not explicitly stated by stakeholders but necessary for the function to work:

Function	Derived Requirement	Derivation Logic
"Point instrument at target"	"AOCS shall provide <= 0.1 deg pointing accuracy during imaging"	From GSD requirement + optical geometry
"Store data onboard"	"OBDH shall provide >= 32 GB solid-state storage"	From data rate x orbit period x missed-pass margin
"Downlink within daily contact window"	"TX shall provide >= 50 Mbps effective data rate"	From daily data volume / total contact time
"Generate power in all modes"	"SA shall produce >= 15 W EOL"	From worst-case sunlit power demand + recharge

[Source: NASA SEH Sec. 4.3.3 -- derived requirements from functional allocation]

---

4. Performance Criteria and Coverage Analysis (15 min)

Teaching Notes

Each leaf function (a function with no sub-functions) must have performance criteria -- quantitative thresholds that define "how well" the function must be performed:

Function	Performance Criteria
Acquire imagery	GSD <= 10 m; SNR >= 100:1; >= 4 spectral bands
Point at target	Accuracy <= 0.1 deg; stability <= 0.01 deg/s; slew rate >= 1 deg/s
Downlink data	Link margin >= 3 dB; daily throughput >= 5 GB
Generate power	Positive margin in all modes; battery DOD <= 30%
Maintain thermal env	All components within operating range with >= 5 degC margin

Performance criteria form the quantitative basis for subsystem-level requirements.

Coverage Analysis

Every leaf function must trace to at least one requirement. If it does not, there is a coverage gap -- the function is defined but has no verification path.

SpaceCDF shows coverage status with badges:

- Green badge: Function has linked requirements (covered)
- Amber badge: "No requirements" -- coverage gap detected
- Red badge: Function conflicts with existing requirements

Real Mission Example: CAPSTONE Coverage Gap

NASA's CAPSTONE mission (2022, Advanced Space) experienced a coverage gap during development: the "maintain attitude during trajectory correction manoeuvre" function initially had no formal pointing requirement during burns. This gap was identified during functional review and led to the derivation of SSR-AOCS-007: "The AOCS shall maintain pointing accuracy <= 2 deg during all propulsive manoeuvres."

[Source: Advanced Space, "CAPSTONE Mission Overview," 2022]

---

5. SpaceCDF Function Tree Exercise (40 min)

Instructions

1. Navigate to the Functions tab in SpaceCDF
2. The tool auto-generates functions based on your selected mission type
3. Review the generated tree: Are the functions appropriate for YOUR mission?
- For comms missions: should show relay/receive/transmit (not multispectral imagery)
- For EO missions: should show acquire/point/store/downlink
- For technology demos: should show demonstrate/characterise/report
4. Edit functions:
- Click edit to change name, domain allocation, or performance criteria
- Click +sub to add sub-functions (decompose further)
- Click x to remove inappropriate functions
5. Add ground segment functions if your mission includes ground processing
6. Check multi-allocation: For any function allocated to multiple domains, verify the interface is captured
7. Coverage check: Are there any leaf functions (amber badge) without linked requirements? Derive requirements for them.

Exercise Tasks

1. Build a function tree with at least 3 levels of decomposition
2. For each leaf function, write one derived requirement with a measurable threshold
3. Create a function-to-requirement traceability table (minimum 8 entries)
4. Identify at least one multi-allocated function and define the interface boundary
5. Complete Worksheet 2.2

Worked Example: 3U EO CubeSat Function Tree

> *Function Tree for Agricultural Monitoring CubeSat*
>
> `
>
> F-001: Acquire multispectral imagery
>
> +-- F-002: Point telescope at target area (AOCS)
>
> +-- F-003: Expose detector and capture image frames (Payload)
>
> +-- F-004: Compress and store image data (OBC)
>
> +-- F-005: Downlink stored data to ground station (Comms + AOCS)
>
> F-010: Generate electrical power (EPS)
>
> +-- F-011: Convert solar energy to electrical (SA)
>
> +-- F-012: Store energy for eclipse (Battery)
>
> +-- F-013: Regulate and distribute power (EPS board)
>
> F-020: Maintain thermal environment (Thermal)
>
> +-- F-021: Reject waste heat from electronics (Radiator)
>
> +-- F-022: Maintain battery temperature during eclipse (Heater)
>
> F-030: Survive launch environment (Structure)
>
> F-040: Communicate with ground for TTC (Comms)
>
> F-050: Dispose at end of life (Propulsion/Ops)
>
> `
>
> Derived requirements from F-002 (Point telescope at target):
>
> - SSR-AOCS-001: "The AOCS shall achieve <= 0.5 deg pointing accuracy during imaging mode"
>
> - SSR-AOCS-002: "The AOCS shall achieve pointing stability <= 0.01 deg/s during imaging"
>
> Multi-allocation for F-005 (Downlink data):
>
> - Comms subsystem: owns RF chain (transponder, antenna, modulation)
>
> - AOCS subsystem: provides antenna pointing towards ground station
>
> - Interface: AOCS must receive pass-schedule triggers from OBC to initiate slew to ground station pointing

Session Summary

Topic	Key Takeaway
Functional decomposition	Bridges requirements (WHAT) to design (HOW) by identifying system actions
Function types	Observe, Communicate, Navigate, Point, Power, Protect, Store, Process, Support
Universal functions	Power, thermal, launch survival, TTC, disposal -- every spacecraft needs all five
Mission-specific trees	EO: acquire/point/store/downlink; Comms: receive/route/transmit; AIS: receive/decode/store
Allocation	Each function assigned to subsystem domain(s); multi-allocation creates interfaces
Derived requirements	Functions generate new requirements not stated by stakeholders
Coverage analysis	Every leaf function must trace to at least one requirement -- no gaps allowed

Session 2.3: Orbit Selection and Mission Architecture

Duration: 2 hours

Prerequisites: Sessions 2.1--2.2 (requirements and functions defined)

SpaceCDF Tabs: Mission Architecture, Orbit Trade Advisor

References

- Vallado, Fundamentals of Astrodynamics and Applications, 4th ed., 2013, Ch. 2--6
- Wertz, Everett & Puschell, Space Mission Engineering: The New SMAD, 2011, Ch. 5--7
- Curtis, Orbital Mechanics for Engineering Students, 4th ed., 2020, Ch. 2--4
- NASA, Systems Engineering Handbook, 2016, Sec. 4.4 (Process 4: Design Solution Definition)
- ECSS, ECSS-U-AS-10C Rev.2: Space Debris Mitigation Requirements, 2023
- IADC, Space Debris Mitigation Guidelines, IADC-02-01 Rev 3, 2021
- FCC, Report and Order FCC 22-74: Space Innovation, 2022

Learning Objectives

By the end of this session, participants will be able to:

1. Define and compute the six Keplerian orbital elements
2. Calculate orbital period, velocity, and eclipse fraction for circular orbits
3. Compute the Sun-synchronous inclination using J2 precession
4. Evaluate orbit trade-offs across altitude, coverage, lifetime, radiation, and debris compliance
5. Apply the Hohmann transfer D V equations for orbit raising and deorbit

6. Use SpaceCDF's orbit trade advisor with mission-appropriate scoring weights

---

1. Keplerian Orbital Elements (25 min)

Teaching Notes

Six parameters (the classical orbital elements) fully describe the size, shape, and orientation of an orbit, plus the satellite's position on it.

[Source: Vallado, Ch. 2; Curtis, Ch. 2]

Element	Symbol	Physical Meaning	Typical Range
Semi-major axis	a	Size of the orbit	6571--42164 km (LEO to GEO)
Eccentricity	e	Shape (0 = circle, 0 < e < 1 = ellipse)	0--0.001 for LEO circular
Inclination	i	Tilt of orbit plane from equatorial plane	0--180 deg
RAAN	\Omega	Orientation of ascending node in equatorial plane	0--360 deg
Argument of perigee	omega	Orientation of perigee within orbit plane	0--360 deg
True anomaly	\nu	Satellite position along the orbit	0--360 deg

For circular LEO missions (the most common CubeSat orbit type), the key design parameters reduce to three: altitude (h), inclination (i), and LTAN (local time of ascending node, for Sun-synchronous orbits).

Figure: Earth

(See digital/markdown version for interactive diagram)

Key Equations

> *Key Equations -- Orbital Mechanics Fundamentals*
>
> *Orbital period (circular orbit):*
>
> *$T = 2pi sqrt(a^3)/(u)$*
>
> *where a = R_E + h is the semi-major axis (m), u = 3.986 x 10^14 m^3/s^2 is Earth's gravitational parameter, and R_E = 6371 km.*
>
> *Orbital velocity (circular):*

> *$v = sqrt(u)/(a)$*
>
> *Eclipse fraction (maximum, circular orbit, cylindrical shadow):*
>
> *$f_eclipse = (1)/(pi) \arccos(\fracsqrta^2 - R_E^2a)$*
>
> *Sun-synchronous inclination (J2-driven RAAN precession = 360 deg/year):*
>
> *$cos(i) = -\frac2 \dot\Omega_req a^7/23 R_E^2 J_2 sqrtu$*
>
> *where J_2 = 1.0826 x 10^-3, \dot\Omega_req = (2pi)/(365.25 x 86400) rad/s.*

---

2. Worked Examples: Orbital Computations (15 min)

> *Worked Example -- 500 km Circular LEO*
>
> *Given: h = 500 km, circular orbit.*
>
> *Step 1 -- Semi-major axis:*
>
> *a = 6371 + 500 = 6871 km = 6.871 x 10^6 m*
>
> *Step 2 -- Orbital period:*
>
> *T = 2pi sqrt((6.871 x 10^6)^3)/(3.986 x 10^14) = 2pi sqrt8.137 x 10^5 = 2pi x 5668 = 5669 s ~= 94.5 min*
>
> *Step 3 -- Orbital velocity:*
>
> *v = sqrt(3.986 x 10^14)/(6.871 x 10^6) = sqrt5.802 x 10^7 = 7617 m/s ~= 7.62 km/s*
>
> *Step 4 -- Eclipse fraction (maximum):*
>
> *f = (1)/(pi) \arccos(\fracsqrt6871^2 - 6371^26871) = (1)/(pi) \arccos((2594)/(6871)) = (1)/(pi) \arccos(0.3774) = (1)/(pi) x 67.8\degree = 0.376 (37.6%)*
>
> *Step 5 -- Eclipse and sunlight duration:*
>
> *t_eclipse = 94.5 x 0.376 = 35.5 min; t_sun = 94.5 - 35.5 = 59.0 min*
>
> *Step 6 -- Sun-synchronous inclination:*
>
> *cos(i) = -\frac2 x 1.991 x 10^-7 x (6.871 x 10^6)^3.53 x (6.371 x 10^6)^2 x 1.0826 x 10^-3 x sqrt3.986 x 10^14*

> *Numerator: ~= -1.301 x 10^17; Denominator: ~= 1.006 x 10^18*
>
> *cos(i) ~= -0.1293 \Rightarrow i ~= 97.4 deg*

---

3. Orbit Selection Trade-Offs (25 min)

Teaching Notes

The orbit is the single most impactful early design decision. It cascades to every subsystem.

[Source: SMAD, Ch. 7; Wertz et al., "Reducing Space Mission Cost," Ch. 3]

Parameter	Lower Altitude (300--400 km)	Higher Altitude (600--800 km)
GSD	Better (shorter range to target)	Worse (longer range)
Orbital lifetime	Short (1--5 years, natural decay)	Long (25--100+ years)
Debris compliance	Easy (FCC 5-year rule met naturally)	May require active deorbit propulsion
Radiation (TID)	Lower (2--5 krad/yr)	Higher (10--20 krad/yr above 700 km)
Launch cost	Lower D V to orbit	Slightly higher
Link budget	Better (shorter slant range)	Worse (longer path loss)
Coverage/swath	Narrower per pass	Wider per pass
Atmospheric drag	Significant (limits lifetime)	Negligible above ~700 km
Eclipse fraction	~35--38%	~33--35%

Orbit Types and Applications

Orbit	Altitude	Inclination	Best For	Real Example
LEO	300--600 km	Any	Technology demos, ISS deployment	Many CubeSats
SSO	400--800 km	97--99 deg	EO (consistent solar illumination)	Planet SuperDove (475 km)
ISS orbit	~410 km	51.6 deg	ISS-deployed CubeSats	NanoRacks deployments
MEO	2000--20200 km	Various	Navigation	GPS (20200 km)
GEO	35786 km	0 deg	Comms, weather	Anik F2 (Telesat)
HEO (Molniya)	500--40000 km	63.4 deg	High-latitude coverage	Meridian (Russia)
Lunar NRHO	1500--70000 km	Lunar	Cislunar operations	CAPSTONE (NASA/Advanced Space)

Debris Compliance Rules

Rule	Requirement	Applies To
IADC guideline (2021)	Post-mission disposal within 25 years	International (voluntary but expected)
FCC rule (2024+)	Post-mission disposal within 5 years	All FCC-licensed LEO satellites
ECSS-U-AS-10C Rev.2	Compliance with IADC + ESA Zero Debris Charter	ESA missions
ISED (Canada)	Currently 25-year rule; tightening under review	Canadian-licensed satellites

Critical altitude boundaries for FCC 5-year compliance:

Altitude	Natural Lifetime	FCC Compliant?	Action Needed
< 450 km	< 5 years	Yes	None (natural decay)
450--550 km	5--20 years	Marginal	Drag augmentation may suffice
550--650 km	20--50 years	No	Active deorbit (propulsion or drag sail)
> 700 km	> 100 years	No	Propulsion mandatory; ESA Zero Debris zone

[Source: IADC-02-01 Rev 3, Sec. 5.3.2; FCC 22-74; ECSS-U-AS-10C Rev.2]

Perturbation Effects

Perturbation	Cause	Effect on Orbit	Design Impact
J_2 (Earth oblateness)	Equatorial bulge	RAAN precession, argument of perigee drift	Enables Sun-synchronous; frozen orbits at omega = 90\degree
Atmospheric drag	Residual atmosphere	Semi-major axis decay, eventual re-entry	Limits lifetime below ~600 km; drives propulsion need
Solar radiation pressure	Photon momentum	Eccentricity oscillations, orbit perturbation	Significant for large A/m ratio (drag sails, large SA)
Third-body (Moon/Sun)	Gravitational pull	Long-period oscillations in e, i	Significant for GEO and HEO; negligible for LEO
Magnetic field	Lorentz force on charged S/C	Very small drag-like effect	Negligible for most missions

---

4. Ground Coverage and Revisit (15 min)

Teaching Notes

> *Key Equations -- Ground Coverage*
>
> *Swath width (nadir-pointing sensor with half-cone angle theta):*
>
> *$W_swath = 2h tan(theta)$*
>
> *Ground track spacing (for a single satellite in LEO):*
>
> *The Earth rotates ~22.9 deg per orbit (for T ~= 95 min). At the equator, this corresponds to:*
>
> *$D_lon = \frac360\degreeT_sidereal x T_orbit ~= 22.9\degree ~= 2550 km (at equator)$*
>
> *Revisit time (approximate, single satellite):*
>
> *$t_revisit ~= \fracD_lonW_swath x T_orbit$*

> *Constellation revisit (N identical satellites in same plane):*
>
> *$t_revisit,constellation ~= \fract_revisit,singleN$*

Example -- SuperDove constellation:

Planet operates ~200 SuperDove satellites. With a ~24 km swath at 475 km and multiple orbital planes, the constellation achieves daily global revisit -- a dramatic improvement over a single satellite's ~7-day revisit.

Ground Station Contact Geometry

Contact time per pass depends on the minimum elevation angle eps_\min (typically 5--10 deg):

$t_contact ~= (T)/(pi) \arccos((cos(rho))/(cos(eps_\min))) - geometric correction$

Simplified rule of thumb for LEO at 500 km, eps_\min = 10\degree:

- Maximum pass duration: ~10 min (overhead pass)
- Average pass duration: ~6--7 min
- Passes per day (mid-latitude station): ~4--6

---

5. Hohmann Transfer and Deorbit (10 min)

Teaching Notes

> *Key Equations -- Hohmann Transfer*
>
> *The minimum-energy transfer between two circular orbits of radii r_1 and r_2:*
>
> *$D V_1 = sqrt(u)/(r_1) (sqrt(2r_2)/(r_1 + r_2) - 1)$*
>
> *$D V_2 = sqrt(u)/(r_2) (1 - sqrt(2r_1)/(r_1 + r_2))$*
>
> *$D V_total = |D V_1| + |D V_2|$*
>
> *Worked Example -- Deorbit from 600 km to 200 km (re-entry perigee)*
>
> *r_1 = 6971 km, r_2 = 6571 km*
>
> *D V_1 = sqrt(3.986 x 10^5)/(6971) (sqrt(2 x 6571)/(6971 + 6571) - 1)*
>
> *= 7.561 x (sqrt0.9705 - 1) = 7.561 x (-0.01498) = -0.1133 km/s*

> *D V_total ~= 113 m/s (only the first burn needed to lower perigee)*

This deorbit D V is a critical input to the propulsion sizing in Session 3.4.

---

6. Radiation Environment (10 min)

Teaching Notes

[Source: ECSS-E-ST-10-04C Rev.1; SMAD, Ch. 8.1]

Total Ionising Dose (TID) by Orbit

Orbit	TID (krad/year behind 2mm Al)	Electronics Class
ISS (410 km, 51.6 deg)	2--5	Commercial COTS OK
SSO 500 km	5--10	Commercial / rad-tolerant
SSO 800 km	10--20	Rad-tolerant recommended
MEO 2000 km	50--100	Rad-hard required
GEO 35786 km	10--20	Rad-hard required

Rule of thumb: Below 600 km in LEO, commercial COTS electronics can survive 3-year missions with modest shielding (2--3 mm Al equivalent). Above 600 km, radiation becomes a significant design driver and component cost escalates.

South Atlantic Anomaly (SAA)

At 200--600 km altitude over South America, trapped protons from the inner Van Allen belt dip to lower altitudes. The SAA causes:

- Single-event upsets (SEU) in memory
- Single-event latch-ups (SEL) in CMOS
- Increased background noise in optical detectors

Mitigation: error-correcting memory (EDAC), watchdog timers, latch-up protection circuits.

1U Worked Example: UniSat-1

Orbit Selection: ISS Orbit (400 km, 51.6 deg) for Rideshare

UniSat-1 selects the ISS orbit not by optimisation but by access: deployment from the ISS via NanoRacks or a similar deployer is the cheapest and most accessible launch opportunity for a university 1U CubeSat.

> *Worked Example -- Orbital Parameters at 400 km, 51.6 deg*
>
> *Step 1 -- Semi-major axis:*
>
> *a = 6371 + 400 = 6771 km = 6.771 x 10^6 m*
>
> *Step 2 -- Orbital period:*
>
> *T = 2pi sqrt((6.771 x 10^6)^3)/(3.986 x 10^14) = 2pi x 5564 = 5565 s ~= 92.4 min*
>
> *Step 3 -- Orbital velocity:*
>
> *v = sqrt(3.986 x 10^14)/(6.771 x 10^6) = 7672 m/s ~= 7.67 km/s*
>
> *Step 4 -- Eclipse fraction (maximum):*
>
> *f = (1)/(pi) \arccos(\fracsqrt6771^2 - 6371^26771) = (1)/(pi) \arccos(0.3423) = (1)/(pi) x 70.0\degree ~= 0.389 (38.9%)*
>
> *Step 5 -- Eclipse and sunlight duration:*
>
> *t_eclipse = 92.4 x 0.389 ~= 36 min; t_sun = 92.4 - 36 = 56 min*

Why this orbit works for UniSat-1:

Factor	400 km / 51.6 deg	Impact on UniSat-1
Launch cost	Lowest (ISS resupply rideshare)	Fits university budget
Orbital lifetime	~1 year (natural decay)	Exceeds 6-month mission; compliant with FCC 5-year rule without propulsion
Radiation	Low (~2--3 krad/yr behind 2 mm Al)	COTS electronics safe for 6-month mission
Inclination	51.6 deg	Adequate latitude coverage for space weather science
Eclipse	~36 min (~39% of orbit)	Manageable with 10 Wh battery
Ground contacts	Mid-latitude stations: ~4--6 passes/day	Sufficient for 9600 bps UHF downlink

What this orbit does NOT provide:

- Sun-synchronous lighting (not needed -- magnetometer is not an optical instrument)
- Polar coverage (acceptable -- 51.6 deg covers the majority of magnetic field variation)
- Long lifetime (acceptable -- 6-month design life is well within ~1-year natural lifetime)

No propulsion trade: At 400 km, atmospheric drag causes natural re-entry within approximately 1 year (depending on solar activity and ballistic coefficient). UniSat-1 therefore needs no propulsion system for either orbit maintenance or debris compliance. This eliminates an entire subsystem -- a major simplification for a 1U mission.

---

7. SpaceCDF Orbit Trade Exercise (20 min)

Instructions

1. Navigate to the Mission Architecture tab in SpaceCDF
2. Open the Orbit Trade Advisor panel
3. Enter your mission parameters:
- GSD target (optical missions) or "N/A" for non-optical
- Revisit target (days)
- Mission lifetime (years)
- Latitude band of interest
- Cost ceiling (MEUR)
- Aperture diameter (if known from optical sizing)
4. Click "Compute Orbit Trade"
5. Review the scored candidates:
- Which orbit scores highest overall?
- Is it debris-compliant (FCC 5-year rule)?
- What is the natural orbital lifetime?
6. Click "Use" on your preferred orbit to populate the design parameters
7. Verify orbit fields updated in the dashboard (look for "Set from advisor" badge)

Discussion Points

- Does the best-scoring orbit match your engineering intuition?
- What happens when you change scoring weights (prioritise GSD over cost, or vice versa)?
- For non-optical missions (comms, AIS), does the orbit scoring still make physical sense?
- If your selected orbit is above 600 km, what deorbit strategy will you use?
- Complete Worksheet 2.3

Session Summary

Topic	Key Takeaway
Keplerian elements	Six parameters define the orbit; for LEO: altitude + inclination + LTAN
Key formulae	T = 2pisqrta^3/u; v = sqrtu/a; eclipse fraction from geometry
Sun-synchronous	Requires specific inclination (~97 deg at 500 km) using J_2 precession
Perturbations	J_2 enables SSO; drag limits lifetime; SRP affects high-A/m satellites
Trade-offs	Lower altitude = better GSD and link but shorter lifetime and more drag
Debris rules	FCC 5-year, IADC 25-year; above ~550 km requires active deorbit
Hohmann transfer	D V_deorbit ~= 100 m/s from 600 km; critical propulsion input
Radiation	Below 600 km: COTS OK; above: rad-tolerant/hard needed; SAA causes SEU
Cascade	Orbit choice affects every subsystem -- highest-leverage design decision

---

Session 2.4: Mission Architecture -- Segments,

Interfaces, and Budgets

Duration: 2 hours

Prerequisites: Sessions 2.1--2.3 (requirements, functions, orbit selected)

SpaceCDF Tabs: Mission Architecture, System Architecture, Interfaces, Dashboard

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References

- NASA, Systems Engineering Handbook, 2016, Sec. 4.4 (Process 4) & Sec. 6.3 (Process 12: Interface Management)
- ECSS, ECSS-E-ST-10-24C: Interface Management, 2015
- ECSS, ECSS-E-HB-10-02A: Verification Guidelines, 2010, Sec. 5.2 (Mass Margins)
- Wertz, Everett & Puschell, Space Mission Engineering: The New SMAD, 2011, Ch. 10--11
- ECSS, ECSS-E-ST-20C: Electrical and Electronic, 2021 (Power Budgets)

Learning Objectives

By the end of this session, participants will be able to:

1. Decompose a space mission into its constituent segments (space, ground, launch, user)
2. Construct a system architecture block diagram with subsystem boundaries
3. Identify and classify all subsystem-to-subsystem interfaces using an N-squared matrix
4. Write formal interface requirements for critical subsystem pairs
5. Construct a mass budget with ECSS margin policy and a mode-based power budget
6. Interpret SpaceCDF's dashboard KPIs and budget displays

---

1. Mission Segment Decomposition (15 min)

Teaching Notes

[Source: NASA SEH Sec. 4.4; ECSS-E-ST-10C Sec. 5.4; SMAD, Ch. 1]

Every space mission decomposes into segments, each with distinct functions and interfaces:

Figure: Space Segment

(See digital/markdown version for interactive diagram)

Segmen	Elements	Key Interfaces
Space	Spacecraft bus, payload, flight software	To ground (RF), to launch (mechanical/electrical)
Ground	Ground station, MCC, data processing	To space (RF), to user (network)
Launch	Launch vehicle, deployer, adapter	To space (mechanical, electrical inhibits)
User	End users, applications, data consumers	To ground (data products)

---

2. System Architecture Block Diagram (20 min)

Teaching Notes

The system block diagram shows the internal architecture of the space segment -- all subsystems and their data/power/mechanical connections.

Figure: OBC / Data Handling

(See digital/markdown version for interactive diagram)

Subsystem Roles (CubeSat Reference)

Subsystem	Abbreviation	Primary Function	Typical Mass Fraction
Payload	PL	Mission-specific sensing or communication	25--40%
Electrical Power System	EPS	Generate, store, distribute electrical power	15--25%
Attitude & Orbit Control	AOCS	Determine and control attitude; orbit knowledge	8--15%
Communications (TTC)	LINK / TTC	Telemetry downlink, telecommand uplink, data downlink	5--10%
On-Board Computer	OBC /	Command execution, data handling, flight software	2--5%
Thermal Control	TCS	Maintain all components within temperature limits	1--5%
Structure	STR	Mechanical support, launch load path, CDS compliance	15--25%

Propulsion

PROP

Orbit manoeuvres, deorbit (if required)

0--15%

Harness	HAR	Electrical interconnections	3--7%

3. The N-Squared Interface Matrix (25 min)

Teaching Notes

[Source: NASA SEH Sec. 6.3 -- Process 12: Interface Management; ECSS-E-ST-10-24C]

Interface problems are the leading cause of integration failures. NASA SEH states: "Most system failures can be traced back to interface problems."

N-Squared Matrix Structure

The N^2 matrix is a standard systems engineering tool for mapping all subsystem interactions:

- Diagonal cells: Subsystems (EPS, AOCS, Comms, Thermal, Structure, Propulsion, OBC, Payload)
- Off-diagonal cells: Interface between the row subsystem and the column subsystem
- Upper triangle: Outputs from row to column (data, power, commands)
- Lower triangle: Outputs from column to row

For 8 subsystems: 8 x 7 / 2 = 28 potential interface pairs. A typical CubeSat has 18--22 active interfaces.

Interface Types

Type	Symbol	Description	Example
Mechani	M	Physical attachment, loads, alignment tolerances	Payload mounting to structure face
Electrical	E	Power connections, bus voltage, switched lines	EPS 5V bus to all subsystems
Thermal	T	Heat transfer paths, thermal coupling, conduction	Transponder waste heat to radiator panel
Data	D	Digital bus (I^2C, SPI, UART, CAN, RS-422)	OBC commands to AOCS controller
RF	R	Electromagnetic coupling or intentional RF paths	TX emissions coupling into payload receiver
Optical	O	Light paths, field-of-view clearance, stray light	Star tracker FOV clearance from solar array

Common CubeSat Interface Concerns

Interface Pair	Types	Key Concern
EPS <-> AOCS	E	Bus voltage compatibility; reaction wheel peak power draw
EPS <-> Comms	E	TX peak power demand (~6--10 W); switched line allocation
EPS <-> Thermal	E, T	SA thermal coupling; radiator vs SA area competition on external faces
EPS <-> Payload	E	Peak power switching; duty cycle coordination
Structure <-> AOCS	M	Reaction wheel and star tracker mounting alignment; vibration isolation
Structure <-> Payload	M, O	Payload alignment stability; optical FOV clearance
OBC <-> AOCS	D	Attitude data for payload pointing; mode transition commands
OBC <-> Comms	D	Telemetry packet routing; telecommand distribution

OBC <-> Payload	D	Science data acquisition trigger; instrument commanding
Comms <-> Payload	R	EMC: TX conducted/radiated emissions vs payload receiver sensitivity
Comms <-> AOCS	R, O	Antenna pattern vs star tracker FOV; antenna pointing coordination
Thermal <-> Payload	T	Detector cooling requirement; operating temperature limits
AOCS <-> Payload	M	Reaction wheel micro-vibration vs payload pointing stability (jitter)

Conflict Detection and Resolution

Interface conflicts arise when two subsystems have incompatible requirements at their shared boundary.

Severity Classification:

Severity	Description	Example	Required Action
Critical	Design cannot close without resolution	EMC: TX radiation prevents payload operation	Must resolve before PDR
Major	Significant impact on design margin	Radiator area competes with SA area	Mitigation plan required by PDR
Minor	Manageable with minor adjustment	Star tracker FOV partially blocked by antenna stow	Accommodation analysis

Resolution Options:

1. Relocate: Move a component to avoid the conflict (e.g., star tracker to different face)
2. Shield/Isolate: Add EMC shielding, vibration isolators, thermal insulation
3. Time-Division: Schedule conflicting activities to avoid simultaneity (e.g., no TX during imaging)
4. Accept Risk: Document residual risk and margin impact in the risk register

Writing Interface Requirements

For each significant interface, write formal requirements:

Example -- EPS <-> All Subsystems:

```
IR-PWR-001: "The EPS shall provide regulated bus voltages of
             3.3 V +/- 0.1 V and 5.0 V +/- 0.25 V to all subsystems."
IR-PWR-002: "Each subsystem shall not exceed its allocated power
             draw without EPS coordination."
```

Example -- Comms <-> Payload (EMC):

```
IR-EMC-001: "TX conducted emissions shall be below -60 dBm in the
             payload receiver band (1.5-1.6 GHz) during imaging mode."
IR-EMC-002: "TX and payload acquisition shall not operate simultaneously
             unless IR-EMC-001 is verified by test."
```

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4. Engineering Budgets: Mass and Power (30 min)

Teaching Notes

[Source: ECSS-E-HB-10-02A Sec. 5.2; SMAD, Ch. 10--11]

Engineering budgets are the quantitative backbone of the design. They answer: "Will this design close?"

Mass Budget

> *Key Equations -- Mass Budget*
>
> *Mass margin:*
> *$Margin_% = \fracM_allocation - M_MEVM_allocation x 100%$*
>
> *where MEV = Maximum Expected Value = CBE + maturity margins.*
>
> *Status thresholds: Green: > 20% | Amber: 10--20% | Red: < 10% | Exceeded: < 0%*

Term	Definition
CBE (Current Best Estimate)	Best estimate of actual mass based on current knowledge
MEV (Maximum Expected Value)	CBE + equipment maturity margin = worst-case expected mass
Equipment maturity margin	Applied per component based on design maturity (TRL)
System margin	Applied at system level as management reserve

ECSS Margin Policy by Phase

[Source: ECSS-E-HB-10-02A Sec. 5.2]

Phase	Equipment Margin	System Margin	Compound
0/A (concept)	20%	20%	~44%
B1 (preliminary)	10%	20%	~32%
B2 (detailed)	5%	15%	~21%
C/D (build/test)	3%	10%	~13%
E (as-built)	0%	5%	~5%

> *Worked Example -- 3U CubeSat Mass Budget (Phase A)*
>
> *| Subsystem | CBE (kg) | Equip. Margin (20%) | MEV (kg) |*
> *|-----------|---------|---------------------|---------|*
> *| Payload | 1.50 | 0.30 | 1.80 |*
> *| EPS | 0.75 | 0.15 | 0.90 |*
> *| AOCS | 0.55 | 0.11 | 0.66 |*
> *| Comms (TTC) | 0.25 | 0.05 | 0.30 |*

> | OBC | 0.08 | 0.02 | 0.10 |
>
> | Thermal | 0.05 | 0.01 | 0.06 |
>
> | Structure | 0.35 | 0.07 | 0.42 |
>
> | Harness | 0.15 | 0.03 | 0.18 |
>
> | Dry Total | 3.68 | | 4.42 |
>
> | System Margin (20%) | | | 0.88 |
>
> | Dry MEV | | | 5.30 |
>
> | Propellant | | | 0.00 |
>
> | Wet Mass | | | 5.30 |
>
> | Launcher Allocation | | | 6.00 (3U limit) |
>
> | Mass Margin | | | 0.70 kg (11.7%) -- Amber |

Mode-Based Power Budget

The power budget is computed per operational mode because not all subsystems draw power simultaneously:

Subsystem	Safe (W)	Idle (W)	Imaging (W)	Downlink (W)	Eclipse (W)
OBC	1.0	1.0	1.0	1.0	1.0
AOCS	0.5	1.0	3.0	2.0	0.5
Payload	0	0	5.0	0	0
Comms (TX)	0.5	0.5	0.5	6.0	0
Thermal	0.5	0.5	0.5	0.5	2.0
Total	2.5	3.0	10.0	9.5	3.5

Orbit-Average Power

> *Key Equations -- Orbit-Average Power*
>
> $P_avg = Sum_modes (P_mode x duty_mode)$
>
> *Example for 95-min orbit (60 min sunlight, 35 min eclipse):*
>
> | Mode | Power (W) | Duty (%) | Contribution (W) |
>
> |------|----------|----------|-------------------|
>
> | Idle | 3.0 | 45% | 1.35 |
>
> | Imaging | 10.0 | 10% | 1.00 |
>
> | Downlink | 9.5 | 8% | 0.76 |
>
> | Eclipse | 3.5 | 37% | 1.30 |

| Total | | 100% | 4.41 W |

Other Budget Types (Preview)

These budgets will be developed in detail during Week 2 Day 3--4 sessions:

Budget	Key Equation	Session
Link	Margin = EIRP - FSPL + G/T - k - 10log_10(R_b) - E_b/N_0	3.3
Pointing	theta_total = sqrtSum theta_i^2 (RSS)	3.2
D V	D V = I_sp * g_0 * ln(m_0/m_f) (Tsiolkovsky)	3.4
Data	Daily Downlink >= Daily Generation	3.3

1U Worked Example: UniSat-1

Simplified Architecture: Only 5 Subsystems

UniSat-1 demonstrates that a 1U CubeSat can be built with a radically simplified architecture compared to a 3U EO mission. Several subsystems are eliminated entirely:

Subsystem	3U EO CubeSat	UniSat-1 (1U)	Rationale for Elimination
EPS	Required	Required	Always needed
OBC	Required	Required	Always needed (minimal: MSP430 or ARM Cortex-M class)
Comms	S-band + X-band	UHF only	9600 bps is sufficient for < 1 kbps payload data
Structur	Required	Required	Always needed (ISIS 1U or Pumpkin Rev C)
Payload	Telescope (complex)	MEMS magnetometer (simple: 50 g, 0.2 W)	
AOCS	Star tracker + RWs + MTQs	Eliminated	Passive magnetic only (permanent magnet + hysteresis rods, treated as structure, not a subsystem)
Thermal	MLI + heaters	Eliminated	Passive coatings only; 400 km LEO thermal environment is benign for 6-month mission
Propulsi	May be needed	Eliminated	400 km orbit decays naturally in ~1 year; no orbit maintenance needed
Harness	Significant	Minimal	Only 3--4 board-to-board connections via PC/104 stack

N-squared matrix -- drastically reduced:

For UniSat-1, the interface matrix shrinks from 28 potential pairs (8 subsystems) to just 10 pairs (5 subsystems). The active interfaces are:

Interface Pair	Types	Key Concern
EPS <-> OBC	E, D	Power regulation, housekeeping telemetry
EPS <-> Comms	E	TX peak power (~0.5 W -- small relative to budget)
EPS <-> Payload	E	Payload switching (0.2 W -- trivial)
OBC <-> Comms	D	TM/TC packet routing
OBC <-> Payload	D	Magnetometer data acquisition (I2C or SPI)
Structure <-> All	M	Mounting, CDS rail compliance

Mass budget (Phase A):

Subsystem	CBE (g)	Equip. Margin (20%)	MEV (g)
Payload (MEMS magnetometer)	50	10	60
EPS (board + battery + body-mounted cells)	250	50	300
OBC (MSP430/Cortex-M board)	30	6	36
Comms (UHF transceiver + antenna)	80	16	96
Structure (1U frame)	200	40	240
Passive magnetic AOCS (magnet + rods)	30	6	36
Harness	50	10	60
Dry Total	690		828
System Margin (20%)			166
Dry MEV			994
CDS Allocation			1330
Mass Margin			336 g (25.3%) -- Green

Power budget (all modes use the same simple duty cycle):

Mode	Power (W)	Duty (%)	Contribution (W)
Idle (OBC + beacon)	0.7	50%	0.35
Science (+ magnetometer)	0.9	10%	0.09
Downlink (+ UHF TX)	1.2	5%	0.06
Eclipse (OBC only)	0.5	35%	0.18
Orbit Average		100%	0.68 W

With ~2 W available from body-mounted cells, the power margin is substantial (~1.3 W, or 66%). This is one of the luxuries of a simple payload on a 1U bus.

Key architectural insight: The UniSat-1 architecture is so simple that a small team (3--5 people) can design, build, test, and operate it. This makes it ideal for a university or educational programme. The CDF process still applies -- but the sessions are shorter and the trade space is narrower.

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5. SpaceCDF Exercise (30 min)

Instructions

1. System Architecture tab: Review or edit the subsystem block diagram for your mission
2. Interfaces tab: Review the N^2 matrix
- Click on 3 interface cells to examine types and concerns
- For any red-bordered cell (conflict), use the "Resolve Conflict" workflow
- Write interface requirements for your most critical pair
3. Dashboard: Examine all KPI cards:
- Mass margin: green/amber/red?
- Power margin per mode
- Link margin (if computed)
- Cost vs ceiling
4. Budget Breakdown: Open per-subsystem mass and power charts
5. Complete Worksheet 2.4

Discussion Questions

- Which budget is tightest (closest to zero or negative margin)?
- What single design change would most improve the tightest budget?
- How does the ECSS margin policy affect your design freedom at Phase A vs Phase C?
- Which interface pair is most likely to cause integration problems?

Session Summary

Topic	Key Takeaway
Mission segments	Space, Ground, Launch, User -- each with defined interfaces
System architecture	Block diagram shows subsystems, data buses, power distribution
N^2 matrix	Maps all 28 potential interface pairs; 6 types (M, E, T, D, R, O)
Interface conflicts	Severity classification (critical/major/minor); 4 resolution options
Interface requirements	Formal boundary agreements; must be verifiable
Mass budget	CBE + equipment margin + system margin = MEV; compare to allocation
Power budget	Mode-based; duty cycling gives orbit-average; SA must cover peak + recharge
ECSS margins	Decrease with maturity: ~44% at Phase A to ~13% at Phase C/D
Budget closure	Negative margin = design does not close; reduce demand or increase allocation

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Session 3.1: Power System and Thermal Control Design

Duration: 2 hours

Prerequisites: Sessions 2.1--2.4 (requirements, functions, orbit, architecture defined)

SpaceCDF Tabs: Dashboard (Power KPI), Engineering Budgets, Timing Budget, Parametric

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References

- Wertz, Everett & Puschell, Space Mission Engineering: The New SMAD, 2011, Ch. 11.4 (EPS), Ch. 11.5 (Thermal)
- ECSS, ECSS-E-ST-20C: Electrical and Electronic, 2021
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- Patel, Spacecraft Power Systems, 2005, Ch. 3--8
- Gilmore, Spacecraft Thermal Control Handbook, Vol. 1, 2002
- GomSpace, P31u EPS Datasheet, 2023

- MMA Design, HaWK Solar Array Datasheet, 2023
- Spectrolab, 30% Triple-Junction Solar Cell Datasheet, 2020
- Ratnakumar et al., Lithium-Ion Batteries for Space, NASA JPL, 2003
- Gilmore, Spacecraft Thermal Control Handbook, Vol. 2: Cryogenics, 2003

Learning Objectives

By the end of this session, participants will be able to:

1. Size a solar array from mission power demand, eclipse profile, and degradation
2. Size a battery from eclipse energy demand, depth-of-discharge, and cycle-life requirements
3. Compute orbit-average power using duty cycle analysis
4. Explain EPS architecture (DET vs PPT, MPPT, bus regulation) and articulate the physics of each
5. Perform first-order thermal balance analysis (hot case and cold case) with full radiative derivation
6. Select thermal control methods and apply ECSS thermal margins
7. Explain MLI construction, heat pipe operation, and heater sizing
8. Verify power and thermal budgets in SpaceCDF

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1. Electrical Power System Architecture (25 min)

Teaching Notes

[Source: SMAD, Ch. 11.4; ECSS-E-ST-20C; Patel, Ch. 3]

The EPS is the "utility company" of the spacecraft. It must continuously supply regulated power to all subsystems through every operational mode, including eclipse. Unlike terrestrial power systems, spacecraft EPS cannot draw from a grid -- the solar array, battery, and power conditioning electronics must form a fully self-contained, autonomous energy system with zero maintenance for the mission lifetime.

EPS Block Diagram

Figure: Solar Array

(See digital/markdown version for interactive diagram)

Solar Cell Physics

A photovoltaic cell converts photon energy into electrical energy via the photovoltaic effect. When a photon with energy $E = h\nu \geq E_g$ (where E_g is the semiconductor bandgap) strikes the cell, it promotes an electron from the valence band to the conduction band, creating an electron-hole pair. The built-in electric field at the p-n junction sweeps carriers to opposite terminals, producing a voltage (V_{oc}) and current (I_{sc}).

Single-junction vs multi-junction cells:

A single-junction cell (e.g., silicon, $E_g = 1.12$ eV) can only absorb photons with $E \geq E_g$. Photons with $E < E_g$ pass through unabsorbed; photons with $E \gg E_g$ lose excess energy as heat (thermalisation loss). The theoretical maximum efficiency for a single-junction cell under AM0 (space) illumination is approximately 31% (Shockley-Queisser limit).

Multi-junction (MJ) cells stack two or three p-n junctions of different semiconductor materials, each tuned to absorb a different portion of the solar spectrum:

Cell Technology	Structure	Bandgaps (eV)	AM0 Efficienc	Temp Coefficient	Flight Heritage
Monocrystalline Si	Single junction	1.12	16--18%	-0.45%/deg	Extensive (ISS, many LEO)
GaAs single-junction	Single junction	1.42	22--24%	-0.21%/deg	Moderate
InGaP/GaAs dual-junction	2-junction	1.86 / 1.42	26--28%	-0.20%/deg C	Moderate
InGaP/GaAs/Ge triple-junction	3-junction (lattice-matched)	1.86 / 1.42 / 0.67	28--30%	-0.19%/deg C	Extensive (>90% of modern S/C)
InGaP/GaAs/InGaAs IMM	3-junction (inverted metamorphic)	1.86 / 1.42 / 1.0	32--33%	-0.18%/deg C	Growing (SolAero ZTJ, Spectrolab XTJ Prime)

[Source: Spectrolab XTJ Prime datasheet; SolAero ZTJ datasheet; Green et al., "Solar Cell Efficiency Tables," Progress in Photovoltaics, v.62, 2024]

Why triple-junction GaAs dominates space: The AM0 solar spectrum (above the atmosphere) is richer in UV and blue photons than the AM1.5 terrestrial spectrum. Triple-junction cells capture this energy across three bandgaps. The top cell (InGaP, $E_g = 1.86$ eV) absorbs blue/UV, the middle cell (GaAs, $E_g = 1.42$ eV) absorbs visible, and the bottom cell (Ge, $E_g = 0.67$ eV) absorbs near-IR. Each junction contributes voltage in series, while current is limited by the lowest-current junction (current matching constraint).

Temperature effects: In orbit, solar cells operate at 40--80 degC depending on mounting and orbit. Cell efficiency decreases with temperature due to increased intrinsic carrier concentration (which reduces V_{oc}). The temperature coefficient for triple-junction GaAs is approximately -0.19%/degC relative --

meaning a cell rated at 29.5% at 28 degC drops to approximately 28.5% at 80 degC. This must be included in power budget calculations:

$eta_cell(T) = eta_ref x [1 + beta (T - T_ref)]$

where beta ~= -0.0019 /degC for triple-junction GaAs and T_ref = 28 degC (standard test conditions).

Degradation mechanisms: Solar cells degrade in orbit due to:

- Radiation damage: Energetic protons and electrons (trapped in the Van Allen belts and from solar particle events) displace atoms in the crystal lattice, creating recombination centres that reduce minority carrier lifetime and thus current. Degradation is characterised as equivalent 1 MeV electron fluence. Typical rates: 2--3%/year in LEO, 5--8%/year in MEO (through proton belt), 1--2%/year in GEO.
- UV darkening: Ultraviolet radiation darkens cover glass adhesive over time, reducing transmission.
- Micrometeoroid erosion: Gradual pitting of cover glass reduces optical transmission.
- Electrostatic discharge (ESD): In GEO or polar orbits, differential charging can cause arcing between cells, permanently damaging interconnects.

Cover glass (typically ceria-doped borosilicate, 100--150 um thick) mitigates radiation and UV effects. The combined degradation factor is:

$L_d = (1 - delta)^n$

where delta = 0.025 (2.5%/year for triple-junction GaAs in LEO with standard cover glass) and n = mission lifetime in years.

Body-Mounted vs Deployable Solar Arrays

Body-mounted cells are bonded directly to the spacecraft's external panels. They are the simplest and most reliable option (no deployment mechanism, no hinges, no drive motors) but are severely area-limited.

Mounting	Advantages	Disadvantages	Typical Power
Body-mounted	No mechanism risk, no power for tracking, low mass, low cost	Limited area (satellite surface only), poor illumination geometry for nadir-pointing S/C, high cell temperature	1U: ~2 W, 3U: ~7 W, 6U: ~12 W
Fixed deployab	2--4x more area, better sun angle, lower cell temp (radiative cooling from back side)	Deployment mechanism (single point of failure), aerodynamic drag increase, structural dynamics	3U: ~15--25 W, 6U:
Tracking deployable	Optimal sun incidence (costheta ~= 1), maximum power	SADM (Solar Array Drive Mechanism) adds mass/cost/complexity, continuous power for motor	Large S/C: 100+ W

Real mission examples:

- Planet SuperDove (3U+): Body-mounted + two fixed deployable wings. ~25 W BOL. The deployables are spring-hinged panels that fold against the 3U body during launch and deploy after ejection from the P-POD.
- Asteria (6U, JPL): Dual deployable arrays, 48 W BOL. Used MMA Design HaWK panels with triple-junction GaAs cells.
- ISS CubeSats (various 1U): Body-mounted only, 2--3 W. Adequate for simple sensor missions with low duty cycles.

Architecture Types

Architecture	How It Works (Physics)	Efficiency	Complexity	Typical Use
DET (Direct Energy Transfe	SA connects directly to bus; a shunt regulator diverts excess current to a resistor bank (dissipated as heat) when SA output exceeds load. No series regulator between SA and bus. Voltage varies with illumination.	80--85 % (shunt losses)	Low	Heritage GEO spacecraft, some
PPT / MPPT (Peak Power	A DC-DC converter (typically boost or buck-boost) between SA and bus continuously adjusts its input impedance to operate the SA at its maximum power point (MPP). Uses perturb-and-observe or incremental conductance algorithm. Extracts 10--15% more power than DET, especially at off-nominal temperatures and end-of-life.	90--95 %	Medium	Most modern CubeSats
Unregulated bus	Battery connects directly to the power bus through protection FETs only. Bus voltage equals battery voltage (varies from 3.0 V at empty to 4.2 V at full per cell, or 6.0--8.4 V for 2S configuration). Subsystems must tolerate voltage variation.	Highest (no regulator	Lowest	Very simple CubeSats, 1U missions
Regulated bus	DC-DC converters create fixed voltage rails (3.3 V, 5 V, 12 V) from battery/SA input. Subsystems see constant voltage regardless of battery state. Adds ~5--10% losses in regulators but greatly simplifies subsystem design.	Good (85--90 % overall)	Medium	Most CubeSat COTS EPS (GomSpace P31u, Endurosat, AAC Clyde)

MPPT physics: A solar cell's I-V curve has a distinct "knee" where the maximum power (P = I x V) is extracted. At open-circuit (I = 0), voltage is maximum but power is zero. At short-circuit (V = 0), current is maximum but power is zero. The MPP sits at approximately 75--80% of V_oc and 90--95% of I_sc. Temperature shifts the I-V curve (higher T moves V_oc left), so the MPP moves. An MPPT controller dynamically tracks this point, typically updating every 0.1--1 s. Common CubeSat MPPT converters achieve 95--97% tracking efficiency.

Battery charge regulation: The EPS must prevent overcharging (which causes lithium plating, gas generation, and thermal runaway in Li-ion cells) and overdischarging (which causes copper dissolution from the negative current collector, permanently damaging the cell). Modern CubeSat EPS boards implement:

- CC-CV charging: Constant-current charging until cell voltage reaches 4.2 V, then constant-voltage taper until current drops below C/20
- Under-voltage lockout: Bus disconnect when cell voltage falls below 3.0 V (or 2.8 V for emergency)
- Cell balancing: For multi-cell series configurations (2S or higher), passive or active balancing circuits ensure cells remain within 50 mV of each other
- Temperature cutoffs: Charging inhibited below 0 degC and above 45 degC (Li-ion charging below 0 degC causes lithium plating)

CubeSat standard: Most commercial EPS boards (GomSpace P31u, Endurosat, AAC Clyde) use MPPT + regulated bus with 3.3 V and 5 V rails, plus an unregulated battery rail (6.0--8.4 V for 2S Li-ion).

---

2. Solar Array Sizing (25 min)

Teaching Notes

Key Equations -- Solar Array Sizing (Full Derivation)

> *Step 1: Orbit-average power demand:*
>
> $P_{avg} = \sum_{modes} P_{mode} \times f_{duty,mode}$
>
> *This is computed from the ConOps mode table. For each mode (imaging, downlink, eclipse/safe, idle), multiply the mode power by the fraction of orbit spent in that mode.*
>
> *Step 2: SA end-of-life power requirement:*
>
> *During sunlight, the SA must simultaneously: (a) power all sunlit loads and (b) recharge the battery for the upcoming eclipse. The recharge power accounts for battery charge/discharge efficiency:*
>
> $P_{SA,EOL} = P_{peak,sunlight} + \frac{P_{eclipse} \times t_{eclipse}}{t_{sunlight} \times \eta_{path,eclipse}}$
>
> *where $\eta_{path,eclipse} = \eta_{charge} \times \eta_{discharge} \times \eta_{regulator}$.*
>
> *For a typical CubeSat EPS: $\eta_{charge} \approx 0.92$ (Li-ion coulombic efficiency x charge regulator), $\eta_{discharge} \approx 0.95$ (battery internal resistance losses), $\eta_{regulator} \approx 0.90$ (DC-DC converter). Combined: $\eta_{path,eclipse} \approx 0.79$.*
>
> *A simpler approximation uses $\eta_{charge} \approx 0.90$ as a lumped path efficiency, which is common in textbooks but slightly optimistic.*
>
> *Step 3: Account for degradation and temperature:*
>
> $P_{SA,BOL} = \frac{P_{SA,EOL}}{L_d \times L_T}$
>
> *where:*
>
> *- $L_d = (1 - \delta)^n$ is the radiation degradation factor ($\delta = 0.025$/yr for TJ GaAs in LEO)*
>
> *- $L_T = 1 + \beta(T_{cell} - T_{ref})$ is the temperature derating factor ($\beta = -0.0019$/degC, $T_{ref} = 28$ degC, T_{cell} typically 60--80 degC in LEO)*
>
> *For a cell operating at 65 degC: $L_T = 1 + (-0.0019)(65 - 28) = 1 - 0.070 = 0.930$ (7% power loss from temperature).*
>
> *Step 4: Compute SA area:*
>
> $A_{SA} = \frac{P_{SA,BOL}}{\eta_{cell} \times S \times \cos(\theta) \times f_{pack} \times f_{cover}}$

where:

- eta_cell = 0.295 (triple-junction GaAs efficiency at AM0, 28 degC -- STC rating)

- S = 1361 W/m^2 (solar constant at 1 AU, per Kopp & Lean 2011)

- theta = sun incidence angle (0 deg for ideal tracking; for body-mounted, use orbit-averaged costheta)

- f_pack = 0.85--0.90 (cell packing factor -- fraction of panel area covered by cells; gaps exist for cell interconnects, edge clearance, and harness routing)

- f_cover = 0.97 (cover glass transmission loss, typically 2--4%)

Step 5: SA mass:

$m_SA = A_SA x sigma_SA + m_mechanism$

where sigma_SA = areal density of the panel:

- Body-mounted (cells on Al substrate): sigma ~= 2.0--2.5 kg/m^2

- Rigid deployable (cells on Al honeycomb/CFRP panel): sigma ~= 1.5--2.0 kg/m^2

- Flexible deployable (roll-out or fold-out): sigma ~= 0.8--1.2 kg/m^2

- m_mechanism = hinge, spring, hold-down mechanism: typically 0.1--0.3 kg per panel for CubeSats

Worked Example -- 3U EO CubeSat (SuperDove-class) Solar Array

Given: P_peak,sunlight = 10.0 W (imaging mode), P_eclipse = 3.5 W, t_eclipse = 35 min, t_sunlight = 60 min, mission lifetime = 3 years, cell temperature = 65 degC, single deployable panel with fixed sun angle theta = 23 deg (average over orbit for SSO with body-fixed panel).

Step 2: Recharge power (using detailed path efficiency):

eta_path = 0.92 x 0.95 x 0.90 = 0.786

P_recharge = (3.5 x 35)/(60 x 0.786) = (122.5)/(47.2) = 2.60 W

P_SA,EOL = 10.0 + 2.60 = 12.60 W

Step 3: BOL accounting for 3-year degradation + temperature:

L_d = (1 - 0.025)^3 = 0.9269

L_T = 1 + (-0.0019)(65 - 28) = 0.930

P_SA,BOL = (12.60)/(0.9269 x 0.930) = (12.60)/(0.862) = 14.62 W

Step 4: SA area:

> *A_SA = (14.62)/(0.295 x 1361 x cos(23\degree) x 0.85 x 0.97)*
>
> *= (14.62)/(0.295 x 1361 x 0.921 x 0.85 x 0.97)*
>
> *= (14.62)/(305.0) = 0.0479 m^2*
>
> *This is approximately 22 cm x 22 cm -- achievable with a single deployable panel on a 3U CubeSat (the MMA HaWK panel provides up to 0.06 m^2 per wing).*
>
> *Step 5: SA mass (deployable, rigid):*
>
> *m_SA = 0.0479 x 1.8 + 0.15 = 0.086 + 0.15 = 0.24 kg (panel + mechanism)*
>
> *Comparison to Planet SuperDove: The actual SuperDove uses body-mounted cells plus two deployable wings, achieving ~25 W BOL. Our calculation (14.6 W BOL from one panel) is consistent -- SuperDove's higher power supports continuous imaging plus S-band downlink simultaneously.*

CubeSat SA Power Reference

Configuration	1U	3U	6U	12U
Body-mounted only	~2 W	~7 W	~12 W	~20 W
Single deployable	~4 W	~15 W	~30 W	~55 W
Dual deployable	--	~25 W	~48 W	~100 W
Quad deployable	--	--	~80 W	~180 W

[Source: GomSpace, ISIS, MMA Design vendor datasheets; ASTERIA 6U confirmed 48 W BOL; Dove/SuperDove confirmed ~25 W]

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3. Battery Sizing (20 min)

Teaching Notes

Li-ion Cell Chemistry and Physics

All modern spacecraft batteries use lithium-ion (Li-ion) chemistry. During discharge, lithium ions migrate from the graphite anode (negative electrode) through an organic electrolyte and polymer separator to the lithium metal oxide cathode (positive electrode), while electrons flow through the external circuit doing work. During charging, the process reverses.

Common cathode chemistries used in space:

Chemistry	Cathode	Nominal Voltage	Energy Density	Cycle Life (30% DOD)	Thermal Stability	Space Heritage
LCO (LiCoO_2)	Cobalt oxide	3.7 V	150--200 Wh/kg	~10,000	Moderate	ISS, many CubeSats (18650 cells)
NMC (LiNiMnCoO_	Nickel-manganese-cobalt	3.7 V	170--250 Wh/kg	~5,000	Moderate	Growing heritage
LFP (LiFePO_4)	Iron phosphate	3.2 V	90--120 Wh/kg	~50,000	Excellent	Niche (when cycle life is paramount)
NCA (LiNiCoAlO_2)	Nickel-cobalt-aluminium	3.6 V	200--260 Wh/kg	~3,000	Lower	Limited space heritage

[Source: Ratnakumar et al., NASA JPL; Saft VES-16 space cell datasheet; Samsung SDI 18650 specifications]

Cell form factors:

- 18650 cylindrical: 18 mm diameter, 65 mm long. The workhorse of CubeSat missions. Common cells: Samsung 25R (2500 mAh, 20A continuous), Panasonic NCR18650B (3350 mAh, moderate rate), Sony VTC6 (3000 mAh, high rate). Energy: 9--12 Wh per cell.
- Pouch cells: Custom dimensions, higher energy density (~250 Wh/kg) but require external structural support. Used in some 6U+ missions and all large spacecraft. GomSpace NanoPower BPX uses pouch cells.
- Prismatic cells: Rigid case, intermediate between cylindrical and pouch. Used in some mission-specific designs.

Cell configuration:

Cells are arranged in series (S) to increase voltage and parallel (P) to increase capacity:
- 1S (single cell): 3.0--4.2 V bus. Used for very simple 1U CubeSats.
- 2S (two in series): 6.0--8.4 V bus. Standard for most CubeSats (matches GomSpace P31u default).
- 2S2P (two series, two parallel): 6.0--8.4 V bus, double capacity. Used for higher-energy missions (e.g., GomSpace BP4 pack: 4 cells, 2S2P, ~38 Wh).

Protection circuits: Every flight battery pack includes:
- Cell voltage monitoring: Per-cell voltage measurement to detect over/under-voltage
- Over-current protection: Current-sense resistors + MOSFET switches to disconnect at overcurrent (prevents short-circuit thermal runaway)
- Temperature monitoring: Thermistors on each cell; inhibit charging below 0 degC and above 45 degC
- Heater circuit: Kapton heater on battery pack, thermostatically controlled, to maintain cells above minimum charging temperature during eclipse

Capacity fade model: Li-ion cells lose capacity over time due to solid electrolyte interphase (SEI) growth on the anode. A simplified calendar + cycling fade model:

$C(t, N) = C_0 x (1 - alpha sqrtt) x (1 - beta * N * DOD^gamma)$

where C_0 = initial capacity, t = time (years), N = number of cycles, alpha ~= 0.02 (calendar fade), beta ~= 3 x 10^-6, gamma ~= 2.1 (cycling fade). For LEO at 30% DOD, this gives approximately 5--8% capacity loss per year -- consistent with flight data from ISS battery replacements and CubeSat fleet telemetry.

> *Key Equations -- Battery Sizing*
>
> *Required battery energy:*
>
> *$E_bat = \fracP_eclipse x t_eclipseDOD x eta_discharge$*
>
> *where:*
>
> *- DOD = maximum depth of discharge*
>
> *- eta_discharge = 0.95 (discharge efficiency, accounting for internal resistance I^2R losses)*
>
> *- t_eclipse in hours*
>
> *Battery mass:*
>
> *$m_bat = \fracE_bate_specific$*
>
> *where e_specific = 150--200 Wh/kg for packaged Li-ion 18650 cells (cell-level energy density is higher, but packaging adds ~30% mass).*
>
> *Cycle life vs DOD relationship (Li-ion 18650, LCO chemistry):*
>
> | DOD | Typical Cycle Life | Suitable Mission Duration | Annual Eclipses (LEO, 15/day) |
> |-----|-------------------|--------------------------|-------------------------------|
> | 80% | ~500 cycles | < 1 month | 450 |
> | 50% | ~2,000 cycles | < 4 months | 1,800 |
> | 30% | ~10,000 cycles | 1--2 years | 5,475 |
> | 20% | ~30,000 cycles | 3--5 years | 10,950 |
> | 10% | ~100,000 cycles | > 7 years | 38,325 |
>
> *Design rule of thumb: For a multi-year LEO mission, start with 20--30% DOD. For a short technology demonstration (< 6 months), 40--50% DOD is acceptable and significantly reduces battery size/mass/cost.*

> *Worked Example -- Battery for 3U EO CubeSat (SuperDove-class)*
>
> *Given: P_eclipse = 3.5 W, t_eclipse = 35 min = 0.583 h, DOD = 0.25 (conservative for 3-year mission), eta = 0.95.*
>
> *Step 1 -- Required battery energy:*
>
> *E_bat = (3.5 x 0.583)/(0.25 x 0.95) = (2.04)/(0.2375) = 8.59 Wh*

Step 2 -- Apply ECSS margin (20% at Phase A/B):

> *E_bat,spec >= 8.59 x 1.20 = 10.3 Wh. Specify minimum 10 Wh, ideally 20 Wh for operational flexibility.*
>
> *Step 3 -- Verify cycle life:*
>
> *3-year mission at 15 orbits/day = 16,425 eclipses. At 25% DOD, Li-ion 18650 cells provide ~20,000 cycles. Margin = 22%. Pass.*
>
> *Including capacity fade: after 3 years, ~15--20% capacity loss from calendar + cycling aging. Effective DOD increases to 0.25 / 0.82 = 0.30 -- still within the 10,000-cycle regime. Acceptable with monitoring.*
>
> *Step 4 -- Battery mass:*
>
> *m_bat = (20)/(170) = 0.118 kg (using 170 Wh/kg for packaged 18650 cells)*
>
> *Step 5 -- Battery volume:*
>
> *Two 18650 cells in 2S1P: 2 x 18mm x 65mm = approximately 34 mL, fitting easily within a 3U stack.*
>
> *Comparison to Planet SuperDove: SuperDove carries approximately 20 Wh in a 2S2P configuration (4 cells), consistent with our sizing. The actual operating DOD is estimated at 10--15% per eclipse, giving substantial cycle-life margin for the multi-year constellation replenishment cadence.*

Failure modes to watch for:

- Thermal runaway: If a cell is overcharged, mechanically damaged, or experiences an internal short, the exothermic decomposition of the cathode material can lead to thermal runaway (self-heating > heat dissipation), potentially reaching 600+ degC. Mitigation: cell-level fuses, per-cell voltage monitoring, thermal cutoffs.
- Lithium plating: Charging below 0 degC causes metallic lithium to deposit on the anode surface rather than intercalating into graphite. This is irreversible, reduces capacity, and can cause internal shorts. Mitigation: battery heater + thermostat + software lockout.
- Capacity imbalance: In series configurations, the weakest cell limits the pack. If cells age at different rates, the weakest cell hits under-voltage lockout while others still have capacity. Mitigation: cell balancing circuits, matched cell lots.

---

4. Thermal Control System (35 min)

Teaching Notes

[Source: ECSS-E-ST-31C; Gilmore, Ch. 1--4; SMAD, Ch. 11.5]

The Physics of Spacecraft Thermal Control

Spacecraft thermal control is fundamentally different from terrestrial thermal engineering because there is no convection in vacuum. The only heat transfer mechanisms are:

1. Conduction: Heat flow through solid material, governed by Fourier's law: \dotQ = -kA (dT)/(dx). Critical within the spacecraft structure and between components and mounting surfaces.
2. Radiation: Heat transfer via electromagnetic radiation, governed by the Stefan-Boltzmann law: \dotQ = \varepsilon sigma A T^4. This is the only mechanism for rejecting heat to the environment.

There is no convective cooling. A component that overheats cannot be cooled by a fan. All waste heat must be conducted to a radiating surface and then radiated to space. This is the central constraint of spacecraft thermal design.

Thermal Environment in LEO

A spacecraft in LEO experiences four thermal inputs and one thermal sink:

Source	Flux	Direction	Variability
Direct solar	S = 1361 +/- 1 W/m^2 (at 1 AU)	Sun-facing surfaces only	Seasonal (+/- 3.3% due to Earth's orbital eccentricity: S_perihelion = 1414 W/m^2 in January, S_aphelion = 1322 W/m^2 in July)
Earth albedo	alpha_E x S ~= 0.30 x 1361 ~= 408 W/m^2	Earth-facing surfaces (nadir)	Varies with cloud cover, surface type (0.06 for ocean to 0.80 for fresh snow); orbit-average range 0.25--0.35
Earth infrared	q_IR ~= 240 W/m^2 (orbit average)	Earth-facing surfaces (nadir)	Range 200--270 W/m^2 depending on latitude, season, cloud cover
Internal dissipatio	Q_int = P_dissipated	From electronics waste heat	Varies with operational mode; nearly all electrical power eventually becomes heat
Deep space (sink)	T_space ~= 2.7 K (CMB)	Zenith-facing radiator surfaces	Effectively 0 K for engineering purposes

View Factors

The fraction of a surface's radiative "view" that sees each thermal source is critical for accurate thermal modelling. For a nadir-pointing spacecraft in LEO:

$F_Earth = (1)/(1 + (h/R_E)^2 + 2(h/R_E))$

where h = altitude (km) and R_E = 6371 km. For a 500 km orbit: F_Earth = 1 / (1 + (500/6371)^2 + 2 x 500/6371) = 1 / 1.163 = 0.860. The nadir face sees 86% Earth and 14% deep space. The zenith face sees 100% deep space (assuming no S/C self-shadowing). Side faces see a mix.

At higher altitudes, F_Earth decreases: at 800 km it is 0.79, at 35,786 km (GEO) it is only 0.018 -- which is why GEO thermal design is dominated by solar flux and internal dissipation, not Earth IR/albedo.

Thermal Balance Equation -- Full Derivation

> *Key Equations -- Thermal Equilibrium*

At steady state, the absorbed heat equals the radiated heat:

> $Q_{in} = Q_{out}$
>
> *Absorbed heat (expanded):*
>
> $Q_{in} = \underbrace{\alpha_s * A_{sun} * S}_{direct\ solar} + \underbrace{\alpha_s * A_{alb} * F_{alb} * \alpha_E * S}_{Earth\ albedo} + \underbrace{\varepsilon * A_{IR} * F_{IR} * q_{IR}}_{Earth\ IR} + \underbrace{Q_{int}}_{internal\ dissipation}$
>
> *Radiated heat:*
>
> $Q_{out} = \varepsilon * \sigma * A_{rad} * T^4$
>
> *where:*
>
> *- α_s = solar absorptance of surface coating (dimensionless, 0--1)*
>
> *- ε = infrared emittance of surface coating (dimensionless, 0--1)*
>
> *- $\sigma = 5.670 \times 10^{-8}\ W/m^2/K^4$ (Stefan-Boltzmann constant)*
>
> *- A_{sun}, A_{alb}, A_{IR}, A_{rad} = projected areas for each flux (m^2)*
>
> *- F_{alb}, F_{IR} = view factors to Earth for albedo and IR surfaces*
>
> *- T = equilibrium temperature (K)*
>
> *Note on α_s vs ε: Solar absorptance (α_s) is measured over the solar spectrum (0.2--2.5 um, peak at 0.5 um visible). Infrared emittance (ε) is measured over the thermal IR spectrum (3--50 um, peak at ~10 um for room-temperature objects). These are different spectral ranges, so $\alpha_s \neq \varepsilon$ for most real surfaces. This decoupling is the basis of all passive thermal control: by choosing the α_s / ε ratio, the designer controls the equilibrium temperature.*
>
> *Solving for equilibrium temperature:*
>
> $T = \left(\frac{Q_{absorbed} + Q_{internal}}{\varepsilon \sigma A_{rad}}\right)^{1/4}$
>
> *This equation is valid only for a single isothermal node (lumped-parameter model). For multi-node thermal models (which are needed for any real spacecraft), the heat balance is solved simultaneously for all nodes using numerical methods (Thermal Desktop, ESATAN, or similar thermal analysis software).*

Hot Case and Cold Case

Hot case	Maximum solar exposure (S = 1414 W/m^2, perihelion), all subsystems active (max Q_int), worst sun angle (max A_sun), BOL coatings (alpha_s at minimum -- fresh white paint), max albedo (0.35)	Components exceed maximum operating temperature
Cold case	Eclipse (no solar), minimum power dissipation (safe mode, min Q_int), EOL coatings (alpha_s increased by UV darkening), minimum Earth IR (200 W/m^2), deep space view	Components fall below minimum operating temperature

Design philosophy: The thermal engineer designs to keep all components within their qualified temperature range under both worst-case hot and worst-case cold conditions, with ECSS-mandated margins applied.

Surface Coatings -- The Passive Thermal Toolbox

Coating	alpha_s (BOL)	alpha_s (EOL, 5 yr	\varepsilon	alpha_s / \varepsilon	Use Case
White paint (AZ-93, S13G-LO)	0.14--0.20	0.25--0.35	0.89--0.92	0.16--0.22	Radiator surfaces (stay cool; low solar absorption, high IR emission)
Black paint (Aeroglaze Z306)	0.95	0.95	0.89	1.07	Internal surfaces (maximise radiative exchange between components)
Gold tape (2 mil Kapton + VDA)	0.22--0.25	0.25--0.30	0.03--0.05	5.0--7.5	MLI outer layer, thermal isolation
Bare aluminium (polished)	0.10--0.15	0.15--0.20	0.03--0.05	2.5--4.0	Reflective surfaces, low emissivity
Alodine (chromate conversion on Al)	0.35--0.40	0.40--0.50	0.12--0.16	2.5--3.0	Moderate thermal control, structural surfaces
Anodised aluminium (clear or black)	0.30--0.50	0.35--0.55	0.75--0.86	0.4--0.65	CubeSat external structure (standard finish per CDS)
MLI blanket (effective)	0.05--0.15	0.10--0.20	0.02--0.05	~3	Thermal isolation of sensitive components
Solar cells (with cover glass)	0.75--0.92	0.78--0.92	0.80--0.85	~1.0	SA surfaces (high absorption is unavoidable; cells get hot)
OSR (Optical Solar Reflector)	0.05--0.08	0.08--0.12	0.78--0.80	0.06--0.10	High-performance radiators (large S/C, GEO)

Key design insight: A surface with low alpha_s / \varepsilon (e.g., white paint: 0.2) stays cool because it reflects most solar energy but efficiently radiates thermal IR. A surface with high alpha_s / \varepsilon (e.g., gold tape: 6.0) stays warm because it absorbs solar energy but barely radiates. This is why white paint is used on radiators and gold/MLI is used for insulation.

Coating degradation: UV radiation darkens most white paints over time, increasing alpha_s while leaving \varepsilon nearly unchanged. This means alpha_s / \varepsilon increases, and the surface gets hotter at EOL. The thermal engineer must design the hot case with BOL coatings (which give the highest alpha_s... wait -- actually BOL white paint has the lowest alpha_s, so the cold case is more conservative at BOL, and the hot case is more conservative at EOL when alpha_s has increased. This is a common source of confusion: BOL coatings give a colder cold case; EOL coatings give a hotter hot case.

MLI (Multi-Layer Insulation) -- Construction and Physics

MLI blankets are the most common thermal insulation on spacecraft. They work by minimising both radiation and conduction heat transfer through multiple reflective layers separated by low-conductance spacers.

Construction (typical MLI blanket):

1. Outer cover: 1 mil (25 um) aluminised Kapton (VDA -- Vapour Deposited Aluminium on one side). Provides mechanical protection and low solar absorptance.
2. Inner reflective layers: 10--20 layers of 0.25 mil (6 um) double-aluminised Mylar (DAM). Each layer reflects IR radiation, and the vacuum gaps between layers have zero convection.
3. Spacer material: Dacron or Nomex netting between each DAM layer, preventing conductive contact between adjacent reflective sheets.
4. Inner cover: 1 mil aluminised Kapton, protecting the inner layers.

Effective emissivity: An ideal MLI blanket with N reflective layers has an effective emissivity of:

$\varepsilon_{eff} = \frac{1}{2/\varepsilon_{inner} + (N-1)(2/\varepsilon_{layer} - 1)}$

For 20 layers of DAM ($\varepsilon_{layer} = 0.03$): $\varepsilon_{eff} \approx 0.002$. In practice, real MLI achieves $\varepsilon_{eff} = 0.01$--0.03 due to seams, penetrations (harness, mounting), and edge effects. The ratio of actual to theoretical performance is typically 2--5x worse.

CubeSat MLI challenges: CubeSats have limited surface area and many penetrations (connectors, antennas, sensors, solar cells), making it difficult to achieve good MLI performance. Most CubeSats in LEO do not use MLI -- they rely on the moderate thermal environment (Earth IR provides a "warm floor") and surface coatings. MLI becomes essential for deep-space CubeSats (e.g., MarCO, CAPSTONE) or missions with sensitive payloads (IR detectors, laser systems).

Heater Sizing

When passive thermal control cannot prevent a component from falling below its minimum temperature (typically during eclipse or safe mode), electrical heaters are required.

> *Key Equations -- Heater Sizing*
>
> *Required heater power (to maintain minimum temperature during worst cold case):*
>
> $P_{heater} = \varepsilon \sigma A_{rad} T_{min}^4 - Q_{environment,cold} - Q_{internal,cold}$
>
> *where T_{min} is the minimum allowable temperature of the component (converted to Kelvin).*
>
> *Heater types for CubeSats:*
>
> *- Kapton foil heaters: Etched-foil resistance elements laminated between Kapton sheets. Flexible, thin (0.2--0.5 mm), lightweight (2--10 g each). Typical power: 0.5--5 W per heater. Bond directly to component surface with pressure-sensitive adhesive.*
>
> *- Cartridge heaters: Cylindrical, inserted into drilled holes. Higher power density but heavier and less common on CubeSats.*
>
> *Thermostat control: Simple bimetallic thermostats (e.g., Honeywell Klixon) switch heaters on/off at set temperatures (e.g., on at -5 degC, off at +5 degC). Mass: ~2 g each. For higher reliability, software-controlled heaters using temperature sensor feedback and EPS switches are preferred on modern CubeSats -- but this*

requires OBC to be running, which may not be the case in safe mode.

Heat Pipes and Thermal Straps

Heat pipes are passive two-phase heat transfer devices that transport large amounts of thermal energy with very small temperature differences. They are widely used on larger spacecraft and are beginning to appear on 6U+ CubeSats.

How a heat pipe works:

1. Working fluid (ammonia, methanol, or water) evaporates at the hot end (evaporator), absorbing latent heat
2. Vapour travels through the hollow pipe core to the cold end (condenser)
3. At the condenser, vapour releases latent heat and condenses back to liquid
4. Liquid returns to the evaporator via capillary action in a wick structure (sintered metal, axial grooves, or screen mesh)
5. The process is continuous, passive (no moving parts, no power), and can transport 10--100 W across 20--50 cm with < 5 degC temperature difference

Thermal conductance of a heat pipe: Effective thermal conductivity is 10,000--100,000 W/m/K (compared to copper at 400 W/m/K and aluminium at 237 W/m/K). A 6 mm diameter ammonia heat pipe can transport ~30 W over 30 cm with < 3 degC gradient.

Thermal straps are flexible conductive links (braided copper, graphite fibre, or pyrolytic graphite sheet) used to conduct heat between components that cannot be rigidly connected (e.g., across a hinge or between a vibration-isolated payload and the spacecraft bus). Typical conductance: 0.5--5 W/K.

Heat Transport Method	Conductance	Mass	Power	Orientation Sensitivity	CubeSat Use
Aluminium conduction	~0.5--2 W/K per path	Part of structure	0 W	None	Always (inherent)
Copper thermal strap	1--5 W/K	10--50 g	0 W	None	Occasional (6U+)
Heat pipe (ammonia)	5--50 W/K	20--100 g	0 W	Gravity-dependent (must test in relevant orientation)	Rare (6U+, some 3U)
Pumped fluid loop	50--500 W/K	500+ g	5--20 W	None	Large S/C only

Thermal Control Methods Summary

Method	Type	Mass Impact	Typical Use	Key Design Parameter
Surface coatings	Passive	Negligible	Always -- select alpha_s/\varepsilon ratio per face	alpha_s/\varepsilon ratio
MLI blankets	Passive	0.5--2.0 kg/m^2	Insulate sensitive components from environment	Number of layers, \varepsilon_eff
Radiators	Passive	Part of structure	Reject internal waste heat to deep space	Radiator area, \varepsilon, view to space
Heaters	Active	0.005--0.02 kg each	Maintain minimum temp during eclipse/safe mode	Power, thermostat set point
Heat pipes	Passive	0.02--0.10 kg each	Transport heat from source to radiator	Working fluid, Q_max, orientation
Thermal straps	Passive	0.01--0.05 kg each	Flexible conductive link across joints/hinges	Conductance (W/K)

Louvers	Active	0.1--0.5 kg	Variable-conductance radiators (rare on CubeSats)	Open/close temperature range

ECSS Thermal Margins

[Source: ECSS-E-ST-31C, Table 5-1]

Phase	Hot Margin (above predicted max)	Cold Margin (below predicted min)
Qualification	Predicted + 15 degC	Predicted - 15 degC
Acceptance	Predicted + 10 degC	Predicted - 10 degC
Operating	Predicted + 5 degC	Predicted - 5 degC

These margins ensure that thermal model uncertainties (typically +/- 5--10 degC for simplified models, +/- 2--5 degC for detailed correlated models) do not cause in-orbit temperature exceedances.

> *Worked Example -- 3U CubeSat Full Thermal Analysis*
>
> *Hot case (sunlit, all systems active, perihelion, EOL coatings):*
>
> *Simplified single-node model. 3U CubeSat, nadir-pointing.*
>
> *- Sun-facing area (+Z, zenith-facing 3U panel): A_sun = 0.034 m^2*
> *- SA surfaces (sun-facing): alpha_s = 0.85, \varepsilon = 0.82 (solar cell properties)*
> *- Nadir face (-Z): A_nadir = 0.034 m^2, anodised Al (alpha_s = 0.45 EOL, \varepsilon = 0.82)*
> *- Side faces (4x): A_side = 4 x 0.01 m^2 = 0.04 m^2, anodised Al*
> *- Internal dissipation (imaging mode): Q_int = 10.0 W*
>
> *Q_solar = 0.85 x 0.034 x 1414 = 40.9 W (ouch -- but much of this is captured by the SA and converted to electricity, so the net thermal input from solar cells is Q_solar,thermal = alpha_s x A x S x (1 - eta_cell) = 0.85 x 0.034 x 1414 x 0.705 = 28.8 W)*
>
> *Q_albedo = 0.45 x 0.034 x 0.35 x 1414 = 7.6 W*
>
> *Q_Earth IR = 0.82 x 0.034 x 270 = 7.5 W (nadir face, hot case Earth IR)*
>
> *Q_int = 10.0 W (but ~12.6 W of the 10 W load power ultimately becomes heat after doing useful work)*
>
> *Total Q_in ~= 28.8 + 7.6 + 7.5 + 10.0 = 53.9 W*
>
> *Total radiating area (all 6 faces, minus SA area which is a net absorber): A_rad ~= 0.066 m^2 (accounting*

for partial Earth blockage on nadir face)

Average emissivity: $\varepsilon_{avg} \approx 0.82$

$T_{hot} = ((53.9)/(0.82 \times 5.67 \times 10^{-8} \times 0.066))^{0.25} = ((53.9)/(3.07 \times 10^{-9}))^{0.25}$

$= (1.756 \times 10^{10})^{0.25} = 364$ K = +91 degC

This exceeds most component limits! However, this simplified calculation overestimates temperature because it treats the satellite as a single isothermal node and includes solar cell thermal absorption on the zenith face. In practice:

- The zenith face (solar cells) runs hotter than the bus

- Internal components are conductively coupled to all faces, including the cold nadir face

- A multi-node model typically predicts peak internal temperatures of +40 to +55 degC for this scenario

Thermal engineer's response: If the single-node calculation exceeds 60 degC, a detailed multi-node thermal model is required. The simplified calculation is a screening tool, not a design tool.

Cold case (eclipse, safe mode, aphelion, BOL coatings):

- No solar flux, no albedo

- Earth IR only: $Q_{Earth\ IR} = 0.82 \times 0.034 \times 200 = 5.58$ W (cold case: 200 W/m^2)

- Internal dissipation (safe mode): $Q_{int} = 1.5$ W (OBC + heater)

- Total $Q_{in} = 5.58 + 1.5 = 7.08$ W

$T_{cold} = ((7.08)/(0.82 \times 5.67 \times 10^{-8} \times 0.070))^{0.25}$

$= ((7.08)/(3.26 \times 10^{-9}))^{0.25} = (2.172 \times 10^{9})^{0.25} = 216$ K = -57 degC

This is too cold for Li-ion batteries (min -20 degC operating, min 0 degC charging) and most COTS electronics (min -40 degC).

Action: Add battery heater. To maintain battery at T_{min} = -10 degC = 263 K:

Need additional heat input: $Q_{heater} = \varepsilon \sigma A_{rad} T_{min}^4 - Q_{other}$

= 0.82 x 5.67 x 10^-8 x 0.070 x 263^4 - 7.08 = 3.26 x 10^-9 x 4.78 x 10^9 - 7.08 = 15.6 - 7.08 = 8.5 W

> *Problem: 8.5 W heater in eclipse exceeds the battery capacity. Resolution: This is an isothermal whole-spacecraft calculation. In reality, the battery is inside the bus, partially insulated by the structure and surrounding boards. A targeted heater of 0.5--1.0 W directly on the battery pack, with some MLI wrapping, is typically sufficient to keep the battery above -10 degC in a 35-minute eclipse. The structure's thermal mass (aluminium at c_p = 900 J/kg/K) provides significant thermal inertia -- a 1 kg 1U CubeSat cooling from +20 degC at 7 W net loss drops only about 16 degC in 35 minutes.*
>
> *Transient check:*
>
> *D T = \fracQ_net x tm x c_p = ((7.08 - 0) x 35 x 60)/(5.0 x 900) = (14,868)/(4500) = 3.3 degC per 35-min eclipse*
>
> *Wait -- in eclipse Q_out > Q_in: net cooling rate = \varepsilon sigma A_rad T^4 - Q_in,eclipse. At T = 293 K (20 degC):*
>
> *Q_out = 0.82 x 5.67 x 10^-8 x 0.070 x 293^4 = 3.26 x 10^-9 x 7.37 x 10^9 = 24.0 W*
>
> *Q_net cooling = 24.0 - 7.08 = 16.9 W*
>
> *D T = (16.9 x 2100)/(5.0 x 900) = (35,490)/(4500) = 7.9 degC drop in 35 minutes*
>
> *So from +20 degC, the satellite cools to about +12 degC after one eclipse -- well within limits. No heater needed for the bus; battery heater only if battery is thermally isolated from bus.*
>
> *ECSS margin check -- Payload CCD:*
>
> *Predicted maximum temperature of payload CCD = 42 degC.*
>
> *- Operating limit = 50 degC. Margin = 50 - 42 = 8 degC > 5 degC. Pass.*
>
> *- Qualification test: must test at 42 + 15 = 57 degC. If qualification limit is 60 degC: Pass.*
>
> *Predicted minimum temperature of battery = -8 degC during worst eclipse.*
>
> *- Operating limit = -10 degC. Margin = -8 - (-10) = 2 degC < 5 degC. Fail -- inadequate margin.*
>
> *- Action: Add heater (0.5 W survival heater with thermostat set to -5 degC), or add MLI around battery pack.*

1U Worked Example: UniSat-1

Power Sizing: Body-Mounted Only

UniSat-1 uses body-mounted solar cells on all five sun-exposed faces (the sixth face mounts the deployment switch interface). With no deployable panels, the power system is simpler, lighter, and cheaper -- but severely power-limited.

> *Worked Example -- UniSat-1 Solar Array Sizing*
>
> *Given: Body-mounted cells on 5 faces of a 1U (100 x 100 mm each). ISS orbit: 400 km, 51.6 deg inclination, 92.4 min period, 56 min sunlight, 36 min eclipse. $P_{eclipse}$ = 0.5 W (OBC only), $P_{peak,sunlight}$ = 1.2 W (science + downlink overlap avoided by scheduling). Mission lifetime = 6 months. Cell temperature = 55 degC (body-mounted cells run cooler on 1U due to better thermal coupling to bus mass).*
>
> *Step 1 -- Effective illuminated area:*
>
> *At any given time in LEO with passive magnetic attitude (slow tumble ~1 deg/s), on average only ~1.5 faces are well-illuminated. Effective average area:*
>
> $A_{eff} \approx 1.5 \times (0.10 \times 0.10) = 0.015\ m^2$
>
> *Step 2 -- SA BOL power (with temperature derating):*
>
> $L_T = 1 + (-0.0019)(55 - 28) = 0.949$
>
> $P_{SA,BOL} = \eta_{cell} \times L_T \times S \times A_{eff} \times f_{pack} = 0.295 \times 0.949 \times 1361 \times 0.015 \times 0.80 = 4.57$ W *(illuminated peak)*
>
> *Step 3 -- Orbit-average power available:*
>
> $P_{avg,avail} = P_{SA,BOL} \times \text{\fract\_sunT} \times \eta_{EPS} = 4.57 \times (56)/(92.4) \times 0.85 = 2.35$ W
>
> *After 6-month degradation ($(1 - 0.025)^{0.5} = 0.987$):*
>
> $P_{avg,EOL} = 2.35 \times 0.987 = 2.32$ W
>
> *Step 4 -- Power demand (orbit-average):*
>
> *From Session 2.4: $P_{avg,demand}$ = 0.68 W.*
>
> *Power margin: 2.32 - 0.68 = 1.64 W (71% margin). Even with conservative geometry assumptions, the link closes comfortably.*
>
> *Note on body-mounted vs tumbling: The key uncertainty in 1U body-mounted power is the attitude. With passive magnetic stabilisation, the satellite aligns roughly with Earth's magnetic field, providing more predictable illumination than a random tumble. However, the effective area varies significantly around the orbit as the B-field direction changes with latitude. The 1.5-face average is conservative. A Monte Carlo*

simulation of illumination geometry over many orbits typically yields a more optimistic 1.7--1.9 effective face average.

> *Worked Example -- UniSat-1 Battery Sizing*
>
> *Given: P_eclipse = 0.5 W, t_eclipse = 36 min = 0.60 h, DOD = 0.50 (acceptable for 6-month mission: ~2,740 cycles), eta = 0.95.*
>
> *E_bat = (0.5 x 0.60)/(0.50 x 0.95) = (0.30)/(0.475) = 0.63 Wh*
>
> *With margin: specify minimum 10 Wh (standard GomSpace NanoPower P31u battery pack -- this is the smallest available COTS battery with flight heritage).*
>
> *Cycle count check: 6 months at 15 orbits/day = 2,740 eclipses. At 50% DOD, Li-ion cells comfortably survive > 2,000 cycles. Pass.*
>
> *Actual operating DOD: With 10 Wh battery and 0.63 Wh per eclipse demand, actual DOD = 0.63 / 10 = 6.3% per eclipse. At this DOD, cycle life exceeds 100,000 cycles. Battery degradation is negligible over the 6-month mission.*
>
> *Key insight: The 10 Wh battery is massively oversized for the actual eclipse demand. This is common in 1U missions -- the minimum COTS battery available provides far more capacity than needed. The excess capacity provides excellent margin and enables recovery from anomalies (multiple missed sunlit periods).*

Thermal: Passive Only

UniSat-1 uses no heaters, no MLI, and no active thermal control. This is justified by three factors:

1. Low altitude (400 km): Strong Earth IR flux (~240 W/m^2) provides a warm floor, preventing extreme cold cases
2. Short mission (6 months): No long-term coating degradation to worry about
3. Tolerant components: COTS electronics typically operate from -20 degC to +60 degC; the 400 km LEO thermal environment stays within -10 degC to +45 degC for a 1U with standard aluminium/anodised surfaces

> *Quick Thermal Check -- UniSat-1 Cold Case (Transient)*
>
> *Worst eclipse, all subsystems off except OBC (0.5 W internal dissipation):*
>
> *- Earth IR absorbed: \varepsilon x A_nadir x q_IR = 0.85 x 0.01 x 240 = 2.04 W*
>
> *- Internal dissipation: 0.5 W*
>
> *- Total heat in: 2.54 W*

Steady-state temperature (if eclipse were infinite):

> *T_steady = (2.54 / (0.85 x 5.67 x 10^-8 x 0.045))^0.25 = 195 K = -78 degC*
>
> *This looks alarming -- but the eclipse is only 36 minutes, and the thermal mass prevents the satellite from reaching steady state.*
>
> *Transient analysis: Starting at T_0 = +15 degC (293 K) at eclipse entry:*
>
> *Net cooling rate at 293 K: Q_rad,out = 0.85 x 5.67 x 10^-8 x 0.045 x 293^4 = 2.17 x 10^-9 x 7.37 x 10^9 = 16.0 W*
>
> *Net cooling: 16.0 - 2.54 = 13.5 W*
>
> *Temperature drop: D T = \fracQ_net x tm x c_p = (13.5 x 36 x 60)/(1.0 x 900) = (29,160)/(900) = 32 degC*
>
> *But this is a linear approximation -- as the satellite cools, the radiation rate drops as T^4, so the actual cooling slows. A more accurate estimate gives D T ~= 20--25 degC, resulting in a minimum temperature of about -5 to -10 degC.*
>
> *Conclusion: The 1U at 400 km reaches approximately -5 to -10 degC during worst-case eclipse, starting from a warm sunlit entry. This is within COTS operating limits (-20 degC to +60 degC for most components) and within battery operating range (-20 degC to +60 degC for discharge). No heaters needed. The ISS orbit's relatively short eclipse (36 min vs 35 min for SSO) and the strong Earth IR flux at 400 km make passive thermal control viable for a 1U mission.*

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5. Real Mission Examples (10 min)

Planet SuperDove EPS

Parameter	Value	Design Rationale
Form factor	3U+, ~5 kg	Flock constellation; P-POD compatible
SA configuration	Body-mounted + two deployable wings	~25 W BOL needed for continuous imaging + S-band downlink
SA cells	Triple-junction GaAs (Spectrolab or SolAero)	Standard space-grade, 29.5% AM0
Battery	Li-ion, ~20 Wh (2S2P 18650)	Supports ~3.5 W eclipse load for 35 min at < 20% DOD
Bus voltage	Unregulated 7.2--8.4 V (2S) + regulated 3.3 V, 5 V	Standard CubeSat EPS architecture
Peak demand	~18 W (imaging mode)	Multi-spectral imager + star tracker + reaction wheels + OBC
Orbit	475 km SSO, ~94 min period	Optimal for EO: sun-synchronous for consistent lighting

Eclipse

~35 min max, ~3 W demand

OBC + AOCS only during eclipse; no imaging or downlink

Thermal	Body-mounted radiator panels + battery heater	Passive thermal control; battery heater for eclipse charging margin

[Source: Planet Labs conference presentations; Salas et al., "SuperDove Constellation," SSC 2021]

CAPSTONE Thermal Design

NASA's CAPSTONE (12U, 25 kg) operates in a near-rectilinear halo orbit (NRHO) around the Moon with extreme thermal cycling:

- Perilune: strong Earth/Moon IR + solar
- Apolune: deep space cold, long shadow periods (up to 12+ hours)
- Thermal control: MLI wrapping (10-layer DAM blankets), heaters on propulsion lines (to prevent propellant freezing), passive radiator panels with white paint (AZ-93)
- Battery heaters: 5 W total, thermostatically controlled, critical for survival during long eclipses
- Operating temperature range: -20 degC to +50 degC for electronics; propulsion lines maintained above +5 degC

[Source: Advanced Space, "CAPSTONE Design Overview," SmallSat Conference 2022]

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6. SpaceCDF Exercise (30 min)

Instructions

1. Run the design in SpaceCDF if not already converged
2. Dashboard -- review power KPIs:
- Is power margin positive in all modes (sunlight, eclipse, safe)?
- What is the orbit-average power demand?
- What SA configuration was selected (body/deployable)?
3. Timing Budget card -- review mode durations:
- Does the duty cycle match your ConOps modes?
- Is eclipse time consistent with your orbit calculation from Session 2.3?
4. Engineering Budgets -- review the power waterfall:
- Which subsystem is the largest power consumer in each mode?
- Where could power be reduced if the budget is tight?
5. Parametric tab -- review thermal predictions:
- Hot case temperature prediction
- Cold case temperature prediction
- Any exceedances?

Worksheet 3.1 Tasks

1. Size the solar array for your mission (show all 5 calculation steps, including temperature derating)

2. Size the battery (show calculation with DOD justification and cycle-life verification)
3. Compute orbit-average power using duty cycle table
4. Identify hot case and cold case conditions for your orbit
5. Check thermal margins against ECSS requirements
6. Identify the dominant thermal concern for your mission (hot case or cold case) and propose a mitigation

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Session Summary

Topic	Key Takeaway
Solar cell physics	Triple-junction GaAs (InGaP/GaAs/Ge) achieves 28--30% AM0; multi-junction stacking captures broader solar spectrum
Temperature effects	Cells lose ~0.19%/degC (relative); operating at 65 degC costs ~7% power vs STC
Degradation	Radiation damage: (1-delta)^n, delta ~= 2.5%/yr LEO; cover glass mitigates proton/electron damage
EPS architecture	MPPT + regulated bus is CubeSat standard; MPPT extracts 10--15% more power than DET
SA sizing	P_SA = P_peak + P_recharge; derate for degradation (1-delta)^n and temperature L_T
SA area	A = P_BOL / (eta * S * costheta * f_pack * f_cover)
Battery chemistry	Li-ion (LCO/NMC): 150--200 Wh/kg packaged; 3.0--4.2 V per cell; CC-CV charging
Battery sizing	E = P_ecl * t_ecl / (DOD * eta); DOD 20--30% for multi-year LEO
Battery failure modes	Thermal runaway (overcharge), lithium plating (cold charge), capacity imbalance (series cells)
SA power reference	Body-mounted: 2--12 W; single deploy: 4--30 W; dual deploy: 25--48 W
Thermal physics	No convection in vacuum; radiation (\varepsilon sigma T^4) is only mechanism for heat rejection
Thermal balance	Q_in = Q_out; solve for T = (Q/\varepsilonsigma A)^1/4 (single-node); multi-node for real design
alpha_s / \varepsilon ratio	Low ratio = cold surface (radiator); high ratio = warm surface (insulation)
MLI construction	VDA Kapton outer + 10--20 DAM layers + Dacron spacers; \varepsilon_eff = 0.01--0.03
Heater sizing	P_heater = \varepsilonsigma A T_min^4 - Q_env - Q_int; Kapton foil heaters, thermostat control
Thermal margins	ECSS: +/-5 degC operating, +/-10 degC acceptance, +/-15 degC qualification
Transient effects	Thermal mass (mc_p) prevents reaching steady state during short eclipses; 1U at 400 km cools ~20--25 degC in 36 min eclipse

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Session 3.2: Attitude and Orbit Control System (AOCS)

Duration: 2 hours

Prerequisites: Sessions 2.1--2.4 and 3.1 (requirements, orbit, power defined)

SpaceCDF Tabs: Dashboard (AOCS KPI), Pointing Budget, Architecture (AOCS)

References

- Wertz, Everett & Puschell, Space Mission Engineering: The New SMAD, 2011, Ch. 11.1 (ADCS)
- Sidi, Spacecraft Dynamics and Control, 1997, Ch. 4--9
- Markley & Crassidis, Fundamentals of Spacecraft Attitude Determination and Control, 2014
- ECSS, ECSS-E-ST-60-10C: Control Performance, 2008
- ECSS, ECSS-E-ST-60-20C: Star Tracker Performance Testing, 2019
- Wertz, Space Mission Analysis and Design, 3rd ed., 1999, Ch. 11 (ADCS)
- Hughes, Spacecraft Attitude Dynamics, 1986
- Blue Canyon Technologies, XACT ADCS Datasheet, 2023
- CubeSpace, ADCS Product Catalogue, 2023

Learning Objectives

By the end of this session, participants will be able to:

1. Explain the distinction between attitude determination and attitude control and the physics of each sensor/actuator type
2. Select AOCS hardware architecture based on pointing requirements
3. Compute a pointing error budget using root-sum-square (RSS) combination
4. Calculate disturbance torques from gravity gradient, aerodynamic drag, solar radiation pressure, and residual magnetic dipole
5. Size actuators (reaction wheels, magnetorquers) for the computed disturbance environment with worked equations
6. Explain the physics of momentum storage, saturation, and desaturation
7. Compare CMGs vs reaction wheels and articulate when each is appropriate
8. Verify AOCS design against requirements using SpaceCDF's pointing budget tool

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1. AOCS Fundamentals (25 min)

Teaching Notes

[Source: SMAD, Ch. 11.1; Markley & Crassidis, Ch. 1--3]

The AOCS performs two distinct functions:

Function	Purpose	Hardware
Attitude Determination	Know the spacecraft's orientation relative to a reference frame	Sensors: star tracker, sun sensors, magnetometer, gyroscope, Earth sensor, GPS receiver
Attitude Control (AC)	Change or maintain the spacecraft's orientation	Actuators: reaction wheels, magnetorquers, thrusters, control moment gyros

The attitude state: A spacecraft's attitude is its orientation in 3D space relative to a reference frame (typically J2000 Earth-centred inertial, or the local-vertical/local-horizontal frame for nadir-pointing missions). The attitude is described by a rotation (e.g., quaternion, direction cosine matrix, or Euler angles) plus angular velocity \vecomega. Euler's equation of rotational motion governs the dynamics:

$I \dot\vecomega + \vecomega x (I \vecomega) = \vecT_external + \vecT_control$

where I is the spacecraft inertia tensor, \vecT_external is the sum of disturbance torques, and \vecT_control is the actuator torque. The \vecomega x (I \vecomega) term is the gyroscopic coupling -- it means that rotating about one axis can induce motion about other axes if the inertia tensor is not spherically symmetric.

Attitude Sensors -- How They Work

Star Trackers

A star tracker is the most accurate attitude sensor available, providing absolute attitude knowledge to arcsecond-level accuracy.

How it works:

1. A CMOS or CCD detector (typically 1024x1024 to 2048x2048 pixels) images a patch of sky through a wide-angle lens (typically 15--25 deg FOV)
2. The onboard processor detects bright point sources (stars) in the image, computing their centroid positions to sub-pixel accuracy using Gaussian fitting
3. The processor matches the observed pattern of star positions against an onboard star catalogue (typically Hipparcos or Tycho-2, containing 3,000--10,000 stars, stored in a k-d tree or hash table for fast lookup)
4. The "lost in space" algorithm identifies the star pattern without any prior attitude knowledge (first acquisition), typically taking 1--5 seconds
5. Once identified, the processor computes a quaternion rotation from the catalogue (inertial) frame to the camera (body) frame
6. In tracking mode, the processor tracks known stars frame-to-frame, providing updates at 1--10 Hz with 3--10 arcsec accuracy (1-sigma, per axis)

Key specifications:

Parameter	Typical CubeSat Star Tracker	Large S/C Star Tracker
Accuracy (1-sigma, boresight)	5--15 arcsec	0.5--3 arcsec
Accuracy (1-sigma, roll)	30--100 arcsec	5--20 arcsec
FOV	10--15 deg circular	15--25 deg circular
Update rate	2--5 Hz	5--20 Hz
Sensitivity	Stars to magnitude 6--7	Stars to magnitude 8--10
Mass	50--350 g	1--5 kg
Power	0.5--2 W	5--15 W
Products	Blue Canyon NST (350g), Sinclair SS-411 (90g), CubeSpace CubeStar (50g)	Sodern Hydra, Leonardo AA-STR

[Source: Blue Canyon NST datasheet; CubeSpace CubeStar datasheet; ECSS-E-ST-60-20C]

Exclusion zones: Star trackers cannot operate when bright objects are in or near the FOV:

- Sun: Exclusion angle typically 25--45 deg (direct sunlight saturates the detector and can cause permanent damage to some sensor types)

- Earth (illuminated limb): Exclusion angle typically 25--35 deg (Earth's brightness overwhelms star signals)
- Moon: Exclusion angle typically 10--15 deg
- Stray light: Internal reflections from nearby spacecraft structure can create false stars

Implication for mission design: A nadir-pointing spacecraft in LEO always has the Earth within ~65 deg of one hemisphere. The star tracker must be mounted on a face that never points toward Earth (typically the zenith or anti-velocity face). If the Sun is near the orbital plane (beta angle near 0), the star tracker may be periodically blinded. Two star trackers mounted on different faces provide redundancy and eliminate single-axis exclusion zone gaps.

Why star trackers are the most accurate: Other sensors measure vectors to specific objects (Sun, Earth, magnetic field) -- each provides only 2 of the 3 attitude degrees of freedom (direction but not roll around that vector). A star tracker measures multiple star directions simultaneously, providing a full 3-axis attitude fix from a single measurement. The accuracy is limited by optical diffraction, centroiding noise, and catalogue accuracy -- all of which are extremely well characterised.

Sun Sensors

Sun sensors determine the direction to the Sun, providing 2-axis attitude information (the Sun vector in the body frame). They are simple, reliable, radiation-tolerant, and low-power.

Coarse sun sensors (photodiode-based):

A coarse sun sensor consists of one or more photodiodes behind a mask or window. The photocurrent is proportional to the cosine of the incidence angle:

$I = I_0 \cos(\theta)$

A set of 6 coarse sun sensors (one per face of the spacecraft) determines the Sun direction to 2--5 deg accuracy by comparing the photocurrents from each face. The face with the highest current is sun-facing; the ratio between adjacent faces gives the angle.

Fine sun sensors (linear array or quadrant detector):

A fine sun sensor uses a slit mask above a linear photodiode array (similar to a miniature sundial). Sunlight passes through the slit and illuminates a specific position on the array, which is proportional to the incidence angle. Fine sun sensors achieve 0.1--1.0 deg accuracy.

Type	Accuracy	FOV	Mass	Power	Products
Coarse (photodiode)	2--5 deg	~hemisph	1--5 g	< 1 mW	NewSpace NCSS-SA05, Solar MEMS
Fine (analog, slit+array)	0.1--0.5 deg	60--120	5--30 g	10--50	NewSpace NFSS-411, Solar MEMS
Digital (APS detector + slit)	0.01--0.05 deg	60--120 deg	30--50 g	50--200 mW	TNO micro digital sun sensor

[Source: NewSpace Systems NFSS-411 datasheet; Solar MEMS nanoSSOC datasheets]

Why every spacecraft needs sun sensors: Sun sensors are the only sensor guaranteed to work in any attitude, at any rate, in any environment (LEO, GEO, deep space). They are the primary sensor for safe mode: when the spacecraft tumbles, the OBC reboots, or the star tracker is blinded, the sun sensors can determine the Sun direction and enable the spacecraft to orient its solar arrays for power generation. Without sun sensors, a spacecraft that enters safe mode may not recover power.

Magnetometers

A magnetometer measures the local geomagnetic field vector \vecB in the body frame. By comparing the measured \vecB to a model of Earth's magnetic field (IGRF -- International Geomagnetic Reference Field) at the known orbit position, the spacecraft attitude can be determined.

How it works:

- Fluxgate magnetometer: A ferromagnetic core is periodically driven into saturation by an excitation coil. The presence of an external magnetic field creates an asymmetry in the saturation waveform, which is detected by a sense coil. Three orthogonal fluxgate elements provide the 3-axis field vector. Resolution: 1--10 nT. Accuracy: +/-200--500 nT (limited by spacecraft residual magnetic dipole contamination).
- Magnetoresistive (AMR/GMR): Thin-film magnetic sensors. Smaller and cheaper than fluxgates but noisier and less stable. Common in COTS CubeSat magnetometers.

Attitude determination from magnetometers:

- A single magnetometer measurement provides the magnetic field direction in the body frame (2 DOF, analogous to a sun sensor providing the Sun direction)
- Comparing with the IGRF model gives attitude, but accuracy is limited by:
- IGRF model uncertainty (~100 nT in LEO)
- Spacecraft residual magnetic dipole contamination (can be 1000+ nT close to electronics)
- Only 2 DOF per measurement (no roll determination around the field vector)
- Magnetometer-only attitude determination: ~5--10 deg accuracy
- Magnetometer + sun sensor (combined): ~1--3 deg accuracy (the two independent vectors provide a full 3-axis solution via TRIAD or q-method)

Residual dipole contamination: Every electronic circuit creates a magnetic field. Current loops, permanent magnets in motors/speakers, magnetised structural components, and battery cells all contribute to the spacecraft's residual magnetic dipole. This contaminates the magnetometer reading. Mitigation: mount the magnetometer on a deployable boom (10--30 cm from the bus), use magnetically clean design practices (twisted pairs, balanced current loops), and perform a residual dipole calibration in orbit by comparing magnetometer readings during eclipse (no solar array current) vs sunlit.

Parameter	Typical CubeSat Magnetometer
Range	+/-60,000 nT (sufficient for LEO: B ~20,000--50,000 nT)
Resolution	10--100 nT
Accuracy (after calibration)	200--1000 nT
Mass	5--30 g
Power	10--50 mW
Products	NewSpace NMAG, PNI RM3100, Honeywell HMC5883L

Gyroscopes (Rate Sensors)

Gyroscopes measure angular velocity \vecomega rather than absolute attitude. They provide high-bandwidth rate information for control loops and bridge the gap between star tracker updates.

Types used on CubeSats:

- MEMS gyroscope: Vibrating structure (tuning fork or ring) whose Coriolis force is proportional to rotation rate. Small, cheap (< 50 EUR per axis), low power. Bias drift: 1--10 deg/hr. Products: InvenSense MPU-6050, Analog Devices ADIS16265.

- MEMS IMU (6-axis): Combined 3-axis accelerometer + 3-axis gyroscope. Mass: 5--15 g. Very common in CubeSats. Products: Sensonor STIM300 (high-end), InvenSense ICM-20948 (COTS).
- Fibre optic gyroscope (FOG): Much better stability (bias drift 0.01--1 deg/hr) but larger, heavier (200+ g), and more expensive. Used on high-end CubeSats and small satellites.

Why gyroscopes drift: MEMS gyroscopes have a non-zero bias (a constant offset in the measured rate even when stationary) that changes with temperature and time. Integrating angular rate to get attitude (theta = \int omega dt) accumulates this bias error linearly. A 1 deg/hr bias drift means the attitude estimate drifts 1 deg per hour. Star trackers provide absolute attitude corrections that reset this drift -- the combination of star tracker (low rate, absolute) + gyroscope (high rate, relative) via a Kalman filter is the standard approach for high-performance attitude determination.

GPS Receivers for Orbit Determination

GPS receivers determine the spacecraft's position and velocity (orbit determination), not attitude (though multi-antenna GPS can provide coarse attitude).

How it works in LEO:
- GPS satellites orbit at ~20,200 km altitude; LEO spacecraft orbit below them at 300--800 km
- GPS signals travel downward through the ionosphere to the LEO receiver
- The receiver must use a specialised correlator that handles Doppler shifts up to +/-40 kHz (LEO relative velocity ~7.5 km/s vs GPS satellite velocity ~3.9 km/s)
- Typical accuracy: 5--20 m position, 0.1--0.5 m/s velocity (single frequency, C/A code)
- Dual-frequency GPS with carrier phase: sub-meter position accuracy

Why GPS matters for AOCS: Accurate orbit knowledge is needed for:
- Nadir pointing (must know where "down" is, which requires knowing position)
- Ground target tracking (must know position to compute pointing angles)
- IGRF evaluation (magnetometer attitude determination needs position input)
- Orbit manoeuvre planning

Parameter	Typical CubeSat GPS Receiver
Position accuracy	5--20 m (C/A code)
Velocity accuracy	0.1--0.5 m/s
Time accuracy	100 ns
Altitude limit	Typically 600 km (must verify ITAR/COCOM limits removed)
Mass	15--30 g
Power	0.5--1.0 W
Products	SkyFox Labs piNAV-NG, NovAtel OEM719, Hemisphere V200

[Source: SkyFox Labs piNAV datasheet; NovAtel OEM7 specifications]

AOCS Architecture Selection by Pointing Requirement

Pointing Requirem	Architecture	Sensors	Actuators	Mass	Power	Cost
> 5 deg	Passive magnetic	None (or 1 magnetometer)	Permanent magnet + hysteresis rods	~0.05 kg	0 W	~2 kEUR
2--5 deg	B-dot detumble + magnetic pointing	3-axis magnetometer + coarse sun sensors	3-axis magnetorquers	~0.10 kg	0.2 W	~8 kEUR

0.1--2 deg	Active 3-axis (RW + sensors)	Fine sun sensors + magnetometer + (optional gyro)	3--4 RW + 3 MTQ for desaturation	~0.50 kg	2--4 W	~35 kEUR
< 0.1 deg	Fine pointing (RW + star tracker)	Star tracker + fine sun sensors + magnetometer + gyro	4 RW + 3 MTQ	~0.80 kg	3--5 W	~55 kEUR
< 0.01 deg	Very fine pointing	Dual star trackers + MEMS gyro + fine sun sensors + magnetometer	4 RW + 3 MTQ	~1.20 kg	4--6 W	~80 kEUR

Real mission examples:

Mission	Form Factor	Pointing Req	Architecture	AOCS Mass	Key Sensor
Astrocast (3U, IoT)	3U	~5 deg	MTQ + sun sensors + magnetometer	0.1 kg	Sun sensors
Planet SuperDove (3U+, EO)	3U+	~0.1 deg	4 RW + ST + 3 MTQ + 6 sun sensors	0.8 kg	Blue Canyon NST star tracker
CAPSTONE (12U, cislunar nav)	12U	~0.05 deg	4 RW + ST + sun sensors + IMU	~1.0 kg	Star tracker + IMU
ASTERIA (6U, exoplanet)	6U	~0.003 deg (10 arcsec)	4 RW + ST + fine guidance sensor	~1.2 kg	Custom fine guidance camera

[Source: Pong et al., "ASTERIA: Achieving 10-arcsecond Pointing on a 6U CubeSat," SSC 2018]

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2. Disturbance Torques (25 min)

Teaching Notes

In LEO, four external torques disturb the spacecraft attitude. The AOCS must counteract them continuously. Understanding the source and magnitude of each disturbance is essential for sizing actuators.

[Source: SMAD, Ch. 11.1; Sidi, Ch. 5; Wertz 1999, Ch. 11]

Gravity Gradient Torque

Physics: A spacecraft in orbit experiences a non-uniform gravitational field -- the side closer to Earth is pulled more strongly than the far side. For an elongated body, this differential pull creates a torque that tends to align the long axis with the local vertical (nadir direction). This is the principle behind gravity gradient stabilisation.

> *Key Equations -- Gravity Gradient Torque*
>
> *$T_gg = (3u)/(2a^3) |I_z - I_x| sin(2theta)$*
>
> *where:*
>

> - $u = 3.986 \times 10^{14}$ m^3/s^2 (Earth's gravitational parameter)
>
> - $a = R_E + h$ (semi-major axis in metres)
>
> - I_z, I_x are principal moments of inertia about the maximum and minimum axes (kg m^2)
>
> - theta = angle between the long axis and the local vertical
>
> - Worst case occurs at theta = 45 deg, where sin(2theta) = 1
>
> For a body with $I_z \approx I_x$ (a cube), $T_{gg} \approx 0$ -- this is why 1U CubeSats experience minimal gravity gradient torque.

Aerodynamic Torque

Physics: At LEO altitudes (200--600 km), residual atmospheric molecules collide with the spacecraft surface. The force acts through the centre of pressure (cp), which generally does not coincide with the centre of mass (cm). The offset creates a torque.

> *Key Equations -- Aerodynamic Torque*
>
> $T_{aero} = \frac{1}{2} \rho v^2 C_D A_{ref} d_{cp-cm}$
>
> where:
>
> - rho = atmospheric density (kg/m^3) -- varies by orders of magnitude with altitude, solar activity (F10.7), and local time:
>
> - 300 km: $\rho \approx 2 \times 10^{-11}$ (solar min) to 3×10^{-10} (solar max)
>
> - 400 km: $\rho \approx 4 \times 10^{-12}$ to 1×10^{-11}
>
> - 500 km: $\rho \approx 6 \times 10^{-13}$ to 5×10^{-12}
>
> - 600 km: $\rho \approx 1 \times 10^{-13}$ to 8×10^{-13}
>
> - v = orbital velocity (~7.6 km/s at 500 km)
>
> - $C_D \approx 2.0$--2.3 (molecular flow drag coefficient; 2.2 is standard for flat plates in free molecular flow)
>
> - A_{ref} = cross-sectional area perpendicular to velocity (m^2)
>
> - d_{cp-cm} = offset between centre of pressure and centre of mass (m); typically 0.5--5 cm for CubeSats depending on deployable configuration

Solar Radiation Pressure Torque

Physics: Sunlight carries momentum. When photons strike a surface, they transfer momentum (p = E/c for absorption, p = 2E/c for specular reflection). The resulting force acts through the centre of solar pressure, which may not coincide with the cm.

> *Key Equations -- SRP Torque*

> *$T_SRP = (S)/(c) A_s (1 + q) d_sp-cm$*
>
> *where:*
> *- S = 1361 W/m^2 (solar constant at 1 AU)*
> *- c = 3 x 10^8 m/s (speed of light)*
> *- S/c = 4.54 x 10^-6 N/m^2 (solar radiation pressure at 1 AU)*
> *- A_s = illuminated area (m^2)*
> *- q = surface reflectance (0 for perfect absorber, 1 for perfect specular reflector)*
> *- d_sp-cm = offset between solar pressure centre and centre of mass (m)*
>
> *Note: SRP torque is tiny in LEO compared to aero and magnetic torques. It becomes the dominant disturbance in GEO and deep space (where there is no atmosphere and the magnetic field is weak).*

Residual Magnetic Dipole Torque

Physics: A spacecraft with a net magnetic dipole moment \vecM (from current loops in wiring, permanent magnets in reaction wheel motors, magnetised ferromagnetic components) interacts with Earth's magnetic field \vecB to produce a torque:

> *Key Equations -- Magnetic Dipole Torque*
>
> *$\vecT_mag = \vecM x \vecB$*
>
> *Magnitude: T_mag = M * B * sin(alpha)*
>
> *where:*
> *- M = spacecraft residual magnetic dipole moment (A m^2)*
> *- B = local geomagnetic field strength (T):*
> *- LEO (400--600 km): B ~= 2--5 x 10^-5 T (varies with latitude; strongest near poles, weakest near equator)*
> *- GEO (35,786 km): B ~= 1 x 10^-7 T*
> *- alpha = angle between \vecM and \vecB*
>
> *Typical CubeSat residual dipole moments:*
>
> *| Source | Dipole Moment (A m^2) | Notes |*
> *|--------|-------------------------|-------|*
>

Reaction wheel motor	*0.005--0.02 per wheel*	*Permanent magnets in brushless motor*
Solar array wiring	*0.001--0.01*	*Current loops from SA to EPS*
Battery cells	*0.001--0.005*	*Nickel in cell casing*
Unshielded cables	*0.005--0.05*	*Depends on routing and length*
Total (typical 3U)	*0.01--0.10*	*Varies significantly with design*

Why magnetic torque dominates for CubeSats: COTS electronics are not designed for magnetic cleanliness. Short wiring runs create small but unbalanced current loops. Reaction wheel motors contain permanent magnets. The result is a residual dipole of 0.01--0.1 A m^2, which in a 30 uT field produces 3×10^{-7} to 3×10^{-6} N m -- often larger than gravity gradient or SRP torques.

Worked Example -- Disturbance Torques for 3U CubeSat at 500 km (SuperDove-class)

Spacecraft properties: 3U (100 x 100 x 340 mm), mass = 5 kg, $I_z = 0.035$ kg m^2 (long axis), $I_x = 0.007$ kg m^2 (short axis), $A_{ref} = 0.034$ m^2 (3U face), $d_{cp-cm} = 0.02$ m (deployable panels offset cm from geometric centre).

Gravity gradient (worst case, theta = 45 deg):

$T_{gg} = (3 \times 3.986 \times 10^{14})/(2 \times (6871 \times 10^3)^3) \times |0.035 - 0.007| \times 1$

$= (1.196 \times 10^{15})/(6.494 \times 10^{20}) \times 0.028 = 1.84 \times 10^{-6} \times 0.028 = 5.2 \times 10^{-8}$ N m

Aerodynamic (at 500 km, solar minimum, $\rho \approx 6 \times 10^{-13}$ kg/m^3):

$F_{aero} = 0.5 \times 6 \times 10^{-13} \times 7617^2 \times 2.2 \times 0.034 = 1.30 \times 10^{-6}$ N

$T_{aero} = F_{aero} \times d_{cp-cm} = 1.30 \times 10^{-6} \times 0.02 = 2.6 \times 10^{-8}$ N m

Note: at solar maximum ($\rho \approx 5 \times 10^{-12}$), this increases by ~8x to 2.1×10^{-7} N m.

Solar radiation pressure:

$F_{SRP} = (1361)/(3 \times 10^8) \times 0.034 \times 1.5 = 2.31 \times 10^{-7}$ N

$T_{SRP} = F_{SRP} \times d_{sp-cm} = 2.31 \times 10^{-7} \times 0.02 = 4.6 \times 10^{-9}$ N m

Residual magnetic dipole (M = 0.05 A m^2, B = 3×10^{-5} T):

$T_{mag} = 0.05 \times 3 \times 10^{-5} = 1.5 \times 10^{-6}$ N m

Summary:

> | Source | Torque (N m) | Rank | Notes |
> |--------|-------------|------|-------|
> | Gravity gradient | 5.2×10^{-8} | 3rd | Small because 3U is not very elongated |
> | Aerodynamic (solar min) | 2.6×10^{-8} | 4th | Increases 8x at solar max |
> | Solar radiation pressure | 4.6×10^{-9} | 5th | Negligible at LEO distances |
> | Residual magnetic dipole | 1.5×10^{-6} | 1st | Dominates by >10x |
> | Total (worst-case sum) | $\approx 1.6 \times 10^{-6}$ | | Conservative estimate |
> | Total (RSS) | $\approx 1.5 \times 10^{-6}$ | | More realistic (uncorrelated sources) |
>
> *Key finding: The residual magnetic dipole dominates for CubeSats due to COTS electronics and short wiring runs. Magnetic cleanliness matters. Reducing the residual dipole from 0.05 to 0.01 A m^2 (achievable with careful wire routing, twisted pairs, and degaussing) would reduce the total disturbance by 5x.*

---

3. Attitude Actuators -- Physics and Sizing (25 min)

Teaching Notes

Reaction Wheels -- Physics of Angular Momentum Storage

How reaction wheels work:

A reaction wheel is a flywheel (typically a brass or steel ring, 20--200 g for CubeSats) spun by a brushless DC motor. By Newton's third law, changing the wheel's angular momentum produces an equal and opposite torque on the spacecraft:

$\vec{H}_{total} = \vec{H}_{spacecraft} + \vec{H}_{wheels} = constant$

If the wheel speeds up ($D H_{wheel} > 0$), the spacecraft receives an equal and opposite angular momentum change ($D H_{SC} = -D H_{wheel}$), causing it to rotate. The control torque is:

$T_{control} = \frac{dH_{wheel}}{dt} = I_{wheel} * \dot{\omega}_{wheel}$

where I_{wheel} is the wheel's moment of inertia and $\dot{\omega}_{wheel}$ is the wheel's angular acceleration.

Momentum storage: The maximum angular momentum a wheel can store is:

$H_{max} = I_{wheel} \times \omega_{max}$

For a Blue Canyon RW210: $I_{wheel} \approx 1.5 \times 10^{-5}$ kg m^2, $\omega_{max} \approx 6000$ RPM = 628 rad/s, giving $H_{max} = 0.0094$ N m s ≈ 10 mN m s.

Saturation: As disturbance torques act on the spacecraft, the reaction wheels absorb angular momentum. Over time, the wheel speed increases until it reaches omega_max (saturation). At saturation, the wheel can no longer absorb momentum in that direction, and control authority is lost. The time to saturation from zero speed is:

$t_sat = \fracH_maxT_disturbance = (10 x 10^-3)/(1.5 x 10^-6) = 6667 s ~= 111 minutes$

This is approximately 1.2 orbits -- so the wheels would saturate after about one orbit without desaturation. This is why magnetorquers are essential companions to reaction wheels.

The zero-crossing problem: When a reaction wheel passes through zero speed (reversing direction), the static friction in the bearings creates a "dead zone" where the wheel cannot produce smooth, continuous torque. This causes a brief loss of control authority and increased jitter. Mitigations:

- Bias momentum: Operate all wheels with a positive bias speed (e.g., 500 RPM), so they never cross zero during normal operations
- 4-wheel pyramid configuration: The skewed geometry means individual wheels reverse less frequently
- Lubrication: Space-rated bearings use solid or vapour-deposited lubricants (MoS_2, Braycote) that minimise static friction

Jitter: Reaction wheel imbalance (mass asymmetry in the flywheel) creates vibrations at the spin frequency and its harmonics. For imaging missions, this jitter degrades image quality. Jitter amplitude depends on wheel speed, imbalance mass, and the spacecraft's structural transfer function. Typical CubeSat reaction wheel jitter: 5--20 arcsec at the payload, depending on isolation.

Reaction Wheel Configurations

Configuration	Description	Pros	Cons	Use
3 orthogonal	One wheel per body axis (X, Y, Z)	Minimum mass, simple control	No redundancy; single wheel failure = loss of 1-axis control	Low-cost missions with short lifetime
3 + 1 skew	3 orthogonal + 1 on a skew axis (e.g., [1,1,1] direction)	Single-fault tolerant; the skew wheel + remaining 2 provide 3-axis control	Slightly more complex control law distribution	Standard for CubeSats
4-wheel pyramid	4 wheels tilted ~20--30 deg from body axes, symmetrically arranged	Optimal torque/momentum distribution; single-fault tolerant; reduced zero-crossings	More complex mounting, heavier	High-performance missions, agile S/C

The distribution matrix maps wheel torques to body-frame torques:

$\vecT_body = D * \vecT_wheels$

For a 4-wheel pyramid with cant angle beta:

$D = \beginbmatrix cosbeta & 0 & -cosbeta & 0 \\ 0 & cosbeta & 0 & -cosbeta \\ sinbeta & sinbeta & sinbeta & sinbeta \endbmatrix$

> *Key Equations -- Reaction Wheel Sizing*
>
> *Torque requirement (to counteract disturbances + provide slewing capability):*
>

$T_{RW,min} >= k \times T_{disturbance,total}$

where $k >= 2$ is the control margin factor (typically 2--5 to provide adequate control bandwidth and slewing performance).

Momentum storage requirement (accumulation between desaturation cycles):

$H_{required} = T_{disturbance} \times \frac{t_{desat}}{2}$

where t_{desat} is the time between magnetorquer desaturation events (typically one half-orbit to one orbit). The factor of 1/2 accounts for the average (sinusoidal disturbance torques average to half their peak over a quarter orbit).

Slew rate (for agile/imaging missions):

$\dot{\theta}_{max} = \frac{H_{RW,max}}{I_{axis}}$

For a 3U CubeSat with RW210 (H = 10 mN m s) and $I_{axis} = 0.035$ kg m^2:

$\dot{\theta}_{max} = 0.010 / 0.035 = 0.286$ rad/s = 16.4 deg/s -- more than adequate for target-to-target slewing.

Slew time for a given angle (acceleration-limited, trapezoidal profile):

$t_{slew} = 2\sqrt{\frac{\theta_{slew} * I_{axis}}{T_{RW}}}$

For a 90 deg (1.57 rad) slew with RW210 (T = 1.0 mN m) and I = 0.035:

$t_{slew} = 2\sqrt{(1.57 \times 0.035)/(0.001)} = 2\sqrt{54.95} = 14.8$ s

Worked Example -- Reaction Wheel Sizing for 3U CubeSat (SuperDove-class)

Given: $T_{disturbance} = 1.5 \times 10^{-6}$ N m (from Section 2), desaturation interval = 1 orbit (5670 s), pointing requirement = 0.1 deg, slew requirement = 90 deg in < 60 s.

Torque requirement:

$T_{RW,min} = 3 \times 1.5 \times 10^{-6} = 4.5 \times 10^{-6}$ N m = 0.0045 mN m

This is a very low torque requirement. The minimum available CubeSat wheel (RW-0.01 at 0.23 mN m) exceeds this by 50x. The sizing driver is actually the slew rate and momentum storage, not the disturbance rejection torque.

Momentum storage:

$H_{required} = 1.5 \times 10^{-6} \times (5670)/(2) = 4.25 \times 10^{-3}$ N m s = 4.25 mN m s

Slew time check with candidate wheels:

> | Product | Torque (mN m) | Momentum (mN m s) | Mass (g) | 90 deg Slew (s) | Momentum Margin | Manufacturer |
> |---------|--------------|-------------------|----------|----------------|-----------------|-------------|
> | RW-0.01 | 0.23 | 1.0 | 30 | 69 s | -3.25 mN m s (FAIL) | Hyperion |
> | RW210 | 1.0 | 10 | 55 | 14.8 s | +5.75 mN m s (135%) | Blue Canyon |
> | RW3-1.0 | 1.0 | 15 | 50 | 14.8 s | +10.75 mN m s (253%) | CubeSpace |
> | RW400 | 4.0 | 40 | 120 | 7.4 s | +35.75 mN m s (841%) | Blue Canyon |
>
> *The RW-0.01 fails the momentum storage requirement (would saturate in < 1 orbit). The RW210 (10 mN m s) provides 135% margin and 14.8 s slew time. Selected: RW210 (or CubeSpace RW3-1.0).*
>
> *Configuration: 4 wheels (3+1 skew) for single-fault tolerance. Total AOCS actuator mass: 4 x 55 g = 220 g.*

Magnetorquers -- Physics of Magnetic Torque Generation

How magnetorquers work:

A magnetorquer (MTQ) is simply a coil of wire (or a ferromagnetic rod wrapped with wire). When current flows through the coil, it creates a magnetic dipole moment:

$\vec{m} = N * I * A * \hat{n}$

where N = number of turns, I = current (A), A = coil cross-sectional area (m^2), and \hatn is the coil normal direction.

This magnetic dipole interacts with Earth's geomagnetic field \vecB to produce a torque:

$\vec{T}_{MTQ} = \vec{m} \times \vec{B}$

The torque is perpendicular to both the dipole moment and the magnetic field. This has a critical implication: a magnetorquer cannot produce torque parallel to the local magnetic field vector. At any instant, only 2 of 3 axes can be torqued. Over a full orbit, as the field direction rotates, all 3 axes become accessible -- but not simultaneously.

Magnetorquer types for CubeSats:

Type	Dipole Moment	Mass	Power	Form Factor	Products
Air-core coil (PCB trace)	0.01--0.05 A m^2	1--5 g	0.1--0.3 W	Flat PCB, integrates into solar panel substrate	ZARM Technik MTC-1, custom
Air-core rod (wound wire)	0.05--0.50 A m^2	10--30 g	0.2--0.5 W	Cylindrical, 60--100 mm long	CubeSpace CubeMAG, NewSpace NTQS
Ferromagnetic core rod	0.2--5.0 A m^2	20--100 g	0.3--1.0 W	Cylindrical with mu-metal core, 60--100 mm long	ZARM Technik MTQ-1, ISIS iMTQ

The ferromagnetic core concentrates the magnetic flux, providing 5--20x more dipole moment per unit current than an air-core coil of the same size. However, ferromagnetic cores can retain residual magnetism after power-off, contributing to the spacecraft's residual magnetic dipole.

Why magnetorquers cannot point (only detumble and desaturate):

The torque \vecT = \vecm x \vecB is always perpendicular to \vecB. This means:

1. You cannot generate torque about the \vecB direction at any given instant
2. The achievable torque direction changes continuously as the spacecraft orbits (because \vecB rotates)
3. Pointing control requires torque in any direction at any time -- magnetorquers cannot provide this

Magnetorquers are excellent for:

- Detumbling: The B-dot controller (\vecm = -k \dot\vecB) brakes spacecraft rotation by opposing the change in the measured B-field. Works regardless of attitude.
- Desaturation: Systematically dumping momentum from reaction wheels by applying the correct dipole moment: \vecm = -k (\vecH_wheel x \hatB)
- Coarse pointing (2--10 deg): Possible over time using model-predictive control that plans the dipole commands over a full orbit, exploiting the field rotation. But accuracy is limited.

> *Key Equations -- Magnetorquer Sizing*
>
> *Desaturation torque:*
>
> *$T_MTQ = m_dipole x B x sin(alpha)$*
>
> *Average torque over an orbit (accounting for varying alpha): T_MTQ,avg ~= 0.7 x m_dipole x B_avg*
>
> *Desaturation time (to dump one wheel from full momentum):*
>
> *$t_dump = \fracH_wheelT_MTQ,avg$*
>
> *Design requirement: t_dump < t_shadow (must complete desaturation during the portion of the orbit where the field geometry is favourable).*

> *Worked Example -- Magnetorquer Sizing for 3U CubeSat*
>
> *Given: RW210 momentum = 10 mN m s, B_avg = 3 x 10^-5 T.*
>
> *Option 1: CubeMAG rod (m = 0.2 A m^2):*
>
> *T_MTQ,avg = 0.7 x 0.2 x 3 x 10^-5 = 4.2 x 10^-6 N m*
>
> *t_dump = (10 x 10^-3)/(4.2 x 10^-6) = 2381 s ~= 40 min*

> *This is approximately 42% of one orbit period. Acceptable -- desaturation can be scheduled once per orbit during the non-imaging portion.*
>
> *Disturbance rejection check: The MTQ average torque (4.2 x 10^-6 N m) is 2.8x the total disturbance torque (1.5 x 10^-6 N m). The MTQ can dump momentum faster than it accumulates. Pass.*
>
> *Configuration: 3 MTQ rods, one per body axis (X, Y, Z). This ensures torque can be generated about any two axes at any given time. Total MTQ mass: 3 x 30 g = 90 g.*

Control Moment Gyroscopes (CMGs) vs Reaction Wheels

CMGs are an alternative momentum exchange device used on large, agile spacecraft. A CMG consists of a spinning flywheel mounted on a gimbal. Instead of changing the wheel speed (as in a reaction wheel), the CMG changes the direction of the angular momentum vector by rotating the gimbal. This produces a gyroscopic output torque:

$T_CMG = H_wheel x \dotdelta$

where H_wheel is the constant wheel momentum and \dotdelta is the gimbal rate.

The torque amplification effect: For a CMG wheel spinning at high speed (H = 1--100 N m s), even a slow gimbal rate (\dotdelta = 1 rad/s) produces a large output torque (T = 1--100 N m). This is orders of magnitude more than a reaction wheel of similar mass. CMGs are "torque machines"; reaction wheels are "momentum machines."

Parameter	Reaction Wheel	CMG (Single Gimbal)
Output torque	T = I_w \dotomega_w (low, 0.001--10 N m)	T = H_w \dotdelta (high, 0.1--1000+ N m)
Control complexity	Simple (speed command)	Complex (gimbal singularity avoidance)
Mass efficiency	Lower torque/kg	Higher torque/kg (10--100x)
Failure mode	Bearing wear, motor failure	Gimbal lock (singularity), bearing wear
Typical use	CubeSats, small satellites, non-agile	Large agile satellites, ISS, Earth observation with rapid retargeting
CubeSat status	Standard (many COTS products)	Emerging (Honeybee Robotics microCMG, some research prototypes)

When to use CMGs:

- Spacecraft requiring rapid slewing (> 3 deg/s for large spacecraft)
- Large moments of inertia where reaction wheel torque is insufficient
- Missions requiring frequent retargeting (e.g., video from orbit, rapid revisit EO)

When to use reaction wheels:

- CubeSats and small satellites (adequate torque, simpler control, more COTS options)
- Missions with modest agility requirements (< 10 deg/s slew for CubeSats)
- Cost-constrained missions (CMGs are significantly more expensive)

The ISS uses four 4600 kg CMGs, each storing 3500 N m s of momentum, to maintain attitude without propellant. The Pleiades Neo Earth observation satellite uses CMGs for rapid retargeting between imaging strips.

4. Pointing Error Budget (25 min)

Teaching Notes

[Source: ECSS-E-ST-60-10C; SMAD, Ch. 11.1]

The pointing error budget combines all independent error sources using root-sum-square (RSS) to determine the total pointing uncertainty. This is a statistical combination assuming errors are uncorrelated and normally distributed.

ECSS pointing performance taxonomy:

Term	Definition	Measured Over
APE (Absolute Performance Error)	Total error between actual pointing and commanded pointing	Single measurement
RPE (Relative Performance Error)	Variation in pointing over a short time (jitter/stability)	Measurement window (e.g., integration time)
MPE (Mean Performance Error)	Systematic bias in pointing	Long-term average

For most CubeSat missions, APE is the primary requirement. RPE matters for long-exposure imaging (e.g., ASTERIA's 10-arcsec stability over 20-minute exposures).

> *Key Equations -- Pointing Error Budget (RSS)*
>
> *$theta_APE = sqrttheta_sensor^2 + theta_actuator^2 + theta_alignment^2 + theta_thermal^2 + theta_jitter^2 + theta_orbit^2 + theta_timing^2$*
>
> *The result must satisfy:*
>
> *$theta_APE <= theta_requirement$*

Error Source Definitions and Physics

Source	Description	Physics	Typical Values (Star Tracker)	Typical Values (Sun Sensor)
Sensor accuracy	Intrinsic measurement noise of attitude sensor	Photon noise, centroiding error, optical distortion	3--15 arcsec (0.001--0.004 deg)	0.5--2 deg
Actuator resolutio	Minimum controllable torque step; control loop dead band	Motor cogging torque, driver quantisation, control bandwidth	2--5 arcsec (0.001 deg)	N/A (MTQ: 1--5 deg)
Alignme nt	Misalignment between sensor boresight and payload boresight; measured during I&T	Mechanical tolerances, shimming, bonding accuracy, measurement uncertainty	30--60 arcsec (0.01--0.02	0.5 deg
Thermal distortion	Structural deformation with temperature changes; orbital thermal cycling	CTE mismatch between materials, temperature gradients across structure	10--30 arcsec (0.003--0.01 deg)	0.1 deg
Jitter	High-frequency vibration from reaction wheels, mechanisms	Wheel imbalance forces at spin frequency and harmonics, structural resonances	5--20 arcsec (0.001--0.006 deg)	N/A

Orbit knowled	Uncertainty in satellite position (affects nadir pointing vector computation)	GPS accuracy, propagation error between GPS fixes	1--5 arcsec (< 0.001 deg)	0.05 deg
Timing	Time-stamping error between sensor read and actuator command	Clock synchronisation, interrupt latency, bus communication delay	1--3 arcsec (< 0.001 deg)	0.01 deg

> *Worked Example -- Pointing Budget for 3U EO CubeSat (Star Tracker + RW)*
>
> *| Error Source | Value (deg) | Value (arcsec) | Value^2 (deg^2) | Notes |*
>
> *|-------------|------------|----------------|---------------------|-------|*
>
> *| Star tracker accuracy | 0.003 | 10.8 | 9.0 x 10^-6 | Blue Canyon NST, 1-sigma boresight |*
>
> *| Reaction wheel resolution | 0.001 | 3.6 | 1.0 x 10^-6 | Motor cogging + control dead band |*
>
> *| Alignment knowledge | 0.020 | 72 | 4.0 x 10^-4 | Dominant -- shimmed to 1 arcmin |*
>
> *| Thermal distortion | 0.010 | 36 | 1.0 x 10^-4 | Al structure, 40 degC orbital range |*
>
> *| RW jitter | 0.005 | 18 | 2.5 x 10^-5 | At 3000 RPM, no isolation mount |*
>
> *| Orbit knowledge (GPS) | 0.001 | 3.6 | 1.0 x 10^-6 | GPS fix every 10 s |*
>
> *| Timing error | 0.001 | 3.6 | 1.0 x 10^-6 | < 1 ms timestamp sync |*
>
> *| RSS Total | sqrt5.37 x 10^-4 = 0.023 deg | 83 arcsec | | |*
>
> *Requirement: 0.1 deg (3-sigma) -- this is typical for a 5 m GSD imager at 500 km (where 0.1 deg corresponds to ~870 m pointing error on ground, or ~175 pixels for a 5 m GSD sensor).*
>
> *Margin: 0.1 - 0.023 = 0.077 deg (77% margin) -- comfortable.*
>
> *Key insight: Alignment knowledge (0.020 deg = 72 arcsec) dominates the budget at 74% of the RSS. Improving the star tracker accuracy from 10 to 3 arcsec would change the RSS total from 83 to 82.3 arcsec -- negligible improvement. Budget-driven design means investing effort in the dominant term: better alignment calibration (e.g., on-orbit calibration using ground targets) would have far more impact than upgrading any sensor.*
>
> *To achieve < 0.01 deg (36 arcsec) pointing, the alignment must be improved to < 0.005 deg (18 arcsec), which requires precision optical alignment during I&T and/or on-orbit alignment calibration.*

---

5. Momentum Management and Desaturation (10 min)

Teaching Notes

Disturbance torques cause angular momentum to accumulate in reaction wheels over time. Without management, wheels saturate and lose control authority. Understanding this cycle is essential for AOCS design.

The momentum lifecycle:

1. Accumulation: External disturbance torques (gravity gradient, aero, SRP, magnetic) act on the spacecraft body. The control loop commands the reaction wheels to counteract these torques, absorbing the angular momentum. Wheel speed increases (or decreases) at a rate of $\dot{H} = T_{disturbance}$.

2. Monitoring: The OBC monitors wheel speeds. When any wheel exceeds a threshold (typically 80% of maximum), a desaturation manoeuvre is triggered.

3. Desaturation: The OBC activates the magnetorquers to generate a torque that opposes the stored wheel momentum. The algorithm computes the optimal dipole command:

$\vec{m}_{cmd} = -k_d (\vec{H}_{wheel} \times \hat{B})$

where k_d is the desaturation gain and $\hat{B}$ is the unit magnetic field vector. This produces a torque $\vec{T} = \vec{m} \times \vec{B}$ that is in the direction to reduce $\vec{H}_{wheel}$.

4. Completion: Wheel speeds return to near-zero (or bias speed). The cycle repeats.

Desaturation constraints:

- MTQs can only generate torque perpendicular to the local magnetic field vector. They cannot dump momentum parallel to $\vec{B}$.
- Near the magnetic equator, $\vec{B}$ is nearly horizontal (north-pointing). MTQs can effectively dump momentum about the pitch and roll axes but not yaw.
- Near the magnetic poles, $\vec{B}$ is nearly vertical. MTQs can dump pitch and yaw but not roll.
- Over a full orbit, the field direction rotates sufficiently to dump all three axes. But at any instant, one axis is poorly controllable.
- A multi-pass desaturation strategy (spreading the dump over a full orbit) is more efficient than a single-point dump.

Typical desaturation frequency for CubeSats:

- At 500 km with 1.5 uN m total disturbance and 10 mN m s wheels: one desaturation per orbit (every ~95 minutes)
- Duration: 5--15 minutes per cycle
- Power: 0.5--1.5 W during desaturation (3 MTQ rods active)
- During desaturation, pointing accuracy degrades slightly (the MTQ torques perturb the attitude). Imaging should be inhibited during desaturation.

Wheel Configurations -- 3+1 Redundancy

The standard 4-wheel configuration provides full 3-axis control with one spare:

- 3 wheels in the body X, Y, Z axes provide minimum control
- 4th wheel on a skew axis (typically [1,1,1] normalised, or cant angle 20--30 deg from each axis) provides redundancy and enhanced torque distribution
- If one wheel fails, the remaining three (including the skew wheel) maintain 3-axis control with reduced but adequate authority

4-wheel torque envelope: With 4 wheels in a pyramid configuration, the maximum torque in any body direction is:

$T_{max,body} = \sqrt{2} * T_{wheel} \approx 1.41 \times T_{wheel}$

for the optimal distribution. This is better than the 3-orthogonal configuration where $T_{max,body} = T_{wheel}$ along any axis.

1U Worked Example: UniSat-1

Passive Magnetic Attitude Stabilisation

UniSat-1 does not have an active AOCS. Instead, it uses passive magnetic stabilisation -- the simplest and cheapest attitude control method, requiring zero power and minimal mass.

How it works -- physics:

1. Permanent magnet: A small bar magnet (typically AlNiCo or NdFeB, ~10--20 g, dipole moment M_p = 0.1--1.0 A m^2) is embedded along one body axis (say, the Z-axis). In Earth's magnetic field $\vec{B}$, the magnet experiences a restoring torque:

$\vec{T}_{restoring} = \vec{M}_p \times \vec{B}$

This torque acts to align the magnet axis with the local field direction, analogous to a compass needle aligning with magnetic north. The restoring torque is maximum when the magnet is perpendicular to the field (alpha = 90 deg) and zero when aligned (alpha = 0).

For M_p = 0.5 A m^2 and B = 3 x 10^{-5} T: T_{max} = 0.5 x 3 x 10^{-5} = 1.5 x 10^{-5} N m. This is ~10x larger than any environmental disturbance torque, ensuring stable alignment.

2. Hysteresis rods: Two or more strips of magnetically soft material (e.g., Permalloy, HyMu-80, ~5--10 g each, dimensions ~60 x 1 x 1 mm) are mounted perpendicular to the permanent magnet. As the satellite oscillates around the field-aligned equilibrium, the external field component along the hysteresis rod alternates, driving the rod material around its B-H hysteresis loop. The area enclosed by the hysteresis loop represents energy dissipated per cycle as heat in the rod material. This extracts kinetic energy from the satellite's oscillation, damping it over time.

The energy dissipated per oscillation cycle is:

$E_{dissipated} = V_{rod} \times \oint H\, dB$

where V_{rod} is the rod volume and $\oint H\, dB$ is the area of the hysteresis loop. For HyMu-80 material, typical energy density is ~100 J/m^3 per cycle.

Damping time constant: From initial tumble (~10 deg/s after deployment) to settled oscillation (~1--5 deg amplitude), the damping process typically takes hours to days, depending on the hysteresis rod material, volume, and the initial tumble rate.

Performance:

Parameter	Passive Magnetic	Active (RW + ST)
Pointing accuracy	~10--15 deg (to local B-field)	< 0.1 deg (to inertial frame)
Settling time	Hours to days after deployment	Minutes after mode transition
Residual tumble rate	~1--5 deg/s (damped from initial ~10 deg/s)	< 0.01 deg/s
Power	0 W	3--5 W
Mass	~30--50 g	500--800 g
Cost	~2 kEUR	~55 kEUR
Failure modes	Demagnetisation (radiation, temperature)	Motor failure, bearing wear, software bugs

Why this works for UniSat-1:

The MEMS magnetometer payload does not require accurate pointing. In fact, it benefits from being in a slowly rotating/tumbling state because this provides magnetic field measurements across multiple directions, improving the scientific data quality. The magnetometer can measure the field vector regardless of spacecraft orientation.

No pointing budget needed: Since there is no payload pointing requirement, there is no need for a pointing error budget. This eliminates the star tracker, reaction wheels, magnetorquers (as actuators), sun sensors, and gyroscopes -- removing the most expensive and power-hungry subsystem from the design.

> *Disturbance environment for 1U at 400 km:*
>
> *| Source | Torque (N m) | Calculation | Notes |*
>
> *|--------|-------------|-------------|-------|*
>
> *| Gravity gradient | ~1 x 10^-8 | (3u)/(2a^3) D I; D I ~= 0.001 kg m^2 for 1U | Small because nearly cubic shape (I_z ~= I_x) |*
>
> *| Aerodynamic | ~3 x 10^-8 | (1)/(2)rho v^2 C_D A d; rho(400km) ~= 5 x 10^-12 | Higher rho at 400 km than 500 km |*
>
> *| Solar radiation pressure | ~1 x 10^-9 | (S)/(c) A (1+q) d; A = 0.01 m^2 | Small area |*
>
> *| Permanent magnet (restoring) | ~1 x 10^-5 | M_p x B x sin(alpha) | Dominant -- this IS the control torque |*
>
> *The permanent magnet restoring torque (~10^-5 N m) is three orders of magnitude larger than all disturbances combined. This ensures the satellite remains approximately field-aligned.*

Limitation: Passive magnetic stabilisation provides alignment to the local magnetic field, which rotates as the satellite orbits. The satellite does not point at nadir, the Sun, or any fixed direction. For missions requiring Earth-pointing or Sun-tracking, active AOCS is mandatory. For UniSat-1's magnetometer mission, this is not a limitation -- it is a feature.

---

6. SpaceCDF Exercise (30 min)

Instructions

1. Architecture tab (AOCS): Select the AOCS architecture appropriate for your pointing requirement
- Review the derived requirements that appear
- Check that the selected hardware fits within your power and mass budgets
2. Pointing Budget card on the Dashboard:
- Review the RSS error tree
- Identify the largest contributor
- Verify the total is within your pointing requirement
3. Dashboard AOCS KPI:
- Pointing accuracy achieved
- Margin to requirement
- AOCS power demand by mode
4. Equipment Browser (if time permits):
- Browse reaction wheels: compare mass, torque, momentum, cost
- Browse star trackers: compare accuracy, mass, FOV, exclusion zones
- Note that star tracker exclusion zones constrain mounting face options

Worksheet 3.2 Tasks

1. Select AOCS architecture and justify based on pointing requirement
2. Calculate all 4 disturbance torques for your orbit and spacecraft configuration
3. Size the reaction wheel (torque, momentum storage, and slew time)
4. Size the magnetorquer for desaturation (verify dump time < 1 orbit)
5. Complete the pointing error budget table (RSS of all 7 sources)
6. Verify margin to pointing requirement and identify the dominant error source

---

Session Summary

Topic	Key Takeaway
AD vs AC	Determination = know orientation (sensors); Control = change orientation (actuators)
Star trackers	Pattern-match stars against catalogue; 3--15 arcsec accuracy; exclusion zones (Sun 25--45 deg, Earth 25--35 deg); most accurate sensor
Sun sensors	Coarse (photodiode, 2--5 deg) or fine (slit+array, 0.1 deg); essential for safe mode; every S/C needs them
Magnetometers	Measure \vecB field; compare with IGRF model; 5--10 deg accuracy alone; residual dipole contamination is key error source
GPS receivers	Orbit determination (position/velocity), not attitude; 5--20 m accuracy in LEO; needed for nadir pointing computation
Gyroscopes	Measure angular rate \vecomega; MEMS drift 1--10 deg/hr; combined with star tracker via Kalman filter for high-bandwidth estimation
Architecture selection	Driven by pointing requirement: passive magnetic for > 5 deg; MTQ for 2--5 deg; RW+ST for < 0.1 deg
Disturbance torques	Gravity gradient, aero, SRP, magnetic dipole; magnetic dipole dominates for CubeSats
Reaction wheel physics	H = I_w omega_w; T = dH/dt; saturation occurs when omega arrow omega_max; zero-crossing jitter

RW sizing

Torque >= 2 x disturbance; momentum >= half-orbit accumulation; check slew time

Magnetorquer physics	\vecT = \vecm x \vecB; torque always perpendicular to \vecB; cannot point, only detumble/desaturate
MTQ sizing	Desaturation torque must exceed disturbance accumulation rate; dump time < 1 orbit
CMGs vs RWs	CMGs: torque amplification (T = H \dotdelta), for large/agile S/C; RWs: simpler, cheaper, standard for CubeSats
Pointing budget	RSS of 7 independent sources; alignment knowledge typically dominates; improve the dominant term
Budget-driven design	Investing in the smallest error source has negligible impact on total; focus on the dominant term
Redundancy	4-wheel (3+1 skew) provides single-fault tolerance; distribution matrix maps wheels to body torques
Momentum management	MTQs dump momentum against Earth's \vecB-field; ~once per orbit; 5--15 min; imaging inhibited during dump

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Session 3.3: Communications and Link Budget Design

Duration: 2 hours

Prerequisites: Sessions 2.1--3.2 (requirements, orbit, power, AOCS defined)

SpaceCDF Tabs: Link Budget, Spectrum Selector, Equipment Browser (Comms)

References

- Wertz, Everett & Puschell, Space Mission Engineering: The New SMAD, 2011, Ch. 13 (Communications)
- ECSS, ECSS-E-ST-50-05C: Radio Frequency and Modulation, 2011
- CCSDS, 131.0-B-4: TM Synchronization and Channel Coding, 2023
- ITU, Radio Regulations, 2020 (Articles 5 and 22)
- Maral & Bousquet, Satellite Communications Systems, 6th ed., 2020, Ch. 5
- Roddy, Satellite Communications, 4th ed., 2006, Ch. 4--6
- Haykin, Communication Systems, 5th ed., 2009
- Sklar, Digital Communications: Fundamentals and Applications, 2nd ed., 2001
- IARU, Amateur Satellite Frequency Coordination, 2023

Learning Objectives

By the end of this session, participants will be able to:

1. Construct a complete link budget from first principles (in decibels)
2. Compute free space path loss for any frequency and slant range
3. Select an appropriate frequency band based on data rate, licensing, and equipment availability

4. Explain the physics of each antenna type and compute antenna gain and beamwidth
5. Choose modulation and coding scheme based on required E_b/N_0 and spectral efficiency
6. Explain the physical basis for coding gain and why FEC is essential for space links
7. Size an antenna (gain, beamwidth, mass) for the selected frequency
8. Determine data throughput and verify the data budget closes
9. Identify ground station options and compute contact geometry
10. Use SpaceCDF's link budget tool and spectrum selector

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1. The Link Budget Concept (15 min)

Teaching Notes

[Source: SMAD, Ch. 13; ECSS-E-ST-50-05C; Roddy, Ch. 4]

The link budget is the accounting statement for the communication link. Every gain and every loss from transmitter to receiver is tallied in decibels (dB) to determine whether the link "closes" -- meaning the received signal is strong enough to decode with acceptable error rate.

The fundamental question: does the received signal have enough energy per bit, relative to the noise, to achieve the required bit error rate? This is quantified by E_b/N_0 -- the ratio of energy per information bit to noise spectral density.

Link Budget Flow

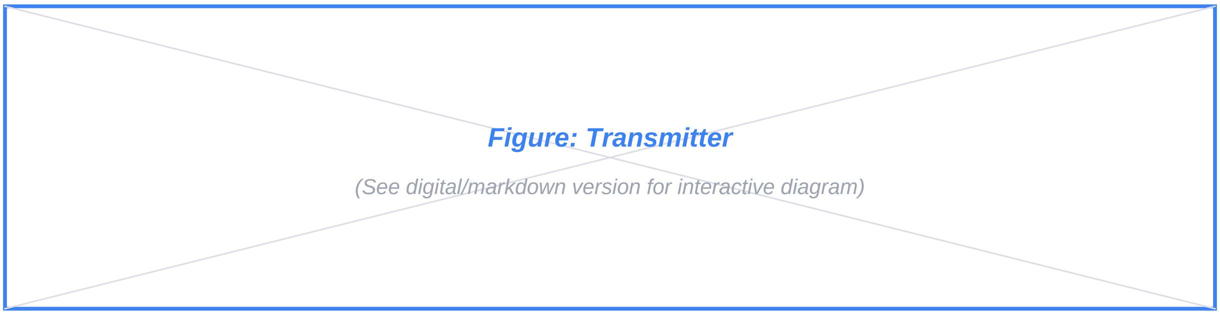

Figure: Transmitter

(See digital/markdown version for interactive diagram)

Decibel Refresher

All link budget terms are in decibels to convert multiplication/division into addition/subtraction:

Conversion	Formula	Example
Watts to dBW	$P_{dBW} = 10 \log_{10}(P_W)$	2 W = +3.0 dBW
Milliwatts to dBm	$P_{dBm} = 10 \log_{10}(P_{mW})$	1 W = +30 dBm
dBW to Watts	$P_W = 10^{P_{dBW}/10}$	-3 dBW = 0.5 W
Ratio to dB	$G_{dB} = 10 \log_{10}(G)$	Gain of 100 = 20 dB

dBW vs dBm	dBm = dBW + 30	0 dBW = 30 dBm

Common power values:

Power	dBW	dBm
0.1 W	-10.0	+20.0
0.5 W	-3.0	+27.0
1 W	0.0	+30.0
2 W	+3.0	+33.0
5 W	+7.0	+37.0
10 W	+10.0	+40.0

---

2. Complete Link Budget Equation (25 min)

Teaching Notes

> *Key Equations -- Link Budget (dB form)*
>
> *EIRP (Effective Isotropic Radiated Power):*
>
> *$EIRP = P_TX + G_TX - L_TX (dBW)$*
>
> *EIRP represents the power that an isotropic antenna would need to radiate to produce the same signal strength in the direction of maximum antenna gain. It combines the transmitter power, antenna gain (directivity), and cable/filter losses.*
>
> *Free Space Path Loss:*
>
> *$FSPL = 20log_10((4pi d)/(lambda)) = 20log_10((4pi d f)/(c)) (dB)$*
>
> *where d = slant range (m), f = frequency (Hz), c = 3 x 10^8 m/s.*
>
> *Physical interpretation: FSPL is not "energy absorption" -- no energy is lost. It is the geometric spreading of radiated power over a sphere of radius d. An isotropic antenna radiates equally in all directions, so the power per unit area at distance d is P/(4pi d^2). The lambda^2 dependence arises because a receive antenna's effective aperture scales as A_eff = Glambda^2/(4pi) -- at higher frequencies the receive antenna "captures" less of the available flux (unless the antenna is physically larger).*
>
> *Receiver figure of merit:*
>
> *$G/T = G_RX - 10log_10(T_sys) (dB/K)$*

G/T is the single most important parameter of any receive station. It combines the antenna gain (signal capture) with the system noise temperature (noise floor). A high G/T means good sensitivity.

> *System noise temperature T_{sys} includes:*
>
> *- Antenna noise temperature T_A (depends on what the antenna "sees": sky ~10--50 K at zenith, ~150--300 K looking at Earth/ground)*
>
> *- Feed/cable losses: $T_{feed} = T_{physical} (L-1)$ where L = loss factor*
>
> *- LNA noise temperature: $T_{LNA} = T_0 (F-1)$ where F = noise figure, T_0 = 290 K*
>
> *- Subsequent stages (reduced by LNA gain)*
>
> *Rule of thumb: T_{sys} ~= 100--200 K for a professional ground station with cryogenic or low-noise LNA; T_{sys} ~= 400--800 K for an amateur station with COTS LNA.*
>
> *Carrier-to-noise density ratio:*
>
> $C/N_0 = EIRP - FSPL - L_{atm} - L_{point} - L_{pol} + G/T - k \text{ (dBHz)}$
>
> *where k = -228.6 dBW/K/Hz (Boltzmann constant: $k = 1.381 \times 10^{-23}$ J/K).*
>
> *Energy per bit to noise density:*
>
> $E_b/N_0 = C/N_0 - 10\log_{10}(R_b) \text{ (dB)}$
>
> *where R_b = data rate (bps). This is the fundamental quality metric: each bit needs a certain amount of energy (E_b) relative to the noise floor (N_0) to be correctly demodulated.*
>
> *Link margin:*
>
> $Margin = E_b/N_{0,available} - E_b/N_{0,required} - L_{implementation} \text{ (dB)}$
>
> *Requirement: Margin >= 3 dB for Phase B+ (per ECSS-E-ST-50-05C). This 3 dB margin covers:*
>
> *- Transmitter power variation (aging, temperature)*
>
> *- Antenna gain uncertainties*
>
> *- Atmospheric scintillation*
>
> *- Pointing error variations*
>
> *- Ground station performance variation*

Complete Link Budget Table

Line	Parameter	Formula / Typical Value	Unit	Physical Meaning
1	TX Power	P_TX (e.g., 2 W = +3.0)	dBW	RF power from amplifier output
2	TX Antenna Gain	G_TX (e.g., +6.0 for patch)	dBi	Directivity relative to isotropic
3	TX Line Losses	L_TX (cables, filters: -1.5)	dB	Ohmic loss in RF cables and connectors

4	EIRP	= P_TX + G_TX - L_TX	dBW	Effective radiated power
5	Free Space Path	FSPL = 20log_10(4pi d f/c)	dB	Geometric spreading + aperture effect
6	Atmospheric Loss	L_atm (-0.3 to -3.0)	dB	Molecular absorption (O_2, H_2O)
7	Pointing Loss	L_point (-0.5 to -3.0)	dB	Signal reduction from antenna mispointing
8	Polarisation Loss	L_pol (-0.1 to -3.0)	dB	Mismatch between TX and RX polarisation
9	RX Antenna Gain	G_RX (e.g., +35 for 3 m dish)	dBi	Ground antenna directivity
10	System Noise Temp	T_sys (e.g., 150 K = 21.8 dBK)	dBK	Total noise temperature of receive chain
11	G/T	= G_RX - 10log_10(T_sys)	dB/K	Receiver figure of merit
12	Boltzmann Constant	k = -228.6	dBW/K/	Thermal noise power spectral density
13	C/N_0	= EIRP - FSPL + G/T - k - L_losses	dBHz	Signal-to-noise density ratio
14	Data Rate	10log_10(R_b)	dBbps	Information throughput
15	E_b/N_0 available	= C/N_0 - 10log_10(R_b)	dB	Available energy per bit
16	E_b/N_0 required	From modulation/coding selection	dB	Minimum needed for target BER
17	Implementation Loss	Typically 1.5--2.5 dB	dB	Real vs ideal demodulator performance
18	LINK MARGIN	= E_b/N_0,avail - E_b/N_0,req - L_impl	dB	Must be >= 3 dB

3. Antenna Types -- Physics and Selection (15 min)

Teaching Notes

[Source: SMAD, Ch. 13; Balanis, Antenna Theory: Analysis and Design, 4th ed., 2016]

The antenna converts guided RF energy (in cables/waveguides) into radiated electromagnetic waves (and vice versa for receive). The key performance parameters are:

- Gain (G): How much more signal the antenna concentrates in its beam direction compared to an isotropic radiator. Higher gain = narrower beam = more signal but requires more precise pointing.
- Beamwidth (theta_3dB): The angular width of the main beam between the -3 dB (half-power) points.
- Polarisation: Linear (vertical/horizontal), circular (RHCP/LHCP), or elliptical.

Fundamental trade-off: G \propto 1/theta_3dB^2. A higher-gain antenna has a narrower beam and requires more precise pointing. For a spacecraft without accurate pointing, a low-gain omnidirectional antenna is mandatory. For a spacecraft with 0.1 deg pointing, a high-gain directional antenna is feasible.

Antenna Types for Spacecraft

Antenna Type	Gain (dBi)	Beamwidth	Pointing Needed?	Mass (CubeSat)	Physics
Monopole /Dipole	0--2	~omnidirectional (toroidal pattern)	No	5--20 g (deployable wire)	Current distribution on a wire radiates; lambda/4 monopole on ground plane or lambda/2 dipole
Turnstile	0--3	Hemispherical	No	10--30 g	Two crossed dipoles fed 90 deg apart; produces circular polarisation
Patch (microstri	5--8	60--90 deg	Loose (~10 deg)	10--50 g	Resonant conducting patch on dielectric substrate over ground plane; lambda/2 patch at resonance
Patch array					

(2x2, 4x4)

10--18	15--40 deg	Moderate (~5 deg)	50--200 g	Multiple patch elements with corporate feed network; gain = G_element + 10log_10(N)

Horn	15--25	10--25 deg	Yes (~2 deg)	100--500 g	Flared waveguide; smooth aperture illumination gives low side lobes
Parabolic reflector	25--45	1--5 deg	Yes (< 1 deg)	500 g -- 5 kg	Feed at focal point illuminates parabolic dish; G = eta_a (pi D/lambda)^2
Phased array	15--35	Electronically steered, 1--30 deg	Electronic (no mechanical)	200 g -- 2 kg	Array of elements with individual phase shifters; beam steered by adjusting relative phases; no moving parts

Monopole/dipole physics: A quarter-wave monopole (L = lambda/4) is the simplest antenna. At 437 MHz (UHF), lambda = 0.686 m, so a monopole is 17.2 cm long -- easily deployable from a CubeSat using a spring-loaded tape measure or nitinol wire. The radiation pattern is omnidirectional in the azimuthal plane (perpendicular to the wire) and null along the wire axis. Gain is approximately 0--2 dBi depending on the ground plane size.

Patch antenna physics: A microstrip patch antenna consists of a conducting patch (typically rectangular, L ~= lambda/2 at the desired frequency) printed on a dielectric substrate (e.g., Rogers RT/duroid, \varepsilon_r = 2.2--10.2, thickness 1--3 mm) above a ground plane. The patch resonates at the frequency where its length equals lambda_eff/2 (where lambda_eff = lambda_0/sqrt\varepsilon_r). At S-band (2.25 GHz), a patch is approximately 40 x 40 mm -- small enough to mount on a CubeSat face. Gain is typically 5--8 dBi with a hemispherical pattern. Circular polarisation is achieved by feeding two orthogonal modes 90 deg apart (dual-feed or corner-truncated patch).

Parabolic reflector physics: A parabolic dish focuses incoming plane waves to its focal point (for receive) or collimates spherical waves from a feed at the focus into a parallel beam (for transmit). The gain depends on the dish diameter D and the wavelength lambda:

> *Key Equations -- Antenna*
>
> *Gain of a parabolic antenna:*
>
> *$G = eta_a ((pi D)/(lambda))^2$*
>
> *In dBi: G_dBi = 10log_10[eta_a ((pi D)/(lambda))^2]*
>
> *where eta_a ~= 0.55--0.65 (aperture efficiency, accounting for illumination taper, spillover, blockage, and surface errors), D = antenna diameter (m), lambda = c/f.*
>
> *Half-power beamwidth (HPBW):*
>
> *$theta_3dB ~= (70lambda)/(D) (degrees)$*
>
> *Patch antenna gain (single element): typically 5--8 dBi, beamwidth ~60--90 deg*
>
> *Patch array gain (N elements): G_array = G_element + 10log_10(N)*
>
> *Pointing loss (when antenna is mispointed by angle Dtheta):*
>
> *$L_point ~= -12 (\fracDthetatheta_3dB)^2 (dB)$*

> *This shows that a narrower beam (smaller theta_3dB) is more sensitive to pointing errors.*

Worked example -- antenna gain at different bands for a 30 cm dish:

Band	Frequency	lambda (mm)	D/lambda	Gain (dBi)	HPBW (deg)	Pointing req
S-band	2.25 GHz	133	2.3	14.3	30	~5 deg
X-band	8.4 GHz	35.7	8.4	25.4	8.3	~1 deg
Ka-band	26 GHz	11.5	26.1	35.2	2.7	~0.3 deg

This table illustrates the fundamental link between frequency, antenna size, gain, and pointing requirement. At Ka-band, even a small dish provides high gain -- but the beamwidth is so narrow that sub-degree pointing accuracy is essential. This is why Ka-band CubeSat links require star tracker + reaction wheel AOCS.

Phased array physics: A phased array consists of multiple antenna elements (patch, dipole, or slot) arranged in a grid. Each element has an individual phase shifter (and sometimes amplitude control). By adjusting the relative phase between elements, the beam can be electronically steered without moving the antenna. The beam direction theta_s for element spacing d and phase increment Dphi:

$sin(theta_s) = (Dphi * lambda)/(2pi d)$

Advantages: no moving parts, fast beam steering (microseconds), multiple simultaneous beams possible. Disadvantages: complex, expensive, high power consumption for active arrays, scan loss at wide angles (costheta_s factor). Phased arrays are used on Starlink satellites (Ka-band), Iridium NEXT (L-band), and are emerging for CubeSats in Ka-band.

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4. Free Space Path Loss by Band (10 min)

Teaching Notes

FSPL increases with both frequency and distance. At a fixed slant range, higher-frequency bands lose more signal -- but this is offset by the ability to use smaller, higher-gain antennas at higher frequencies.

> *Key Equations -- FSPL (expanded form)*
>
> *$FSPL (dB) = 20log_10(4pi) + 20log_10(d) + 20log_10(f) - 20log_10(c)$*
>
> *$= 21.98 + 20log_10(d_m) + 20log_10(f_Hz) - 169.54$*
>
> *Practical form (with km and GHz):*
>
> *$FSPL (dB) = 92.45 + 20log_10(d_km) + 20log_10(f_GHz)$*

Slant Range Geometry

The slant range d from a LEO spacecraft to a ground station depends on the orbit altitude h and the elevation angle \varepsilon above the horizon:

$d = R_E [sqrt((R_E + h)/(R_E))^2 - cos^2(\varepsilon) - sin(\varepsilon)]$

For typical LEO orbits:

Altitude	Elevation 90 deg (nadir)	Elevation 30 deg	Elevation 10 deg	Elevation 5 deg
400 km	400 km	723 km	1150 km	1500 km
500 km	500 km	875 km	1300 km	1650 km
600 km	600 km	1020 km	1460 km	1820 km

Design rule: Always compute the link budget at the minimum elevation angle (worst case), typically 5--10 deg. Below 5 deg, atmospheric losses increase sharply, ground clutter enters the antenna sidelobes, and the link is generally unusable.

FSPL by Band and Geometry

Band	Centre Frequency	FSPL at 500 km (nadir)	FSPL at 1300 km (10 deg elev)	Difference
VHF	146 MHz	139.0 dB	147.3 dB	8.3 dB
UHF	437 MHz	148.3 dB	157.6 dB	9.3 dB
S-band	2250 MHz	162.5 dB	170.8 dB	8.3 dB
X-band	8200 MHz	173.8 dB	182.1 dB	8.3 dB
Ka-band	26 GHz	183.8 dB	192.1 dB	8.3 dB

The elevation angle penalty: Going from nadir to 10 deg elevation increases FSPL by ~8 dB (distance increases by ~2.6x; FSPL scales as 20log_10(2.6) = 8.3 dB). This is a significant loss and is why contact time at high elevations is much more valuable than contact time at low elevations.

---

5. Frequency Band Selection and Licensing (15 min)

Teaching Notes

[Source: ITU Radio Regulations, Articles 5 and 22; ISED RSS-SAT; FCC Part 25; IARU Satellite Frequency Coordination]

Band selection is a design constraint that affects data rate, antenna size, atmospheric losses, equipment availability, licensing cost, and data policy. The choice of band is one of the earliest and most consequential decisions in mission design.

Band Comparison Table

Band	Frequency	Allocation	Max BW	Practical Data Rate	Atmospheric Loss (10 deg)	Rain Fade	Licensing	Equipment Availability
VHF	144--146 MHz	Amateur	15 kHz	< 9.6 kbps	0.1 dB	None	IARU coord (free, 3--6 mo)	Many COTS, low cost
UHF	435--438 MHz	Amateur	20 kHz	< 19.2 kbps	0.2 dB	Negligible	IARU coord (free, 3--6 mo)	Many COTS, low cost
S-band	2200--2290	Space research/	5 MHz	0.1--10 Mbps	0.5 dB	< 0.5 dB	ISED/FCC ($30--45K, 6--12 mo)	Many COTS
X-band	8025--8400	EES	375 MHz	10--400 Mbps	1.0 dB	1--3 dB	ISED/FCC + ITU ($50--80K, 12+ mo)	Growing COTS
Ka-band	25.5--27.0 GHz	EES/FSS	1.5 GHz	100--2000 + Mbps	2--5 dB	3--15 dB (location-dependent)	Complex ITU ($100K+, 18+ mo)	Emerging COTS

EES = Earth Exploration Satellite; FSS = Fixed Satellite Service

Atmospheric attenuation physics: The atmosphere absorbs and scatters RF energy. Molecular oxygen has a strong absorption line at 60 GHz (used for inter-satellite links where atmospheric penetration is not needed). Water vapour absorbs at 22.2 GHz (near Ka-band). At S-band and below, atmospheric losses are minimal (< 1 dB). At Ka-band, losses of 2--5 dB are typical at 10 deg elevation, and rain fade can add 3--15 dB in tropical regions.

Rain fade: Raindrops scatter and absorb RF energy. The effect scales approximately as f^2 -- negligible below 4 GHz, significant above 10 GHz, and severe above 20 GHz. Rain fade is characterised by the rain rate (mm/hr) and the path length through rain. In tropical regions, rain rates of 50+ mm/hr can cause 10+ dB additional loss at Ka-band. Mitigation: adaptive data rate (reduce rate during rain), site diversity (multiple ground stations), power control (increase TX power during fade). For Ka-band links, a rain fade margin of 5--10 dB is typically included.

Amateur band regulations (IARU):

- Non-commercial, educational, and experimental use only
- Open data policy: all transmissions must be unencrypted and the protocol must be published
- No commercial data or imagery downlink
- Coordination through IARU (International Amateur Radio Union): free but takes 3--6 months
- Very popular for university CubeSats (low cost, no license fees, large amateur community provides free ground station support via SatNOGS network)

Commercial band licensing:

- Requires national filing (FCC in US, ISED in Canada, Ofcom in UK, CNES/ANFR in France) + ITU coordination for frequencies above 1 GHz
- Costs: 30--45K for S-band (typical),50--80K for X-band, $100K+ for Ka-band
- Timeline: 6--12 months for S-band, 12--18 months for X-band, 18+ months for Ka-band
- Commercial data can be encrypted and proprietary
- Bandwidth allocation may be limited by the national authority

Band Selection Decision Tree

```
Required data rate?
  <= 9.6 kbps AND non-commercial/educational -> VHF/UHF amateur (IARU, free, 3-6 mo)
  <= 10 Mbps                                 -> S-band commercial (ISED/FCC, $30-45K, 6-12 mo)
  <= 400 Mbps                                -> X-band (ISED/FCC + ITU, $50-80K, 12+ mo)
  > 400 Mbps                                 -> Ka-band (complex ITU, $100K+, 18+ mo, rain fade)
```

6. Modulation and Coding -- Physics and Selection (20 min)

Teaching Notes

[Source: CCSDS 131.0-B-4; Sklar, Ch. 7--8; Haykin, Ch. 10; DVB-S2 standard]

Modulation -- How Information Becomes RF

Modulation encodes digital data onto an RF carrier by varying the carrier's amplitude, frequency, or phase.

Phase Shift Keying (PSK): The most common modulation family for space communications. The carrier phase is shifted by discrete amounts to represent different bit patterns:

- BPSK (Binary PSK): 2 phase states (0 deg and 180 deg). Each symbol carries 1 bit. Spectral efficiency: 1 bps/Hz. Most robust to noise.
- QPSK (Quadrature PSK): 4 phase states (0, 90, 180, 270 deg). Each symbol carries 2 bits. Spectral efficiency: 2 bps/Hz. Requires the same E_b/N_0 as BPSK but doubles the data rate for the same bandwidth. The standard choice for most space links.
- 8PSK: 8 phase states, 3 bits/symbol, spectral efficiency 3 bps/Hz. Requires ~3.5 dB more E_b/N_0 than QPSK for the same BER.
- 16APSK: 16 states (amplitude + phase), 4 bits/symbol, spectral efficiency 4 bps/Hz. Requires ~7 dB more E_b/N_0 than QPSK.

Frequency Shift Keying (FSK): The carrier frequency is shifted between discrete values. Less spectrally efficient than PSK but more tolerant of amplifier nonlinearity. Variants:

- FSK: Simple frequency switching.
- GFSK (Gaussian FSK): Gaussian filter smooths frequency transitions, reducing spectral spreading. Used by many CubeSat UHF radios (AX.25 protocol at 9600 bps).
- MSK (Minimum Shift Keying): A special case of FSK with minimum frequency deviation that still allows coherent detection. Constant envelope (important for nonlinear amplifiers). Spectral efficiency ~1 bps/Hz.

Why QPSK is preferred over BPSK for space links: QPSK transmits 2 bits per symbol while requiring the same energy per bit as BPSK. This means QPSK achieves twice the data rate for the same bandwidth and the same E_b/N_0 performance. The only additional complexity is that the receiver must resolve 4 phase states instead of 2, which is trivial for modern digital receivers.

Forward Error Correction (FEC) -- Coding Gain

FEC adds redundant bits to the data stream before transmission. The receiver uses these redundant bits to detect and correct bit errors without retransmission. The improvement in E_b/N_0 requirement (compared to uncoded) is called the coding gain.

Why coding is essential for space links: Space links operate at very low received signal power (femtowatts to picowatts). Without coding, the required E_b/N_0 for BER 10^-6 is 10.5 dB. With LDPC coding, this drops to 2.0 dB -- a coding gain of 8.5 dB. This is equivalent to either: increasing the TX power by 7x, or increasing the antenna diameter by 2.7x, or reducing the data rate by 7x. Coding achieves the same benefit at the cost of only a few watts of digital processing power.

Code types used in space communications:

Code Type	Code Rate	Coding Gain at BER 10^-6	Decoding Complexity	Standard	Use
Convolutional (K=7)	1/2	~5.5 dB	Low (Viterbi decoder, hardware)	CCSDS	Legacy telecommand, AX.25
Convolutional + RS (concatenated)	~0.44	~7.5 dB	Medium	CCSDS	Standard CCSDS telemetry (many heritage missions)
Turbo code	1/2 to 1/6	~8--10 dB	High (iterative decoder)	CCSDS	Deep space links (Mars, Jupiter)
LDPC (Low-Density Parity Check)	1/2	~8.5 dB	Medium-High (iterative BP decoder)	CCSDS, DVB-S2	Modern space downlinks, recommended for CubeSats
LDPC	3/4	~6.5 dB	Medium-High	CCSDS, DVB-S2	Balanced performance and throughput
LDPC	7/8	~5.0 dB	Medium-High	DVB-S2	Maximum throughput, strong signal

[Source: CCSDS 131.0-B-4; DVB-S2 ETSI EN 302 307]

Code rate r: The ratio of information bits to total transmitted bits. A rate-1/2 code transmits 1 information bit for every 2 transmitted bits (50% overhead). This means the channel data rate must be R_channel = R_info / r -- a rate-1/2 code requires twice the channel bandwidth for a given information rate.

Complete Modulation + Coding Table

Modulation + Coding	E_b/N_0 Required (BER 10^-6)	Spectral Efficiency (bps/Hz)	Typical Use	Implementation
GMSK uncoded	10.5 dB	~1.0	AX.25 amateur, legacy	Simple radio IC
BPSK uncoded	10.5 dB	1.0	Legacy telecommand	Simple
QPSK uncoded	10.5 dB	2.0	Simple telemetry	Moderate
QPSK + conv (r=1/2, K=7)	5.0 dB	1.0	Standard CCSDS TM	Hardware Viterbi
QPSK + conv + RS	3.0 dB	0.88	Heritage CCSDS	Hardware
QPSK + LDPC (r=1/2)	2.0 dB	1.0	High-efficiency downlink	FPGA-based
QPSK + LDPC (r=3/4)	4.0 dB	1.5	Balanced --	FPGA-based
QPSK + LDPC (r=7/8)	5.5 dB	1.75	Bandwidth-limited	FPGA-based
8PSK + LDPC (r=3/4)	6.5 dB	2.25	High-rate downlink	FPGA-based
16APSK + LDPC (r=3/4)	8.5 dB	3.0	Maximum throughput	FPGA-based

Design guidance for CubeSats:

- UHF amateur: GMSK or AFSK (AX.25 protocol) at 1.2--9.6 kbps channel rate. Add convolutional coding if link margin is tight.
- S-band: QPSK + LDPC (r=1/2 or r=3/4). This is the sweet spot: 5.5--8.5 dB coding gain with manageable complexity. Most COTS S-band CubeSat transmitters (Endurosat, NanoAvionics, AAC Clyde) support DVB-S2 or CCSDS LDPC natively.
- X-band / Ka-band: 8PSK or 16APSK + LDPC for maximum spectral efficiency when bandwidth is limited.

7. Ground Stations (10 min)

Teaching Notes

The ground station is half of the communication link. Its performance (G/T) directly determines the achievable data rate. Upgrading the ground station is often the cheapest way to improve link performance (compared to upgrading the spacecraft transmitter or antenna).

Ground Station Types

Type	Antenna	G/T (S-band)	Cost	Availability	Examples
Amateur (SatNOGS)	10--15 dBi Yagi	-15 to -10 dB/K	Free (volunteer-operated)	Global network, 200+ stations	SatNOGS network
University	2--3 m dish	+10 to +15 dB/K	$50--200K (build)	Limited availability	Many universities have S-band stations
Commercial (small)	3--5 m dish	+15 to +25 dB/K	500/pass or5--20K/month	KSAT Lite, AWS Ground Station	KSAT, Amazon, Leaf Space
Commercial (large)	5--13 m dish	+25 to +35 dB/K	1000/pass or20--50K/month	SSC, KSAT, ATLAS	SSC (Esrange), KSAT (Svalbard), ATLAS (Fairbanks)
Deep Space Network	34--70 m dish	+45 to +60 dB/K	NASA-funded only	DSN (3 sites globally)	Goldstone, Canberra, Madrid

[Source: KSAT Lite pricing 2024; AWS Ground Station pricing; SSC SmallSat ground segment]

Contact geometry and pass duration:

A LEO satellite is in view of a ground station for a limited time per orbit. The pass duration depends on the orbit altitude and the maximum elevation angle:

$t_pass ~= (2)/(n) \arccos(\fraccos(\varepsilon_max)cos(\varepsilon_min))$

For a simplified estimate:

Altitude	Min Elevation	Max Pass Duration	Typical Usable Duration	Passes/Day (mid-latitude)
400 km	10 deg	~7 min	~5 min	3--4
500 km	10 deg	~8 min	~6 min	4--5
600 km	10 deg	~9 min	~7 min	4--5
800 km	10 deg	~11 min	~9 min	5--6

The "usable duration" is shorter than the total pass because the link only closes above the minimum elevation angle, and the first/last 30--60 seconds are used for signal acquisition and link setup.

Ground station selection criteria:

- Location: Polar stations (Svalbard at 78 degN, McMurdo at 78 degS) see every orbit of a polar/SSO satellite, providing 12+ contacts per day. Mid-latitude stations see only 3--5 passes. Equatorial stations see even fewer passes of polar satellites.
- G/T: Determines the achievable data rate. A 3 dB improvement in G/T allows doubling the data rate.

- Licensing: Must be compatible with the spacecraft frequency allocation
- Cost: Ranges from free (SatNOGS, university) to $1000+/pass (commercial)
- Reliability: SLA (service-level agreement) for commercial stations; university stations may have limited operator availability

EIRP (ground station transmit, for uplink): For telecommand uplink, the ground station transmits to the spacecraft. Ground station EIRP is typically 40--60 dBW for commercial stations, sufficient for robust command links even with low-gain spacecraft receive antennas.

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8. Worked Examples: Complete Link Budgets (15 min)

3U EO CubeSat -- S-band Downlink

> *Worked Example -- S-band Downlink for 3U EO CubeSat (SuperDove-class)*
>
> *Scenario: 500 km SSO, S-band (2250 MHz), 1 Mbps downlink, 10 deg minimum elevation (slant range 1300 km), 3 m ground station dish (T_sys = 150 K, G_RX = 35 dBi at S-band).*
>
> | Line | Parameter | Value | Unit | Calculation/Source |
> |------|-----------|-------|------|--------------------|
> | 1 | TX Power (2 W) | +3.0 | dBW | COTS S-band TX (Endurosat) |
> | 2 | TX Antenna Gain (patch) | +6.0 | dBi | Single-element S-band patch |
> | 3 | TX Line Losses | -1.5 | dB | 15 cm cable + connector |
> | 4 | EIRP | +7.5 | dBW | 3.0 + 6.0 - 1.5 |
> | 5 | FSPL (2250 MHz, 1300 km) | -170.8 | dB | 92.45 + 20log_10(1300) + 20log_10(2.25) |
> | 6 | Atmospheric Loss | -0.5 | dB | S-band, 10 deg elevation |
> | 7 | Pointing Loss | -1.0 | dB | 5 deg mispoint, 80 deg beamwidth patch |
> | 8 | Polarisation Loss (RHCP-RHCP) | -0.3 | dB | Minor axial ratio mismatch |
> | 9 | RX Antenna Gain (3 m dish) | +35.0 | dBi | 10log_10(0.6 x (pi x 3.0 / 0.133)^2) |
> | 10 | System Noise Temp (150 K) | 21.8 | dBK | Professional LNA + sky temp |
> | 11 | G/T | +13.2 | dB/K | 35.0 - 21.8 |
> | 12 | Boltzmann Constant | +228.6 | dBW/K/Hz | -k in link equation |
> | 13 | C/N_0 | +76.7 | dBHz | 7.5 - 170.8 - 0.5 - 1.0 - 0.3 + 13.2 + 228.6 |
> | 14 | Data Rate (1 Mbps) | 60.0 | dBbps | 10log_10(10^6) |
> | 15 | E_b/N_0 available | +16.7 | dB | 76.7 - 60.0 |
> | 16 | E_b/N_0 required (QPSK + LDPC r=3/4) | 4.0 | dB | From modulation/coding table |

| 17 | Implementation Loss | 2.0 | dB | Real demodulator vs ideal |
| 18 | LINK MARGIN | +10.7 | dB | 16.7 - 4.0 - 2.0 |

> *Result: Link closes with 10.7 dB margin (requirement: >= 3 dB). Pass.*
>
> *Design insight: The generous 10.7 dB margin suggests the link is over-designed for 1 Mbps. The team could increase the data rate:*
>
> *Maximum data rate at 3 dB margin:*
>
> *E_b/N_0 available at max rate = 4.0 + 2.0 + 3.0 = 9.0 dB*
>
> *C/N_0 = 76.7 dBHz, so $R_{b,max} = 10^{(76.7 - 9.0)/10} = 10^{6.77} \approx 5.9$ Mbps at 3 dB margin.*
>
> *Alternatively, at 5 Mbps ($10\log_{10}(5 \times 10^6) = 67.0$ dBbps):*
>
> *E_b/N_0 = 76.7 - 67.0 = 9.7 dB. Margin = 9.7 - 4.0 - 2.0 = 3.7 dB. Pass (barely).*

1U Worked Example: UniSat-1

UHF Link Budget: 437 MHz at 9600 bps

UniSat-1 uses the UHF amateur band at 437 MHz with a ground station equipped with a 10 dBi Yagi antenna. This is the lowest-cost and simplest communication architecture available to CubeSat missions.

> *Worked Example -- UHF Downlink Link Budget for UniSat-1*
>
> *Scenario: 400 km orbit, UHF (437 MHz), 9600 bps downlink (GMSK), 10 deg minimum elevation angle, amateur ground station with 10 dBi Yagi antenna, T_{sys} = 600 K (COTS LNA, coax cable losses, sky noise near horizon).*
>
> *Slant range at 10 deg elevation:*
>
> *From 400 km altitude, the worst-case slant range at 10 deg elevation is approximately 1150 km.*
>
> | Line | Parameter | Value | Unit | Notes |
> |------|-----------|-------|------|-------|
> | 1 | TX Power (0.5 W) | -3.0 | dBW | Standard CubeSat UHF radio |
> | 2 | TX Antenna Gain (monopole) | 0.0 | dBi | Quarter-wave monopole, ~omnidirectional |
> | 3 | TX Line Losses | -0.5 | dB | Short cable to antenna |
> | 4 | EIRP | -3.5 | dBW | Low EIRP is the fundamental UHF challenge |

| 5 | FSPL (437 MHz, 1150 km) | -155.5 | dB | Lower than S-band (good) |

| 6 | Atmospheric Loss | -0.3 | dB | UHF atmospheric loss is minimal |

| 7 | Pointing Loss (omni antenna) | -0.5 | dB | Omni pattern -- negligible |

| 8 | Polarisation Loss (linear-linear) | -3.0 | dB | Major loss -- Faraday rotation in ionosphere rotates polarisation randomly |

| 9 | RX Antenna Gain (10 dBi Yagi) | +10.0 | dBi | 5-element Yagi, manually tracked |

| 10 | System Noise Temp (600 K) | 27.8 | dBK | Amateur station: warm LNA + cable loss |

| 11 | G/T | -17.8 | dB/K | Low G/T is the ground station limitation |

| 12 | Boltzmann Constant | +228.6 | dBW/K/Hz | |

| 13 | C/N_0 | +48.0 | dBHz | |

| 14 | Data Rate (9600 bps) | 39.8 | dBbps | |

| 15 | E_b/N_0 available | +8.2 | dB | |

| 16 | E_b/N_0 required (GMSK uncoded) | 10.5 | dB | |

| 17 | Implementation Loss | -2.0 | dB | |

Margin = 8.2 - 10.5 - 2.0 = -4.3 dB. The link does NOT close!

Fix 1: Add FEC. Convolutional coding (r=1/2, K=7):

- E_b/N_0 required drops to 5.0 dB (5.5 dB coding gain)

- Channel rate remains 9600 bps, but useful throughput is 4800 bps (half is redundancy)

- Margin = 8.2 - 5.0 - 2.0 = +1.2 dB -- still below 3 dB requirement.

Fix 2: Upgrade ground antenna to cross-Yagi (13 dBi) with circular polarisation:

- RX gain: +13.0 dBi (was +10.0) -> +3.0 dB improvement

- Polarisation loss: -0.5 dB (was -3.0 dB, now RHCP-to-RHCP) -> +2.5 dB improvement

- Net improvement: +5.5 dB

- New C/N_0: 53.5 dBHz

- New E_b/N_0 available: 53.5 - 39.8 = 13.7 dB

- Margin = 13.7 - 5.0 - 2.0 = +6.7 dB -- Pass (> 3 dB).

Polarisation loss physics: At UHF (437 MHz), the ionosphere causes Faraday rotation of linearly polarised signals. The rotation angle depends on the total electron content (TEC) along the path and varies with time of day, solar activity, and signal path. If the satellite transmits linear polarisation and the ground station receives with linear polarisation, the polarisation planes may be orthogonal at times, causing up to complete signal loss (theoretically infinite loss, practically 20+ dB fades). Using circular polarisation on at least one end (preferably both) eliminates Faraday rotation loss, reducing it to ~0.3--0.5 dB axial ratio mismatch. This is why circular polarisation is mandatory for reliable UHF satellite links.

> *Final link budget summary (with FEC + cross-Yagi):*
>
> | Parameter | Value |
> |-----------|-------|
> | TX power | 0.5 W |
> | TX antenna | Monopole (0 dBi) |
> | Frequency | 437 MHz |
> | Channel rate | 9600 bps |
> | Useful throughput | 4800 bps (with r=1/2 FEC) |
> | Ground antenna | 13 dBi cross-Yagi, RHCP |
> | Link margin | +6.7 dB |

Data throughput for UniSat-1: At 4800 bps useful throughput, a 7-minute pass delivers:
V_{pass} = 4800 x 420 x 0.85 = 1.71 Mbit = 214 kB per pass.

The 0.85 factor accounts for protocol overhead (packet headers, acknowledgements, retransmissions, link setup time).

With 4 passes/day: V_{daily} = 856 kB/day ~= 0.84 MB/day. The magnetometer generates < 1 kbps x 600 s/orbit x 15 orbits = 9 Mbit/day = 1.13 MB/day. This is marginal -- the team may need to prioritise data or add a second ground station. Using the SatNOGS network (200+ volunteer stations worldwide) could provide 10+ additional contacts per day at no cost.

Key lesson from UniSat-1 link budget: UHF links are power-starved compared to S-band or X-band. The lower FSPL at 437 MHz (~15 dB less than S-band) does not compensate for the low TX power (0.5 W vs 2 W = 6 dB less), low antenna gain (0 dBi vs 6 dBi = 6 dB less), and higher system noise temperature of amateur stations (600 K vs 150 K = 6 dB worse). FEC coding and circular polarisation are essential for closing a UHF CubeSat link.

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9. Data Budget (10 min)

Teaching Notes

The data budget determines whether the communication system can deliver all mission data to the ground. Even if the link budget closes, the mission fails if the total data volume exceeds the downlink capacity.

> *Key Equations -- Data Budget*
>
> *Daily data generation:*
>

> $V_{gen} = R_{payload} \times t_{imaging} \times N_{orbits} \times f_{compression}$
>
> *Daily downlink capacity:*
>
> $V_{DL} = R_{downlink} \times t_{contact} \times N_{passes} \times \eta_{protocol}$
>
> *where eta_protocol = 0.80--0.90 accounts for packet overhead, retransmissions, link setup time, and handshaking.*
>
> *Data budget closure:*
>
> $V_{DL} >= V_{gen}$ *(data budget closes)*
>
> *Backlog clearance time: If V_DL < V_gen per day, data accumulates in onboard storage. The backlog clearance time is:*
>
> $t_{clear} = \frac{V_{stored}}{V_{DL} - V_{gen}}$
>
> *If V_DL < V_gen, the backlog grows forever -- the mission cannot sustain its data generation rate.*
>
> *Worked Example -- Data Budget for 3U EO CubeSat*
>
> *Generation: 240 Mbps raw imaging data x 5 min/orbit x 15 orbits/day x 0.25 (4:1 JPEG2000 compression)*
>
> *= 240 x 300 x 15 x 0.25 = 270,000 Mbit/day = 33.75 GB/day*
>
> *Wait -- that's extremely high. Let's be more realistic about imaging time. Not every orbit has a target. Assume 4 imaging passes per day, 5 minutes each:*
>
> *V_gen = 240 x 10^6 x 300 x 4 x 0.25 = 72,000 Mbit = 9.0 GB/day*
>
> *Still high. Planet SuperDove images approximately 1--2 minutes per orbit over priority targets, compresses heavily (10:1+), and downlinks selectively.*
>
> *Revised: 240 Mbps x 1 min/pass x 4 passes/day x 0.10 (10:1 compression) = 240 x 60 x 4 x 0.10 = 5,760 Mbit = 720 MB/day*
>
> *Downlink (S-band at 5 Mbps): 5 Mbps x 6 min/pass x 5 passes/day x 0.85 = 153 Mbit/pass x 5 = 765 Mbit = 95.6 MB/day x 5 = 478 MB/day*
>
> *Hmm, let's recompute carefully:*
>
> *V_DL = 5 x 10^6 x 360 x 5 x 0.85 = 7,650 Mbit = 956 MB/day*

> *Result: 956 MB/day > 720 MB/day. Data budget closes with 33% margin.*
>
> *Sensitivity: If imaging duty cycle doubles (2 min/pass), generation rises to 1440 MB/day > 956 MB/day. Options: (a) X-band for higher data rate, (b) additional ground stations, (c) more aggressive compression, (d) onboard data prioritisation/selection.*

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10. SpaceCDF Exercise (25 min)

Instructions

1. Spectrum Selector (Dashboard): Select your license type and frequency band
2. Link Budget tab:
- Enter TX power, antenna type, frequency, ground station parameters
- Review the computed link margin
- Verify it meets >= 3 dB requirement
3. Compare to your hand calculation from the worked example
4. Equipment Browser: Select a transponder and antenna that match your band choice
- Note RF compatibility warnings (band mismatch between transponder and antenna)
5. Complete Worksheet 3.3

Discussion Questions

- What is the most impactful parameter in your link budget? (Usually FSPL or G/T)
- How does doubling the data rate affect link margin? (Reduces by 3 dB -- because 10log_10(2) = 3 dB)
- Could you use a lower-power transmitter and still close the link? What is the minimum TX power?
- Does your data budget close? If not, what is the cheapest fix? (Usually: more ground station passes, or lower imaging duty cycle)
- What is the effect of Faraday rotation on your UHF link? (If applicable)

Session Summary

Topic	Key Takeaway
Link budget	Margin = E_b/N_0,avail - E_b/N_0,req - L_impl >= 3 dB
EIRP	EIRP = P_TX + G_TX - L_TX -- transmitter's effective power in beam direction
FSPL physics	Geometric spreading (1/d^2) + aperture scaling (lambda^2); not energy absorption
FSPL formula	92.45 + 20log_10(d_km) + 20log_10(f_GHz); increases 6 dB per doubling of frequency or distance
G/T	Receiver figure of merit: antenna gain minus noise temperature; most important ground station parameter

Antenna types	Monopole (0 dBi, omni) to parabolic (25--45 dBi, narrow beam); gain vs pointing trade-off
Antenna gain	G = eta_a (pi D/lambda)^2; patch ~6 dBi; 3 m dish: 35 dBi at S-band, 25 dBi at X-band
Pointing loss	-12(Dtheta/theta_3dB)^2 dB; narrower beam = more sensitive to pointing errors
Modulation	QPSK: 2 bits/symbol, same E_b/N_0 as BPSK but 2x spectral efficiency; standard choice
FEC coding	LDPC (r=3/4): E_b/N_0 = 4.0 dB, coding gain ~6.5 dB; essential for space links
Coding gain	5--10 dB improvement for free (just digital processing); equivalent to 3--10x power increase
UHF	Low EIRP, Faraday rotation (use circular polarisation), high ground station noise; FEC mandatory
Band selection	UHF for < 9.6 kbps (free, amateur); S-band for < 10 Mbps; X-band for < 400 Mbps; Ka-band for > 400 Mbps
Rain fade	Negligible below 4 GHz; 1--3 dB at X-band; 3--15 dB at Ka-band; add margin or adaptive rate
Ground	G/T: amateur -15 dB/K, university +12 dB/K, commercial +25 dB/K; polar stations see every orbit
Data budget	Daily downlink capacity must exceed daily data generation; compression ratio is a key lever
Licensing	Amateur (free, IARU, 3--6 mo); S-band (30--45K, 6--12 mo); X-band (50--80K, 12+ mo)

Session 3.4: Structure, Propulsion, and Equipment Selection

Duration: 2 hours

Prerequisites: Sessions 2.1--3.3 (all subsystem sizing complete)

SpaceCDF Tabs: Equipment Browser, Dashboard, Trade Studies, Budget Breakdown

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References

- Cal Poly, CubeSat Design Specification (CDS) Rev 14.1, February 2022
- Wertz, Everett & Puschell, Space Mission Engineering: The New SMAD, 2011, Ch. 11.3 (Structure), Ch. 17 (Propulsion)
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- Sutton & Biblarz, Rocket Propulsion Elements, 9th ed., 2017, Ch. 2--4
- Enpulsion, NANO R3 Thruster Datasheet, 2023
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- Tyvak, Structure Specifications, 2023
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Learning Objectives

By the end of this session, participants will be able to:

1. Verify CubeSat Design Specification (CDS) compliance for 1U--12U form factors
2. Explain the structural load environment (quasi-static, vibration, shock) and its physical origins
3. Compute structural margin of safety for quasi-static launch loads
4. Estimate the fundamental frequency requirement and verify it against deployer specifications
5. Apply the Tsiolkovsky rocket equation to compute propellant mass for a given D V
6. Explain the physics of each propulsion technology and select based on mission requirements
7. Select OBC architecture based on processing, radiation, and interface requirements
8. Size onboard data storage from the data budget
9. Select flight hardware using SpaceCDF's equipment browser with budget tracking

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1. CubeSat Structure and CDS Compliance (30 min)

Teaching Notes

[Source: Cal Poly CDS Rev 14.1, February 2022; ECSS-E-ST-32C]

CDS Dimensional Specifications

Form	Dimensions (mm)	Max Mass (kg)	Internal Volume (cm^3)	Typical Deployer
1U	100 x 100 x 113.5	2.0	~1000	ISIPOD, P-POD
1.5U	100 x 100 x 170.2	3.0	~1500	ISIPOD
2U	100 x 100 x 227.0	4.0	~2000	ISIPOD, P-POD
3U	100 x 100 x 340.5	6.0	~3000	P-POD, ISIPOD, NanoRacks NRCSD
6U	100 x 226.3 x 340.5	12.0	~6000	6U deployer (Exolaunch, D-Orbit)
12U	226.3 x 226.3 x 340.5	24.0	~12000	12U deployer (Exolaunch)

Structural Materials

CubeSat rail material: Aluminium 7075-T6

This is the most commonly specified structural aluminium alloy for CubeSat rails. The CDS mandates hard-anodised aluminium for the rails (the four load-bearing edges that slide along the deployer guide channels).

Property	Al 7075-T6	Al 6061-T6	Ti-6Al-4V	CFRP (quasi-isotropic)
Density (kg/m^3)	2810	2700	4430	1600
Yield strength sigma_y (MPa)	503	276	880	N/A (use ultimate)
Ultimate strength sigma_u (MPa)	572	310	950	500--800
Young's modulus E (GPa)	71.7	68.9	114	70--150 (direction-dependent)
CTE (ppm/degC)	23.6	23.1	8.6	0--2 (tuneable)
Thermal conductivity (W/m/K)	130	167	6.7	3--10

[Source: MMPDS / ASM Handbook; Hexcel HexPly datasheets]

Why Al 7075-T6 for rails:

- High strength-to-weight ratio (superior to 6061-T6)
- Hard anodisation provides a durable, low-friction surface finish for deployer guide channel contact (reduces galling, provides electrical insulation)
- Good machinability
- Extensive flight heritage (virtually every CubeSat ever launched)
- CTE-matched to Al deployer structure (prevents differential thermal expansion binding)

Why NOT other materials for rails:

- Titanium: Excellent strength but poor thermal conductivity (thermal hot spots), difficult to machine, risk of galling against aluminium deployer
- CFRP: Cannot be anodised; CTE mismatch with Al deployer causes binding at temperature extremes; poor electrical conductivity (grounding/bonding issues)
- Stainless steel: Too heavy; poor CTE match

Anodisation physics: Anodisation is an electrochemical process that grows a hard aluminium oxide (Al_2O_3) layer on the surface. Hard anodisation (Type III) produces a 25--75 um thick oxide layer with hardness of 60--70 HRC (harder than most steel). This layer provides: wear resistance against deployer contact, electrical insulation (prevents arcing between satellite and deployer), corrosion resistance, and controlled surface optical properties (alpha_s ~= 0.3--0.5, \varepsilon ~= 0.8--0.85 for clear anodise; alpha_s ~= 0.9, \varepsilon ~= 0.85 for black anodise).

Key CDS Requirements

Requirement	Specification	Physical Rationale
Rail material	Hard anodised aluminium (7075-T6 or 6061-T6)	Wear resistance, CTE match to deployer, electrical isolation
Rail cross-section	8.5 x 8.5 mm minimum contact area	Adequate bearing area for launch loads; prevents rail yielding under quasi-static acceleration
Surface finish	All external surfaces anodised or non-outgassing coating	Prevent contamination of other payloads on launch vehicle (molecular outgassing deposits on optics)
Deployment switches	Minimum 1 on each accessible rail face (+X, -X)	Inhibit all spacecraft activity until fully deployed from deployer (prevents inadvertent deployment, RF emissions in fairing)
Remove Before Flight	Required; physically disables all power systems	Final safety inhibit; removed at launch pad after integration; ensures zero RF emissions and zero deployment actuator current until intentional removal
Protrusions	None beyond rail envelope in stowed configuration	Ensures clean ejection from deployer; prevents snagging on guide rails or adjacent CubeSat
Centre of gravity	Within 2 cm of geometric centre (per deployer ICD)	Prevents wobble during deployment ejection; ensures all CubeSats eject with similar tip-off rates
Fundamental frequency	> 40 Hz first mode (typical deployer requirement)	Prevents dynamic coupling between satellite and launch vehicle structural modes (which cluster at 10--30 Hz)

PC/104 Stack Architecture

Most CubeSat avionics use the PC/104-compatible stack architecture, a heritage from the industrial embedded computing standard adapted for space:

Physical specifications:

- Board size: 96 x 90 mm (standard) or 90 x 96 mm
- Connector: 104-pin stack-through header (2 x 52 pins, 2.54 mm pitch) -- original PC/104 pinout

carries power + I2C + SPI + UART + GPIO
- Stack spacing: Typically 10--15 mm between boards (constrained by component height and connector mating height)
- Stack capacity: 1U accommodates ~4 boards; 3U accommodates ~12 boards (340 mm / ~28 mm per board slot)

What rides on the stack:

- EPS board (battery management, MPPT, power distribution)
- OBC board (processor, memory, interfaces)
- Communications board (UHF radio, or S-band transponder)
- AOCS board (if integrated -- some vendors combine IMU + magnetorquer driver + RW interface on one PCB)
- Payload interface board (ADC, sensor interfaces)

Mechanical concerns:

- Solder joints are the weakest point; random vibration causes fatigue cracking at heavy component leads (especially tall electrolytic capacitors, large connectors, and crystal oscillators)
- Board-to-board connectors must be properly preloaded (too loose = intermittent contact; too tight = difficult assembly/disassembly during I&T)
- Standoffs and spacers must be correctly torqued; Loctite 222 (low-strength threadlocker) is standard

Launch Load Environment

The launch environment subjects the satellite to loads from engine thrust, aerodynamic buffeting, stage separation, and pyrotechnic events. The satellite must survive all of these without structural failure or functional degradation.

Load Type	Physical Source	Typical Level	Duration	Frequency	Verification Method
Quasi-static acceleration	Engine thrust + aeroloading	6--12 g axial, 2--4 g lateral	Seconds to minutes	0 (static equivalent)	Analysis + sine vibration test
Sine vibration	Low-frequency vehicle dynamics	0.5--3 g (5--100 Hz)	Minutes	5--100 Hz	Sine sweep test (3 axes)
Random vibration	Acoustic noise + turbulent boundary layer	5--15 grms (20--2000 Hz)	60--120 s per	20--2000 Hz	Random vibration test (3 axes)
Shock	Pyrotechnic separation events (stage sep, fairing sep, deployer spring release)	500--2000 g at separation (high	< 10 ms	100--10,000 Hz	Shock response spectrum (SRS) test
Acoustic	Sound pressure from engine exhaust, aerodynamic noise	120--140 dB (20--10,000 Hz)	Minutes	20--10,000 Hz	Usually covered by random vib for CubeSats

[Source: ECSS-E-ST-10-03C; NASA GEVS (GSFC-STD-7000B); Falcon 9 Payload User's Guide; PSLV User's Guide]

Random vibration PSD profile (typical CubeSat deployer level):

Frequency (Hz)	ASD Level (g^2/Hz)	Notes
20	0.01	Low-frequency start (ramp up)
50	0.04	Ramp up at +6 dB/oct
100	0.04	Flat region start

800

0.04

Flat region end

2000	0.01	Roll off at -6 dB/oct
Overall	~7 grms	Typical for CubeSat deployer qualification level

The flat region at 0.04 g^2/Hz from 100--800 Hz is where most structural damage occurs, because this is where PCB resonances and solder joint fatigue are excited.

What fails during vibration testing:

1. Solder joints: Heavy components (connectors, tall capacitors, transformers) with long lever arms crack at their solder joints. Mitigation: use surface-mount components, stake tall components with adhesive (Loctite 4860 or similar), use conformal coating.
2. Deployable mechanisms: Antenna hinges, solar panel hold-down mechanisms, and deployment springs can fail if not properly constrained. Mitigation: adequate preload on hold-down mechanisms, shock testing of pyrotechnic release devices.
3. Optical components: Lenses and mirrors can shift or crack if not properly mounted with strain-relief. Mitigation: RTV potting, flexure mounts.
4. Wire harness: Chafing against structure edges. Mitigation: edge radii > 1 mm, harness tie-downs every 50 mm, protective sleeving.

Structural Margin of Safety

> *Key Equations -- Structural Margin of Safety*
>
> *$MoS = \fracsigma_allowablesigma_design x FoS - 1$*
>
> *where:*
> *- sigma_allowable = material yield or ultimate strength (MPa)*
> *- sigma_design = computed stress under design loads (MPa)*
> *- FoS = factor of safety*
>
> *Requirement: MoS >= 0 for all load cases.*
>
> *Factors of safety (ECSS-E-ST-32C):*
>
> | Material / Joint | Yield FoS | Ultimate FoS | Rationale |
> |-----------------|----------|-------------|-----------|
> | Metallic (Al 7075-T6) | 1.25 | 1.5 | Standard structural metals |
> | Composite (CFRP) | 1.5 | 2.0 | Higher variability in laminate properties |
> | Bonded joints | 1.5 | 2.0 | Bond strength is highly process-dependent |
> | Pressurised systems | 1.5 | 2.0 | Burst hazard to other payloads |
> | Mechanisms (single-use) | -- | 2.0 | Must work first time; no test opportunity |

> *Design loads: The design load includes the quasi-static acceleration (from the launch vehicle user guide), multiplied by a dynamic amplification factor (DAF ~= 1.25--1.5) if the satellite's natural frequency is near any launch vehicle forcing frequency.*

> *Worked Example -- Axial Load on 3U CubeSat Rail (SuperDove-class)*
>
> *Given: 3U CubeSat, mass = 5 kg, axial launch load = 9 g (Falcon 9 typical), 4 rails (load shared equally), rail cross-section = 8.5 x 8.5 mm, material = Al 7075-T6 (sigma_y = 503 MPa, sigma_u = 572 MPa).*
>
> *Step 1 -- Design load per rail:*
>
> *F = (m x n x g_0)/(4) = (5 x 9 x 9.81)/(4) = (441.5)/(4) = 110.4 N*
>
> *Step 2 -- Compressive stress:*
>
> *sigma = \fracFA_rail = (110.4)/(8.5 x 10^-3 x 8.5 x 10^-3) = (110.4)/(7.225 x 10^-5) = 1.53 MPa*
>
> *Step 3 -- Margin of safety (yield):*
>
> *MoS_y = (503)/(1.53 x 1.25) - 1 = (503)/(1.91) - 1 = 262 \gg 0 Pass (by a very large margin)*
>
> *Step 4 -- Margin of safety (ultimate):*
>
> *MoS_u = (572)/(1.53 x 1.5) - 1 = (572)/(2.30) - 1 = 248 \gg 0 Pass*
>
> *Key insight: For CubeSats, quasi-static axial stress on the rails is never the critical load case. The rails are massively over-designed for direct compression. The critical structural design drivers are usually:*
>
> *1. Stiffness (fundamental frequency > 40 Hz) -- driven by internal board/component mounting, not rail strength*
>
> *2. Random vibration fatigue on PCB solder joints -- the real failure mode*
>
> *3. Deployment mechanism reliability -- spring force, latch engagement, alignment tolerances*
>
> *4. CG location -- difficult to achieve with asymmetric payloads or propulsion tanks*

Fundamental Frequency

> *Key Equations -- Fundamental Frequency (simplified beam model)*
>
> *For a cantilevered beam (simplified CubeSat model, clamped at deployer interface):*
>
> *$f_1 = (1.875^2)/(2pi L^2) sqrt\fracEIrho A_cross$*
>
> *where E = Young's modulus (Pa), I = second moment of area (m^4), rho = linear density (kg/m), A_cross = cross-section area (m^2), L = length (m).*

> *Requirement: $f_1 > 40$ Hz (from deployer ICD). Some deployers require > 90 Hz (e.g., NanoRacks NRCSD).*
>
> *For a 3U CubeSat modelled as a cantilevered Al box beam:*
>
> *- $L = 0.34$ m, $E = 72$ GPa, box wall thickness $t = 1.5$ mm*
>
> *- $I \approx (b^4 - (b-2t)^4)/(12) = (0.10^4 - 0.097^4)/(12) \approx 1.36 \times 10^{-6}$ m^4*
>
> *- Linear mass: $\rho_L = m/L = 5/0.34 = 14.7$ kg/m*
>
> *$f_1 = (3.516)/(2\pi \times 0.34^2) \sqrt{(72 \times 10^9 \times 1.36 \times 10^{-6})/(14.7)} = (3.516)/(0.726) \sqrt{6666} = 4.84 \times 81.6 = 395$ Hz*
>
> *This easily exceeds 40 Hz. The structure itself is very stiff. However, the actual first mode is usually determined by: (a) the heaviest internal component on its mounting bracket (e.g., a 350 g star tracker cantilevered on a bracket), or (b) a deployable mechanism in its stowed configuration (e.g., a folded solar panel constrained only by a hold-down pin). These local modes, not the overall structural mode, are typically the design concern.*

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2. Propulsion System Design (30 min)

Teaching Notes

[Source: SMAD, Ch. 17; Sutton & Biblarz, Ch. 2--4; Goebel & Katz, Fundamentals of Electric Propulsion, 2008]

When Propulsion is Required

Need	Typical D V	Example Scenario	Timeline
Orbit maintenance (drag compensation)	5--15 m/s per year	LEO below 400 km in solar maximum	Continuous low-thrust
Deorbit (active disposal)	50--150 m/s	Active disposal from > 600 km (FCC 5-year rule)	End of mission
Collision avoidance	1--5 m/s per event	Conjunction avoidance, 2--5 events per year for LEO	On-demand, within hours
Constellation phasing	10--50 m/s	Spreading satellites into operational orbit slots	Weeks to months
Orbit raising	50--200 m/s	Transfer from deployment orbit to operational orbit	Weeks to months
Station-keeping	1--10 m/s per year	Maintain orbit altitude and phase	Periodic

When NO Propulsion is Needed

- Orbit < 500 km: Natural atmospheric decay provides FCC 5-year deorbit compliance (depends on ballistic coefficient and solar activity)
- Low-cost technology demonstration: Limited lifetime acceptable, no orbit maintenance needed
- Constellation using differential drag for phasing (e.g., Planet SuperDove adjusts its cross-section area to create differential drag, enabling free phasing manoeuvres)
- Budget-constrained missions where propulsion cost/risk/mass exceeds benefit

The Tsiolkovsky Rocket Equation -- Physics

The rocket equation is the fundamental relationship governing all propulsive manoeuvres. It derives from conservation of momentum: the momentum of the exhaust equals the momentum change of the spacecraft.

> *Key Equations -- Tsiolkovsky Rocket Equation*
>
> *Starting from F = \dotm v_e (thrust = mass flow rate x exhaust velocity) and integrating:*
>
> *$D V = v_e ln((m_0)/(m_f)) = I_sp * g_0 * ln((m_0)/(m_f))$*
>
> *Rearranged for propellant mass:*
>
> *$m_propellant = m_dry x (e^D V / (I_sp * g_0) - 1)$*
>
> *where:*
>
> *- v_e = I_sp x g_0 = effective exhaust velocity (m/s)*
>
> *- I_sp = specific impulse (s) -- the "fuel efficiency" of the thruster. Physically: how many seconds a thruster can produce 1 N of thrust from 1 kg of propellant under standard gravity. Higher I_sp = less propellant needed.*
>
> *- g_0 = 9.80665 m/s^2 -- standard gravitational acceleration (conversion factor)*
>
> *- m_0 = initial (wet) mass (kg) = m_f + m_propellant*
>
> *- m_f = final (dry) mass (kg) = spacecraft mass after all propellant is consumed*
>
> *The tyranny of the rocket equation: The propellant mass grows exponentially with D V / v_e. For D V = v_e (one exhaust velocity worth of D V), 63% of the initial mass must be propellant. For D V = 2 v_e, 86% must be propellant. This is why high-I_sp systems are so valuable for large D V missions -- they move the v_e in the denominator, dramatically reducing the mass ratio.*

CubeSat Propulsion Technologies -- Physics and Comparison

Cold Gas Propulsion

Physics: Compressed gas (N_2, xenon, R-236fa refrigerant, or other) is stored in a tank at 1--30 MPa. When a valve opens, gas expands through a nozzle, converting thermal/pressure energy to kinetic energy.

No combustion, no heating, no chemical reaction.

Thrust: $F = \dot{m} v_e + (p_e - p_a) A_e$. For a small converging nozzle, $v_e \approx \sqrt{2 c_p T_0}$ where T_0 is the tank temperature. For N_2 at 300 K: $v_e \approx$ 500--750 m/s, giving $I_{sp} \approx$ 50--75 s.

Advantages: Simplest system (no ignition, no power except valve solenoid), fast response (ms-level valve opening), high reliability, no plume contamination concerns.

Disadvantages: Low I_{sp} (large propellant mass for given D V), bulky high-pressure tank, limited total impulse.

Products: VACCO MiPS (R-236fa, I_{sp} = 40 s, 4 thrusters, 0.3 kg dry), VACCO ArgoMoon (Xe cold gas), Bradford ECAPS cold gas.

Missions using cold gas: MarCO (6U, JPL -- used R-236fa cold gas for trajectory correction and attitude control en route to Mars), many ISS-deployed CubeSats for collision avoidance.

Resistojet / Warm Gas

Physics: Similar to cold gas, but the propellant is electrically heated before expansion through the nozzle. Heating increases the gas temperature T_0, which increases the exhaust velocity ($v_e \propto \sqrt{T_0}$). Common propellants: butane (C_4H_{10}), water (H_2O), ammonia (NH_3).

Butane propulsion: Butane is stored as a liquid at its saturation pressure (~2 atm at 20 degC). When heated to 200--400 degC and expanded through a nozzle, it achieves $I_{sp} \approx$ 80--100 s. The liquid storage is much denser than compressed gas, allowing more propellant in a smaller tank.

Advantages: Higher I_{sp} than cold gas (~2x), dense liquid storage, moderate complexity.

Disadvantages: Requires electrical power for heating (5--15 W during firing), lower thrust than cold gas, potential for nozzle clogging if propellant decomposes.

Products: Busek BGT-X5 (butane, I_{sp} = 80 s), NanoAvionics EPSS (butane, I_{sp} = 85 s), Pale Blue water resistojet (I_{sp} = 70--80 s).

Green Monopropellant

Physics: A liquid propellant is injected into a catalyst bed where it decomposes exothermically, producing hot gases that expand through a nozzle. "Green" propellants are alternatives to hydrazine (N_2H_4) that are less toxic and easier to handle.

Propellant	Chemical	I_{sp} (s)	Density (kg/m^3)	Toxicity	TRL	Heritage
Hydrazine (N_2H_4)	Monopropellant, Shell 405 catalyst	220--230	1010	Extremely toxic (carcinogen)	9	50+ years, thousands of missions
AF-M315E (ASCENT)	HAN-based ionic liquid	235--250	1460	Low toxicity	8	GPIM demo (2019, NASA)
LMP-103S	ADN-based ionic liquid	225--235	1240	Low toxicity	8	PRISMA demo (2010, SSC), SkySat
HTP (H_2O_2 90%)	Hydrogen peroxide + silver catalyst	150--165	1400	Moderate (oxidiser)	7	Various small satellites

[Source: Masse et al., "GPIM AF-M315E Propulsion System," AIAA 2019; Anflo et al., "Flight Demonstration of LMP-103S," AIAA 2011]

Advantages: High thrust (0.1--1 N for CubeSat systems), good I_sp (220+ s), proven technology (heritage from hydrazine systems).

Disadvantages: Requires catalyst preheating (2--10 W, 10--30 min warmup), higher system mass (tank + catalyst bed + valves + feed system), propellant handling safety requirements (even "green" propellants require PPE), higher cost ($100K+ for flight units).

Products: Aerojet MPS-130 (AF-M315E, 1 N thrust, I_sp = 235 s, 3 kg system mass), Bradford HPGP (LMP-103S, 1 N, I_sp = 230 s).

Electric Propulsion -- Electrospray (FEEP)

Physics: Field Emission Electric Propulsion (FEEP) uses a strong electric field (~10^9 V/m) at the tip of a needle or along the edge of a slit to ionise and extract metal atoms (typically indium or gallium) or ionic liquid droplets. The ions are accelerated by an electric field to high velocities (10--50 km/s), producing very high I_sp.

How it works (indium FEEP):

1. Solid indium is heated to just above its melting point (157 degC) to form a liquid reservoir
2. Capillary action draws liquid indium to an array of sharp emitter tips
3. A high voltage (1--10 kV) between the emitter tips and an extractor grid creates an intense electric field at the tip apex
4. The field ionises individual indium atoms via field evaporation
5. The ions are accelerated through the extractor grid, creating a beam of In^+ ions at 20--40 km/s
6. A neutraliser (typically a carbon nanotube or thermionic emitter) emits electrons to neutralise the beam and prevent spacecraft charging

Performance: I_sp = 500--5000 s (adjustable by varying acceleration voltage), thrust = 0.01--1 mN, power = 20--60 W.

Advantages: Extremely high I_sp (minimal propellant consumption), no pressurised tanks, compact solid propellant storage, precise thrust control (useful for formation flying).

Disadvantages: Very low thrust (months-long burn times for significant D V), requires significant power (20--60 W for ~0.5 mN thrust), plume contamination from metal ions (indium deposition on surfaces), limited heritage.

Products: Enpulsion NANO R3 (indium FEEP, 0.35 mN, I_sp = 1000--5000 s, 0.9 kg dry mass, < 40 W), Accion TILE (ionic liquid electrospray, 0.1 mN, I_sp = 1500 s).

Missions using FEEP/electrospray: LISA Pathfinder (ESA, precision formation flying, used colloid thrusters), SSTL NovaSAR-1 (Enpulsion NANO for orbit maintenance).

Electric Propulsion -- Hall Effect Thruster

Physics: A Hall effect thruster uses crossed electric and magnetic fields to ionise a neutral propellant gas (xenon, krypton, or iodine) and accelerate the resulting ions to high velocity.

How it works:

1. Neutral propellant gas (Xe or I_2) is injected into an annular discharge channel
2. A radial magnetic field (from permanent magnets or electromagnets) traps electrons in a Hall current loop, preventing them from reaching the anode
3. The trapped electrons collide with neutral gas atoms, ionising them
4. The ions, being much heavier, are not significantly deflected by the magnetic field and are accelerated axially by the electric field (100--500 V) between anode and cathode
5. An external cathode (hollow cathode or RF cathode) provides electrons for beam neutralisation and to sustain the discharge

Performance: I_sp = 800--3000 s, thrust = 1--50 mN for CubeSat-scale systems, power = 50--300 W.

Advantages: Higher thrust than FEEP (N range for larger systems), excellent I_sp, well-proven technology (GEO station-keeping heritage: Aerojet PPS-1350, Busek BHT-200).

Disadvantages: Requires significant power (> 50 W for CubeSat-scale), heavy cathode assembly, xenon storage requires high-pressure tanks (100--300 bar), channel erosion limits lifetime.

Products: Exotrail ExoMG-nano (40 mN, I_sp = 800 s, 1.5 kg, 60 W), Busek BHT-200 (13 mN, I_sp = 1370 s, 1.0 kg, 200 W), Enpulsion MICRO (Hall thruster, iodine propellant, 1 mN, I_sp = 1000 s).

Iodine propulsion: Iodine (I_2) is emerging as an alternative to xenon for Hall thrusters and gridded ion engines. Iodine is solid at room temperature (stored without a pressure vessel), has a density of 4940 kg/m^3 (vs xenon at ~1600 kg/m^3 at 100 bar), and has similar atomic mass (127 vs 131). The Busek BIT-3 (iodine Hall thruster) and ThrustMe NPT30-I2 have demonstrated iodine propulsion in orbit.

No Propulsion -- Passive Deorbit Strategies

For missions below ~500 km where propulsion is not needed for operations, passive deorbit can satisfy the FCC 5-year or IADC 25-year guidelines:

Method	Mechanism	Mass	Volum	Effectiveness	TRL
Atmospheric drag (natural)	Below 500 km, atmospheric drag naturally decays the orbit	0	0	Depends on ballistic coefficient and solar cycle	9
Drag sail	Deployable membrane increases cross-section area by 10--100x	0.1--0.5 kg	0.25--1 U	Very effective above 600 km	7--8
Drag chute (tether)	Electrodynamic tether interacts with geomagnetic field to decelerate	0.2--0.5 kg	0.5U	Moderate effectiveness; depends on orbital inclination	6--7

[Source: Cranfield Icarus drag sail, 0.1 kg; NanoSail-D2, NASA; InflateSail, SSC]

Propulsion System Comparison Table

Parameter	Cold Gas (N_2)	Warm Gas (Butane)	Green Monoprop (AF-M315E)	Electrospray (FEEP)	Hall Effect (Xe)
I_sp (s)	40--75	80--100	230--250	500--5000	800--3000
Thrust	10--100 mN	5--50 mN	0.1--1 N	0.01--1 mN	1--50 mN
Propellant mass (100 m/s, 5 kg S/C)	0.87 kg	0.44 kg	0.18 kg	0.042 kg	0.055 kg
System dry mass	0.3 kg	0.5 kg	3.0 kg	0.9 kg	1.5 kg

Total system mass

1.17 kg

0.94 kg

3.18 kg

0.94 kg

1.55 kg

Burn time (100 m/s)	Minutes	Minutes	Seconds	Months	Days--weeks
Power during firing	< 1 W	5--15 W	2--10 W (preheat)	20--60 W	50--300 W
Complexity	Low	Low-medium	High	Medium	High
Cost	~15 kEUR	~30 kEUR	~120 kEUR	~50 kEUR	~80 kEUR
TRL	9	7--8	7--8	7--8	6--8 (CubeSat)

> *Worked Example -- Propellant Mass Comparison for 100 m/s Deorbit*
>
> *Scenario: 3U CubeSat, m_dry = 5.0 kg, deorbit from 600 km (D V = 113 m/s).*
>
> *Cold gas (I_sp = 60 s, v_e = 589 m/s):*
> *m_prop = 5.0 x (e^113/589 - 1) = 5.0 x (e^0.192 - 1) = 5.0 x 0.212 = 1.06 kg*
>
> *Total system: 1.06 + 0.3 = 1.36 kg = 23% of 6 kg CubeSat mass limit*
>
> *Green monopropellant (I_sp = 235 s, v_e = 2305 m/s):*
> *m_prop = 5.0 x (e^113/2305 - 1) = 5.0 x (e^0.0490 - 1) = 5.0 x 0.0502 = 0.251 kg*
>
> *Total system: 0.251 + 3.0 = 3.25 kg = 54% of mass limit (system is heavy even though propellant is light)*
>
> *Electrospray (FEEP) (I_sp = 1200 s, v_e = 11,772 m/s):*
> *m_prop = 5.0 x (e^113/11772 - 1) = 5.0 x (e^0.00960 - 1) = 5.0 x 0.00965 = 0.048 kg*
>
> *Total system: 0.048 + 0.9 = 0.95 kg = 16% of mass limit (lightest total, but takes months to execute)*
>
> *Trade-off summary: The electrospray system is the lightest overall because the high I_sp minimises propellant mass, and the dry mass is moderate. Cold gas uses the most propellant but has the lowest dry mass. Green monopropellant is dominated by its heavy feed system. The optimal choice depends on the mission timeline: if deorbit must happen quickly (days), cold gas or monoprop; if months are acceptable, electric propulsion wins on mass.*

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3. On-Board Data Handling (20 min)

Teaching Notes

OBC Architecture -- Processor Selection

The OBC is the spacecraft's brain, managing all data handling, commanding, telemetry generation, and FDIR (Fault Detection, Isolation, and Recovery). Processor selection involves a fundamental trade between radiation tolerance, processing power, power consumption, and cost.

Flight-Heritage Processors

Processor	Architecture	Clock (MHz)	RAM	Rad Tolerance	Power (W)	TRL (Space	Cost	Typical Use
TI MSP430	16-bit RISC	16--25	2--10 kB	Moderate (COTS, tested to	0.005--0.01	9	< 10 EUR	Ultra-low-power housekeeping, safe mode OBC
ARM Cortex-M4 (STM32F4)	32-bit ARM	168	192 kB + ext	Low-moderate (COTS, 10--30	0.05--0.20	8--9	10--20 EUR	Standard CubeSat OBC, TM/TC handling, ADCS control loop
ARM Cortex-M7 (STM32H7)	32-bit ARM	400--480	1 MB + ext	Low-moderate (COTS, 10--20	0.1--0.5	7--8	15--30 EUR	Higher-performance CubeSat OBC, onboard image processing
ARM Cortex-A (Linux-capable, e.g., NXP i.MX6)	32/64-bit ARM	500--1200	256 MB--1 GB	Low (COTS, < 10 krad)	1--3	6--7	20--50 EUR	Payload processing, AI/ML inference, Linux OS
Xilinx Zynq (SoC: ARM + FPGA)	ARM Cortex-A 9 +	667 + programmabl	512 MB + FPGA	Moderate (Zynq-7000) to High (Kintex	2--5	7--8	100--500 EUR	High-throughput data processing, SDR (software-defined radio), image
LEON3/4 (rad-hard	32-bit SPARC	50--250	External	High (100--300 krad, SEL	1--3	9	10K--50K	ESA heritage missions, GEO, deep space
RAD750 (BAE Systems)	32-bit PowerPC	200	128 MB	Very high (1 Mrad, SEL immune)	5--10	9	200K+ EUR	NASA flagship missions (MRO, Curiosity, JWST)

[Source: ST Microelectronics STM32 datasheets; Xilinx Zynq-7000 datasheet; Cobham Gaisler LEON3 datasheet]

RTOS vs Bare-Metal vs Linux

Approach	OS	Pros	Cons	When to Use
Bare-metal	None (custom event	Minimum overhead, deterministic timing, smallest code size	Hard to maintain, no task isolation, no file system	Ultra-simple 1U missions (MSP430)
RTOS (FreeRTOS, ChibiOS,	Real-time OS	Deterministic scheduling, task isolation, mature ecosystem, small footprint (10--50 kB)	More complex than bare-metal; requires task priority design	Standard for CubeSat C&DH: ADCS loop, TM/TC, mode
Linux (Yocto, Buildroot)	Full OS	Rich ecosystem (Python, networking, file system, device drivers), easy development	Non-deterministic (not suitable for hard real-time), large footprint (50+ MB), power-hungry processor	Payload data processing, AI/ML, onboard image analysis

Radiation effects on processors:

The space radiation environment causes two categories of effects:

1. Total Ionising Dose (TID): Accumulated radiation damage from trapped protons/electrons and solar particles. Measured in rad(Si) or gray. Causes threshold voltage shifts in CMOS transistors, increasing leakage current and eventually causing functional failure.

- LEO (500 km, 51.6 deg): ~1--5 krad/year behind 2 mm Al shielding
- LEO polar/SSO (800 km): ~5--10 krad/year
- MEO (through proton belt): ~50--100 krad/year
- GEO: ~10--30 krad/year

2. Single Event Effects (SEE): A single energetic particle (proton or heavy ion) deposits enough charge in a transistor to flip a bit (SEU -- Single Event Upset), latch a transistor (SEL -- Single Event Latchup, potentially destructive), or burn out a power device (SEB -- Single Event Burnout).

- SEU rate in LEO: ~1--10 bit flips per day per GB of SRAM (highly variable with orbit and shielding)
- SEL mitigation: Current-limiting resistors on power lines, latchup detection circuits, watchdog resets
- SEU mitigation: Error Detection and Correction (EDAC) on memory (Hamming codes, TMR -- Triple Modular Redundancy)

CubeSat approach to radiation: Most CubeSats in LEO (< 600 km, < 3-year mission) use COTS processors with EDAC on memory and a watchdog timer. Total dose over a 3-year LEO mission is typically 3--15 krad, which most COTS ARM Cortex-M processors survive (tested and characterised, even if not guaranteed). For longer missions, higher orbits, or critical applications, rad-tolerant or rad-hard processors are needed.

Data Storage Sizing

> *Key Equations -- Data Storage*
>
> $S_{required} = V_{daily} \times N_{days} \times f_{safety}$
>
> *where V_daily = daily data generation, N_days = days between full downlinks (typically 1--3 for LEO), f_safety = 2 (to handle missed passes, ground station outages, and safe mode periods).*
>
> *Storage technologies:*
>
> | Technology | Capacity | Write Speed | Radiation Tolerance | Power | CubeSat Use |
> |---|---|---|---|---|---|
> | NOR flash | 4--256 MB | 1--5 MB/s | Moderate (10--50 krad) | Low | Code storage, boot ROM, critical parameters |
> | NAND flash (SLC) | 1--128 GB | 10--50 MB/s | Low-moderate (5--20 krad) | Low | Primary data storage |
> | NAND flash (MLC/TLC) | 32--512 GB | 50--200 MB/s | Low (< 10 krad) | Low | Maximum capacity (use EDAC and scrubbing) |
> | SD card (industrial) | 4--128 GB | 10--50 MB/s | Low (< 5 krad) | Very low | Budget missions (risk: wear levelling + radiation = data loss) |
> | MRAM | 1--64 MB | 10--50 MB/s | High (> 100 krad) | Very low | Critical parameters, non-volatile log |

> *Worked Example -- Storage for 3U EO CubeSat*
>
> *Given: Daily generation = 720 MB (from Session 3.3 data budget), daily downlink = 480 MB, days to clear backlog = 720/480 = 1.5 days.*

> *S_required = 720 x 3 x 2 = 4320 MB ~= 4.3 GB*
>
> *Specify: >= 8 GB NAND flash storage (next standard size). The GomSpace A3200 OBC includes 4 GB NAND flash; adding an 8 GB SD card or additional NAND chip provides adequate capacity.*
>
> *For a high-resolution imager generating 4.5 GB/day, storage requirement is: 4500 x 3 x 2 = 27 GB. Specify >= 32 GB flash storage.*

Flight Software Functions

Function	Description	Typical Execution	Criticality
Mode manage	Transition between Safe, Idle, Imaging, Downlink, Eclipse modes based on state machine	Event-driven	Critical (incorrect transition = mission loss)
ADCS control	Read sensors (star tracker, gyro, magnetometer), compute attitude estimate (Kalman filter), command actuators (PID controller)	1--10 Hz	Critical (loss of pointing = loss of mission)
TM/TC handling	Generate CCSDS telemetry packets, parse and execute telecommands	1 Hz (TM), event-driven (TC)	Critical (loss of communication = loss of mission)
Data handling	Payload data acquisition, compression (JPEG2000, CCSDS 122.0), buffering, downlink queue management	On-demand	Important
FDIR	Fault Detection, Isolation, and Recovery: watchdog timer, over-current protection, sensor consistency checks, autonomous safe mode trigger	1--10 Hz	Critical (must detect and recover from faults autonomously)
Housekeeping	Monitor temperatures, voltages, currents, wheel speeds; log to non-volatile memory	0.1--1 Hz	Important
Scheduling	Time-tagged command execution: autonomous imaging over target, downlink preparation before ground pass, desat scheduling	1 Hz	Important
Thermal control	Read temperature sensors, control heaters (on/off thermostat or PID)	0.1 Hz	Moderate (for heater-equipped missions)

---

4. Equipment Selection Exercise (45 min)

Instructions

This is the primary hands-on session for Day 4 of the design week. Teams select actual hardware components.

1. Open the Equipment Browser (button in header bar)
2. The sidebar shows categories annotated by need:
- Blue dot = Required for your mission
- Circle = Optional
- Dimmed = Not applicable

3. For each required category, select a component:
- Check the quantity needed (e.g., 4 reaction wheels, 3 magnetorquers)
- Note any RF compatibility warnings (transponder band must match antenna band)
- Watch the live budget bar showing running mass / power / cost totals
4. For each selection, verify:
- Does it fit within the subsystem mass allocation?
- Is power draw within the power budget for its operational mode?
- Is the interface compatible (PC/104? I^2C? SPI? CAN?)
- Is the component qualified for the launch vibration environment?
5. Review the Budget Breakdown on the Dashboard:
- Has per-subsystem mass changed from the parametric estimate?
- Is the overall mass margin still positive?
- Is the power budget still positive in all modes?

Component Trade Study

For at least one subsystem, select 2--3 candidate components and run a formal tabular trade:

1. Navigate to the Trade Studies tab
2. Load or create a "Component Selection Trade" study
3. Define criteria: mass, power, cost, TRL, heritage, performance
4. Score each candidate (1--5 scale)
5. Apply weights and compute weighted scores
6. Document the winner and rationale

Real Mission Example: Iridium NEXT Equipment Selection

Iridium NEXT (Thales Alenia Space, 2017--2019) serves as a large-scale example of rigorous equipment selection. For the phased-array antenna:

Criterion	Weight	Candidate A (Thales)	Candidate B (Raytheon)
Performance	0.30	4.5	4.0
Mass	0.20	3.5	4.0
Cost	0.20	3.0	4.5
TRL	0.15	5.0	4.0
Schedule	0.15	4.0	3.5
Weighted		3.95	4.00

The selection was ultimately Candidate A (Thales) due to contractual considerations beyond the numerical trade -- illustrating that trade studies inform but do not dictate decisions.

---

5. SpaceCDF Budget Closure Check (15 min)

Instructions

After equipment selection, perform a final budget health check:

1. Dashboard KPIs: Record all margins
- Mass margin (%) -- green/amber/red?
- Power margin per mode (W)
- Link margin (dB)
- Cost vs ceiling (MEUR)
- Pointing accuracy vs requirement (deg)
2. Budget Comparison: Compare parametric estimates to equipment-based totals
3. Identify any negative margins -- these must be resolved before proceeding to integration (Week 3)

If a Budget Does Not Close

Budget	Common Fix	Impact	Typical Mass/Power/Cost Trade
Mass (negative)	Remove propulsion; select lighter components; reduce redundancy; move to larger form factor	Risk / performanc	Removing propulsion saves 0.5--1.5 kg
Power (negative)	Add deployable SA; reduce payload duty cycle; select lower-power AOCS; schedule operations to avoid simultaneous loads	Cost / schedule	Deployable panel adds ~15 W but costs 0.3 kg + 25 kEUR
Link (negative)	Increase TX power; use higher-gain antenna; reduce data rate; upgrade coding; upgrade ground station	Mass / power	3 dB gain from coding is "free"; 3 dB from bigger antenna costs mass
Cost (over ceiling)	Use COTS instead of rad-hard; remove propulsion; reduce ground segment; use SatNOGS instead of commercial ground	Risk / capability	COTS vs rad-hard saves 10--100x on processor cost
Pointing (insufficient)	Upgrade star tracker; improve alignment calibration; add vibration isolation for RW; reduce thermal gradients	Cost / complexity trade	Alignment improvement is usually cheapest

---

1U Worked Example: UniSat-1

CDS Compliance for 1U Form Factor

The CubeSat Design Specification (CDS Rev 14) defines the 1U envelope:

Parameter	1U Specification	UniSat-1 Design	Compliance
Dimensions	100.0 x 100.0 x 113.5 mm	100.0 x 100.0 x 113.5 mm (ISIS 1U frame)	Pass
Maximum mass	2.0 kg (CDS Rev 14, ISIPOD)	1.0 kg target (50% margin to 2 kg limit)	Pass
Rail material	Hard anodised Al 7075-T6	Standard (part of ISIS frame, 7075-T6, Type III anodise)	Pass
Rail cross-section	8.5 x 8.5 mm minimum	Standard (ISIS: 8.5 x 8.5 mm)	Pass
Deployment switches	Min 1 per accessible face	2 switches (ISIS standard, on +X/-X rail faces)	Pass
RBF pin	Required	Included (ISIS standard, on -Z face)	Pass
CG offset	<= 2 cm from geometric centre	< 1 cm (symmetric PCB stack layout, battery centred)	Pass
Protrusions	None beyond rail envelope (stowed)	UHF monopole antenna stowed along rail (spring-loaded, within envelope)	Pass
Fundamental frequency	> 40 Hz first mode	~600 Hz (1U Al structure is extremely stiff)	Pass

Note on CDS mass limit: The CDS Rev 14 specifies 2.0 kg as the 1U deployer limit for the ISIPOD.

However, many deployer providers (e.g., NanoRacks, Exolaunch) specify 1.33 kg for 1U. Always check the specific deployer ICD. UniSat-1 targets 1.0 kg, well within either limit.

No propulsion: At 400 km altitude, atmospheric drag provides natural deorbit. The ballistic coefficient for a 1U is:

BC = (m)/(C_D x A) = (1.0)/(2.2 x 0.01) = 45.5 kg/m^2

This gives an orbital lifetime of approximately 8--14 months depending on solar activity (F10.7 index). At solar maximum (F10.7 > 200), the denser atmosphere deorbits the satellite in ~6 months. At solar minimum (F10.7 ~ 70), lifetime extends to ~18 months. Both are within the FCC 5-year rule and IADC 25-year guideline without any propulsion system.

OBC selection rationale: UniSat-1 uses a custom board based on the TI MSP430 (safe mode / housekeeping) + STM32F4 Cortex-M4 (main OBC). The MSP430 runs bare-metal firmware handling the watchdog, power monitoring, and safe-mode recovery. The STM32F4 runs FreeRTOS handling TM/TC, magnetometer data acquisition, and scheduling. Dual-processor architecture provides redundancy: if the STM32F4 fails (SEU, latchup), the MSP430 can maintain safe-mode operations and respond to ground commands.

Data storage: With 0.84 MB/day of magnetometer data and 4800 bps downlink, onboard storage is not a bottleneck. A 4 MB NOR flash is more than adequate (stores ~4 days of data as buffer). No NAND flash, no SD card needed.

Complete 1U Equipment List:

#	Category	Component	Mass (g)	Power (W)	Cost (kEUR)	Qty	Interface
1	Structure	ISIS 1U CubeSat structure (Al 7075-T6)	200	--	4.0	1	Mechanical
2	EPS	GomSpace NanoPower P31us (EPS board + 2S Li-ion battery, 10 Wh)	200	0.3 (quiescent)	12.0	1	I^2C
3	Solar cells	Body-mounted triple-junction GaAs cells (5 faces)	50	-- (generates power)	8.0	5	Direct to EPS
4	OBC	Custom MSP430 + STM32F4 board (with 4 MB NOR flash)	30	0.3	3.0	1	I^2C, SPI, UART
5	Comms	UHF transceiver (NanoCom AX100, 0.5 W TX, 9600 bps)	60	0.5 (TX) / 0.1 (RX)	8.0	1	SPI
6	Antenna	UHF monopole (deployable, lambda/4 = 17 cm nitinol)	20	--	2.0	1	RF coax
7	Payload	MEMS magnetometer (custom PCB, PNI RM3100 sensor)	50	0.2	5.0	1	SPI
8	AOCS (passive)	Permanent magnet (AlNiCo, 0.5 A m^2) + 2 hysteresis rods (HyMu-80)	30	0	1.0	1	None (passive)
9	Harness	Internal cables, connectors, PC/104 stack header	50	--	1.0	1	Various
	TOTAL		690	~1.3 (peak TX)	~44		

Mass budget:

Level	Mass (g)
CBE (Current Best Estimate)	690
+ 20% equipment margin	828
+ 20% system margin	994
MEV (Maximum Expected Value)	994
Deployer limit (ISIPOD)	2000
Margin to limit	1006 g (50%)

Power budget (worst case: TX mode in sunlight):

Load	Power (W)
OBC (STM32F4 + MSP430)	0.3
EPS quiescent	0.3
UHF TX	0.5
Magnetometer	0.2
Total peak	1.3
SA available (orbit avg, EOL)	2.3
Margin	1.0 W (43%)

Key insight: The entire UniSat-1 BOM is 5 COTS components plus 2 custom boards (OBC and magnetometer PCB). Total hardware cost is ~44 kEUR -- an order of magnitude less than a typical 3U mission. With labour, I&T, launch, and operations, the total mission cost is 80--150 kEUR. This demonstrates that a useful space mission can be built for less than the cost of a mid-range car.

Worked Example: Complete 3U EO CubeSat Equipment List

Category	Component	Mass (kg)	Power (W)	Cost (kEUR)	Qty	Interface
EPS Board	GomSpace P31u (MPPT, 3.3V/5V/batt rails)	0.10	0.5	8	1	I^2C
Battery	GomSpace BP4 (2S2P, 38 Wh, Li-ion 18650)	0.20	--	5	1	I^2C (telemetry)
Solar Panels	MMA HaWK deployable (TJ GaAs, ~12 W/panel BOL)	0.45	--	25	2	Direct to EPS
OBC	GomSpace A3200 (ARM Cortex-A, Linux, 4 GB NAND)	0.08	1.0	12	1	I^2C, SPI, UART

> *| Reaction Wheel | Blue Canyon RW210 (1 mN m torque, 10 mN m s momentum) | 0.055 | 0.6 | 8 | 4 | SPI |*
>
> *| Magnetorquer | CubeSpace CubeMAG (0.2 A m^2 dipole) | 0.03 | 0.1 | 3 | 3 | I^2C |*
>
> *| Star Tracker | Blue Canyon NST (10 arcsec accuracy, 2 Hz update) | 0.35 | 1.5 | 35 | 1 | SPI/UART |*
>
> *| Sun Sensor | NewSpace NFSS-411 (0.5 deg accuracy, fine analog) | 0.005 | 0.01 | 1 | 6 | Analog/I^2C |*
>
> *| Transponder | Endurosat S-band TX/RX (2 W, QPSK+LDPC, 1--5 Mbps) | 0.10 | 6.0 (TX) | 15 | 1 | SPI |*
>
> *| Antenna | Endurosat S-band patch (6 dBi, RHCP) | 0.02 | -- | 3 | 1 | RF coax |*
>
> *| Payload | Custom telescope (multispectral, 5 m GSD) | 1.50 | 5.0 | 150 | 1 | LVDS/SPI |*
>
> *| Structure | ISIS 3U frame (Al 7075-T6, hard anodised) | 0.30 | -- | 8 | 1 | Mechanical |*
>
> *| Harness | Custom cables, connectors, PC/104 stack | 0.15 | -- | 5 | 1 | Various |*
>
> *| TOTAL | | 3.57 | ~10 (imaging mode) | ~290 | | |*
>
> *Parametric estimate from Session 2.4: 3.68 kg CBE. Equipment total: 3.57 kg. Difference: -3% (within expected accuracy of parametric models).*
>
> *Mass budget:*
>
> *- CBE: 3.57 kg*
>
> *- + 20% equipment margin: 4.28 kg*
>
> *- + 20% system margin: 5.14 kg (MEV)*
>
> *- CDS limit: 6.0 kg*
>
> *- Margin to limit: 0.86 kg (14%) -- amber, acceptable for Phase A but tight for Phase B+.*

Session Summary

Topic	Key Takeaway
CDS compliance	Standard dimensions, Al 7075-T6 anodised rails (8.5 mm), deployment switches, RBF pin, CG limits
Structural materials	Al 7075-T6: sigma_y = 503 MPa, E = 72 GPa; anodisation for wear/insulation; CFRP not suitable for rails
Launch loads	6--12 g axial QS, 7 grms random vib (20--2000 Hz), 500--2000 g shock; PCB solder joints are the weak point
Structural MoS	MoS = sigma_allow/(sigma_design x FoS) - 1 >= 0; FoS 1.25 yield / 1.5 ultimate (metallic)
Frequency req	First mode > 40 Hz; CubeSat Al structures easily exceed this; local modes (PCBs, deployables) are the risk
Tsiolkovsky	m_prop = m_dry x (e^D V/(I_sp g_0) - 1); exponential growth with D V / v_e
Cold gas	I_sp 40--75 s; simple, fast, reliable; heavy propellant penalty for D V > 30 m/s
Green monoprop	I_sp 225--250 s; high thrust (0.1--1 N); heavy feed system; AF-M315E and LMP-103S flight-proven
Electrospray (FEEP)	I_sp 500--5000 s; minimal propellant; months-long burns; 20--60 W power; indium or ionic liquid
Hall thruster	I_sp 800--3000 s; moderate thrust (1--50 mN); 50--300 W; iodine emerging as Xe alternative
Propulsion trades	High-I_sp: less propellant, more dry mass, long burns; Low-I_sp: more propellant, lighter system, fast burns
When to skip propulsion	Below 500 km (natural deorbit); tech demo; differential drag constellation
OBC processors	MSP430 (0.01 W, safe mode), Cortex-M4 (0.2 W, standard CubeSat), Zynq (3 W, high-throughput), LEON3 (rad-hard, ESA)
RTOS vs Linux	FreeRTOS for real-time C&DH (standard); Linux for payload processing (data-intensive)

Radiation effects	TID: 1--10 krad/yr LEO; SEU: 1--10 bit flips/day/GB; mitigate with EDAC, watchdog, redundancy
Data storage	S >= 2 x daily generation; SLC NAND flash for primary storage; NOR flash for code/critical params
Equipment selection	Live budget tracking; RF compatibility check; interface compatibility; trade study for contested selections
Budget closure	All margins must be positive before proceeding to integration week; mass margin > 10% at Phase A

Session 4.1: Equipment Selection & Bill of Materials

Duration: 2 hours

Prerequisites: Day 3 complete (subsystems sized, components identified via parametric agents)

References: ECSS-Q-ST-20C (Quality Assurance), ECSS-E-ST-10-24C (Interfaces), ITAR/EAR (22 CFR 120-130 / 15 CFR 730-774), CDS Rev 14.1, NASA SEH Rev 2 section 6.8

Learning Objectives

By the end of this session, participants will be able to:

1. Apply a structured make/buy/reuse decision framework to component selection
2. Evaluate COTS components against requirements using TRL and heritage criteria
3. Construct a Bill of Materials (BOM) with full traceability to requirements
4. Identify export control constraints (ITAR, EAR, Canadian Controlled Goods) for selected equipment
5. Verify interface compatibility (RF, electrical, mechanical) during component selection
6. Track cumulative mass, power, and cost budgets as selections are made

1. The Make/Buy/Reuse Decision (20 min)

Teaching Notes

Before selecting any hardware, the team must decide the procurement strategy for each subsystem. This is a fundamental systems engineering decision that affects cost, schedule, risk, and performance.

[Source: NASA SEH Rev 2 section 6.8 "Decision Analysis"; ECSS-M-ST-10C Rev.1 section 5.4]

Decision Framework

```
For each subsystem or component:
  1. Does a COTS product exist that meets requirements?
     -> Yes: BUY (lowest risk, fastest schedule)
     -> No: Continue
  2. Does flight-proven hardware from a previous mission exist?
     -> Yes: REUSE (low risk, may need delta-qualification)
     -> No: Continue
  3. Can the requirement be met by modifying existing hardware?
     -> Yes: MODIFY (moderate risk, moderate schedule)
     -> No: MAKE (highest risk, longest schedule, most expensive)
```

Make/Buy/Reuse Trade Matrix

Factor	Buy (COTS)	Reuse (Heritage)	Modify	Make (Custom)
Cost (NRE)	None	None-Low	Moderate	High
Cost (Recurring)	Vendor price	Reproduction cost	Vendor + delta	Full development
Schedule	4-16 weeks lead	8-24 weeks	12-36 weeks	12-48 months
Risk	Low (if TRL >= 7)	Low (flight-proven)	Medium	High
Performance	Fixed by vendor	Fixed by heritage	Tuneable	Fully customisable
IP ownership	Vendor retains	May be shared	Negotiated	Full ownership
Qualification	Vendor-provided data	Delta-qual only	Partial re-qual	Full qualification

Key Equations

> *Non-Recurring Engineering (NRE) Cost Estimate:*
>
> *NRE = Labour_hours x Hourly_rate + Material_cost + Facility_cost + Testing_cost*
>
> *For COTS: NRE is near zero (vendor absorbs development cost).*
>
> *For custom: NRE can be 3-10x the unit recurring cost.*

Worked Example

Problem: A 6U CubeSat mission needs a fine sun sensor with 0.1 degree accuracy. Three options:

Option	Type	Accurac	Mass	Cost	TRL	Lead Time
Bradford SSOC-D60	COTS	0.1 deg	35 g	EUR 15K	9	8 weeks
In-house photodiode array	Custom	0.05 deg	20 g	EUR 8K + 400 hr NRE	4	12 months
Modified heritage sensor	Reuse	0.08 deg	40 g	EUR 6K + 80 hr delta-qual	7	16 weeks

Decision: The COTS option (Bradford SSOC-D60) meets the requirement, has TRL 9, lowest schedule risk, and total cost of EUR 15K vs EUR 8K + ~EUR 40K NRE for custom. Buy.

2. Technology Readiness Level (TRL) Assessment (20 min)

Teaching Notes

TRL is the standard metric for technology maturity. It was developed by NASA in the 1970s and is now used universally in space programmes.

[Source: NASA NPR 7123.1D Appendix E; ECSS-E-HB-11A "Technology Readiness Level (TRL) Guidelines"]
[URL: https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels/]

TRL Scale with Decision Criteria

TRL	Definition	Evidence Required	CubeSat Decision
1	Basic principles observed	Published research	Do not select
2	Technology concept formulated	Analytical studies	Do not select
3	Experimental proof of concept	Lab measurements	Do not select
4	Component validated in lab	Breadboard tested	High risk -- avoid unless no alternative
5	Component validated in relevant environment	Engineering model tested	Acceptable for technology demonstrator missions
6	System/subsystem model demonstrated in relevant environment	Prototype in relevant environment	Acceptable with risk mitigation
7	System prototype demonstrated in operational environment	Prototype tested in space	Preferred minimum for CubeSats
8	Actual system completed and qualified	Flight-qualified hardware	Strong preference
9	Actual system flight-proven	Successful on-orbit operation	Lowest risk

TRL and Risk Relationship

> *Risk Reduction Factor (empirical):*
>
> *Risk_factor = 10^(-(TRL - 1) / 3)*
>
> *| TRL | Risk Factor | Interpretation |*
> *|-----|------------|----------------|*
> *| 3 | 0.22 | High probability of failure/redesign |*
> *| 5 | 0.046 | Moderate risk |*
> *| 7 | 0.010 | Low risk |*
> *| 9 | 0.002 | Very low risk |*

> *This empirical relationship (from SMAD4 Table 20-12) illustrates why TRL >= 6 is the typical threshold for mission-critical components.*

Real Mission Example: MarCO (Mars Cube One)

NASA's MarCO A and B (launched May 2018) were the first interplanetary CubeSats. They used:

- Iris transponder (JPL, TRL 6 at selection -- first deep-space CubeSat radio)
- COTS reaction wheels (Blue Canyon Technologies, TRL 9)
- Custom deployable reflectarray antenna (JPL, TRL 5 at selection)

The custom antenna was the highest-risk item. It required extensive development testing and was the last component to reach TRL 6 for CDR. This delayed the schedule by 4 months but ultimately succeeded.

Lesson: If you must fly a low-TRL component, it will dominate your schedule and risk register.

[Source: "Mars Cube One (MarCO) -- Lessons Learned", A. Klesh et al., 33rd Annual Small Satellite Conference, 2019]

---

3. Bill of Materials (BOM) Construction (20 min)

Teaching Notes

The BOM is the definitive list of all hardware, software, and consumables required to build the satellite. It is the bridge between design and procurement.

BOM Structure

A properly structured BOM follows the WBS hierarchy:

```
BOM Level 0: Spacecraft (S/C-001)
  BOM Level 1: Subsystem (e.g., EPS-000)
    BOM Level 2: Assembly (e.g., EPS-SA-000 Solar Array Assembly)
      BOM Level 3: Component (e.g., EPS-SA-001 Solar Cell String)
        BOM Level 4: Part (e.g., EPS-SA-001-01 Azur 3G30C cell)
```

BOM Fields

Field	Description	Example
Item ID	Unique hierarchical identifier	EPS-BAT-001
Description	Component name and model	GomSpace NanoPower P31u
Manufacturer	Vendor name	GomSpace A/S
Part Number	Manufacturer part number	P31U-9-30
Quantity	Number required	1

Unit Mass (g)	Per-item mass	94
Unit Power (W)	Peak / average power draw	0.5 / 0.2
Unit Cost (EUR)	Procurement cost	8,500
TRL	Technology readiness level	9
Heritage	Previous mission(s) flown	GOMX-3, GOMX-4
Lead Time	Procurement lead time	12 weeks
ECCN/USML	Export classification	EAR99
Status	Selected / Ordered / Received / Tested	Selected

SpaceCDF BOM Generation

In SpaceCDF, the BOM is built automatically from the Equipment Browser selections:

1. Each component selected in the Equipment Browser creates a BOM entry
2. Quantities are set using the quantity selector (e.g., x4 for reaction wheels)
3. The Exports tab generates a formatted BOM spreadsheet
4. The BOM includes both parametric estimates and actual COTS data for comparison

Key Equations

> *BOM Mass Total:*
>
> *M_BOM = Sum_i (m_i x q_i) + M_harness + M_fasteners + M_margin*
>
> *Where:*
>
> *- m_i = unit mass of component i*
>
> *- q_i = quantity of component i*
>
> *- M_harness = harness mass (typically 5-8% of dry mass for CubeSats)*
>
> *- M_fasteners = mechanical fasteners, standoffs, thermal hardware (~3-5%)*
>
> *- M_margin = system margin (typically 20% at Phase A)*

Worked Example

3U CubeSat BOM Summary:

Subsystem	Components	Total Mass (g)	Total Power (W)	Total Cost (kEUR)
Structure	Rails, panels, fasteners	350	0	8.0
EPS	SA + battery + board	420	0.5	18.0
OBC	Flight computer + storage	80	1.2	10.0
AOCS	Star tracker + magnetorquers + RWs	550	4.5	45.0
TTC	S-band transceiver + patch antenna	180	8.0 (TX)	22.0
Thermal	MLI + heaters	60	2.0 (peak)	3.0
Payload	Multispectral imager	800	12.0	65.0
Harness	Cables, connectors	180	0	2.5
Total		2620	28.2 (peak)	173.5
Allocation (6U)		12000	40.0 (SA EOL)	250.0
Margin		9380 (78%)	11.8 (30%)	76.5 (31%)

1U Worked Example: UniSat-1

Complete Bill of Materials

UniSat-1's BOM is remarkably short -- only 5--7 line items plus harness. This simplicity is a major advantage for university teams with limited procurement experience.

> *UniSat-1 BOM (Phase B -- vendor quotes obtained):*
>
> | Item ID | Component | Manufacturer | Part Number | Qty | Unit Mass (g) | Unit Cost (kEUR) | TRL | ECCN | Lead (wks) |
> |---|---|---|---|---|---|---|---|---|---|
> | STR-001 | 1U CubeSat Structure | ISIS | ISIS-1U-STR | 1 | 200 | 4.0 | 9 | EAR99 | 8 |
> | EPS-001 | NanoPower P31us (EPS + 10Wh battery) | GomSpace | P31US-10 | 1 | 200 | 12.0 | 9 | EAR99 | 12 |
> | EPS-SA-001 | Body-mounted GaAs solar cells | AzurSpace | 3G30C | 5 | 10 | 1.5 | 9 | EAR99 | 10 |
> | OBC-001 | Custom flight computer (Cortex-M) | In-house | UNISAT-OBC-01 | 1 | 30 | 3.0 | 5 | EAR99 | -- |
> | COM-001 | UHF Transceiver | GomSpace | NanoCom AX100 | 1 | 55 | 8.0 | 8 | EAR99 | 12 |
> | COM-ANT-001 | UHF Deployable Antenna | Endurosat | UHF-ANT-S | 1 | 25 | 2.5 | 8 | EAR99 | 8 |
> | PL-001 | MEMS Magnetometer Board | In-house | UNISAT-MAG-01 | 1 | 50 | 5.0 | 4 | EAR99 | -- |
> | AOCS-001 | Passive magnetic kit (magnet + rods) | NewSpace | PMAG-1U | 1 | 30 | 1.0 | 9 | EAR99 | 6 |
> | HAR-001 | Internal harness | Custom | -- | 1 | 50 | 1.0 | N/A | EAR99 | -- |
> | | TOTALS | | | | 690 g | ~44 kEUR | | | |

Cost summary (total mission, hardware + services):

WBS Element	Cost (kEUR)	Notes
Hardware (BOM)	44	All COTS except OBC and payload
OBC software	5	Student labour (costed at stipend rate)
Payload calibration	3	University magnetometer lab
I&T	8	Assembly + vibration test (university facility)
Ground station	5	SatNOGS (free) + dedicated Yagi antenna purchase
Launch (ISS deploy)	15	NanoRacks 1U deployment fee
PM/SE/QA	5	Faculty supervision
TOTAL	~85 kEUR	

Comparison to 3U EO mission:

Metric	UniSat-1 (1U)	3U EO CubeSat
BOM line items	9	~15--20
Hardware cost	~44 kEUR	~290 kEUR
Total mission cost	~85 kEUR	~490 kEUR
Development time	6--12 months	18--24 months
Team size	3--5 people	8--15 people

Export control: All UniSat-1 components are classified EAR99 (no licence required). There are no

ITAR-controlled items because the mission uses no star trackers, no radiation-hardened processors, and no propulsion with ITAR-restricted technology. This is a significant advantage for international university collaborations.

Make/Buy/Reuse decisions:

Componen	Decision	Rationale
Structure	Buy (COTS)	ISIS 1U frame is flight-proven, TRL 9, low cost
EPS	Buy (COTS)	GomSpace P31us is the de facto standard, TRL 9
OBC	Make	Minimal board using university lab; lower cost than COTS OBC for this simple application
Comms	Buy (COTS)	GomSpace AX100, TRL 8, well-documented
Payload	Make	Novel MEMS sensor -- this IS the technology demonstration
Passive AOCS	Buy (COTS)	Standard magnetic stabilisation kit

4. Export Control and Procurement (25 min)

Teaching Notes

Export control is one of the most commonly overlooked aspects of CubeSat missions, especially for international teams. Violations carry severe penalties (criminal prosecution, programme cancellation).

[Source: ITAR -- 22 CFR Parts 120-130; EAR -- 15 CFR Parts 730-774; Canadian Controlled Goods Program -- Defence Production Act]

[URL: https://www.pmddtc.state.gov/ddtc_public (US ITAR); https://www.bis.doc.gov (US EAR)]

Export Control Regimes

Regime	Governing Law	Applies To	Key Concern for CubeSats
ITAR	US Arms Export Control Act	Defence articles (USML Categories IV, XI, XV)	Star trackers, rad-hard processors, some GPS receivers
EAR	Export Administration Act	Dual-use items (CCL)	Most COTS space components (ECCN 9A515)
Canadian CGP	Defence Production Act	Controlled goods in Canada	Handling US-origin ITAR items in Canadian facilities
Wassenaar	Wassenaar Arrangement	Multilateral export controls	Encryption, high-accuracy GNSS, imaging sensors

Component Classification Decision

```
Is the component on the US Munitions List (USML)?
  -> Yes: ITAR-controlled. Need DSP-5 license for export.
     Categories: IV (launch vehicles), XI (military electronics),
                 XV (spacecraft systems)
  -> No: Check Commerce Control List (CCL)
     Is it classified under ECCN 9A515 (spacecraft)?
       -> Yes: EAR-controlled. May need BIS license.
```

```
         -> No: Likely EAR99 (no license required for most destinations)
```

Common ITAR-Controlled CubeSat Components

Component Type	Why Controlled	Alternative
Radiation-hardened processors	Military-grade rad tolerance	COTS processors with software mitigation
High-accuracy star trackers (< 1 arcsec)	Missile guidance applicability	Lower-accuracy models (> 5 arcsec)
Certain GPS/GNSS receivers	Above COCOM limits (>60,000 ft, >1000 kt)	Space-rated receivers with COCOM compliance
Propulsion systems (some)	Missile technology	Cold-gas or water-based systems
Encryption modules	Signals intelligence	Open-source encryption (may still need EAR review)

Procurement Workflow

```
  1. Requirements -> Derive component specification
  2. Market survey -> Identify candidate COTS products
  3. Export classification -> Request ECCN from vendor
  4. Trade study -> Score and rank candidates
  5. Request for Quote (RFQ) -> Obtain pricing and lead times
  6. Purchase Order (PO) -> Commit to procurement
  7. Incoming inspection -> Verify against PO and datasheet
  8. Integration -> Install and functionally test
```

Real Mission Example: Export Control Impact

The BRITE-Constellation (Austria/Canada/Poland) was a series of nanosatellites for stellar photometry. The Canadian BRITE satellites (UniBRITE and BRITE-Austria, built by UTIAS/SFL) required careful export control management:

- US-origin ITAR components required Technical Assistance Agreements (TAAs)
- Controlled Goods registration required for the Canadian team
- Export permits needed for shipping between Canada, Austria, and the US launch site
- Total regulatory compliance effort: ~6 person-months and 12+ months lead time

Lesson: Start export classification immediately when components are identified. A single ITAR component can add 6-12 months to the schedule.

[Source: Sarda, K. et al., "BRITE-Constellation Mission and Spacecraft", AIAA/USU Small Satellite Conference, 2014]

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5. Interface Compatibility Verification (20 min)

Teaching Notes

As components are selected, every interface must be verified for compatibility. The three interface categories are RF, electrical, and mechanical.

RF Chain Compatibility

The RF chain (transponder, cable, antenna) must be frequency-matched. This is the most common equipment incompatibility for CubeSat newcomers.

Rule	Correct Example	Incorrect Example
Band match	S-band transponder + S-band patch antenna	S-band transponder + X-band horn
Impedance	50 ohm transponder + 50 ohm cable + 50 ohm antenna	Mixed impedances cause reflections
Connector	SMA on transponder + SMA-SMA cable + SMA on antenna	SMA to N-type needs adapter (loss)
Polarisation	RHCP antenna + RHCP ground station	RHCP to LHCP causes > 20 dB cross-pol loss

> *Impedance Mismatch Loss:*
>
> *Return_Loss (dB) = -20 * log10(|Gamma|)*
>
> *Where Gamma = (Z_load - Z_source) / (Z_load + Z_source)*
>
> *For a 50 ohm source into a 75 ohm load:*
> *Gamma = (75 - 50) / (75 + 50) = 0.2*
> *Return_Loss = -20 * log10(0.2) = 14 dB*
> *Mismatch_Loss = -10 log10(1 - |Gamma|^2) = -10 log10(1 - 0.04) = 0.18 dB*

SpaceCDF checks RF compatibility automatically: if you select a transponder in one band and an antenna in another, a warning dialog appears.

Electrical Interface Verification

Parameter	Typical CubeSat	What to Verify
Bus voltage	3.3V, 5V, or unregulated (6-8.4V Li-ion)	Component input voltage range covers bus voltage
Peak current	Per switched line limit (typically 1-3A)	Component inrush current does not exceed limit
Data protocol	I2C, SPI, UART, CAN, RS-422	All devices on same bus use compatible protocol
Connector	PC/104, Hirose, Harwin	Physical connector type matches or adapter planned

Mechanical Interface Verification

Parameter	CDS 3U Requirement	Verification
Dimensions	100.0 +/- 0.1 mm x 100.0 +/- 0.1 mm x 340.5 +/- 0.5 mm	Caliper measurement
Rail profile	8.5 x 8.5 mm +/- 0.1 mm	Profile gauge

Component stack height	<= 83 mm internal width per U	3D model check
CG location	Within 2 cm of geometric centre	Mass properties measurement

6. Equipment Selection Exercise (35 min)

Instructions

Part A: Equipment Selection (20 min)

1. Open the Equipment Browser in SpaceCDF
2. For each required category (blue dot), select at least one component:
- Start with EPS (batteries + solar panels + EPS board)
- Then OBC, AOCS sensors/actuators, TTC, structure
- Use quantity selectors (e.g., x4 for reaction wheels, x3 for magnetorquers)
3. Watch the live budget bar as you select -- keep mass under allocation
4. When selecting TTC: verify the spectrum band matches your earlier selection
5. After all selections, review the Budget Breakdown on the Dashboard

Part B: BOM Review and Export Check (15 min)

1. Navigate to the Exports tab
2. Generate the BOM -- review all entries
3. For each component, check: Is the ECCN listed? Is it EAR99, ECCN 9A515, or ITAR?
4. Flag any components that may require export licences
5. Compute: BOM total mass vs parametric estimate vs launcher allocation

Worksheet 4.1 Tasks

1. Complete the full BOM table (component, manufacturer, part number, mass, power, cost, TRL, ECCN)
2. Document the make/buy/reuse decision for each subsystem with rationale
3. Compute total BOM mass vs parametric estimate vs allocation -- state margin
4. Identify any export-controlled components and note the required licensing action
5. Describe one interface incompatibility found during selection and how it was resolved

Cited References

#	Reference	URL
1	NASA Technology Readiness Levels	https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels/

2	ECSS-Q-ST-20C Quality Assurance	https://ecss.nl/standard/ecss-q-st-20c-rev-2-quality-assurance-1-march-2023/
3	US DDTC (ITAR)	https://www.pmddtc.state.gov/
4	US BIS (EAR)	https://www.bis.doc.gov/
5	CubeSat Design Specification Rev 14.1	https://www.cubesat.org/s/CDS-REV14_1-2022-02-09.pdf
6	SMAD4 Chapter 20 (Cost Modelling)	Wertz, Everett, Puschell (eds.), Space Mission Engineering, Microcosm 2011
7	MarCO Lessons Learned (Klesh et al.)	SSC19-WKII-07, 33rd Small Sat Conference, 2019

Session Summary

Topic	Key Takeaway
Make/Buy/Reuse	Buy COTS first; Reuse heritage second; Make custom only when necessary
TRL	Minimum TRL 6 for mission-critical components; TRL 7+ preferred for CubeSats
BOM	Hierarchical list: Subsystem -> Assembly -> Component -> Part, with full traceability
Export control	Classify every component (EAR99 / ECCN / ITAR); start early -- delays programme
RF compatibility	Band, impedance, connector, polarisation must all match across RF chain
Electrical	Bus voltage, data protocol, connector type verified per component
Budget tracking	Live totals during selection; stop if allocation exceeded

---

Session 4.2: Verification & Validation Matrix and Compliance

Duration: 2 hours

Prerequisites: Session 4.1 (equipment selected, BOM constructed)

References: ECSS-E-ST-10-02C Rev.1 (Verification), ECSS-E-ST-10-03C (Testing), NASA SEH Rev 2 sections 5.3-5.4, MIL-STD-1540E, MIL-STD-461G

Learning Objectives

By the end of this session, participants will be able to:

1. Distinguish between verification and validation using the ECSS/NASA definitions
2. Assign appropriate verification methods (IADT) to each requirement using a decision logic
3. Define environmental test levels derived from launch vehicle ICD specifications
4. Construct a compliance verification matrix mapping every requirement to its method, level, and

phase
5. Identify when waivers are appropriate and document the waiver rationale

---

1. Verification vs Validation: Precise Definitions (15 min)

Teaching Notes

[Source: ECSS-E-ST-10-02C Rev.1 section 4; NASA SEH Rev 2 section 5.3 (Process 7: Product Verification) and section 5.4 (Process 8: Product Validation)]

	Verification	Validation
Question	"Did we build the system right?"	"Did we build the right system?"
Checks	Requirements (shall statements)	Stakeholder needs and operational expectations
Standard	ECSS-E-ST-10-02C	ECSS-E-ST-10-06C
Performed by	Engineering team	Users, operators, stakeholders
When	Throughout development (Phases B-D)	End of development + early operations (Phase D-E)
Methods	IADT (Inspection, Analysis, Demonstration, Test)	Operational demonstration, user acceptance testing
Output	Verification Control Document (VCD)	Validation report

The Critical Distinction

A system can be verified (every requirement marked "pass") but still fail validation if the requirements were written incorrectly. This happens when:

- Requirements were derived from misunderstood stakeholder needs
- The operational environment differs from assumptions
- Edge cases were not captured in requirements

Example

- Requirement: "The system shall achieve GSD <= 10 m from 500 km altitude"
- Verification (Test): Calibration target imagery at 500 km -> measured GSD = 9.2 m -> verified
- Validation: Users attempt crop assessment with 9.2 m imagery -> "We can see field boundaries but cannot distinguish crop types; we needed GSD <= 5 m" -> not validated

The requirement was met, but the stakeholder need was not satisfied because the requirement was insufficiently stringent.

---

2. IADT Verification Methods in Detail (30 min)

Teaching Notes

ECSS-E-ST-10-02C Rev.1 section 5.3 defines four verification methods. Note: ECSS uses "IADT" ordering; some NASA documents use "ATRI" -- the methods are identical.

[Source: ECSS-E-ST-10-02C Rev.1 section 5.3; freely available from https://ecss.nl]

Inspection (I)

Definition: Visual or physical examination of the product, without operation or stimulation, to verify conformance to requirements.

When to use:

- Physical characteristics (dimensions, mass, surface finish, labelling)
- Manufacturing workmanship (solder joints, cable routing, MLI installation)
- Markings, identification, and safety labels
- Mechanical fit (deployer rail compliance)

Evidence produced: Inspection report with photographs, measurements, pass/fail checklist.

Examples for CubeSats:

- Mass measurement on calibrated scale -> verifies "M_dry <= 4.0 kg"
- Caliper measurement of rail profile -> verifies CDS dimensional compliance
- Visual inspection of antenna stowage -> verifies no protrusions beyond deployer envelope
- Workmanship inspection of PCB solder joints -> verifies IPC-A-610 Class 3

Analysis (A)

Definition: Mathematical models, simulations, statistical analysis, or heritage comparison demonstrate compliance with acceptable confidence.

When to use:

- Physical testing is impossible or impractical (orbital lifetime, radiation total ionising dose)
- Early in development (Phase B) before hardware exists
- To demonstrate margin (link budget, thermal model, structural FEA)
- Statistical parameters (reliability prediction, debris casualty risk)
- Requirements involving orbital mechanics or mission-level performance

Evidence produced: Analysis report with model description, assumptions, inputs, results, uncertainty assessment, and sensitivity analysis.

Examples for CubeSats:

- Orbital mechanics analysis -> verifies revisit time requirement
- Thermal model (finite difference/finite element) -> verifies temperature within limits
- Link budget calculation -> verifies margin >= 3 dB at worst case
- Structural FEA -> verifies positive Margin of Safety under launch loads

> *Margin of Safety (structural):*
>

> *MoS = (Allowable_stress / (Factor_of_Safety x Applied_stress)) - 1*
>
> *MoS must be >= 0 for all load cases.*
>
> *Factor_of_Safety: 1.25 (yield), 1.5 (ultimate) per ECSS-E-ST-32-10C.*

Demonstration (D)

Definition: The system is operated under controlled conditions to show that it performs as required, without requiring precise measurement of performance parameters.

When to use:

- Functional requirements ("the system shall deploy the antenna within 30 minutes of separation")
- Operational procedures ("the system shall respond to a mode change command within 5 seconds")
- Software functional requirements ("FDIR shall autonomously switch to safe mode on loss of attitude")

Evidence produced: Demonstration report with procedure, observations, photographs/video, pass/fail.

Examples for CubeSats:

- Antenna deployment test -> demonstrates deployment within time limit
- Safe mode entry test -> demonstrates autonomous FDIR response
- Ground station commanding test -> demonstrates full command chain

Test (T)

Definition: Physical hardware subjected to controlled conditions; measured performance compared quantitatively to requirement thresholds.

When to use:

- Physical properties must be proven with quantitative measurement
- Environmental survivability must be demonstrated (vibration, thermal-vacuum, EMC)
- End-to-end functional chains must be proven under representative conditions
- Performance must be measured, not just observed

Test types by purpose:

Test Type	Purpose	Level	Phase	Loads/Conditions
Development	Explore design space, find problems early	Unit/BB	B	As needed
Qualification	Prove design withstands worst-case + margin	Unit/Syste	C	Qual levels (MPE + margin)
Acceptance	Prove flight hardware is defect-free	System	D	Acceptance levels (MPE)
Proto-flight	Combined qual + acceptance (single unit)	System	C/D	Qual levels, acceptance duration

Proto-Flight Approach

For CubeSats, the proto-flight approach is standard because building a separate qualification unit is prohibitively expensive. The flight unit is tested to qualification levels but for acceptance duration.

[Source: ECSS-E-ST-10-03C section 5.4.2.3; MIL-STD-1540E section 6.2.4]

Rationale: Qualification duration is longer and more stressful -- this is acceptable for a dedicated qualification model that will not fly. For proto-flight, the shorter duration reduces fatigue life consumption while still screening for workmanship defects.

Parameter	Qualification	Acceptance	Proto-flight
Vibration level	MPE + 3 dB	MPE	MPE + 3 dB
Vibration duration	2 min/axis	1 min/axis	1 min/axis
TVAC temperature range	Predicted +/- 15 C	Predicted +/- 10 C	Predicted +/- 10 C
TVAC cycles	8	4	4-8

---

3. Environmental Test Programme (30 min)

Teaching Notes

[Source: ECSS-E-ST-10-03C (Testing); Launch vehicle Payload User's Guides; MIL-STD-1540E]

Standard Environmental Test Sequence

The test sequence is designed so that each test stresses the hardware in a specific way, and functional tests between environmental tests detect any damage.

```
+-------------------------+
| Receive Flight Hardware  |
+-----------+-------------+
            |
+-----------v-------------+
| Initial Functional Test  |  <- Reference baseline
+-----------+-------------+
            |
+-----------v-------------+
| Sine Vibration (3 axes)  |  <- Low-frequency loads
+-----------+-------------+
            |
+-----------v-------------+
| Random Vibration (3 axes)|  <- Broadband acoustic loads
+-----------+-------------+
            |
+-----------v-------------+        +-------------------+
| Post-Vibe Functional Test| ------>| Anomaly?          |
+-----------+-------------+  Fail  | Investigate &     |
            | Pass                 | resolve           |
```

```
+-----------v-------------+       +------------------+
| Thermal Vacuum Testing   |  <- 4-8 cycles, func at extremes
+-----------+-------------+
            |
+-----------v-------------+
| Post-TVAC Functional Test|
+-----------+-------------+
            |
+-----------v-------------+
| EMC Test (if required)   |
+-----------+-------------+
            |
+-----------v-------------+
| Mass + CG Measurement    |
+-----------+-------------+
            |
+-----------v-------------+
| Deployer Fit Check       |
+-----------+-------------+
            |
+-----------v-------------+
| Pack & Ship to Launch    |
+-------------------------+
```

Sine Vibration Test Levels

[Source: Typical Falcon 9 Transporter Rideshare PUG]

Sine vibration simulates the low-frequency launch vehicle structural modes (typically 5-100 Hz). These are quasi-static loads.

Frequency Range (Hz)	Level	Notes
5 - 8	12.5 mm displacement (0-peak)	Displacement-controlled
8 - 100	1.25 g (0-peak)	Acceleration-controlled
Sweep rate	2 octaves/minute	Standard sweep rate
Axes	3 (X, Y, Z)	Sequential, one axis at a time

> *Sine Vibration Acceleration at Crossover Frequency:*
>
> *a = (2 pi f)^2 * d*
>
> *At f = 8 Hz, d = 12.5 mm = 0.0125 m:*
>
> *a = (2 pi 8)^2 0.0125 = (50.27)^2 0.0125 = 2526.6 * 0.0125 = 31.6 m/s^2 = 3.22 g*
>
> *Note: The transition from displacement to acceleration control occurs where the displacement limit would exceed the acceleration limit. The actual crossover depends on the specific profile.*

Random Vibration Test Levels

Random vibration simulates the broadband acoustic and mechanical environment during launch. Levels are specified as Power Spectral Density (PSD) in g^2/Hz.

Frequency (Hz)	PSD Level (g^2/Hz)	Notes
20	0.01	Start of profile
20 - 50	+3 dB/octave ramp	Rising
50 - 800	0.04 (flat)	Maximum level plateau
800 - 2000	-6 dB/octave ramp	Falling
2000	0.004	End of profile
Overall gRMS	~7.0 gRMS	Qualification level

> *Computing Overall gRMS from PSD Profile:*
>
> *gRMS = sqrt(integral from f1 to f2 of PSD(f) df)*
>
> *For a flat PSD region: gRMS_flat = sqrt(PSD_flat x (f2 - f1))*
>
> *Example: PSD = 0.04 g^2/Hz from 50-800 Hz:*
> *gRMS_flat = sqrt(0.04 x 750) = sqrt(30) = 5.48 gRMS*
>
> *The ramp sections and flat section are summed in quadrature:*
> *gRMS_total = sqrt(gRMS_ramp1^2 + gRMS_flat^2 + gRMS_ramp2^2)*

Thermal Vacuum Test Profile

Parameter	Proto-flight Level	Notes
Hot case temperature	Predicted max + 10 C	e.g., if thermal model predicts +55 C -> test to +65 C
Cold case temperature	Predicted min - 10 C	e.g., if thermal model predicts -20 C -> test to -30 C
Number of cycles	4 minimum (proto-flight)	Start and end at hot; functional test at each extreme
Vacuum level	< 10^-5 mbar	Space-representative vacuum
Dwell time at extreme	>= 1 hour	Allow thermal stabilisation (< 1 C/hr gradient)
Temperature transition rate	1-3 C/min	Limited by chamber capability
Functional test	Full performance at hot and cold extremes	Subset functional at intermediate transitions

> *TVAC Cycle Duration Estimate:*
>
> *t_cycle = 2 x (Delta_T / ramp_rate) + 2 x t_dwell + 2 x t_functional*
>
> *Example: Delta_T = 95 C (from -30 to +65 C), ramp = 2 C/min, dwell = 1 hr, functional = 2 hr:*
> *t_cycle = 2 x (95/2 min) + 2 x 60 min + 2 x 120 min = 95 + 120 + 240 = 455 min = 7.6 hr*

Total TVAC campaign (4 cycles + setup): ~4 x 7.6 + 8 = 38.4 hr ~ 5 working days

EMC Testing

[Source: MIL-STD-461G; ECSS-E-ST-20-07C]

Test	Standard	Purpose
CE102 Conducted Emissions	MIL-STD-461	Measure spurious conducted emissions on power lines
RE102 Radiated Emissions	MIL-STD-461	Measure unintentional RF radiation from satellite
CS101 Conducted Susceptibility	MIL-STD-461	Verify immunity to power line transients
RS103 Radiated Susceptibility	MIL-STD-461G	Verify immunity to external RF fields

Note: EMC testing is not always required for CubeSats. It depends on the launch provider and co-manifested payloads. Check the rideshare ICD.

---

4. Compliance Verification Matrix Construction (20 min)

Teaching Notes

The Verification Matrix (also called Compliance Matrix or Verification Cross-Reference Matrix, VCRM) is the central document linking every requirement to its verification evidence.

[Source: ECSS-E-ST-10-02C Rev.1 section 6; DRD in Annex A]

Matrix Structure

Req ID	Requirement Text	Method	Phase	Level	Responsible	Facility	Status	Evidence
SYS-00	M_dry <= 4.0 kg	I	D	System	Structures	Clean	Planne	Mass measurement report
SYS-00	Survive 7 gRMS random	T	C/D	System	Structures	Vibe lab	Planne	Vibration test report
SYS-00	GSD <= 10 m at 500 km	A	B	System	Payload	N/A	Planne	Optical analysis report
SYS-00	Link margin >= 3 dB	A	B	System	Comms	N/A	Planne	Link budget analysis
SYS-005	Antenna deploys < 30 min	D	D	System	Mechanisms	Clean room	Planned	Deployment demonstration

IADT Assignment Decision Logic

```
Is it a physical property (mass, dimensions, surface finish)?
  -> Inspection (I), Phase D, System level

Can it only be proven by quantitative measurement under controlled conditions?
  -> Test (T), Phase C/D, Unit or System level

Can it be demonstrated by operating the system (functional, not precision)?
```

```
  -> Demonstration (D), Phase D, System level

Can it be shown with acceptable confidence through modelling?
  -> Analysis (A), Phase B/C, System level

Is it a document, process, or plan?
  -> Review of Design (R*), Phase B
  (* Note: ECSS uses IADT; "Review" is sometimes used informally but is not
     a formal ECSS method. Reviews are part of the verification process but
     requirements verified by examining design documentation fall under
     "Analysis" in strict ECSS usage.)
```

Multiple Methods

Some requirements need both analysis (early confidence) and test (final proof):

Requirement	Analysis (Phase B)	Test (Phase C/D)
Structural survival under launch loads	FEA with positive MoS	Vibration test -- no damage
Pointing accuracy <= 0.1 deg	Error budget analysis	On-orbit calibration measurement
Thermal within limits (-20 to +55 C)	Thermal model simulation	TVAC functional test at extremes
RF link margin >= 3 dB	Link budget calculation	RF compatibility test (if feasible)

In the matrix, list the primary method and note "confirmed by [secondary method] in Phase [X]".

Waiver Process

When a requirement cannot be verified by the prescribed method, a waiver is needed:

Waiver Type	When Used	Approval Level
Non-compliance	Requirement cannot be met; accept deviation	Project manager + customer
Method deviation	Different verification method used (e.g., analysis instead of test)	Systems engineer + quality
Tailoring	Standard requirement not applicable to this mission	Systems engineer

Waiver documentation must include: Requirement ID, deviation description, technical justification, risk assessment, and compensating measures.

---

5. SpaceCDF V&V Matrix Exercise (25 min)

Instructions

1. Navigate to the V&V Matrix tab in SpaceCDF
2. Requirements are auto-populated from the design
3. For each requirement, assign:

- Method: I, A, D, or T (use the decision logic above)

- Phase: When will verification occur (B, C, D)?
- Level: Unit, subsystem, or system level?
- Status: All should be "Planned" at this stage

4. Use the filter buttons to view by method type:

- Count: How many are Analysis? Test? Demonstration? Inspection?
- Are there any requirements with no method assigned?

5. Identify requirements that need two methods (analysis in Phase B + test in Phase D)
6. Discuss with your team: which tests are most critical? Which drive the test campaign schedule?

Worksheet 4.2 Tasks

1. Assign IADT method to 10 key requirements (justify each choice)
2. Identify 3 requirements that need both analysis AND test
3. Specify the complete environmental test sequence for your mission
4. Determine which test levels apply from your selected launch vehicle PUG
5. Identify any requirements that may need waivers -- document the rationale

---

Cited References

#	Reference	URL
1	ECSS-E-ST-10-02C Rev.1 (Verification)	https://ecss.nl/standard/ecss-e-st-10-02c-rev-1-verification/
2	ECSS-E-ST-10-03C (Testing)	https://ecss.nl/standard/ecss-e-st-10-03c-testing/
3	NASA SEH Rev 2 sections 5.3-5.4	https://www.nasa.gov/reference/systems-engineering-handbook/
4	MIL-STD-1540E (Test Requirements for Launch/Upper Stage/Space Vehicles)	https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=36197
5	MIL-STD-461G (EMI/EMC)	https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=35789
6	CubeSat Design Specification Rev 14.1	https://www.cubesat.org/s/CDS-REV14_1-2022-02-09.pdf

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Session Summary

Topic	Key Takeaway
V vs V	Verification = meets requirements (IADT); Validation = meets stakeholder needs
IADT	Inspection (physical), Analysis (models), Demonstration (functional), Test (quantitative)
Proto-flight	CubeSats: single flight unit tested to qualification levels, acceptance duration
Sine vibration	5-100 Hz, 1.25 g, quasi-static launch loads
Random vibration	20-2000 Hz, ~7 gRMS overall, broadband acoustic environment
TVAC	Predicted extremes +/- 10 C, 4+ cycles, functional at each extreme
Compliance matrix	Per requirement: method + phase + level + responsible + status + evidence
Waivers	Document deviation, justification, risk, and compensating measures

---

Session 4.3: Risk, Interfaces & FMECA

Duration: 2 hours

Prerequisites: Sessions 4.1-4.2 (equipment selected, V&V methods assigned)

References: ECSS-M-ST-80C (Risk Management), ECSS-Q-ST-30-02C (FMEA/FMECA), ECSS-E-ST-10-24C (Interface Management), NASA SEH Rev 2 section 6.4, NPR 8000.4B

Learning Objectives

By the end of this session, participants will be able to:

1. Construct and populate a risk register using a 5x5 likelihood-consequence matrix
2. Build an N-squared (N^2) interface matrix and identify interface conflicts
3. Perform a simplified FMEA/FMECA for critical subsystems
4. Identify single-point failures and compute series/parallel reliability
5. Classify risks and select appropriate mitigation strategies (accept, mitigate, transfer, avoid)

---

1. Risk Management Process (20 min)

Teaching Notes

[Source: ECSS-M-ST-80C (Risk Management); NASA SEH Rev 2 section 6.4 (Process 13: Technical Risk Management)]

[URL: https://ecss.nl/standard/ecss-m-st-80c-risk-management-31-july-2008/]

The Four-Step Process

Risk management is continuous throughout the project lifecycle. The four steps are:

1. Identify -- What can go wrong? (Brainstorming, checklists, historical data, expert judgment)
2. Assess -- How likely is it? How severe would the consequence be? (Scoring)
3. Mitigate -- What can we do to reduce likelihood or consequence? (Strategy selection)
4. Monitor -- Is the risk increasing, decreasing, or stable? (Tracking, triggers, escalation)

Risk Sources

Source	Example Risks
Technical	Component failure, interface mismatch, performance shortfall, software defect

Schedule	Component delivery delay, test facility unavailable, regulatory delay
Cost	Component price increase, scope creep, test campaign overrun
Programmatic	Funding cut, personnel turnover, partner withdrawal
External	Launch delay/failure, spectrum interference, export control denial

---

2. The 5x5 Risk Matrix (25 min)

Teaching Notes

The 5x5 matrix is the standard risk scoring tool used across ESA, NASA, and commercial space programmes.

Likelihood Scale

Level	Likelihood	Description	Probability Range
1	Remote	Very unlikely; no precedent in similar missions	< 5%
2	Unlikely	Could happen but improbable; has occurred rarely	5 - 20%
3	Possible	Has happened on similar missions; credible scenario	20 - 50%
4	Likely	Expected to occur at least once during the mission	50 - 80%
5	Almost certain	Will almost certainly occur; multiple precedents	> 80%

Consequence Scale

Level	Consequence	Technical Impact	Cost Impact	Schedule Impact
1	Negligible	No performance impact	< 1% overrun	< 1 week slip
2	Minor	Minor performance degradation; workaround exists	1 - 5% overrun	1 - 4 week slip
3	Moderate	Significant performance loss; mission degraded	5 - 15% overrun	1 - 3 month slip
4	Major	Mission capability severely degraded; partial loss	15 - 30%	3 - 6 month slip
5	Catastrophic	Mission failure; total loss	> 30% overrun	> 6 month slip or cancellation

Risk Score and Classification

> *Risk Score:*
>
> *R = L x C*
>
> *Where L = Likelihood (1-5), C = Consequence (1-5)*

5x5 Risk Matrix (Colour-Coded)

```
              Consequence ->
              1       2       3       4       5
    L   5 |   5(M) | 10(H) | 15(H) | 20(C) | 25(C) |
    i   4 |   4(L) |  8(M) | 12(H) | 16(C) | 20(C) |
    k   3 |   3(L) |  6(M) |  9(M) | 12(H) | 15(H) |
    e   2 |   2(L) |  4(L) |  6(M) |  8(M) | 10(H) |
    l   1 |   1(L) |  2(L) |  3(L) |  4(L) |  5(M) |
    i
    h
    o
    o
    d

    L = Low (1-4):     Accept and monitor
    M = Medium (5-9): Mitigate if cost-effective
    H = High (10-15): Active mitigation required
    C = Critical (16-25): Redesign or descope required
```

CubeSat-Specific Risk Examples

Risk	L	C	Score	Category	Typical Mitigation
Deployment mechanism failure (antenna, SA)	3	4	12	High	Redundant mechanisms; 100+ ground test cycles
Communication loss after separation	2	5	10	High	Beacon mode; timer-based antenna deploy; multiple GS
ADCS does not achieve pointing specification	3	3	9	Medium	Margin in pointing budget; on-orbit calibration plan
Power budget negative during eclipse	2	4	8	Medium	Conservative duty cycling; 20% battery margin
COTS component radiation failure (SEU/SEL)	2	4	8	Medium	Watchdog resets; latchup protection circuits; EDAC
Software bug causing spurious safe mode entry	4	2	8	Medium	Extensive software testing; staged upload; safe mode must work independently
Thermal exceedance (hot case)	2	3	6	Mediu	Additional radiator area; duty cycle limit
Launch delay (vehicle failure)	3	2	6	Mediu	Manifest on multiple vehicles; schedule buffer
Spectrum licensing delay	3	2	6	Medium	Start filing 12+ months before planned launch

Mitigation Strategies

Strategy	Description	When to Use	Cost Impact
Accept	Risk is within tolerance; monitor only	Score 1-4 (Low)	None
Mitigate	Reduce L or C through design changes, testing, or procedures	Score 5-15	Moderate -- cost of mitigation
Transfer	Pass risk to another party (insurance, supplier warranty)	Financial risk; operational risk	Premium or contract cost
Avoid	Change design to eliminate the risk entirely	Score 16-25 (Critical)	May affect performance or cost

3. N-Squared (N^2) Interface Matrix (25 min)

Teaching Notes

The N^2 matrix is the standard systems engineering tool for identifying and managing interfaces between subsystems. It was popularised by NASA and ESA CDF practice.

[Source: ECSS-E-ST-10-24C (Interface Management); NASA SEH Rev 2 section 6.3 (Process 12: Interface Management)]

What is an N^2 Matrix?

For a system with N subsystems, the N^2 matrix is an N x N grid where:

- Diagonal cells contain the subsystem names
- Off-diagonal cells contain the interfaces between subsystems
- Cell (i, j) = outputs FROM subsystem i TO subsystem j (read across the row)
- Cell (j, i) = outputs FROM subsystem j TO subsystem i (read down the column)

N^2 Matrix Example (6 Subsystems)

```
         TO ->
         EPS      OBC      AOCS     TTC      Payload  Structure
FROM
EPS      [EPS]    28V bus  28V bus  28V bus  28V bus   Mounting
         ------   5V reg   5V reg   5V reg   5V reg   bolts
                  I2C HK   I2C HK   I2C HK
OBC      Pwr cmd  [OBC]    Cmd      TC data  Cmd      ---
         Telem    ------   I2C/SPI  UART     SPI/UART
AOCS     Pwr req  Att data [AOCS]   Att for  Att data Mount
         ---      AOCS HK  ------   antenna  pointing vibration
TTC      Pwr req  Rx data  ---      [TTC]    ---      Antenna
         ---      Cmd fwd           ------            mount
Payload  Pwr req  Sci data Pointing ---      [PYLD]   FOV
         ---      PL HK    request           ------   clearance
Struct   ---      ---      Sensor   Antenna  Payload  [STRUCT]
                           mounting mounting mounting  ------
```

Interface Types

Colour-code the N^2 matrix by interface type:

Type	Colour	Examples
Electrical power	Red	28V bus, 5V regulated, switched lines
Data	Blue	I2C, SPI, UART, CAN, RS-422
RF	Green	Coaxial cable, waveguide
Mechanical	Orange	Mounting bolts, thermal straps, alignment pins
Thermal	Yellow	Conductive paths, radiative coupling

Software	Purple	Command interfaces, telemetry packets, mode transitions

Interface Conflict Detection

An interface conflict exists when:

- Subsystem A expects to send 28V but Subsystem B only accepts 5V
- Subsystem A uses I2C but Subsystem B only has SPI
- Subsystem A is in S-band but Subsystem B antenna is X-band
- Subsystem A mounts on the +Z face but Structure has no mounting provision on +Z

SpaceCDF detects many of these automatically through the constraint engine and displays conflicts as warning badges on the Dashboard.

Worked Example: Detecting a Conflict

Problem: The OBC sends commands to the reaction wheels via I2C. The selected reaction wheel unit (RWP100 from Blue Canyon) uses RS-422. This is a data protocol mismatch.

Resolution options:

1. Select a different reaction wheel that supports I2C
2. Add an I2C-to-RS-422 bridge (additional component, mass, cost, failure point)
3. Select a different OBC that supports RS-422

In SpaceCDF: This conflict would appear as a warning in the Equipment Browser when the reaction wheel is selected, because the system checks data protocol compatibility.

---

4. FMEA/FMECA (25 min)

Teaching Notes

[Source: ECSS-Q-ST-30-02C (Failure Mode, Effects, and Criticality Analysis); ECSS-Q-ST-30C (Dependability)]

[URL: https://ecss.nl/standard/ecss-q-st-30-02c-failure-mode-effects-and-criticality-analysis-fmeca-6-march-2009/]

Definitions

- FMEA (Failure Mode and Effects Analysis): Identifies failure modes, their causes, and effects on the system
- FMECA (Failure Mode, Effects, and Criticality Analysis): FMEA + criticality ranking

FMEA/FMECA Table Structure

Item	Function	Failure Mode	Cause	Local Effect	System Effect	Severity	Detection	Compensating Provision	Criticality
OBC	Process commands, run FSW	Process or lockup	SEU, firmware bug	No command processing	Loss of mission control	5	HK timeout	Watchdog reset; redundant OBC (if fitted)	1 (SPF)
Battery	Store energy	Cell short	Manufacturing	Reduced capacity	Shortened eclipse	3	Voltage monitori	Cell balancing; margin in capacity	2
Reaction Wheel	Provide torque	Bearing seizure	Lubrication failure	Loss of one axis control	Degraded pointing	4	Current anomaly	4-wheel config (3+1 redundancy)	3 (with redundancy)
Antenna	Radiate RF	Deployment failure	Mechanism jam	No antenna deployed	No communication	5	Beacon absence	Redundant deployment; burn wire + spring	1 (SPF)

Criticality Categories

Category	Definition	Action Required
1 (Catastrophic)	Single failure causes mission loss; no compensation	Redesign to add redundancy or accept with justification
2 (Critical)	Single failure causes significant degradation	Mitigate (redundancy, operational workaround)
3 (Major)	Single failure causes moderate degradation	Monitor; plan operational workaround
4 (Minor)	Single failure has negligible mission impact	Accept

Single-Point Failure (SPF) Analysis

A single-point failure is any single component whose failure alone causes loss of mission. Identifying SPFs is a critical output of the FMECA.

Typical CubeSat Single-Point Failures:

Component	Why it is an SPF	Common Mitigation
OBC (single processor)	No command processing -> no mission	Watchdog timer + autonomous safe mode + EDAC memory
Battery (single pack)	No stored energy -> no eclipse survival	Cell-level monitoring; conservative DoD limit (< 20%)
Antenna (non-redundant deploy)	No RF link -> no commanding/telemetry	Redundant deployment mechanisms (burn wire + spring)
Solar array (deployment)	No power generation -> mission loss within hours	Redundant deployment; hinge spring + motor backup
EPS main board	No power distribution	Typically no mitigation (accepted SPF in CubeSats)

Key Equations: Reliability

> *Series Reliability (all components must work):*
>
> *R_series = Product of R_i for i = 1 to n*

For n components each with reliability R_i:

> *R_series = R_1 x R_2 x ... x R_n*
>
> *Example: 5 components each with R = 0.99:*
>
> *R_series = 0.99^5 = 0.951*
>
> *Parallel Reliability (at least one must work -- redundancy):*
>
> *R_parallel = 1 - Product of (1 - R_i) for i = 1 to n*
>
> *For n identical redundant units each with R:*
>
> *R_parallel = 1 - (1 - R)^n*
>
> *Example: 2 redundant deployment mechanisms each with R = 0.95:*
>
> *R_parallel = 1 - (1 - 0.95)^2 = 1 - (0.05)^2 = 1 - 0.0025 = 0.9975*
>
> *Mean Time Between Failures (MTBF):*
>
> *MTBF = Total_operating_hours / Number_of_failures*
>
> *For a component with failure rate lambda (failures/hour):*
>
> *MTBF = 1 / lambda*
>
> *Reliability over time (exponential model):*
>
> *R(t) = e^(-t / MTBF) = e^(-lambda * t)*
>
> *Example: MTBF = 50,000 hours, mission duration = 8,760 hours (1 year):*
>
> *R = e^(-8760/50000) = e^(-0.1752) = 0.839*
>
> *System Availability:*
>
> *A = MTBF / (MTBF + MTTR)*
>
> *Where MTTR = Mean Time To Repair (or recover, for spacecraft)*
>
> *For a spacecraft with MTBF = 50,000 hr and MTTR = 2 hr (safe mode recovery):*
>
> *A = 50000 / (50000 + 2) = 0.99996*

Worked Example: Reaction Wheel Redundancy

Problem: A fine-pointing mission requires 3-axis attitude control. Each reaction wheel has R = 0.98 over the 2-year mission.

Configuration A -- 3 wheels (no redundancy):

R_system = R^3 = 0.98^3 = 0.941

Configuration B -- 4 wheels in 3-of-4 redundancy (1 spare):

R_system = 4 x R^4 - 3 x R^3 ... Using the binomial:

P(>= 3 working) = C(4,4) x R^4 + C(4,3) x R^3 x (1-R)^1

= 0.98^4 + 4 x 0.98^3 x 0.02

= 0.9224 + 4 x 0.9412 x 0.02

= 0.9224 + 0.0753

= 0.9977

Adding one spare wheel improves reliability from 0.941 to 0.998 -- a significant improvement for modest mass and cost increase.

Real Mission Example: Hitomi (ASTRO-H)

JAXA's Hitomi X-ray observatory was lost on 26 March 2016, just 37 days after launch, due to a cascading failure in the AOCS subsystem:

1. An incorrect parameter in the star tracker caused an erroneous attitude estimate
2. The reaction wheels applied incorrect torques based on the bad estimate
3. Thrusters fired to "correct" the (phantom) attitude error, causing rapid spin
4. The satellite spun up beyond structural limits, and the extensible optical bench broke off

Root cause: Software parameter error (unit conversion) + inadequate FDIR logic + no cross-check between attitude sensors.

Lesson: FMECA must consider cascading failures and common-cause failures, not just single-component failures. Software errors are failure modes too.

[Source: JAXA Hitomi Investigation Report, 2016, available at https://global.jaxa.jp/press/2016/05/20160531_hitomi.html]

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5. Risk & Interface Exercise (25 min)

Instructions

1. Dashboard -- Check the reliability score and conflict count
2. Risk Register Construction (Worksheet 4.3):

- Identify 5 technical risks for your mission design
- Score each on the 5x5 matrix (L x C)
- Define mitigation strategy for any scoring >= 10
- Assign an owner (which CDF position is responsible)

3. N^2 Matrix -- Build a simplified N^2 matrix for your mission:

- List 6 subsystems on the diagonal
- Fill in the interfaces (power, data, RF, mechanical)
- Identify at least 2 interface conflicts

4. SPF Analysis -- Identify all single-point failures in your design:

- For each SPF: Is it acceptable? If not, what redundancy is needed?
- Compute the reliability improvement from adding redundancy to one SPF

Discussion Prompts

- "What is the highest-risk item in your design? What would it cost to mitigate?"
- "Do you have any interface conflicts that cannot be resolved without changing a component selection?"
- "Which single-point failure are you most concerned about? Would the customer accept it?"

Worksheet 4.3 Tasks

1. Complete the risk register (5 risks minimum, scored)
2. Build a 6x6 N^2 interface matrix with colour-coded interface types
3. Complete the FMECA table for 4 critical components
4. List all single-point failures with accept/mitigate decision
5. Calculate series reliability for your mission's critical chain

---

Cited References

#	Reference	URL
1	ECSS-M-ST-80C (Risk Management)	https://ecss.nl/standard/ecss-m-st-80c-risk-management-31-july-2008/
2	ECSS-Q-ST-30-02C (FMEA/FMECA)	https://ecss.nl/standard/ecss-q-st-30-02c-failure-mode-effects-and-criticality-analysis-fmeca-6-march-2009/
3	ECSS-E-ST-10-24C (Interface Management)	https://ecss.nl/standard/ecss-e-st-10-24c-interface-management/
4	NASA SEH Rev 2, section 6.4	https://www.nasa.gov/reference/systems-engineering-handbook/
5	JAXA Hitomi Investigation Report	https://global.jaxa.jp/press/2016/05/20160531_hitomi.html
6	ECSS-Q-ST-30C (Dependability)	https://ecss.nl/standard/ecss-q-st-30c-dependability/

Session Summary

Topic	Key Takeaway
Risk process	Identify -> Assess -> Mitigate -> Monitor (continuous throughout lifecycle)
5x5 matrix	L x C = Risk Score; Low (1-4), Medium (5-9), High (10-15), Critical (16-25)
Strategies	Accept (low), Mitigate (medium-high), Transfer (financial), Avoid (critical)
N^2 matrix	N x N grid mapping all subsystem interfaces; colour-code by type
Interface conflicts	Protocol mismatch, voltage mismatch, band mismatch -- detect early
FMECA	Failure mode -> cause -> local effect -> system effect -> severity -> detection -> mitigation
SPF	Single-point failures must be identified, assessed, and accepted or mitigated
Reliability	R_series = Product(R_i); R_parallel = 1 - Product(1-R_i); R(t) = e^(-lambda*t)

Session 4.4: Cost Estimation & Design Review

Duration: 2 hours

Prerequisites: Sessions 4.1-4.3 (equipment selected, V&V planned, risks assessed)

References: SMAD4 Ch.20, NASA Cost Estimating Handbook (CEH) v4.0, Aerospace Corp SSCM, NPR 7120.5F (WBS), ECSS-M-ST-60C (Cost Management), NPR 7123.1D Appendix G (Reviews)

Learning Objectives

By the end of this session, participants will be able to:

1. Estimate mission cost using parametric (CER), analogy, and bottom-up methods
2. Structure a Work Breakdown Structure (WBS) for a CubeSat project
3. Apply learning curve effects for constellation cost estimation
4. Assess cost risk using confidence levels (P50/P70/P80)
5. Prepare for and conduct a design review (SRR/PDR/CDR) with gate criteria
6. Present design evidence clearly and respond to review board questions

---

1. Cost Estimation Methodologies (25 min)

Teaching Notes

[Source: NASA CEH v4.0 Appendix C; SMAD4 Chapter 20; Aerospace Corp SSCM]

[URL: https://www.nasa.gov/offices/ocfo/references-and-tools/ (NASA CEH)]

Three Primary Methods

Method	Description	Accuracy (1-sigma)	When to Use	Data Required
Parametric	Cost Estimating Relationships (CERs) from historical database	+/- 30-50%	Phase 0/A (few details known)	Mass, power, mission type
Analogy	Compare to similar past mission, adjust for differences	+/- 20-40%	Phase A/B (reference mission available)	Detailed knowledge of reference
Bottom-up	Sum actual vendor quotes + labour estimates + facilities	+/- 10-20%	Phase B/C (detailed design)	BOM, vendor quotes, labour rates

Parametric Cost Estimating Relationships (CERs)

CERs express cost as a function of technical parameters, derived from regression analysis of historical data.

> *General CER Form:*
>
> *Cost = a x (Parameter)^b x Complexity_factor*
>
> *Where:*
> *- a = coefficient (from regression)*
> *- Parameter = usually dry mass (kg), power (W), or data rate*
> *- b = exponent (typically 0.5-1.0)*
> *- Complexity_factor = adjustment for mission difficulty*

USCM (Unmanned Space Vehicle Cost Model) CERs

[Source: SMAD4 Table 20-8; US Air Force USCM database]

Subsystem	CER (FY2010 $K)	Parameter	Typical b
Structure	157 x M_struct^0.83	Dry mass (kg)	0.83
Thermal	394 x M_therm^0.635	Dry mass (kg)	0.635
EPS	62.7 x M_EPS^1.00	Dry mass (kg)	1.00
TTC	545 x M_TTC^0.761	Dry mass (kg)	0.761
AOCS	464 x M_AOCS^0.867	Dry mass (kg)	0.867
Propulsion	17.8 x M_prop^0.75	Dry mass (kg)	0.75
Integration & Test	10.4 x M_dry^0.907	Total dry mass (kg)	0.907
Program Management	12.3% of hardware cost	N/A	N/A
Systems Engineering	14.2% of hardware cost	N/A	N/A

SSCM (Small Satellite Cost Model)

The Aerospace Corporation SSCM was specifically calibrated for satellites < 500 kg. It provides more accurate estimates for CubeSats than USCM.

[Source: Aerospace Corp, "Small Satellite Cost Model (SSCM)", available through Aerospace TOR]

Key SSCM adjustments for CubeSats:

- COTS hardware costs are catalogue prices, not mass-based CERs
- Labour costs dominate for university/small-team projects
- NRE is heavily dependent on mission uniqueness
- Standard CERs (designed for > 100 kg) over-predict by 2-6x for nano/micro class

CubeSat-Calibrated Pricing (SpaceCDF)

SpaceCDF uses a hybrid approach: COTS flat pricing for nano/micro class, CERs for larger spacecraft.

Subsystem	CubeSat COTS Cost (kEUR)	USCM CER (kEUR, for CubeSat mass)	Ratio
EPS	15	60	CER 4x higher
AOCS (fine)	40	66	CER 1.6x higher
TTC (S-band)	20	25	Similar
OBC	10	12	Similar
Structure	8	3	CER lower (min order effect)
Payload	50 (variable)	300	CER 6x higher for COTS payloads

This confirms that for CubeSats, COTS pricing is more reliable than parametric CERs for hardware cost.

Worked Example: Parametric vs Bottom-Up

3U Earth observation mission:

WBS Element	Parametric (kEUR)	Bottom-Up (kEUR)	Notes
Structure	8	8.5	ISIS 3U structure
EPS	18	16.2	GomSpace P31u + SA
OBC	12	10.0	NanoAvionics SatBus
AOCS	45	42.0	BCT XACT-15
TTC	22	20.5	Endurosat S-band
Thermal	5	3.0	Passive only
Payload	65	58.0	Simera Sense xScape
Harness	3	2.5	Custom cables
Hardware subtotal	178	160.7	
I&T (12%)	21	19.3	
Software (8%)	14	12.9	
PM/SE/MA (13%)	23	20.9	
Launch	200	195.0	SpaceX Transporter
Ground (5%)	9	8.0	SatNOGS + dedicated
Operations (3 yr)	45	40.0	0.5 FTE
TOTAL	490	456.8	

The parametric estimate is ~7% higher than bottom-up. This is expected: parametric includes inherent uncertainty and contingency.

---

2. Cost Breakdown Structure (WBS) (20 min)

Teaching Notes

The Work Breakdown Structure (WBS) is the hierarchical decomposition of all work required to complete the mission. It is the foundation for cost estimation, scheduling, and management.

[Source: NPR 7120.5F Appendix G; ECSS-M-ST-60C; NASA CEH v4.0 Appendix B]

Standard CubeSat WBS

```
WBS Level 1: Mission Total
  1.0  Programme Management              5%    (oversight, reviews, reporting)
  2.0  Systems Engineering               5%    (budgets, interfaces, trade studies)
  3.0  Mission Assurance                 3%    (quality, reliability, parts)
  4.0  Payload                          20%    (instrument + calibration)
    4.1  Payload instrument hardware
    4.2  Payload software
    4.3  Payload calibration & characterisation
  5.0  Spacecraft Bus Hardware          30%    (all subsystems)
    5.1  Structure & mechanisms
    5.2  EPS (solar array + battery + board)
    5.3  AOCS (sensors + actuators)
    5.4  TTC (transponder + antenna)
    5.5  OBC & data handling
    5.6  Thermal control
    5.7  Propulsion (if applicable)
    5.8  Harness & cabling
  6.0  Integration & Test               12%    (assembly, env. testing)
    6.1  Assembly & integration
    6.2  Environmental test campaign
    6.3  Test facilities rental
  7.0  Software (Flight + Ground)        8%    (FSW, GSW, mission planning)
    7.1  Flight software
    7.2  Ground segment software
    7.3  Mission planning tools
  8.0  Launch Services                  10%    (vehicle, deployer, integration)
  9.0  Ground Segment                    5%    (antennas, MCS, networks)
  10.0 Operations (mission lifetime)     5%    (staff, consumables, maintenance)
```

Cost by Phase

The distribution of cost across lifecycle phases is important for budgeting:

Phase	% of Total	Activities	Peak Staffing
0/A (Concept)	5-10%	Studies, trade-offs, requirements	Low
B (Preliminary Design)	15-20%	PDR, detailed analysis, long-lead procurement	Growing
C (Detailed Design)	30-40%	CDR, manufacturing, software development	Peak
D (Integration & Test)	20-25%	Assembly, environmental testing, commissioning	High
E (Operations)	10-20%	Routine operations, anomaly management	Low-steady
F (Disposal)	1-3%	Decommissioning, deorbit	Minimal

3. Learning Curve for Constellations (15 min)

Teaching Notes

[Source: SMAD4 section 20.3; Wright's Learning Curve Theory (1936)]

When building multiple identical units, the cost per unit decreases due to manufacturing efficiency, reduced test time, bulk purchasing, and labour learning.

Wright's Learning Curve

> *Nth Unit Cost:*
>
> *C_N = C_1 x N^b*
>
> *Where b = ln(learning_rate) / ln(2)*
>
> *| Learning Rate | b | Cost of 10th Unit (% of 1st) |*
> *|--------------|-----|-----|*
> *| 95% | -0.074 | 77% |*
> *| 90% | -0.152 | 60% |*
> *| 85% | -0.234 | 47% |*

> *Total Cost for N Units (cumulative):*
>
> *C_total = C_1 x Sum from i=1 to N of (i^b)*
>
> *Or approximately: C_total ~ C_1 x N^(1+b) / (1+b) (continuous approximation)*

Simplified Rule of Thumb

> *At a 90% learning rate, every time you double the number of units, the unit cost drops by 10%.*
>
> *Unit 1: EUR 800K*
> *Unit 2: EUR 720K (800 x 0.90)*
> *Unit 4: EUR 648K (720 x 0.90)*
> *Unit 8: EUR 583K (648 x 0.90)*
> *Unit 16: EUR 525K (583 x 0.90)*

Worked Example: 20-Satellite Constellation

First unit cost (bus + payload + I&T): EUR 800K. Learning rate: 90%.

Units	b	Avg Unit Cost	Total Hardware	Calc
1	-0.152	EUR 800K	EUR 800K	First unit
5	-0.152	EUR 659K	EUR 3,295K	5 x 800 x 5^(-0.152)
10	-0.152	EUR 577K	EUR 5,770K	Cumulative sum
20	-0.152	EUR 505K	EUR 10,100K	Cumulative sum

Total constellation estimate:

- 20 satellites hardware: EUR 10.1M
- 2 spare units (10%): EUR 1.0M
- Launch (20 sats x EUR 200K rideshare): EUR 4.0M
- Ground segment: EUR 1.0M
- Operations (3 years): EUR 0.9M
- PM/SE/MA (10%): EUR 1.7M
- Total: ~EUR 18.7M

---

4. Cost Risk and Confidence Levels (10 min)

Teaching Notes

Point estimates are misleading. Every cost estimate has uncertainty. The standard practice is to express cost as a probability distribution.

[Source: NASA CEH v4.0 section 2.3; JPL parametric estimation practice]

Uncertainty by Cost Element

Cost Element	Distribution	Uncertainty (1-sigma)	Rationale
COTS hardware	Normal	+/- 10%	Known pricing from vendor quotes
Custom hardware	Lognormal	+/- 30%	Development uncertainty skews high
Software	Triangular	+/- 40%	Hardest to estimate; frequent overruns
Launch	Normal	+/- 15%	Published pricing; contract negotiation
Operations	Uniform	+/- 25%	Staffing level uncertainty
I&T	Lognormal	+/- 25%	Test anomalies cause schedule/cost growth

Confidence Levels

Percentile	Meaning	Use
P50	50% probability of being at or below this cost	Project baseline; "expected" cost
P70	70% probability	NASA standard commitment level (NPR 7120.5F)

P80

80% probability

Conservative planning; typical for proposals

> *Rule of Thumb for CubeSat Missions:*
>
> *P70 ~ P50 x 1.2*
>
> *P80 ~ P50 x 1.3*
>
> *Example: If P50 = EUR 500K, then P80 ~ EUR 650K.*
>
> *This approximation assumes moderate complexity and well-understood COTS hardware. For missions with custom payloads or new technology, use P80 ~ P50 x 1.5.*

1U Worked Example: UniSat-1

Cost Breakdown: Simple WBS, Mostly COTS

UniSat-1's cost structure is fundamentally different from larger missions because (a) nearly all hardware is COTS, so NRE is near zero, and (b) the team is small and university-based, so labour costs are low.

> *UniSat-1 WBS Cost Estimate (Parametric vs Bottom-Up):*
>
> | WBS Element | Parametric (kEUR) | Bottom-Up (kEUR) | Notes |
> |---|---|---|---|
> | 1.0 Programme Management | 4 | 5 | Faculty oversight, 0.1 FTE x 12 months |
> | 2.0 Systems Engineering | 3 | 3 | Student team lead |
> | 3.0 Mission Assurance | 1 | 1 | Minimal QA for university mission |
> | 4.0 Payload | 8 | 8 | MEMS sensor PCB + calibration |
> | 5.0 Bus Hardware | 38 | 36 | See BOM from Session 4.1 |
> | 5.1 Structure | 4 | 4 | ISIS 1U frame |
> | 5.2 EPS + SA | 20 | 19.5 | P31us + body-mounted cells |
> | 5.3 AOCS (passive) | 1 | 1 | Magnet + hysteresis rods |
> | 5.4 Comms (UHF) | 11 | 10.5 | AX100 + antenna |
> | 5.5 OBC | 3 | 3 | Custom Cortex-M board |
> | 6.0 I&T | 8 | 8 | University clean room + vibe test facility |
> | 7.0 Software | 5 | 5 | FSW (FreeRTOS) + GSW |
> | 8.0 Launch | 15 | 15 | NanoRacks 1U ISS deployment |
> | 9.0 Ground Segment | 5 | 5 | Yagi antenna + SatNOGS network |
> | 10.0 Operations (6 months) | 3 | 3 | Student operators, 0.2 FTE |

> *| TOTAL (P50) | ~90 | ~85 | |*
>
> *| P80 (x 1.3) | ~117 | ~111 | Conservative estimate |*

Key cost observations for 1U missions:

1. Hardware is cheap: Total COTS hardware cost is ~36--44 kEUR. This is less than a single star tracker for a 3U mission.
2. NRE is minimal: Only the OBC and payload require custom development. NRE is estimated at ~8 kEUR (payload calibration + OBC board layout), compared to ~50--150 kEUR for custom payloads on larger missions.
3. Launch cost is proportionally large: At 15 kEUR, the launch represents ~18% of total cost. For a 3U mission at ~200 kEUR launch cost, launch is ~40% of total. The 1U launch cost is low in absolute terms but still a significant fraction.
4. Labour dominates: For a university team, the "free" student labour is the hidden cost. If students were costed at professional rates (~50 EUR/hr), the total labour cost would be ~100--200 kEUR, far exceeding the hardware cost. This is typical for educational missions.
5. No cost drivers from complexity: There is no AOCS software development, no deployable mechanism qualification, no propulsion system integration, no thermal vacuum testing of heaters -- all of which add 10--50 kEUR each on a 3U mission.

Learning curve applicability: If a university builds a series of 1U demonstrators (UniSat-1, UniSat-2, UniSat-3...), the 90% learning rate applies:

Unit	Hardware Cost (kEUR)	Total Cost (kEUR)
UniSat-1	44	85
UniSat-2	40	77
UniSat-4	36	69
UniSat-8	32	62

The asymptotic floor is dominated by launch cost (15 kEUR) and irreducible ground segment + operations costs (~8 kEUR), giving a minimum mission cost of ~40--50 kEUR for repeat builds.

---

5. Design Review Process (25 min)

Teaching Notes

Design reviews are formal decision gates where the project demonstrates readiness to proceed to the next lifecycle phase.

[Source: NPR 7123.1D Appendix G; ECSS-M-ST-10C Rev.1 section 6; NASA SEH Rev 2 section 3.7]
[URL: https://www.nasa.gov/reference/systems-engineering-handbook/]

Review Sequence

Review	Phase Transition	Key Question	Entry Criteria
MCR	Pre-A -> A	Is the mission need justified?	Problem statement, stakeholders, objectives defined
SRR	A -> B	Are requirements complete, consistent, traceable?	Requirements baselined, ConOps defined, feasibility shown
PDR	B -> C	Does preliminary design meet requirements with margin?	All budgets close, interfaces defined, risks identified
CDR	C -> D	Is detailed design complete and ready to build?	All drawings released, test plan approved, suppliers under contract
TRR	Pre-test	Is the system ready for environmental testing?	Assembly complete, procedures approved, facility booked
QR	Post-test	Has the system passed all tests?	All test reports approved, NCRs closed, V&V matrix complete
FRR	Pre-launch	Is everything ready for launch?	Shipping approved, launch manifest confirmed, operations ready

Gate Criteria for SRR (Phase A -> B)

#	Criterion	Priority	Evidence
1	All Level 0/1 requirements baselined	Must pass	Requirements document signed
2	ConOps defined (all mission phases)	Must pass	ConOps document
3	Mission architecture trades completed	Must pass	Trade study reports with rationale
4	Feasibility confirmed (all budgets positive)	Must pass	Mass, power, link budgets with margin
5	Risk register established with mitigations	Must pass	Risk register with scores and plans
6	Preliminary V&V approach defined	Should pass	V&V matrix with methods assigned
7	Schedule and cost estimate (P50/P80)	Must pass	Cost estimate with WBS
8	Interface requirements identified	Should pass	N^2 matrix or ICD outline

Gate Criteria for PDR (Phase B -> C)

#	Criterion	Priority	Evidence
1	All requirements allocated to subsystems	Must pass	Requirements allocation matrix
2	Preliminary design complete for all subsystems	Must pass	Design documents, block diagrams
3	All budgets close with >= 20% margin	Must pass	Mass, power, link, data, pointing budgets
4	Equipment selected (BOM)	Must pass	BOM with TRL, heritage, cost, lead time
5	All interfaces defined (N^2 matrix)	Must pass	Interface control documents
6	V&V matrix complete with methods assigned	Must pass	V&V matrix
7	Risk register updated; no Critical risks unmitigated	Must pass	Risk register
8	Test plan outline approved	Should	Environmental test plan
9	Software architecture defined	Should	Software design document
10	Cost estimate updated (bottom-up)	Must pass	Cost estimate with vendor quotes

Presentation Skills for Reviews

Do:

- Lead with the conclusion ("Mass budget closes with 22% margin")
- Show evidence, not just assertions ("Link budget analysis shows 4.2 dB margin at worst case")
- Acknowledge risks honestly ("Antenna deployment is our highest risk at L3 x C4 = 12")

- Answer questions directly; say "I don't know, we'll take an action" if needed

Do not:

- Read slides aloud
- Hide problems (review boards always find them)
- Present analysis without assumptions stated
- Skip backup slides -- have detailed data ready

---

6. Cost & Review Exercise (25 min)

Instructions

Part A: Cost Estimation (15 min)

1. Dashboard -- Check the Cost KPI card (total cost in MEUR)
2. Cost Breakdown tab -- Review the breakdown by subsystem
3. Exports tab -- Generate BOM and sum COTS component costs
4. On Worksheet 4.4:

- Fill in the WBS cost table using both parametric and bottom-up methods
- Compute P50 and P80 estimates
- If constellation: apply 90% learning curve

Part B: Design Review Preparation (10 min)

1. Open the Gate Review tab in SpaceCDF
2. Check all criteria: which are Pass/Fail/Manual?
3. For any failing criteria, click "Go fix" and resolve
4. Prepare a 3-minute summary of your design for peer review

Worksheet 4.4 Tasks

1. Build a complete WBS cost table (parametric and bottom-up columns)
2. Compute P50 and P80 total cost estimates
3. If constellation: compute total cost with learning curve applied
4. Identify top 3 cost drivers and propose 20% cost reduction
5. List the gate criteria for SRR/PDR and assess your readiness

---

Cited References

#	Reference	URL
1	NASA Cost Estimating Handbook v4.0	https://www.nasa.gov/offices/ocfo/references-and-tools/

2	SMAD4 Chapter 20 (Cost)	Wertz, Everett, Puschell (eds.), Space Mission Engineering, Microcosm 2011
3	Aerospace Corp SSCM	https://www.aerospace.org/capabilities/small-satellite-cost-model
4	NPR 7120.5F (WBS)	https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&c=7120&s=5F
5	ECSS-M-ST-60C (Cost Management)	https://ecss.nl/standard/ecss-m-st-60c-cost-and-schedule-management/
6	NPR 7123.1D (SE Processes) Appendix	https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&c=7123&s=1D
7	NASA SEH Rev 2, section 3.7	https://www.nasa.gov/reference/systems-engineering-handbook/

Session Summary

Topic	Key Takeaway
Cost methods	Parametric (early, CERs), Analogy (reference mission), Bottom-up (vendor quotes)
CubeSat costs	COTS pricing often lower than CER predictions; use hybrid approach
WBS	Standard 10-element structure: PM, SE, MA, Payload, Bus, I&T, SW, Launch, Ground, Ops
CERs	Cost = a x M^b; USCM/SSCM databases; CubeSats need calibrated CERs
Learning curve	90% rate: each doubling of quantity reduces unit cost by 10%
Cost risk	P50 (baseline), P70 (NASA commitment), P80 (conservative); P80 ~ P50 x 1.3
Reviews	SRR, PDR, CDR: formal gates with exit criteria; must pass before proceeding
Presentation	Lead with conclusions, show evidence, acknowledge risks, answer directly

Session 5.1: Ground Segment & Operations Architecture

Duration: 2 hours

Prerequisites: Week 2 complete (full design cycle through cost estimation)

References: ECSS-E-ST-70C (Ground Systems), ECSS-E-ST-70-01C (Ground Segment), CCSDS 131.0-B-4 (TM Coding), CCSDS 231.0-B-4 (TC Coding), CCSDS 133.0-B-2 (Space Packet Protocol), NASA DSN Handbook (810-005)

Learning Objectives

By the end of this session, participants will be able to:

1. Describe the components of a ground segment architecture (antennas, MCS, FDS, networks)
2. Calculate ground station contact time and data volume per pass
3. Explain the CCSDS protocol stack for space communication
4. Design a ground station network for a given orbit and data downlink requirement
5. Construct a mission operations timeline with key milestones

1. Ground Segment Architecture (25 min)

Teaching Notes

The ground segment is everything on the ground that supports the space mission. It is frequently underestimated in early mission design but accounts for 5-15% of total mission cost and is critical to mission success.

[Source: ECSS-E-ST-70C (Ground Systems and Operations); ECSS-E-ST-70-01C (Ground Segment)]
[URL: https://ecss.nl/standard/ecss-e-st-70c-ground-systems-and-operations/]

Ground Segment Components

```
+------------------------------------------------------------------+
|                        GROUND SEGMENT                            |
|                                                                  |
|  +-------------------+    +-------------------+    +-----------+  |
|  | Ground Station    |    | Mission Control   |    | Flight    |  |
|  | Network           |    | System (MCS)      |    | Dynamics  |  |
|  |                   |    |                   |    | System    |  |
|  | - Antenna(s)      |    | - TM processing |    | (FDS)     |  |
|  | - RF front-end    |    | - TC generation   |    |           |  |
|  | - Modem/baseband |    | - Scheduling      |    | - Orbit   |  |
|  | - Tracking        |    | - Monitoring      |    |   determ. |  |
|  | - Data capture    |    | - Anomaly mgmt   |    | - Maneuver|  |
|  +--------+---------+    +--------+---------+    |   planning|  |
|           |                       |               +-+---+-----+   |
|           |    Data Network       |                 |   |         |
|           +----------+------------+-----------------+   |         |
|                      |                                  |         |
|  +-------------------+v--+    +--------------------+     |         |
|  | Mission Planning      |    | Data Processing &  |     |         |
|  | System                |    | Archiving          |     |         |
|  | - Pass scheduling     |    | - Level 0-2 data   |     |         |
|  | - Resource planning   |    | - Archive/catalog  |     |         |
|  | - Conflict resoln.    |    | - Distribution     |     |         |
|  +-----------------------+    +--------------------+     |         |
+------------------------------------------------------------------+
```

Ground Station Components

Component	Function	Typical Specification
Antenna	Transmit uplink commands, receive downlink telemetry	3-13 m dish (S/X-band); Yagi/turnstile (UHF)
RF front-end	Low-noise amplifier (LNA), power amplifier, filters	LNA NF < 1 dB; PA 50-500 W
Modem/baseba	Modulation/demodulation, coding/decoding	CCSDS-compliant; BPSK/QPSK/8PSK
Tracking	Antenna pointing, Doppler tracking, ranging	Autotrack or program-track
Data capture	Record raw baseband data for post-processing	High-speed disk array
Timing	GPS-disciplined oscillator for time synchronisation	< 1 microsecond accuracy

Mission Control System (MCS)

Function	Description	Tools
Telemetry processing	Decode, decommutate, display, and archive TM packets	COSMOS, YAMCS, EGOS (ESA)
Telecommand generation	Create, validate, encode, and uplink TC packets	Same MCS tools
Pass scheduling	Schedule antenna time, plan contact windows	STK, GMAT, custom schedulers
Monitoring & control	Real-time health monitoring, limit checking, alarming	MCS dashboards
Anomaly management	Detect, diagnose, and resolve anomalies	Procedures + expert judgment

Flight Dynamics System (FDS)

Function	Description	Inputs
Orbit determination	Compute current orbit from tracking data	Range, Doppler, GPS (if onboard)
Orbit prediction	Propagate orbit forward for pass planning	Current state vector, perturbation models
Manoeuvre planning	Compute delta-V for orbit maintenance	Target orbit, propulsion model
Conjunction assessment	Predict close approaches with debris/other objects	TLE catalogue, own orbit
End-of-life planning	Compute deorbit manoeuvre or passivation plan	Remaining propellant, target orbit

---

2. Ground Station Contact Analysis (25 min)

Teaching Notes

Understanding how much data can be transferred per pass is fundamental to mission design. The data budget depends on the contact geometry, data rate, and pass duration.

Contact Geometry

For a circular LEO orbit, the visibility of a ground station is determined by the minimum elevation angle constraint.

> *Maximum Slant Range at Minimum Elevation:*
>
> *rho = R_E x (sqrt((h/R_E + 1)^2 - cos^2(epsilon)) - sin(epsilon))*
>
> *Where:*
>
> *- R_E = Earth radius (6371 km)*
>
> *- h = orbital altitude*
>
> *- epsilon = minimum elevation angle (typically 5-10 degrees)*

For h = 500 km, epsilon = 5 degrees:

> *rho = 6371 x (sqrt((500/6371 + 1)^2 - cos^2(5)) - sin(5))*
>
> *rho = 6371 x (sqrt(1.0785^2 - 0.9962^2) - 0.0872)*
>
> *rho = 6371 x (sqrt(1.1632 - 0.9924) - 0.0872)*
>
> *rho = 6371 x (sqrt(0.1708) - 0.0872)*
>
> *rho = 6371 x (0.4133 - 0.0872)*
>
> *rho = 6371 x 0.3261*
>
> *rho = 2077 km*

> *Maximum Pass Duration (overhead pass):*
>
> *T_max = 2 x R_E x arccos(R_E x cos(epsilon) / (R_E + h)) / V_ground_track*
>
> *Simplified approximation for LEO:*
>
> *T_max ~ (2 x rho_max) / V_orbital x (R_E / (R_E + h))*
>
> *For 500 km SSO:*
>
> *V_orbital = sqrt(mu / (R_E + h)) = sqrt(3.986e14 / 6.871e6) = 7613 m/s*
>
> *T_max ~ 2 x 2077e3 / 7613 x (6371/6871) ~ 505 s ~ 8.4 minutes*
>
> *Typical average pass duration (not all passes are overhead): ~6 minutes*

> *Data Volume per Pass:*
>
> *V_data = R_data x T_contact x eta_protocol*
>
> *Where:*
>
> *- R_data = downlink data rate (bps)*
>
> *- T_contact = contact duration (seconds)*
>
> *- eta_protocol = protocol efficiency (typically 0.85-0.95 for CCSDS)*
>
> *Example: R = 2 Mbps, T = 360 s (6 min average), eta = 0.9:*
>
> *V_data = 2e6 x 360 x 0.9 = 648 Mbit = 81 MB per pass*

Contact Frequency

Orbit	GS Latitude	Passes per Day	Avg Duration	Daily Data Volume (2 Mbps)
500 km SSO	52 N (e.g., Waterloo)	3-5	6 min	243-405 MB
500 km SSO	69 N (e.g., Kiruna)	6-8	7 min	544-726 MB
500 km, 51.6 (ISS)	52 N	2-4	5 min	130-260 MB
500 km SSO	SatNOGS network (global)	10-15	5 min	648-972 MB

[Source: STK simulations; validated against SatNOGS observation logs]

Worked Example: Data Budget Closure

Problem: An Earth observation mission generates 500 MB/day of imagery. Is a single ground station at 52 N latitude sufficient with 2 Mbps S-band downlink?

Calculation:

- Passes per day: ~4 (500 km SSO from 52 N)
- Avg pass duration: 6 min = 360 s
- Data per pass: 2e6 bps x 360 s x 0.9 / 8 = 81 MB
- Daily capacity: 4 x 81 = 324 MB
- Required: 500 MB/day
- Shortfall: 176 MB/day -- budget does not close!

Solutions:

1. Add a second ground station at high latitude (e.g., Kiruna -> +6 passes -> +486 MB)
2. Increase data rate to X-band (10 Mbps -> 405 MB/pass -> easily sufficient)
3. Reduce data generation (lower duty cycle or reduce image resolution)
4. Add onboard data compression (2:1 lossless -> 250 MB/day required)
5. Use SatNOGS network for additional UHF passes (housekeeping only)

---

3. CCSDS Protocol Stack (20 min)

Teaching Notes

The Consultative Committee for Space Data Systems (CCSDS) defines the standard protocols for space communication, analogous to TCP/IP for ground networks.

[Source: CCSDS 131.0-B-4 (TM Synchronization and Channel Coding); CCSDS 231.0-B-4 (TC Synchronization and Channel Coding); CCSDS 133.0-B-2 (Space Packet Protocol)]

[URL: https://public.ccsds.org/Pubs/131x0b4.pdf (freely available)]

Protocol Layers

Layer	CCSDS Standard	Function	Ground Analogy
Applicatio	CCSDS 133.0 (Space Packet)	Mission data and housekeeping packets	HTTP/application data
Network	CCSDS 732.0 (AOS) or 132.0 (TM)	Virtual channels, multiplexing	IP routing

Data Link	CCSDS 131.0 (TM coding) / 231.0 (TC coding)	FEC, frame sync, CRC	Ethernet
Physical	CCSDS 401.0 (RF & Modulation)	Modulation, frequency, power	Physical layer

Telemetry (TM) Frame Structure

```
+--------+-----------------+----+
| Header | Data Field      | EC |
| 6 bytes| 1-1019 bytes    | 2B |
+--------+-----------------+----+
  |
  +-- Spacecraft ID (10 bits)
  +-- Virtual Channel ID (3 bits)
  +-- Frame Counter (8 bits)
  +-- Frame Length
```

Forward Error Correction (FEC) Options

Code	Rate	Coding Gain (dB)	Use Case
Convolutional (7, 1/2)	1/2	5.5	Legacy CubeSats
Reed-Solomon (255, 223)	0.87	3.5	Combined with convolutional
Turbo (rate 1/2)	1/2	7.5	High-performance CubeSats
LDPC (rate 7/8)	7/8	6.0	High data rate, bandwidth efficient

Modern CubeSats typically use LDPC coding for downlink (high data rate) and convolutional + RS for uplink (robustness priority).

---

4. Operations Timeline Construction (25 min)

Teaching Notes

The operations timeline (or operations concept timeline) defines all major activities from launch to end of life. It is part of the ConOps and drives ground segment design.

Mission Phases (Operations Perspective)

Phase	Duration	Key Activities	Staffing
LEOP	0-72 hours	Separation, beacon acquisition, SA/antenna deploy, initial health check	24/7 (3 shifts)
Commissioning	1-4 weeks	Subsystem checkout, ADCS calibration, payload first light	16/7 (2 shifts)
Early Operations	1-3 months	Performance characterisation, procedure refinement	8/5 (1 shift)
Nominal Operations	Mission lifetime	Routine science/service, orbit maintenance	8/5 or automated
Extended	If approved	Degraded mode operations, reduced data	Part-time
End of Life	1-4 weeks	Passivation, deorbit manoeuvre, final telemetry	8/5

Operations Timeline (Gantt-Style)

```
Week:  1    2    3    4    5    6    7    8   ...  52
       |LEOP|
       |    |Commissioning         |
       |    |    |    |    |Early Operations    |
       |    |    |    |    |    |    |    |Nominal Operations ------>
       |    |    |    |    |    |    |    |    |    |    |    |    |
24/7:  XXXX
16/7:       XXXXXXXXXXXXXXX
8/5:                        XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX...
```

LEOP Timeline Detail

Time (after separation)	Activity	Success Criterion
T+0 s	Separation from deployer	Deployment switches released
T+0 to T+30 min	Deployment timer countdown (no RF)	Timer runs; all inhibits clear
T+30 min	Antenna deployment	Beacon detected by ground station
T+30 to T+60 min	Beacon acquisition	Carrier lock; beacon decoded
T+1 hr	First telemetry downlink	Housekeeping data received and validated
T+1 to T+6 hr	Initial health assessment	All subsystems reporting nominal HK
T+6 to T+12 hr	Solar array deployment (if separate)	Power generation confirmed
T+12 to T+24 hr	ADCS initialisation	Attitude determination active
T+24 to T+48 hr	ADCS calibration (magnetometer, sun sensor)	Pointing within coarse spec
T+48 to T+72 hr	Communication chain validation (full duplex)	Uplink and downlink at operational rate

SpaceCDF Operations Planning

SpaceCDF's ConOps Editor allows you to:

- Define mission phases with start/end conditions
- Specify operational modes per phase (safe, nominal, science, downlink)
- Set ground contact requirements per phase (frequency, duration)
- The system validates that the ground segment design supports the operations concept

---

5. Ground Segment Design Exercise (25 min)

Instructions

1. Data Budget tab -- Review the daily data generation vs downlink capacity
- Is the data budget closing? If not, identify the bottleneck
- How many ground station passes per day does your orbit provide?

2. ConOps Editor -- Define mission phases:
- LEOP (duration, staffing, contact requirements)

- Commissioning (activities, success criteria)
- Nominal operations (duty cycle, data volume, contact frequency)

3. On Worksheet 5.1:
- Calculate data volume per pass for your mission
- Determine number of ground stations needed to close the data budget
- Construct a LEOP timeline
- Draft a simplified operations timeline (Gantt chart)

Discussion Prompts

- "What happens if your first pass after deployment has no signal?"
- "How would you handle a safe mode entry at 3 AM on a Saturday?"
- "What is the minimum ground segment that could support your mission?"

Worksheet 5.1 Tasks

1. Calculate ground station contact time and data volume per pass
2. Design a ground station network (locations, antenna sizes, data rates)
3. Construct a LEOP timeline (first 72 hours, hourly resolution)
4. Draft a mission operations timeline (Gantt chart, weekly resolution)
5. Estimate ground segment cost (antennas + MCS + FDS + operations staff)

Cited References

#	Reference	URL
1	ECSS-E-ST-70C (Ground Systems and Operations)	https://ecss.nl/standard/ecss-e-st-70c-ground-systems-and-operations/
2	CCSDS 131.0-B-4 (TM Synchronization and Channel Coding)	https://public.ccsds.org/Pubs/131x0b4.pdf
3	CCSDS 133.0-B-2 (Space Packet Protocol)	https://public.ccsds.org/Pubs/133x0b2.pdf
4	CCSDS 231.0-B-4 (TC Synchronization and Channel Coding)	https://public.ccsds.org/Pubs/231x0b4.pdf
5	SatNOGS Network	https://network.satnogs.org/
6	NASA DSN Handbook (810-005)	https://deepspace.jpl.nasa.gov/dsndocs/810-005/
7	YAMCS Mission Control Software	https://yamcs.org/

Session Summary

Topic	Key Takeaway
Ground segment	Antenna network + MCS + FDS + mission planning + data processing
Contact analysis	Pass duration ~6 min (LEO/SSO); data volume = rate x time x efficiency

Data budget	Must close: daily generation <= daily downlink capacity; add GS or increase rate
CCSDS	Standard protocol stack: Space Packet, AOS/TM frames, FEC coding, RF modulation
Operations timeline	LEOP (24/7) -> Commissioning (16/7) -> Nominal (8/5 or automated)
LEOP	First 72 hours critical: antenna deploy, beacon acquisition, health check
SpaceCDF	ConOps Editor defines phases; Data Budget validates ground segment sizing

Session 5.2: Mission Operations Concepts

Duration: 2 hours

Prerequisites: Session 5.1 (ground segment architecture understood)

References: ECSS-E-ST-70-11C (Space Segment Operability), ECSS-E-ST-70-32C (Procedures), ECSS-E-ST-70-41C (Packet Utilisation Standard), NPR 7120.5F, NASA Fault Management Handbook (NASA-HDBK-1002)

Learning Objectives

By the end of this session, participants will be able to:

1. Define a Concept of Operations (ConOps) with operational modes and transitions
2. Design an FDIR (Fault Detection, Isolation, Recovery) architecture
3. Write operational procedures for routine and contingency operations
4. Describe anomaly response process with escalation levels
5. Plan spacecraft operational modes and their entry/exit criteria

1. Concept of Operations (ConOps) (25 min)

Teaching Notes

The ConOps is the bridge between engineering design and operational reality. It describes HOW the system will be used, not just WHAT it can do.

[Source: ECSS-E-ST-70-11C (Space Segment Operability); IEEE 1362 (Concept of Operations Document)]

ConOps Structure

A complete ConOps document covers:

1. Mission overview -- Objectives, stakeholders, success criteria
2. System description -- Space segment, ground segment, user segment
3. Operational scenarios -- Nominal operations, contingency operations
4. Operational modes -- Definition, transitions, constraints
5. Resource management -- Power, data, propellant budgets over time
6. Communication plan -- Contact schedule, data flow, latency requirements
7. Staffing plan -- Personnel, shifts, training requirements
8. Maintenance and logistics -- Software updates, ground equipment maintenance

Operational Modes

Every spacecraft has a defined set of operational modes. Each mode specifies which subsystems are active, power consumption, and data rates.

Mode	Description	Power (W)	Data Rate	Typical Duration	Entry Condition
Safe	Minimum functionality; survival only	5-10	Beacon only (1 bps)	Until ground intervenes	Autonomous fault detection
Detumble	Stop rotation after deployment	8-12	HK only (100 bps)	Minutes to hours	Post-separation; high angular rate
Standby	Sun-pointing; all subsystems ready	12-18	HK telemetry (1 kbps)	Between activities	After detumble; idle periods
Science	Payload operating; attitude controlled	25-40	Science data (variable)	Target-dependent	Scheduled; attitude achieved
Downlink	High-rate data transfer to ground	20-30	Full rate (2-10 Mbps)	During GS contact	GS in view; link established
Manoeu	Orbit adjustment or desaturation	15-25	HK only	Minutes	Scheduled; constraints met
Eclipse	Reduced operations during eclipse	8-15	HK only (100 bps)	~35 min (LEO)	Sun not in view

Mode Transition State Machine

```
                     +----------+
       Separation    |          |    Fault
    +--------------->| DETUMBLE |<-----------+
    |                |          |            |
    |                +----+-----+            |
    |                     | Angular rate     |
    |                     | < threshold      |
    |                +----v-----+            |
    |                |          |    Fault   |
    |     +--------->| STANDBY  |----------->+--------+
    |     |          |          |<-----+     |        |
    |     |          +--+----+--+      |     |  +-----v----+
    |     |             |    |         |     |  |          |
    |     |    Schedule |    | GS      |     +--+  SAFE    |
    |     |             |    | contact |        |  MODE    |
    |     |       +-----v-+  +---v----++        |          |
    |     |       |       |  |        |         +-----+----+
    |     +-------+SCIENCE|  |DOWNLINK|               |
    |    Complete |       |  |        |      Ground    |
    |             +-------+  +--------+      command   |
    |                                        recovery  |
```

```
+---------------------------------------------------+
```

Real Mission Example: PROBA-2 Operations

ESA's PROBA-2 (launched 2009, still operational in 2026) demonstrates mature CubeSat-class operations:

- 5 operational modes: Standby, Science, Manoeuvre, Safe, Off
- Autonomous operations: 95% of routine activities automated via onboard timeline
- Ground contacts: 4-6 passes/day via ESA Redu station (S-band)
- Anomaly rate: ~2-3 per year requiring ground intervention (after initial commissioning)
- Key lesson: Investing in onboard autonomy dramatically reduces operations cost

[Source: ESA PROBA-2 Operations Team, "PROBA-2: Over a Decade of Operations", SpaceOps 2020]

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2. FDIR Architecture (30 min)

Teaching Notes

FDIR (Fault Detection, Isolation, Recovery) is the spacecraft's autonomous ability to detect failures, determine which component failed, and take corrective action -- all without ground intervention.

[Source: NASA Fault Management Handbook (NASA-HDBK-1002); ECSS-E-ST-70-11C section 5.3]

[URL: https://standards.nasa.gov/standard/NASA/NASA-HDBK-1002]

Why FDIR is Critical for CubeSats

- Limited ground contact: LEO CubeSats are out of view ~85% of the time
- No crewed intervention: Unlike ISS, there is no astronaut to "fix" problems
- Thermal constraints: A safe mode must maintain thermal limits within minutes
- Power constraints: Battery can discharge completely in one eclipse without load shedding

FDIR Hierarchy (Levels)

Level	Detection	Response	Latency	Example
0 --	Built-in watchdog	Component reset	Milliseconds	Processor watchdog timer
1 -- Unit	Unit-level monitoring	Unit reconfiguration	Seconds	EPS over-current trip
2 --	Subsystem health check	Subsystem fallback mode	Seconds-minu	Switch to backup star tracker
3 -- System	System-level anomaly detection	Mode change (e.g., enter Safe)	Minutes	Transition to Safe Mode
4 -- Ground	Human-in-the-loop diagnosis	Ground command recovery	Hours-days	Anomaly investigation + patch

FDIR State Machine

```
                  NOMINAL OPERATIONS
                         |
            Fault detected (Level 2-3)
                         |
                  +------v------+
                  |  FAULT      |
                  |  DETECTED   |
                  +------+------+
                         |
            +------------+------------+
            |                         |
      Fault isolated            Cannot isolate
            |                         |
      +-----v-----+           +-------v------+
      | ISOLATION  |           | SAFE MODE    |
      | SUCCESS    |           | (Level 3)    |
      +-----+------+           +-------+------+
            |                          |
      Recovery action            Wait for ground
            |                    (Level 4)
      +-----v-----+                   |
      | RECOVERY   |           +------v-------+
      | ATTEMPT    |           | GROUND       |
      +-----+------+           | RECOVERY     |
            |                  +------+-------+
       Success / Fail                 |
            |                    Command sequence
      +-----v-----+                   |
      | RETURN TO  |<----------------+
      | NOMINAL    |
      +------------+
```

Common CubeSat FDIR Rules

Fault	Detection Method	Threshold	Response
Processor lockup	Watchdog timer	No heartbeat for 60 s	Hardware reset (Level 0)
Battery under-voltage	EPS voltage monitor	V_bat < 6.0V	Load shedding; enter Safe Mode
Battery over-temperature	Temperature sensor	T_bat > 45 C	Disable charging; reduce loads
Attitude loss	No attitude solution for 5 min	ADCS health flag timeout	Enter Detumble Mode
Communication loss	No uplink for 48 hr	Timer-based	Reset TTC; revert to beacon mode
Memory corruption	EDAC error count	> 10 uncorrectable errors/hr	Memory scrub; reboot from backup
Solar array current anomaly	Current sensor	I_SA < expected - 30%	Check attitude; enter Standby
Reaction wheel over-speed	Wheel speed monitor	Omega > 6000 RPM	Reduce momentum; desaturation

Key Design Principles

1. Safe Mode must always work -- It is the last resort; it must be tested exhaustively
2. Fail operational, then fail safe -- Try to continue the mission before giving up
3. No single fault should cause mission loss -- Cross-reference with FMECA
4. Ground override capability -- Every autonomous action must be commandable from ground
5. Logging -- Record all fault events in non-volatile memory for post-analysis

3. Operational Procedures (20 min)

Teaching Notes

[Source: ECSS-E-ST-70-32C (Procedure Definition Language); ECSS-E-ST-70-01C section 5.6]

Procedure Types

Type	Purpose	Execution	Example
Nominal	Routine scheduled operations	Automated (timeline) or manual	Science observation sequence
Contingen	Response to known anomaly types	Semi-automated; operator decision	Safe mode recovery
Emergenc	Response to critical unexpected events	Manual; real-time decision	Communication loss recovery
Maintenan ce	Periodic calibration or updates	Scheduled manual	Magnetometer calibration

Procedure Structure (ECSS-E-ST-70-32C)

Each procedure has:

1. Identifier -- Unique procedure number (e.g., NOM-001)
2. Purpose -- What the procedure accomplishes
3. Preconditions -- System state required before starting
4. Steps -- Ordered list of actions, verifications, decisions
5. Expected results -- What should happen at each step
6. Recovery actions -- What to do if expected results are not observed
7. Post-conditions -- System state after successful completion

Example: Science Observation Procedure

```
Procedure: NOM-SCI-001 "Earth Observation Target Acquisition"
Preconditions: Mode = Standby, Battery SoC > 60%, Target in FOV within 10 min

Step 1: Verify ADCS mode = Fine Pointing
  Expected: ADCS status = FINE, pointing error < 0.1 deg
  If not: Wait 60 s, retry. If still not achieved -> ABORT, remain in Standby

Step 2: Command payload power ON
  Expected: Payload HK reports power = ON within 5 s
  If not: Retry power command. If 3 fails -> ABORT, log anomaly

Step 3: Wait for payload thermal stabilisation (90 s)
  Expected: Payload temperature within operational range

Step 4: Command payload to imaging mode
  Expected: Payload status = IMAGING, frame counter incrementing

Step 5: Wait for target acquisition (ground-predicted start time)
  Expected: Imaging starts at predicted time +/- 5 s
```

```
Step 6: Collect imagery for scheduled duration

Step 7: Command payload to standby
  Expected: Payload status = STANDBY

Step 8: Command payload power OFF (if no more targets this orbit)
  Expected: Payload HK reports power = OFF

Post-conditions: Mode = Standby, imagery stored in mass memory
```

---

4. Anomaly Response Process (20 min)

Teaching Notes

[Source: NASA-HDBK-1002 (Fault Management Handbook); ESA Anomaly Review Board procedures]

Anomaly Escalation Levels

Level	Severity	Response Time	Authority	Example
Green	Informational	Next business day	Operations engineer	Minor parameter out of expected range
Yellow	Warning	Within 4 hours	Lead operator	Subsystem performance degraded
Orange	Serious	Within 1 hour	Flight director	Mode change required; mission impact
Red	Critical	Immediate	Project manager + FD	Mission-threatening; safe mode entered

Anomaly Response Workflow

```
1. DETECT: Telemetry alarm or operator observation
2. ASSESS: Is this a known anomaly type? Check procedures library
   -> Known: Execute contingency procedure
   -> Unknown: Continue to step 3
3. DIAGNOSE: Gather additional telemetry, correlate events, form hypothesis
4. PLAN: Develop response plan (may require Anomaly Review Board)
5. EXECUTE: Implement recovery commands (verify each step)
6. VERIFY: Confirm system returned to expected state
7. DOCUMENT: Record anomaly, root cause, response, and lessons learned
```

Anomaly Report Form Fields

Field	Description
Anomaly ID	Unique identifier (e.g., ANO-2026-042)
Date/Time (UTC)	When the anomaly was detected
Severity	Green / Yellow / Orange / Red
Affected subsystem	EPS, AOCS, TTC, Payload, etc.

Observation	What was observed (TM values, trends)
Diagnosis	Root cause analysis
Action taken	Commands sent, procedures executed
Result	System state after response
Status	Open / Resolved / Under investigation
Lessons learned	Preventive measures for future

Real Mission Example: Kepler Reaction Wheel Failure

NASA's Kepler space telescope lost reaction wheel #2 in July 2012, followed by wheel #4 in May 2013. The anomaly response:

1. Detection: Elevated friction torque in wheel #2 HK telemetry
2. Assessment: Known degradation signature; monitoring intensified
3. Failure: Wheel #2 ceased operation. Mission continued on 3 wheels.
4. Second failure: Wheel #4 failed 10 months later. Only 2 wheels remaining -- insufficient for fine pointing.
5. Recovery: Engineering team developed the "K2" mission concept using solar radiation pressure as a "virtual third wheel"
6. Result: K2 operated for 4+ additional years, discovering 2,700+ exoplanet candidates

Lesson: Creative operational workarounds can save missions. FDIR should be designed with graceful degradation, not just "safe mode or nothing."

[Source: Howell et al., "The K2 Mission: Characterization and Early Results", PASP, 2014]

---

5. Operations Concepts Exercise (25 min)

Instructions

1. ConOps Editor in SpaceCDF:
- Define all operational modes for your mission (at least 5)
- Set entry/exit conditions for each mode
- Verify power budget closes in each mode (especially Safe and Eclipse)

2. FDIR Design:
- Define 5 FDIR rules for your mission (fault, detection, threshold, response)
- Verify that Safe Mode power budget is sustainable indefinitely
- Check: Can every FDIR response be overridden from ground?

3. Worksheet 5.2:
- Complete the operational modes table
- Write one nominal procedure (science observation or downlink)
- Write one contingency procedure (safe mode recovery)
- Fill in the FDIR rules table

Discussion Prompts

- "What is the worst anomaly that could happen to your satellite? How would you respond?"
- "If the satellite enters safe mode on Friday evening and the next ground contact is Monday, will it survive?"
- "How much autonomy should the spacecraft have? Where is the line between onboard and ground decision-making?"

Worksheet 5.2 Tasks

1. Complete the operational modes table (5+ modes with power, data rate, entry/exit)
2. Design the FDIR state machine (at least 5 rules)
3. Write one nominal and one contingency procedure
4. Complete the anomaly response form for a hypothetical scenario
5. Estimate operations staffing requirements by mission phase

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Cited References

#	Reference	URL
1	ECSS-E-ST-70-11C (Space Segment Operability)	https://ecss.nl/standard/ecss-e-st-70-11c-space-segment-operability/
2	ECSS-E-ST-70-32C (Procedures)	https://ecss.nl/standard/ecss-e-st-70-32c-test-and-operations-procedure-language/
3	NASA Fault Management Handbook (NASA-HDBK-1002)	https://standards.nasa.gov/standard/NASA/NASA-HDBK-1002
4	ESA PROBA-2 Operations, SpaceOps	https://www.spaceops.org/
5	Howell et al., K2 Mission, PASP 2014	https://doi.org/10.1086/676406
6	ECSS-E-ST-70-41C (Packet Utilisation Standard)	https://ecss.nl/standard/ecss-e-st-70-41c-telemetry-and-telecommand-packet-utilization/

Session Summary

Topic	Key Takeaway
ConOps	Defines HOW the system is operated: modes, transitions, schedules, staffing
Operational modes	Safe, Detumble, Standby, Science, Downlink, Manoeuvre, Eclipse (minimum set)
Mode transitions	Defined by entry/exit conditions; Safe Mode always reachable from any mode
FDIR	5 levels: Hardware -> Unit -> Subsystem -> System -> Ground; Safe Mode is last resort
Procedures	Nominal (routine), Contingency (known faults), Emergency (unknown), Maintenance
Anomaly response	Detect -> Assess -> Diagnose -> Plan -> Execute -> Verify -> Document
Design principles	Safe Mode must always work; fail operational before fail safe; ground override always

---

Session 5.3: Mission Simulation Day

Duration: 2 hours

Prerequisites: Sessions 5.1-5.2 (ground segment designed, operations concepts defined)

References: ECSS-E-ST-70-11C (Operability), ECSS-E-ST-70-32C (Procedures), ESA OPS-G Training Manual, NASA Mission Operations Directorate Training Handbook

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Learning Objectives

By the end of this session, participants will be able to:

1. Execute a simulated LEOP sequence from separation through first telemetry
2. Perform commissioning activities using operational procedures
3. Respond to injected anomalies using the FDIR architecture and contingency procedures
4. Document anomaly response in real-time using structured forms
5. Conduct nominal science operations with live data budget tracking

---

1. Simulation Overview & Roles (10 min)

Teaching Notes

This session is a hands-on simulation of mission operations. The facilitator acts as the "Universe" -- injecting events, anomalies, and time jumps. The team operates the mission using SpaceCDF and their procedures from Session 5.2.

Role Assignments

Role	Responsibility	SpaceCDF Feature Used
Flight Director (FD)	Overall decision authority; coordinates team	Dashboard, all tabs
Spacecraft Controller (SC)	Sends commands; monitors housekeeping telemetry	ConOps Editor, mode control
Ground Station Operator (GSO)	Manages antenna scheduling; link acquisition	Data Budget, Link Budget
Payload Operator (PO)	Plans and executes science observations	Payload parameters, data budget
Flight Dynamics (FD-nav)	Monitors orbit; plans manoeuvres (if propulsion)	Orbit parameters, sustainability
Anomaly Logger	Documents all events, anomalies, and decisions	Worksheet 5.3

For smaller teams, combine roles: SC+GSO and PO+FD-nav are natural pairings.

Simulation Timeline

Sim Time	Real Time	Phase	Events
T+0 to T+1 hr	0-20 min	LEOP	Separation, beacon acquisition, deploy, health check
T+1 hr to T+24 hr	20-35 min	Early LEOP	SA deploy, ADCS init, comms chain validation
T+1 day to T+7 days	35-50 min	Commissioning	Subsystem checkout, first payload activation
T+7 days to T+30 days	50-65 min	Early Operations	Nominal science, orbit maintenance
Anomaly injection	65-85 min	Contingency	Anomaly response exercise
T+30 days to T+1 year	85-100 min	Nominal Operations	Routine operations, data review
Debrief	100-120 min	Wrap-up	Lessons learned, documentation review

2. LEOP Simulation (20 min)

Teaching Notes

The facilitator narrates the simulation. Time is compressed. Each team executes their LEOP procedure.

Facilitator Script -- LEOP

[T+0 -- Separation]

"Your satellite has separated from the deployer. Deployment switches have released. The 30-minute timer has started. You are in radio silence -- no RF emissions allowed until the timer expires. What is your spacecraft doing right now?"

Expected answers: Timer counting down. Battery providing power. No subsystems active except OBC running timer.

[T+30 min -- Timer Expires]

"The 30-minute timer has expired. Your OBC has commanded antenna deployment. What do you expect to see?"

Expected: Antenna deployment command sent. Beacon should begin transmitting.

[T+35 min -- First Pass Over Ground Station]

"Your ground station at [location] has line of sight. The antenna operator reports: carrier detected at [frequency] with Doppler of -25 kHz. Signal strength is -110 dBm. Is this what you expected?"

Teams should verify:

- Carrier frequency matches expected (accounting for Doppler)
- Signal strength is consistent with link budget at current range
- Beacon data rate is as designed

[T+40 min -- First Telemetry]

"Beacon lock achieved. You are receiving housekeeping telemetry. I am going to read you the first HK frame:"

Parameter	Value	Expected Range	Status

Bus voltage	7.8 V	7.0 - 8.4 V	Nominal
Battery SoC	72%	> 50%	Nominal
Solar array current	0.45 A	0.3 - 0.6 A	Nominal
OBC temperature	22 C	-10 to +50 C	Nominal
Battery temperature	19 C	0 to 45 C	Nominal
ADCS mode	Detumble	Expected	Nominal
Angular rate	3.2 deg/s	Decreasing	Nominal
Antenna status	Deployed	Expected	Nominal
Beacon mode	Active	Expected	Nominal
Uplink status	Not yet attempted	--	--

"All parameters nominal. Proceed with LEOP sequence."

[T+1 hr -- Uplink Validation]

"You now attempt your first uplink command. Send a 'NOP' (no operation) command to verify the uplink chain."

Teams should: compose command, verify encoding, transmit, wait for acknowledgement in telemetry.

---

3. Commissioning Simulation (15 min)

Facilitator Script -- Commissioning

[Time Jump: T+24 hr]

"Your satellite has completed detumbling. ADCS is in sun-pointing mode. All HK parameters are nominal after 8 ground passes. You are ready to begin commissioning. What is your first activity?"

Expected: ADCS calibration (magnetometer bias estimation, sun sensor alignment verification).

[T+3 days -- Payload First Light]

"ADCS is calibrated and achieving 0.3 degree pointing. You are ready for payload first light. Walk me through your procedure."

Teams execute their science observation procedure from Worksheet 5.2.

[T+5 days -- First Data Download]

"Payload has captured 150 MB of imagery. Your next pass is 12 minutes long over [station]. Can you download all the data in one pass?"

Teams should calculate:

- Data rate x pass duration x protocol efficiency = available capacity
- Compare to 150 MB -- determine if multiple passes needed
- Plan the download sequence

---

4. Anomaly Injection Scenarios (30 min)

Teaching Notes

The facilitator selects 2-3 anomalies from the list below, appropriate to the mission design. Teams respond using their FDIR rules and contingency procedures.

Anomaly Menu (Facilitator selects 2-3)

Anomaly A: Battery Under-Voltage

"During eclipse, you receive an alert: battery voltage has dropped to 6.2V, below your 6.5V warning threshold. Battery temperature is 5 C. SoC is estimated at 18%."

Expected response:

1. Verify FDIR has triggered load shedding (payload off, non-essential heaters off)
2. Check power budget: is consumption > generation?
3. Possible causes: SA degradation, unexpected load, battery degradation, long eclipse
4. Immediate action: Verify safe mode; disable non-essential loads manually
5. Investigation: Review power trends over last 24 hours; compare to power budget

Anomaly B: Communication Loss

"You have not received telemetry for the last 3 scheduled passes (18 hours). The ground station reports pointing and frequency are correct but no carrier detected."

Expected response:

1. Verify ground station equipment (antenna, LNA, modem) -- rule out ground fault
2. Check with other ground stations (SatNOGS) for any signal detection
3. Consider causes: TX failure, antenna deployment failure, attitude loss (antenna not pointing)
4. If 48-hour timer exists: satellite may reset TTC and revert to beacon mode
5. Plan: Wait for timer-based reset; use wide-beam ground antenna for beacon search

Anomaly C: ADCS Anomaly -- Loss of Fine Pointing

"Pointing error has increased from 0.1 deg to 5 deg. Star tracker reports 'no solution'. Reaction wheels are at 80% capacity (near saturation)."

Expected response:

1. Diagnose: Star tracker failure? Optical contamination? Sun in FOV?
2. Check magnetometer attitude -- is coarse attitude correct?
3. Immediate: Command desaturation using magnetorquers
4. If star tracker failed: switch to coarse pointing mode (sun sensors + magnetometer)
5. Impact assessment: Can mission science continue in coarse pointing mode?

Anomaly D: Unexpected Safe Mode Entry

"Your satellite has autonomously entered Safe Mode. Last telemetry before entry showed: OBC reboot counter incremented by 3 in 10 minutes. All other parameters were nominal."

Expected response:

1. Investigate: Multiple reboots suggest SEU (Single Event Upset) or firmware bug
2. Check radiation environment: Was the satellite in SAA (South Atlantic Anomaly)?
3. Review FDIR logs (stored in non-volatile memory) for fault triggers
4. Recovery plan: Command exit from Safe Mode; monitor for recurrence
5. If recurring: Upload software patch to increase SEU robustness

Anomaly E: Thermal Exceedance

"Payload temperature has reached 58 C during a science observation, exceeding the 55 C operational limit. Battery temperature is 38 C (approaching 45 C limit)."

Expected response:

1. Immediate: Disable payload to stop heat generation
2. Check attitude: Is the hot radiator face pointing toward the sun?
3. Review thermal model: Was this orbit geometry (high beta angle) predicted?
4. Mitigation: Adjust duty cycle; plan observations for cooler orbit phases
5. If structural: May need to update thermal model and operational constraints

Anomaly Documentation

For each injected anomaly, teams complete the Anomaly Response Form on Worksheet 5.3:

Field	What to Record
Time (sim)	When the anomaly was detected
Observation	Exact telemetry values and symptoms
Initial assessment	Severity level (Green/Yellow/Orange/Red)
Diagnosis	Suspected root cause
Action taken	Commands sent, procedures executed
Result	System state after response
Follow-up required	Additional investigation, procedure updates

5. Nominal Operations & Debrief (25 min)

Nominal Operations (10 min)

[Time Jump: T+30 days to T+1 year]

"Your mission has been operational for one month. All commissioning activities are complete. You are now in nominal operations. Let's review your operational metrics."

Teams report:

- Total data downlinked in 30 days vs plan
- Number of science observations completed vs plan
- Any anomalies encountered (from injection exercise)
- Current budget status: power margin, data margin, propellant remaining (if applicable)
- Orbit status: any conjunction alerts? Debris compliance on track?

Debrief (15 min)

1. What went well?
- Which procedures worked as written?
- Which FDIR rules triggered correctly?
- Where did the team communicate effectively?

2. What could be improved?
- Which procedures needed real-time modification?
- Were there gaps in the FDIR design?
- Where did decision-making slow down?

3. Lessons for the design:
- Did the simulation reveal any design weaknesses?
- Should any requirements be changed based on operational experience?
- What additional autonomy would help?

Facilitator Notes

Preparation

- Review each team's ConOps, FDIR rules, and procedures from Session 5.2
- Select anomalies appropriate to each team's mission (e.g., don't inject propulsion anomaly for a mission without propulsion)
- Prepare HK data tables for each phase (modify the values in Section 2 to match the team's mission parameters)
- Have backup anomalies ready if teams resolve injections quickly

Pacing

- LEOP simulation should feel time-pressured (short passes, decisions needed quickly)
- Commissioning can be more relaxed
- Anomaly injection should be challenging but solvable with the team's procedures
- Allow extra time for debrief -- this is where the deepest learning occurs

Assessment Criteria

Observe and note (for feedback, not formal grading):

- Did the team follow their procedures or improvise?
- Did the Flight Director coordinate effectively?
- Was the anomaly response systematic (detect-diagnose-plan-execute-verify)?
- Did the team document events in real-time?
- Did anyone identify a design flaw during operations?

Session Summary

Topic	Key Takeaway
LEOP	First 72 hours are critical; every step is time-constrained
Commissioning	Systematic subsystem checkout before payload operations
Anomaly response	Follow the process: detect, assess, diagnose, plan, execute, verify, document
Documentation	Real-time logging is essential -- memory is unreliable under stress
FDIR validation	Simulation reveals gaps in FDIR design that analysis alone cannot find
Team coordination	Clear roles, communication protocols, and decision authority are essential
Design feedback	Operational experience should feed back into design (requirements, FDIR, procedures)

Session 5.4: Final Review & Presentations

Duration: 2 hours

Prerequisites: All previous sessions (complete design through simulation)

References: ECSS-M-ST-10C Rev.1 section 6 (Reviews), NPR 7123.1D Appendix G, NASA SEH Rev 2 section 3.7, ECSS-E-ST-10C section 4 (Technical Dossier)

Learning Objectives

By the end of this session, participants will be able to:

1. Prepare a complete design documentation package for review
2. Conduct a design review presentation with evidence-based arguments
3. Evaluate a peer team's design against gate criteria
4. Provide constructive technical feedback using a structured rubric
5. Identify lessons learned and articulate next steps for Phase B

1. Design Documentation Review (20 min)

Teaching Notes

Before presenting, each team must verify that their design documentation is complete and internally consistent.

Documentation Checklist

Document	Source in SpaceCDF	Status Check
Mission Requirements Document (MRD)	Exports -> ECSS Documents	All requirements baselined?
Technical Specification (TS)	Exports -> ECSS Documents	All budgets captured?
Concept of Operations (ConOps)	Exports -> ECSS Documents	All phases defined?
Verification Plan (VP)	V&V Matrix tab	All requirements have methods?
Bill of Materials (BOM)	Exports -> Design Data	All equipment listed with TRL?
Risk Register	Worksheet 4.3	All risks scored and mitigated?
Interface Matrix (N^2)	Worksheet 4.3	Interfaces identified? Conflicts resolved?
Cost Estimate	Worksheet 4.4	WBS complete? P80 computed?
Operations Concept	ConOps Editor	Modes, FDIR, procedures defined?
Sustainability Assessment	Sustainability Card	Debris compliance score?

Internal Consistency Checks

Before presenting, verify:

Check	How to Verify	Common Error
Mass budget closes	Dashboard mass KPI: margin > 0%	Equipment added without updating budget
Power budget closes	Dashboard power KPI: SA > loads in all modes	Eclipse mode power not checked
Link budget closes	Link Budget tool: margin >= 3 dB	Rain attenuation not included
Data budget closes	Data Budget: daily downlink >= daily generation	Assumed too many passes per day
Cost is within ceiling	Cost tab: total <= allocated budget	Forgot launch cost or operations
Requirements traceable	Every requirement maps to an objective	Orphan requirements (no parent objective)
V&V complete	Every requirement has a method assigned	"TBD" entries remaining
No unresolved conflicts	Dashboard: conflict count = 0	Interface mismatches not addressed

SpaceCDF Document Generation

Navigate to Exports tab and generate all documents:

1. ECSS Documents: MRD, TS, VP, ConOps, SEMP, IRD
2. Design Data: BOM, Parametric Model Data
3. Regulatory Filings: ITU API, RSSSA, Export Assessment, COPUOS, EOL Report
4. Presentation: Summary slide data (auto-populated)

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2. Presentation Structure (15 min)

Teaching Notes

Each team presents a 12-minute design review followed by 5 minutes of Q&A from the peer review board.

[Source: ECSS-M-ST-10C Rev.1 section 6; NPR 7123.1D Appendix G (Review Process)]

Presentation Outline (12 minutes)

Section	Duratio	Content	Evidence
1. Mission Need	2 min	Problem statement; stakeholders; objectives with MoPs	Mission Need step
2. Why Space?	2 min	Trade study results; why space beats alternatives	Mission Trade results
3. System Design	3 min	Architecture; orbit; key parameters; budget status	Dashboard, budgets
4. Equipment & Verification	2 min	Key equipment; BOM summary; V&V approach; test plan	BOM, V&V Matrix
5. Risk & Cost	2 min	Top 5 risks; cost estimate (P50/P80); schedule	Risk register, cost tab
6. Operations & Lessons	1 min	ConOps summary; FDIR; what would you change	ConOps, simulation debrief

Presentation Best Practices

Slide design:

- One key message per slide
- Data, not prose (budgets, trade tables, not paragraphs)
- Include backup slides with detailed analyses for Q&A

Delivery:

- Lead with conclusions: "Our mass budget closes with 22% margin"
- Show evidence for every claim: "Link budget analysis shows 4.2 dB margin at worst case"
- Acknowledge risks honestly: "Antenna deployment is our highest risk at 12 (L3 x C4)"
- For questions you cannot answer: "Good question -- we will take that as an action item"

Common pitfalls to avoid:

- Reading slides verbatim
- Hiding known problems (review boards always find them)
- Presenting analysis without stating assumptions
- Spending too long on the mission description and rushing through technical results

---

3. Peer Review Process (10 min)

Teaching Notes

Each team acts as a review board for another team. This mirrors the real design review process where an independent board evaluates the project.

Review Board Roles

Role	Responsibility	Key Questions to Ask
Chair	Manages review; ensures all criteria covered	"Let's systematically check each gate criterion"
Systems reviewer	Evaluates overall architecture and budgets	"Do all budgets close? What is the minimum margin?"
Technical reviewer	Evaluates subsystem design decisions	"Why did you choose this component over alternatives?"
Risk reviewer	Evaluates risk management and V&V	"What is your highest risk? Is the mitigation adequate?"
Operations reviewer	Evaluates ConOps, ground segment, regulatory	"What happens if you lose contact for 48 hours?"

Gate Criteria Evaluation

The review board evaluates the presenting team against PDR-level gate criteria:

#	Criterion	Score (0-2)	Notes
1	Mission need clearly justified	0 = missing, 1 = partial, 2 = strong	Is the problem real? Is space the right answer?
2	Requirements complete and traceable	0/1/2	All requirements linked to objectives? Verifiable?
3	All budgets close with adequate margin	0/1/2	Mass, power, link, data, pointing -- all positive?
4	Equipment selected with justified trades	0/1/2	BOM complete? TRL adequate? Trade studies documented?
5	Interfaces defined and conflicts resolved	0/1/2	N^2 matrix? No unresolved conflicts?
6	V&V approach defined for all requirements	0/1/2	Methods assigned? Test plan outlined?
7	Risks identified with mitigations	0/1/2	Risk register with scores? Top risks mitigated?
8	Cost estimate with WBS and P80	0/1/2	Both parametric and bottom-up? Learning curve applied?
9	ConOps and ground segment designed	0/1/2	Modes defined? FDIR architecture? Ground stations?
10	Sustainability compliance	0/1/2	Debris compliance? 25-year rule? Passivation plan?

Review Board Questions Bank

Standard questions for the review board to ask:

- "Why is space the right answer? What alternatives were considered?"
- "What is your mass margin? What drives it? What if the payload is 20% heavier than estimated?"
- "Walk me through your link budget at worst case. What is the minimum margin?"
- "What are your top 3 risks? What is the residual risk after mitigation?"
- "Show me your data budget. Does it close with a single ground station?"
- "What happens during a safe mode entry in eclipse? Does the thermal budget survive?"
- "Which component has the lowest TRL? What is your qualification plan?"
- "Have you started spectrum licensing? What is on the critical path?"
- "What is your single-point failure list? Which ones have you accepted?"
- "If you had 6 more months and 20% more budget, what would you change?"

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4. Team Presentations (45 min)

Instructions

Presentation Schedule: 12 minutes presentation + 5 minutes Q&A per team.

For 4 teams: allocate 17 min x 4 = 68 min. Adjust timing based on actual number of teams. If fewer than 4 teams, allow 15 min presentation + 8 min Q&A.

For the presenting team:

1. Present the 6-section outline above
2. Use SpaceCDF live for demonstrations where helpful (show Dashboard, budgets)
3. Respond to board questions with evidence

For the review board:

1. Complete the gate criteria evaluation form on Worksheet 5.4
2. Ask at least 3 substantive questions
3. Provide a GO / NO GO / GO WITH ACTIONS recommendation
4. Document action items with responsible person and deadline

After each presentation:

- Chair announces the board's recommendation
- Action items are recorded
- Brief applause and transition to next team

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5. Course Wrap-Up & Lessons Learned (30 min)

Teaching Notes

Individual Reflection (5 min)

Each participant completes the self-assessment rubric on Worksheet 5.4, reflecting on their learning across all three weeks.

Team Lessons Learned (10 min)

Each team discusses and reports:

1. Most challenging design decision: What trade-off was hardest to resolve? Why?
2. Biggest surprise: What aspect of mission design was most different from expectations?
3. Tool feedback: What SpaceCDF feature was most valuable? What was missing?
4. Process insight: How did concurrent design change your approach to engineering?
5. If starting over: What would you do differently in Week 1?

Key Takeaways from the Course (10 min)

Week	Theme	Core Lesson
Week 1	Requirements & Architecture	Start with the problem, not the solution. Requirements define WHAT, not HOW.
Week 2	Subsystem Design & Budgets	Engineering budgets are the language of systems engineering. Everything trades against everything.
Week 3	Integration, Verification & Operations	Building the right thing (V&V) is as important as designing it. Operations reveal what analysis cannot.

The Systems Engineering V-Model -- Completed

```
 You started here:                          You finished here:

   Mission Need                               Mission Validated
      |                                              ^
      v                                              |
   Requirements                              Requirements Verified
      |                                              ^
      v                                              |
   System Design                             System Integrated (sim)
      |                                              ^
      v                                              |
   Subsystem Design                          Subsystem V&V Planned
      |                                              ^
      v                                              |
   Equipment Selected  ------>  Equipment Verified (BOM + compliance)

   Week 1-2 (left side)         Week 3 (right side)
```

You have traversed the complete V-model: from mission need to verified, validated design.

Next Steps for Participants

- SpaceCDF access continues after the course
- Apply the methodology to your own missions or projects
- Use the tool for CDF-style studies with your team
- Refer to the Facilitator's Book for session plans and appendices
- Share feedback to improve the course for future cohorts

Course Evaluation

Distribute the course evaluation form. Key questions:

- Which session was most valuable? Least valuable?
- What content should be added? Removed?
- How effective was the simulation day?
- Would you recommend this course to colleagues?
- What is one thing you will do differently in your engineering practice as a result?

Cited References

#	Reference	URL
1	ECSS-M-ST-10C Rev.1 (Project Planning and Implementation)	https://ecss.nl/standard/ecss-m-st-10c-rev-1-project-planning-and-implementation/
2	NPR 7123.1D Appendix G (Review Process)	https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&c=7123&s=1D
3	NASA SEH Rev 2, section 3.7 (Technical Reviews)	https://www.nasa.gov/reference/systems-engineering-handbook/
4	ECSS-E-ST-10C (System Engineering)	https://ecss.nl/standard/ecss-e-st-10c-rev-1-system-engineering-general-requirements/

Session Summary

Topic	Key Takeaway
Documentation	Complete package: MRD, TS, ConOps, VP, BOM, Risk Register, Cost, Regulatory
Consistency	All budgets must close; no unresolved conflicts; requirements fully traceable
Presentation	6 sections: need, justification, design, V&V, risk+cost, operations+lessons
Peer review	Independent board evaluates against gate criteria; GO/NO GO/GO with actions
Best practices	Lead with conclusions, show evidence, acknowledge risks, answer directly
V-model	Course traverses full V: needs -> requirements -> design -> V&V -> validation
Lessons learned	Document what worked, what failed, and what to change -- this is the real output

Appendix: Position-Specific Deep Dives

This appendix provides detailed guidance for each CDF engineering position, including responsibilities, key parameters owned, decision authority, common pitfalls, and references.

Appendix A: Systems Engineer

Responsibility: Overall system architecture, budget management, cross-domain conflict resolution.

Key Parameters Owned: Mass margin, power margin, system-level cost, composite TRL, health score.

Key Activities:

- Maintain all engineering budgets (mass, power, cost, ?V, data, pointing)
- Apply ECSS margin policy by project phase (44% Phase A -> 13% Phase C/D)

- Resolve cross-domain conflicts (convene affected positions, facilitate trade)
- Prepare gate review evidence packages
- Own the requirements baseline and traceability matrix
- Conduct design reviews and track action items

Decision Authority: System-level trade-offs, margin allocation, requirement waivers.

Common Pitfalls:

- Allowing individual subsystems to consume their margin allocation (no system reserve)
- Not challenging HOW requirements that should be WHAT
- Accepting conflicts without resolution timelines
- Not tracking parametric estimate vs selected equipment delta

References: NASA SEH §2, §6; ECSS-E-ST-10C Rev.1 §5; SMAD4 Ch.3

Appendix B: Mission Analyst

Responsibility: Orbit design, coverage analysis, ground station access, constellation architecture.

Key Parameters Owned: Altitude, inclination, orbit type, eclipse fraction, revisit time, contact time, LTAN.

Key Activities:

- Run orbit trade studies (altitude/inclination/lifetime/coverage/cost)
- Compute coverage and revisit for target latitude bands
- Analyse ground station access (elevation angle, contact duration, pass scheduling)
- Assess debris compliance (lifetime vs 25yr/5yr rules)
- For constellations: Walker delta design, phasing, coverage optimisation

Key Formulae:

- Period: T = 2pi?(a^3/mu)
- Eclipse fraction: f = (1/pi)arccos(?(1-(R_E/a)^2))
- SSO inclination: cos(i) = f(a, J2)
- Hohmann ?V: from vis-viva equation
- Coverage radius: from ground station elevation geometry

References: SMAD4 Ch.5-7; Vallado §2-9; ECSS-U-AS-10C Rev.2

Appendix C: Payload Lead

Responsibility: Instrument performance, data generation, payload accommodation.

Key Parameters Owned: GSD (or equivalent MoP), data rate, pointing requirement, payload mass, power, duty cycle, FOV.

Key Activities:

- Size the payload from performance requirements (GSD -> aperture for optical; data rate -> antenna for comms; resolution -> antenna area for SAR)
- Define payload data volume and duty cycle
- Specify thermal environment needs (detector cooling, operational temperature)
- Coordinate with AOCS for pointing and stability requirements
- Coordinate with comms for downlink capacity matching data generation

Key Formulae:

- Optical: GSD = max(1.22lambdah/D, ph/f)
- Payload mass: M ~ 20D^1.5 + 2 (heritage CER)
- Data rate: R = N_pixels x N_bands x bit_depth x line_rate
- SAR: antenna_length >= 2 x resolution

References: SMAD4 Ch.9; ECSS-E-ST-10-06C; specific instrument handbooks

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Appendix D: Power Engineer

Responsibility: Solar array, battery, EPS architecture, power distribution, duty cycling.

Key Parameters Owned: SA area, SA power (BOL/EOL), battery capacity, bus voltage, power margin per mode.

Key Activities:

- Construct power budget per operational mode (safe/idle/imaging/downlink/eclipse)
- Size solar array from peak sunlight demand + battery recharge
- Size battery from eclipse energy / maximum DoD
- Define switched power line allocation per subsystem
- Assess SA degradation over mission lifetime
- Coordinate with thermal on SA/radiator area competition

Key Formulae:

- P_SA = P_peak_sun + (P_eclipse x t_ecl)/(t_sun x eta_charge)
- P_SA_BOL = P_SA_EOL / (1 - degradation)^years
- A_SA = P_BOL / (eta_cell x S x cos(theta) x eta_pack)
- C_bat = (P_ecl x t_ecl) / (DoD x eta_discharge)

References: SMAD4 Ch.11.4; ECSS-E-ST-20C; vendor datasheets (GomSpace NanoPower)

Appendix E: AOCS Engineer

Responsibility: Attitude determination and control, pointing accuracy, momentum management.

Key Parameters Owned: Pointing accuracy, stability, slew rate, wheel momentum, actuator mass/power.

Key Activities:

- Select AOCS architecture (passive magnetic -> MTQ -> RW -> RW+ST) based on pointing requirement
- Construct pointing error budget (RSS of all sources)
- Size reaction wheels for momentum storage and torque
- Size magnetorquers for momentum dumping
- Define safe mode attitude (sun-pointing) for power survival
- Assess disturbance torques (gravity gradient, magnetic, SRP, aero drag)

Key Formulae:

- RSS pointing: theta = ?(? theta?^2)
- Gravity gradient torque: T = (3mu/2R^3) x (I_z - I_x) x sin(2theta)
- Magnetic torque: T = m x B (dipole moment x field)
- Wheel momentum: H = T x t_accumulation

References: SMAD4 Ch.11.1; ECSS-E-ST-60-10C; Wertz §11.1

Appendix F: Thermal Engineer

Responsibility: Temperature control (hot/cold case), radiators, heaters, MLI.

Key Parameters Owned: Max/min predicted temperatures, radiator area, heater power, thermal margin.

Key Activities:

- Define hot case (max solar + internal dissipation) and cold case (eclipse, min power)
- Select thermal control approach (passive coatings -> MLI -> heaters -> heat pipes)
- Size radiators to reject waste heat in hot case
- Size heaters to maintain minimum temperature in cold case (eclipse)
- Verify ECSS thermal margins (±5°C operating, ±10°C acceptance, ±15°C qualification)
- Coordinate with power on heater power budget and with structure on radiator mounting

Key Formulae:

- Stefan-Boltzmann: Q = ?sigmaAT? (radiative heat transfer)
- Equilibrium: Q_absorbed + Q_internal = Q_radiated
- Solar absorptance/emittance ratio (?/?) determines equilibrium temperature

References: SMAD4 Ch.11.5; ECSS-E-ST-31C; Gilmore, Spacecraft Thermal Control Handbook

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Appendix G: Communications Engineer

Responsibility: Link budget, transponder/antenna selection, frequency licensing, ground station network.

Key Parameters Owned: Link margin, data rate, frequency band, EIRP, antenna gain, modulation/coding.

Key Activities:

- Construct complete link budget (EIRP, FSPL, G/T, C/N0, Eb/N0, margin)
- Select frequency band based on data rate need and licensing constraints
- Select transponder and antenna (RF chain compatibility: band, impedance, polarisation)
- Coordinate with AOCS on antenna pointing accuracy
- Define ground station requirements (antenna size, G/T, location, contact schedule)
- Manage frequency licensing process (IARU/ISED/ITU as appropriate)

Key Formulae:

- FSPL = 20log10(4pid/lambda) dB
- EIRP = P_TX + G_TX - L_TX dBW
- Link margin = Eb/N0_avail - Eb/N0_req - L_impl dB

References: SMAD4 Ch.13; ECSS-E-ST-50-05C; ITU Radio Regulations; CCSDS 131.0-B

Appendix H: Propulsion Engineer

Responsibility: ?V budget, propulsion system selection, propellant management.

Key Parameters Owned: Total ?V, Isp, propellant mass, thrust level, total impulse.

Key Activities:

- Construct ?V budget (orbit insertion, maintenance, collision avoidance, deorbit)
- Select propulsion technology (cold gas, electric, chemical) based on ?V and power
- Size propellant mass using Tsiolkovsky equation
- Coordinate with structure on tank mounting and plume impingement
- Assess debris compliance (deorbit capability vs natural lifetime)

Key Formulae:

- Tsiolkovsky: ?V = Isp x g0 x ln(m_initial/m_final)
- Propellant mass: m_p = m_dry x (e^(?V/(Isp·g0)) - 1)
- Total impulse: I_total = m_p x Isp x g0

References: SMAD4 Ch.17; Sutton & Biblarz, Rocket Propulsion Elements; vendor data

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Appendix I: Structures Engineer

Responsibility: Primary structure, mechanisms, CDS compliance, launch loads, integration.

Key Parameters Owned: Structure mass, natural frequency, margin of safety, deployer compatibility.

Key Activities:

- Select CubeSat structure (form factor, material, vendor)
- Verify CDS compliance (dimensions, rails, switches, RBF, CG)
- Analyse launch loads (quasi-static, random vibration, shock)
- Compute structural margin of safety (MoS)
- Design deployment mechanisms (antenna, solar panels, payload)
- Plan integration sequence (component -> board -> stack -> structure -> deployer)

Key Formulae:

- MoS = (Allowable / (Design x FoS)) - 1
- Natural frequency: f = (1/2pi)?(k/m)

References: CDS Rev 14.1; SMAD4 Ch.11.3; ECSS-E-ST-32C Rev.1; NASA GEVS

Appendix J: Cost Engineer

Responsibility: Cost estimation, WBS, learning curves, risk-adjusted cost.

Key Parameters Owned: Total cost (MEUR), per-WBS element cost, launch cost, operations cost.

Key Activities:

- Construct WBS-level cost estimate (parametric early, bottom-up later)
- Apply CubeSat-specific COTS pricing where applicable
- Estimate learning curve effects for constellations
- Perform Monte Carlo cost risk analysis (P50/P70/P80)
- Track cost against programmatic ceiling
- Identify cost drivers and propose de-scope options

References: SMAD4 Ch.20; Aerospace Corp SSCM; NASA CEH; ECSS-M-ST-60C

Appendix K: Compliance / Regulatory Engineer

Responsibility: ECSS standard compliance, frequency licensing, export control, debris mitigation.

Key Parameters Owned: Standard applicability matrix, filing status, export classification, debris compliance score.

Key Activities:

- Determine applicable ECSS/NASA standards and prepare tailoring matrix
- Manage frequency licensing process (IARU/ISED/ITU filings)
- Assess export control classification for all components (ITAR/EAR/CGP)
- Verify debris compliance (25yr IADC, 5yr FCC, casualty risk)
- Prepare RSSSA filing if mission has remote sensing capability

- File COPUOS registration post-launch

References: ECSS-S-ST-00-02C (tailoring); CPC-2-6-02 (ISED); RSSSA; ITU RR Article 9

Appendix L: Ground Segment Engineer

Responsibility: Ground station network, MCS, data processing pipeline, ops concept.

Key Parameters Owned: Contact time per day, ground station locations, processing latency, data throughput.

Key Activities:

- Select ground station network (own station, KSAT, SatNOGS, DSN)
- Define MCS architecture (COSMOS, OpenMCT, Yamcs)
- Design data processing pipeline (L0 -> L1 -> L2 -> archive -> distribution)
- Plan operations concept (staffing, automation level, anomaly response)
- Coordinate with comms engineer on frequency bands and antenna requirements

References: SMAD4 Ch.14-15; ECSS-E-ST-70C; CCSDS standards

Appendix M: Software Engineer

Responsibility: Flight software architecture, FDIR, TC/TM definition, ground software.

Key Parameters Owned: FSW architecture, mode definitions, command dictionary, telemetry list.

Key Activities:

- Define FSW architecture (mode manager, ADCS control, TM/TC handler, FDIR)
- Implement FDIR (Fault Detection, Isolation, Recovery) rules
- Define telecommand dictionary (all commands with parameters)
- Define telemetry dictionary (all HK and science packets)
- Define autonomous operations (time-tagged commands, on-board scheduling)
- Develop ground control software and procedures

References: ECSS-E-ST-70-01C; ECSS-E-ST-70-41C (PUS); CCSDS 133.0-B (Space Packet Protocol)

Appendix N: Mission Operations Engineer

Responsibility: Operations concept, ground procedures, LEOP planning, anomaly response, scheduling.

Key Parameters Owned: Pass schedule, contact time, command load timing, housekeeping budget, anomaly recovery time.

Key Activities:

- Define operations concept (number of operators, shifts, automation level)
- Plan LEOP sequence (first contact, deployment, commissioning timeline)
- Define ground station network requirements (contact minutes per orbit, latency)
- Write nominal operations procedures (imaging, downlink, orbit maintenance)
- Define anomaly response procedures (safe mode recovery, contingencies)
- Plan scheduling and resource allocation (data volume vs contact time)

Decision Authority: Operations approach, automation vs manual, pass scheduling priority.

Common Pitfalls:

- Underestimating LEOP complexity (first hours are most critical)
- Not planning for degraded operations (single ground station, partial hardware)
- Assuming 24/7 staffing when budget supports only office hours
- Not accounting for seasonal ground station visibility

References: ECSS-E-ST-70C; ECSS-M-ST-10C Annex B (Operations); NASA SEH Appendix T (Phase E)

Appendix O: Project Manager

Responsibility: Schedule, budget, team coordination, review preparation, stakeholder management.

Key Parameters Owned: Project schedule, cost breakdown, FTE allocation, risk register, milestone status.

Key Activities:

- Maintain project schedule with milestones and dependencies
- Track cost against budget (EAC, ETC, cost-to-complete)
- Coordinate between positions (ensure interface agreements are met)
- Prepare review data packages (SRR, PDR, CDR, FRR)
- Manage risk register (convene risk review boards)
- Interface with customer/stakeholder on requirements changes

Decision Authority: Schedule priorities, resource allocation, risk acceptance (with engineering concurrence).

Common Pitfalls:

- Not protecting schedule margin (consumed by early phases)
- Accepting scope creep without schedule/cost impact assessment
- Not tracking earned value (work done vs time spent)
- Holding reviews before engineering products are mature

References: ECSS-M-ST-10C (Project Management); NPR 7120.5 (NASA PM Requirements); PMI PMBOK

Appendix P: User Representative

Responsibility: End-user needs, data product requirements, service level agreements, utilisation planning.

Key Parameters Owned: Data product specifications, latency requirements, coverage needs, user interface requirements.

Key Activities:

- Define data product requirements (format, resolution, accuracy, timeliness)
- Specify user access methods (API, web portal, direct download)
- Define service level agreements (availability, response time, data freshness)
- Validate that system design meets stakeholder expectations
- Represent user community in trade study decisions
- Plan user training and documentation

Decision Authority: Data product format, delivery mechanism, user interface design.

Common Pitfalls:

- Not distinguishing between "nice to have" and actual requirements
- Specifying implementation details instead of user needs
- Not considering different user types (research, commercial, public)
- Assuming unlimited bandwidth for data delivery

References: NASA SEH §4.1 (Stakeholder Expectations); ECSS-E-ST-10C §5.2 (Requirements); ISO 9241 (Usability)